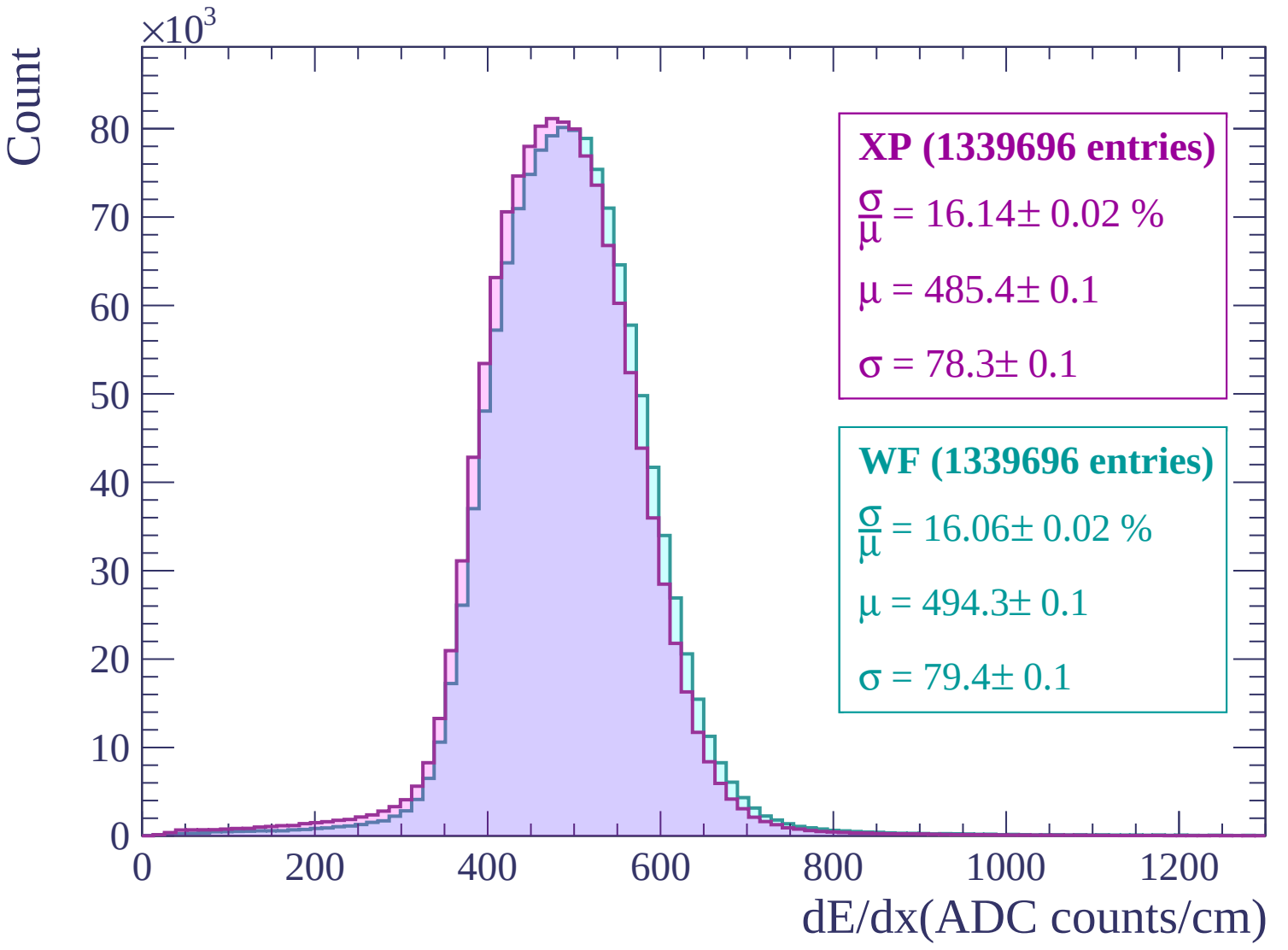
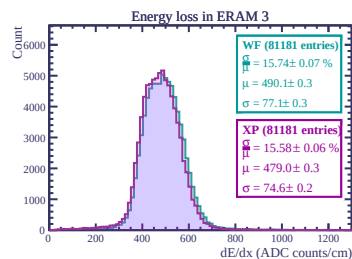
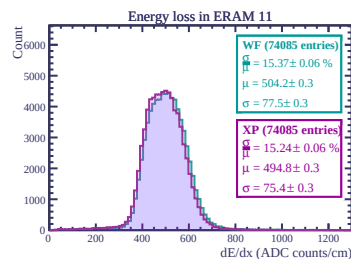
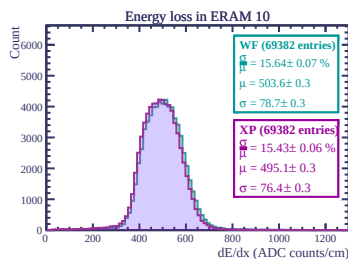
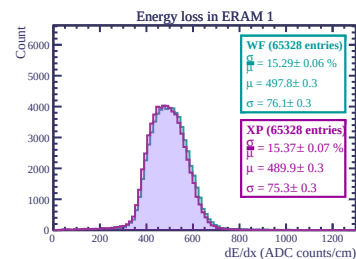
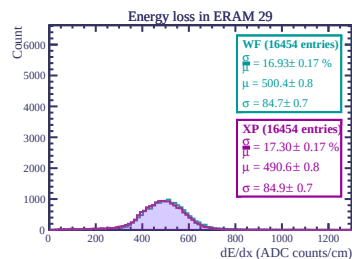
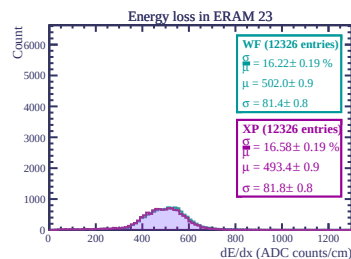
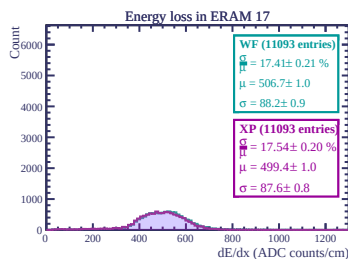
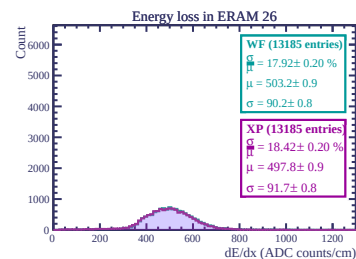
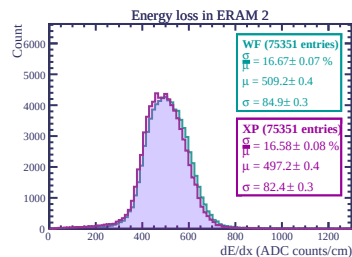
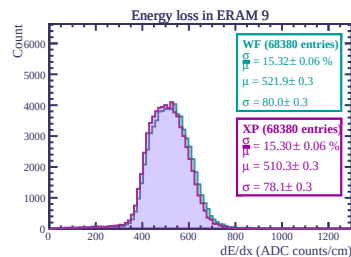
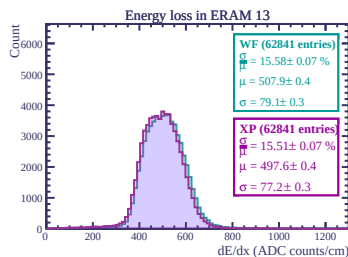
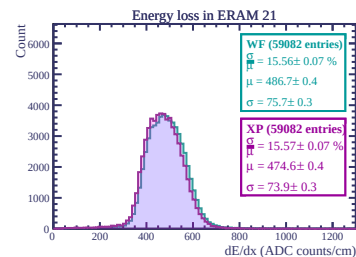
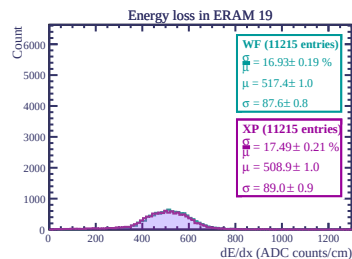
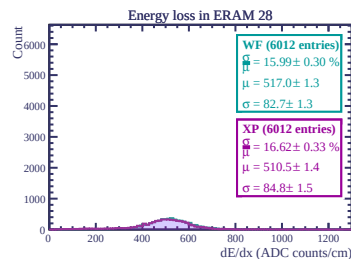
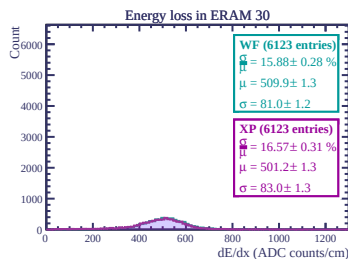
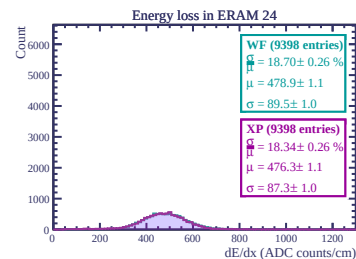
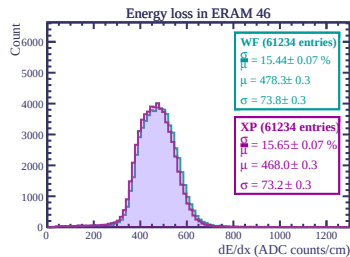
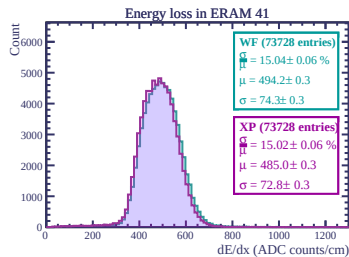
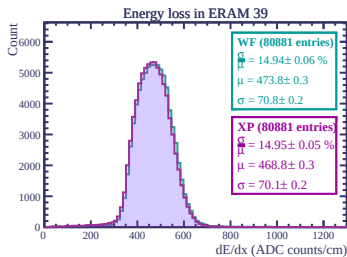
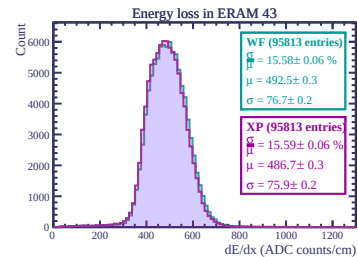
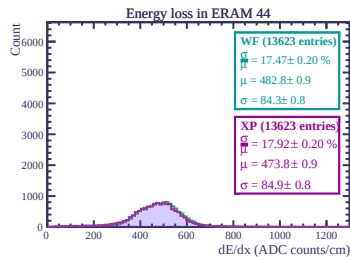
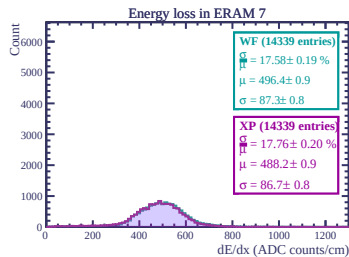
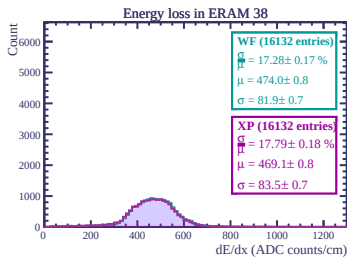
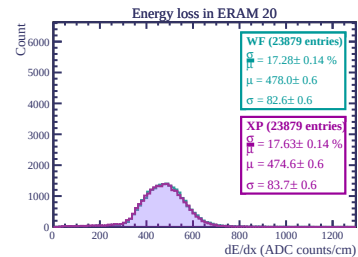
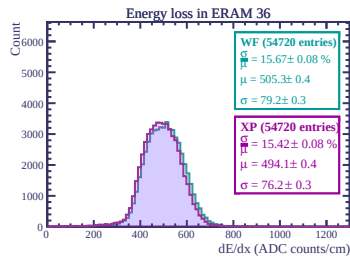
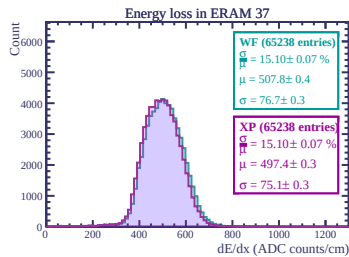
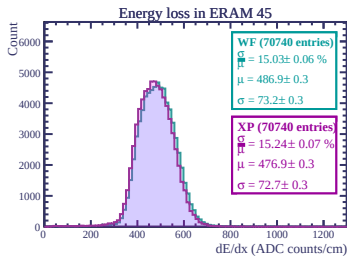
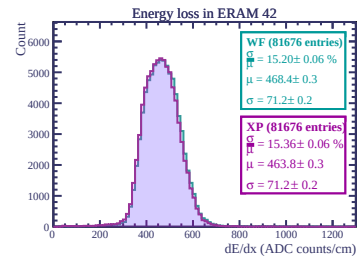
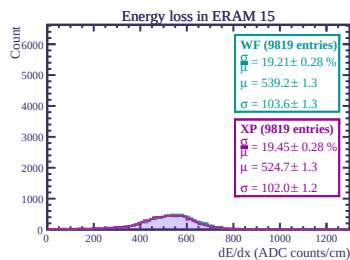
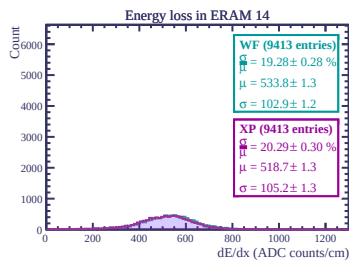
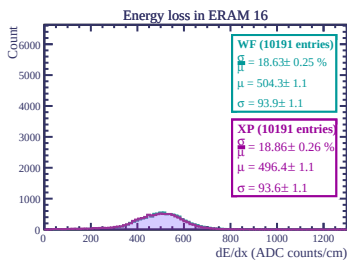
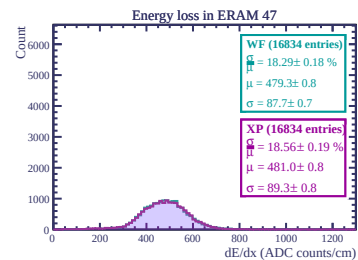
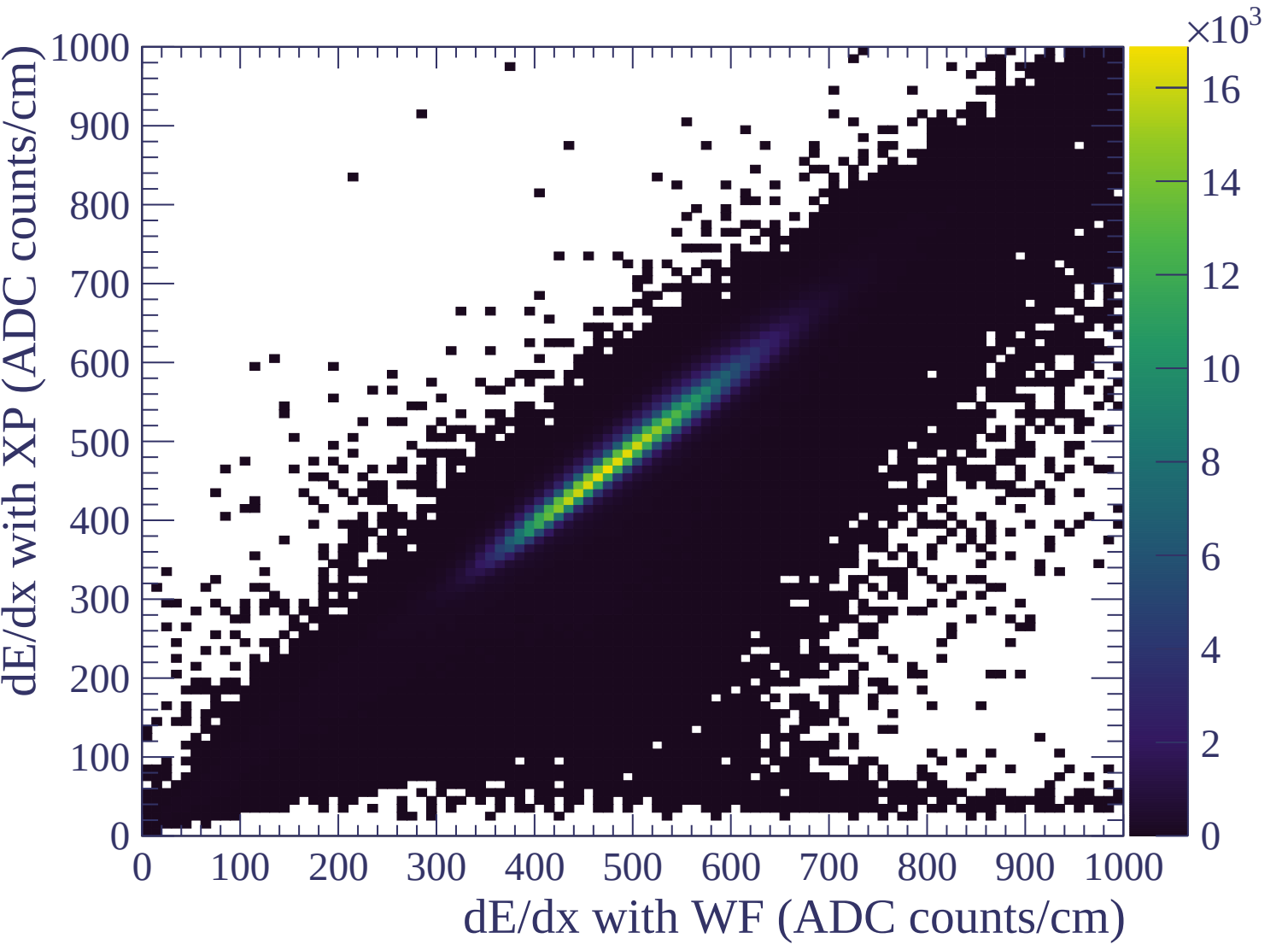


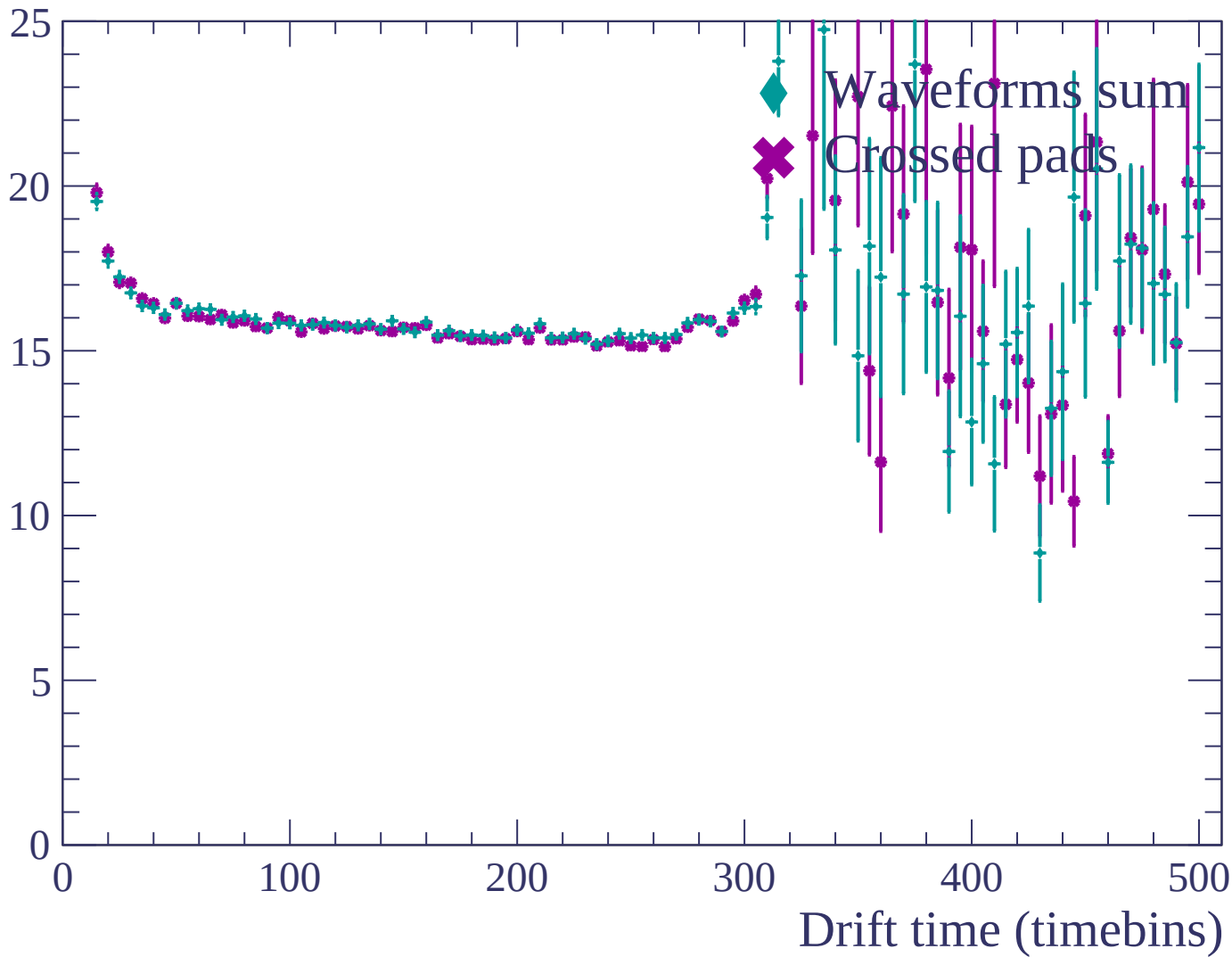


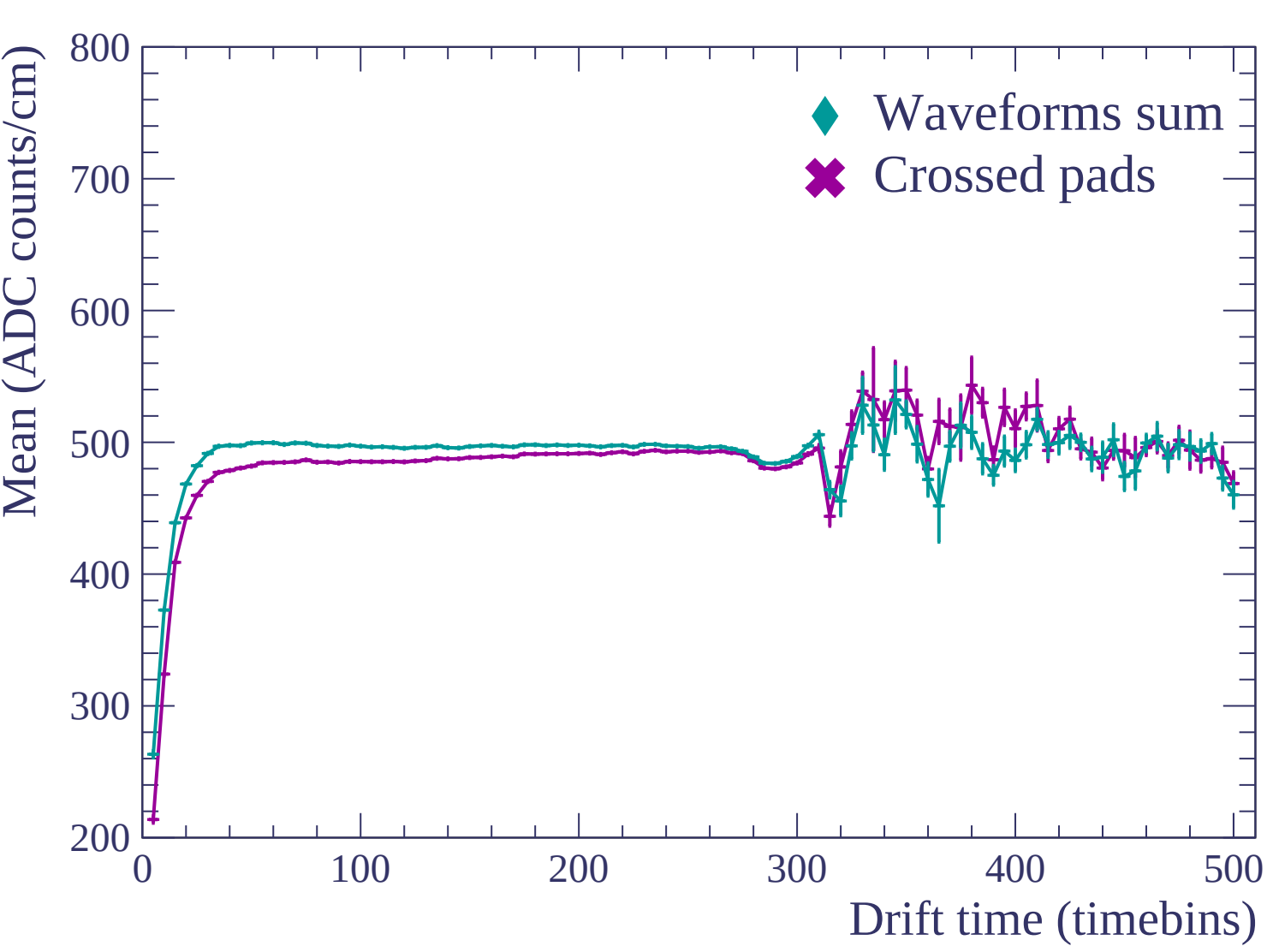


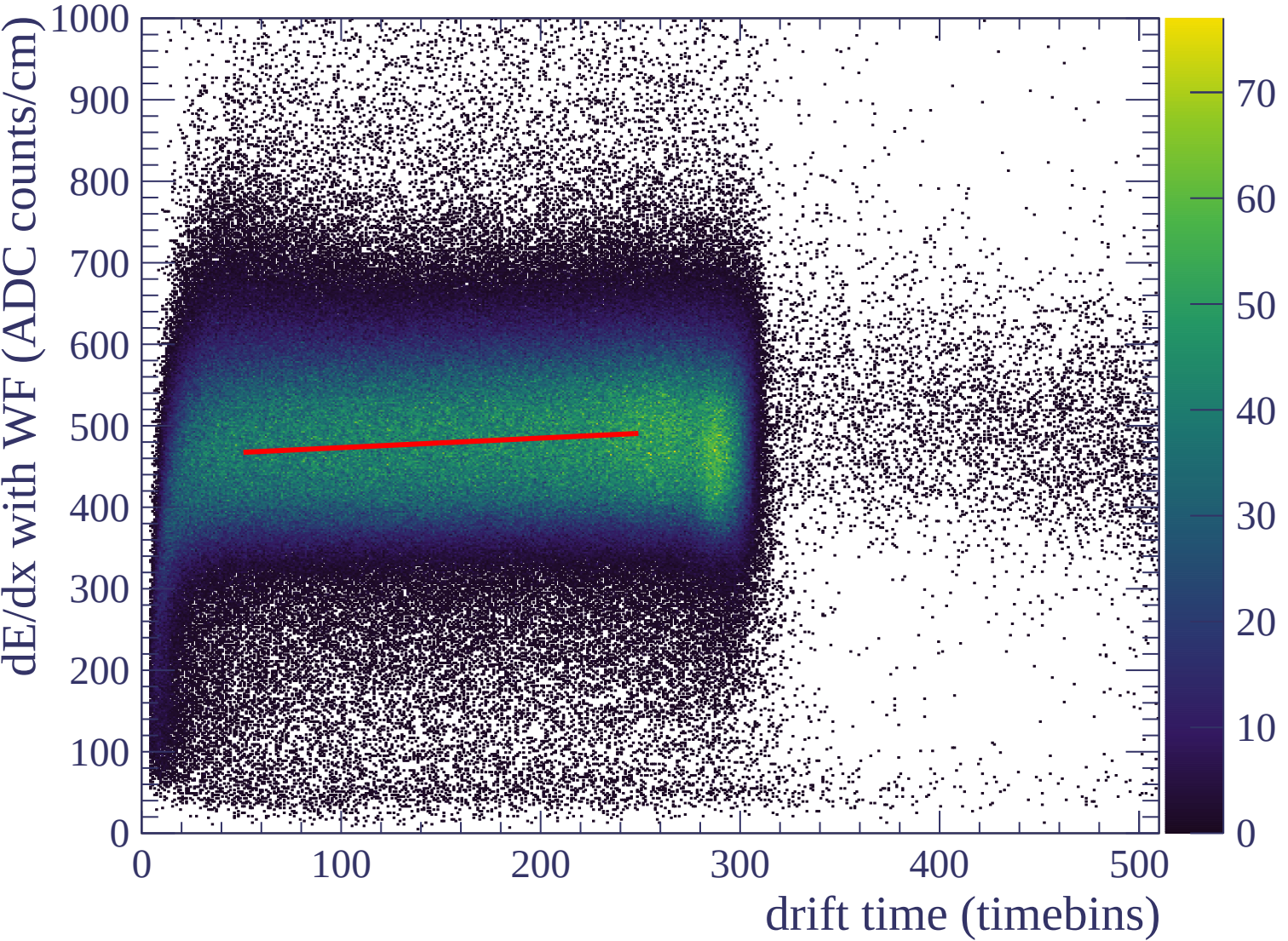


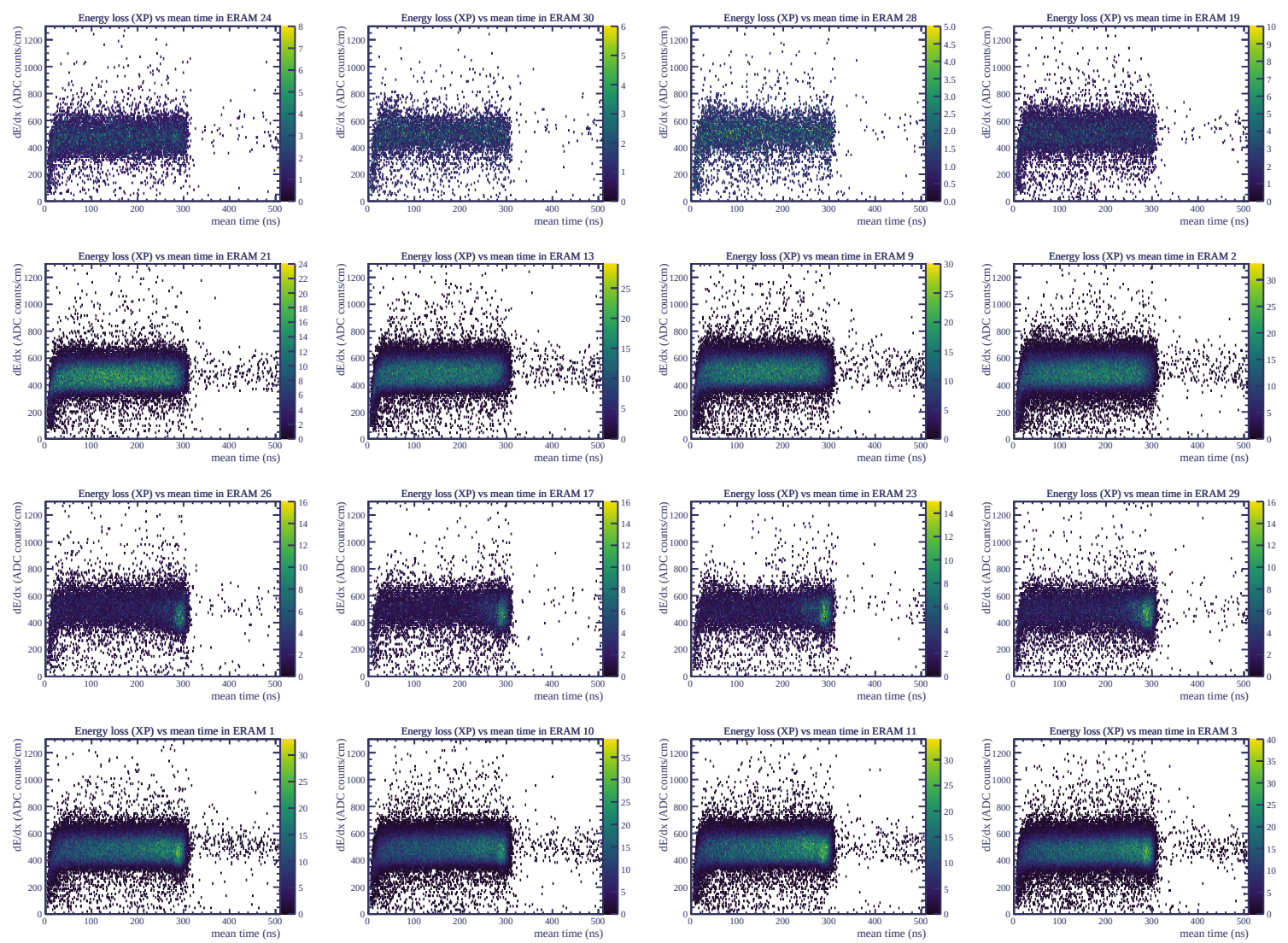


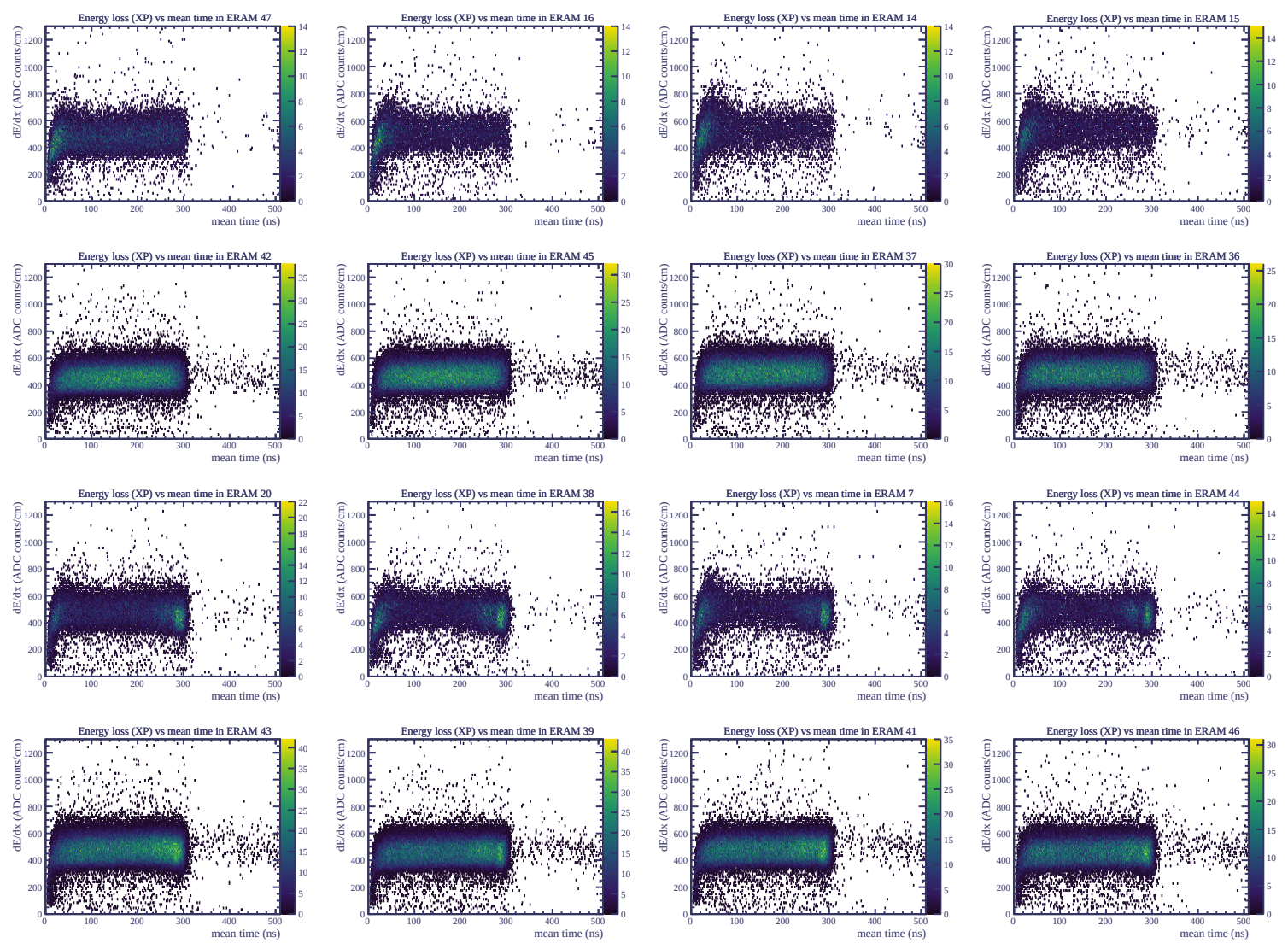
Resolution (%)

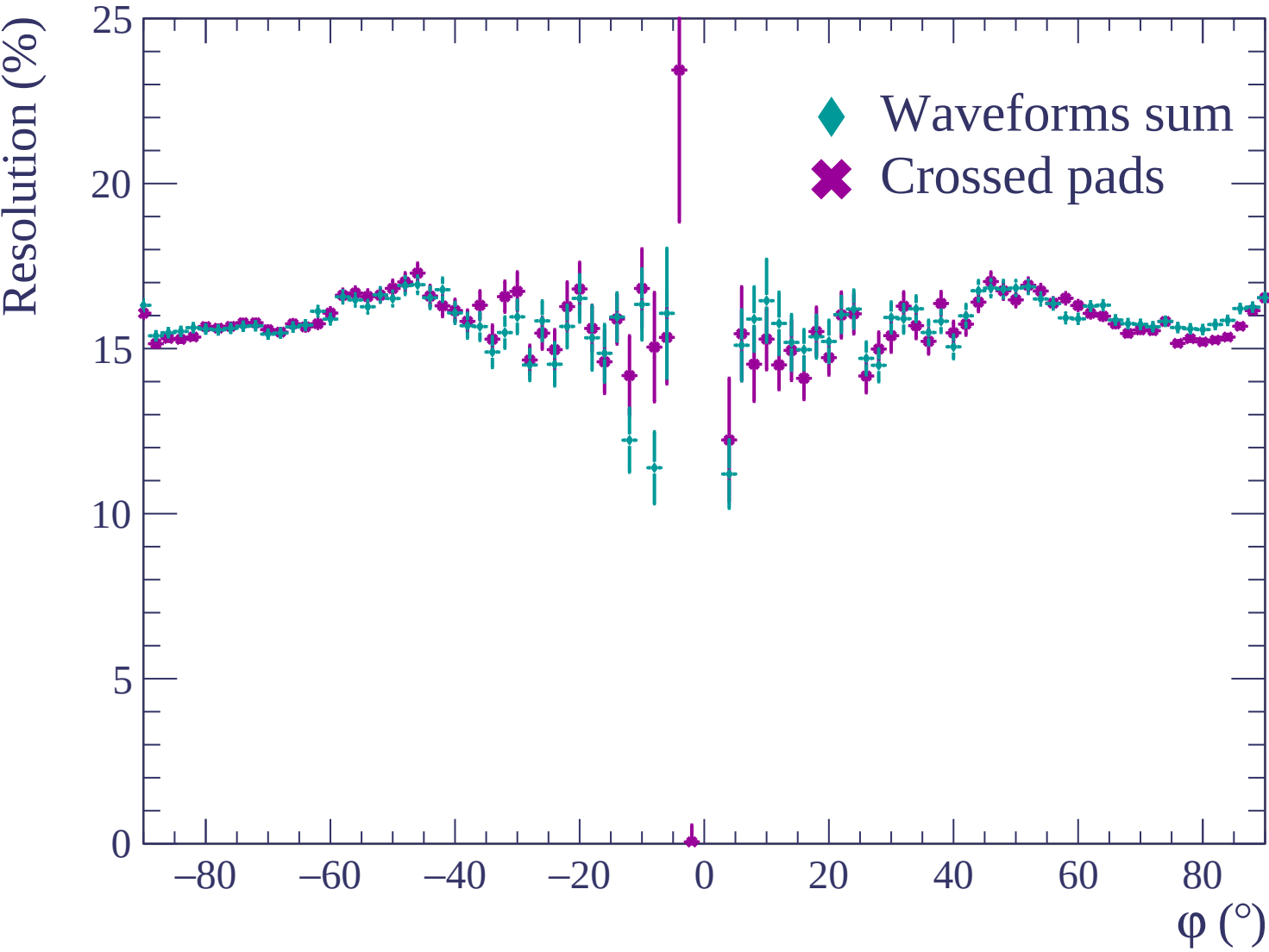


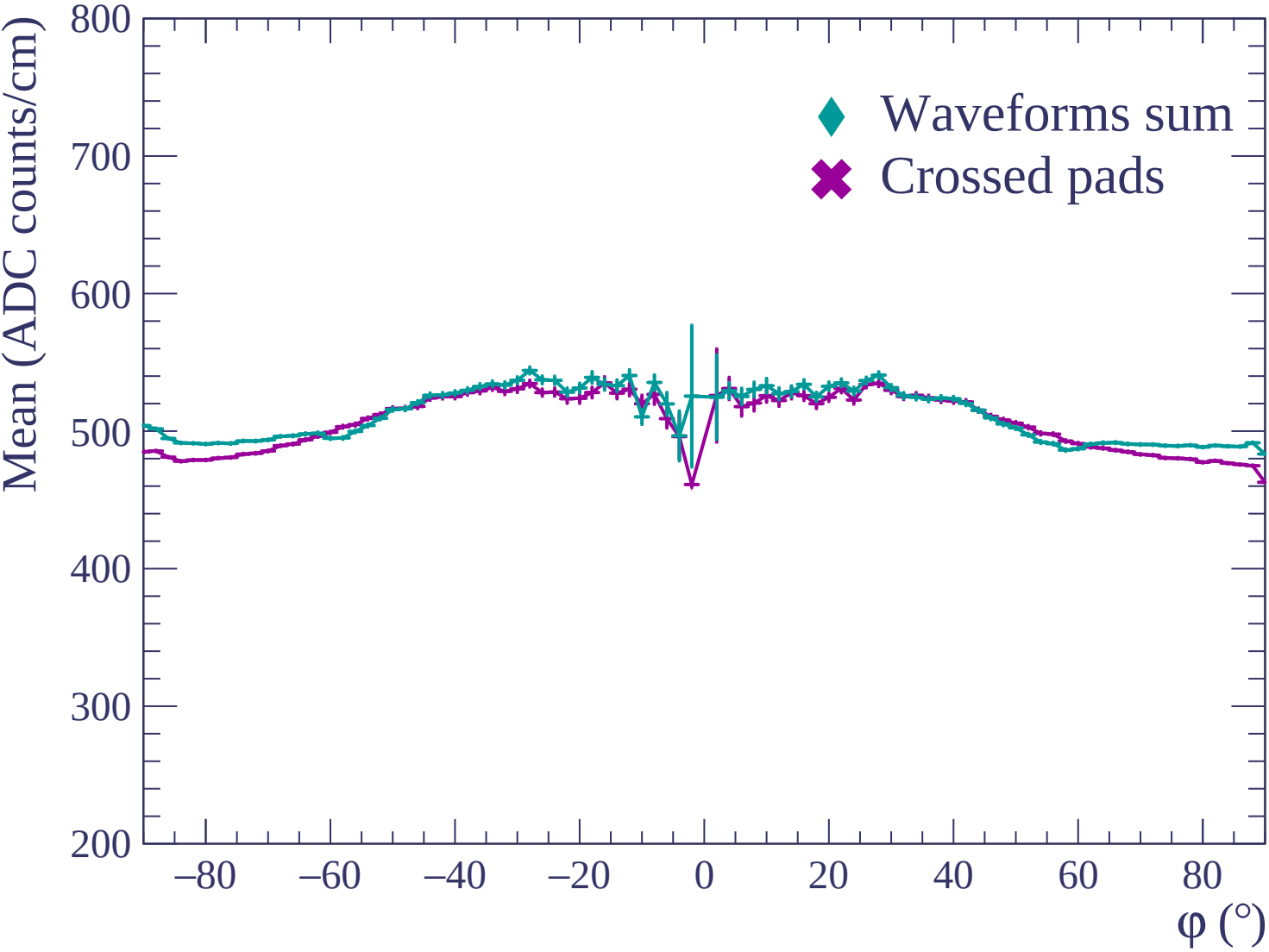


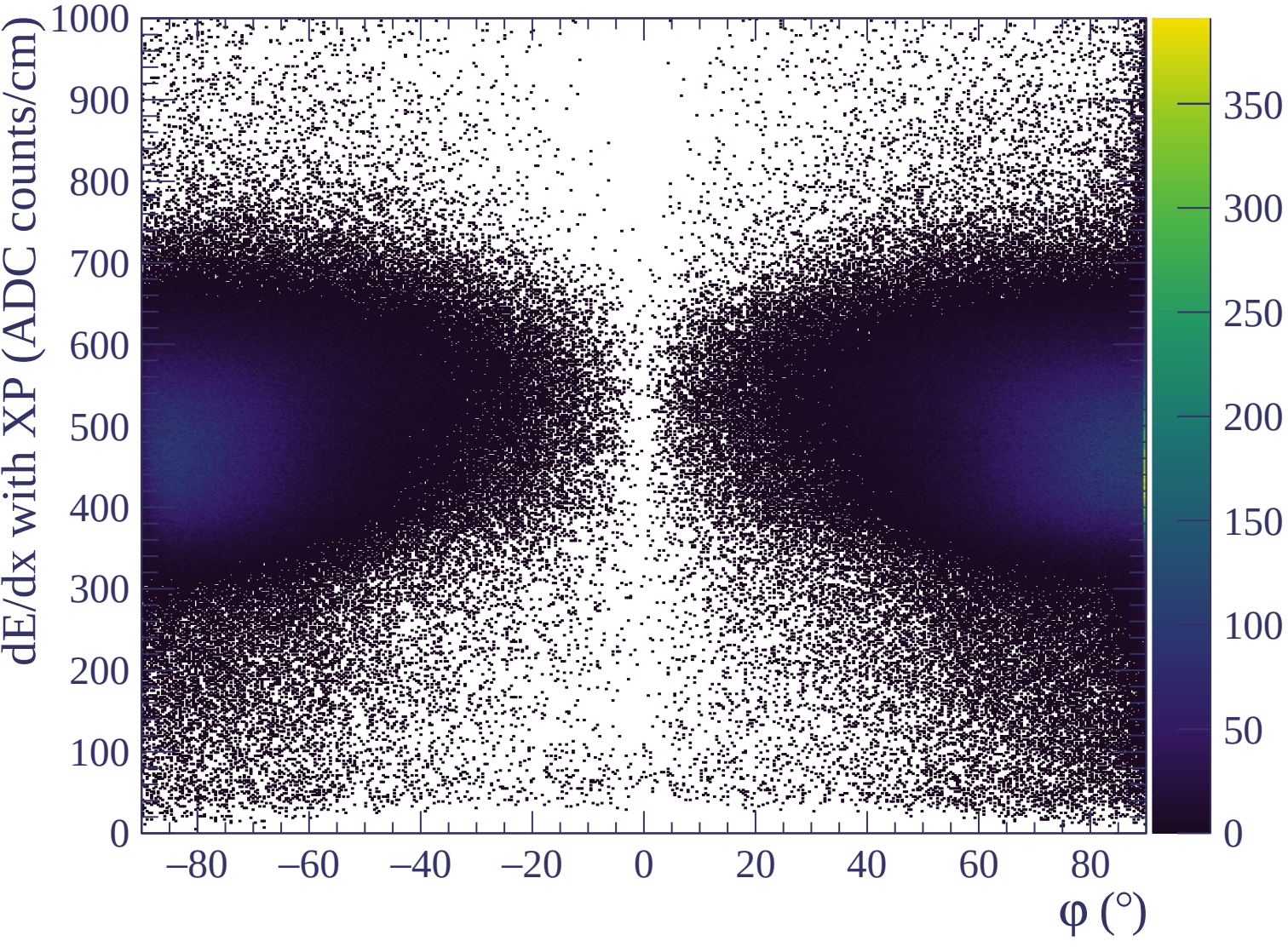


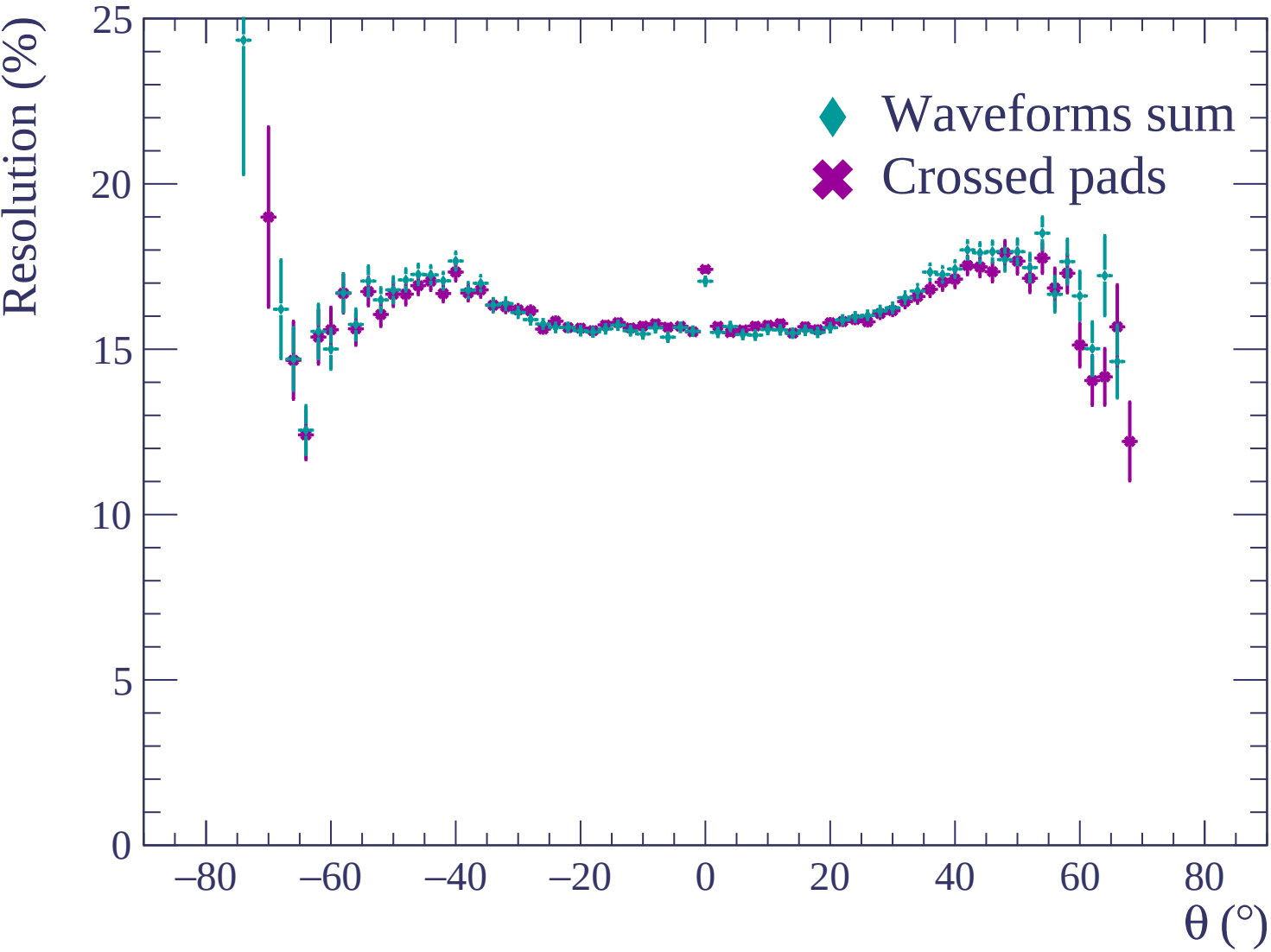


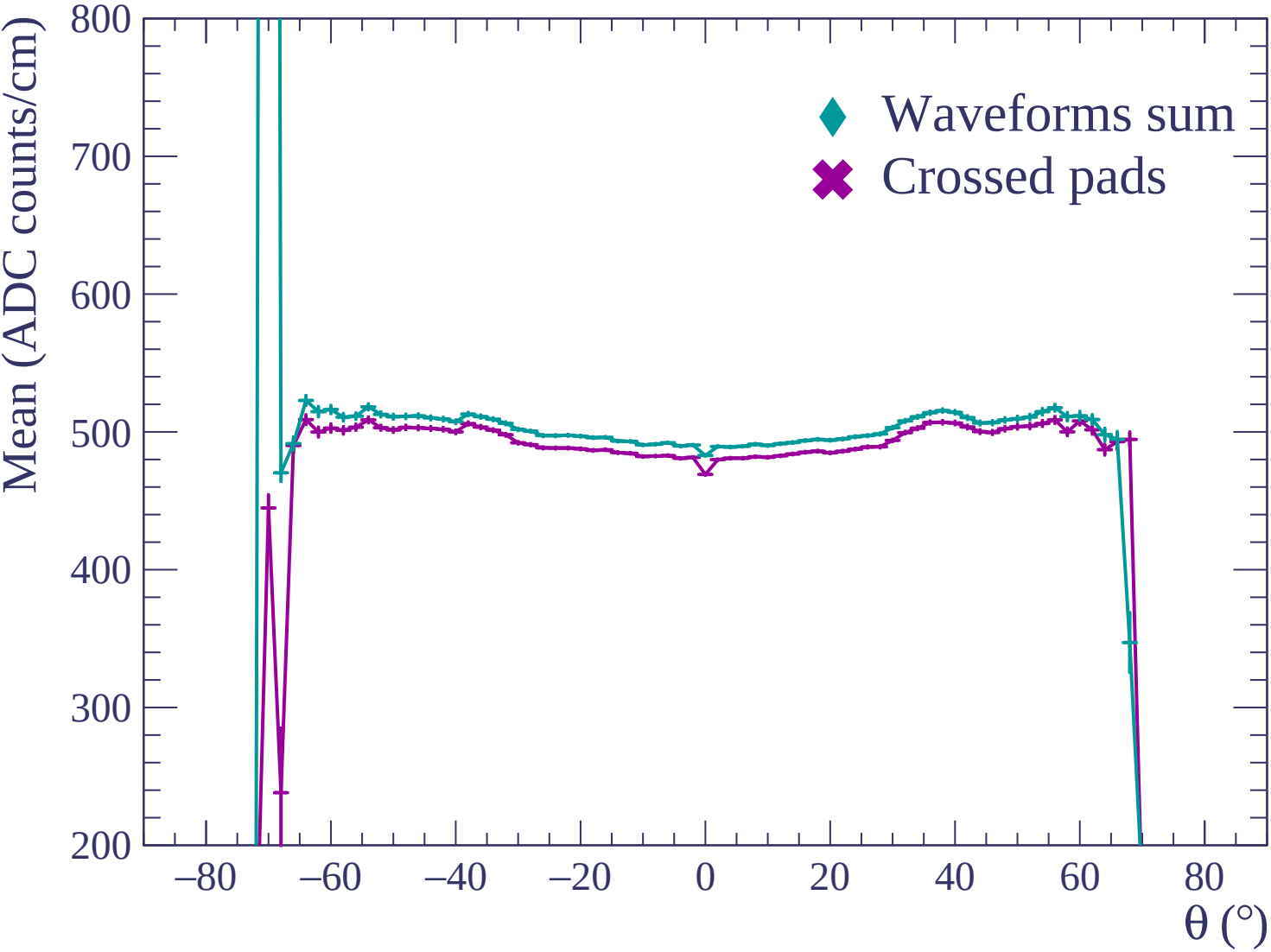


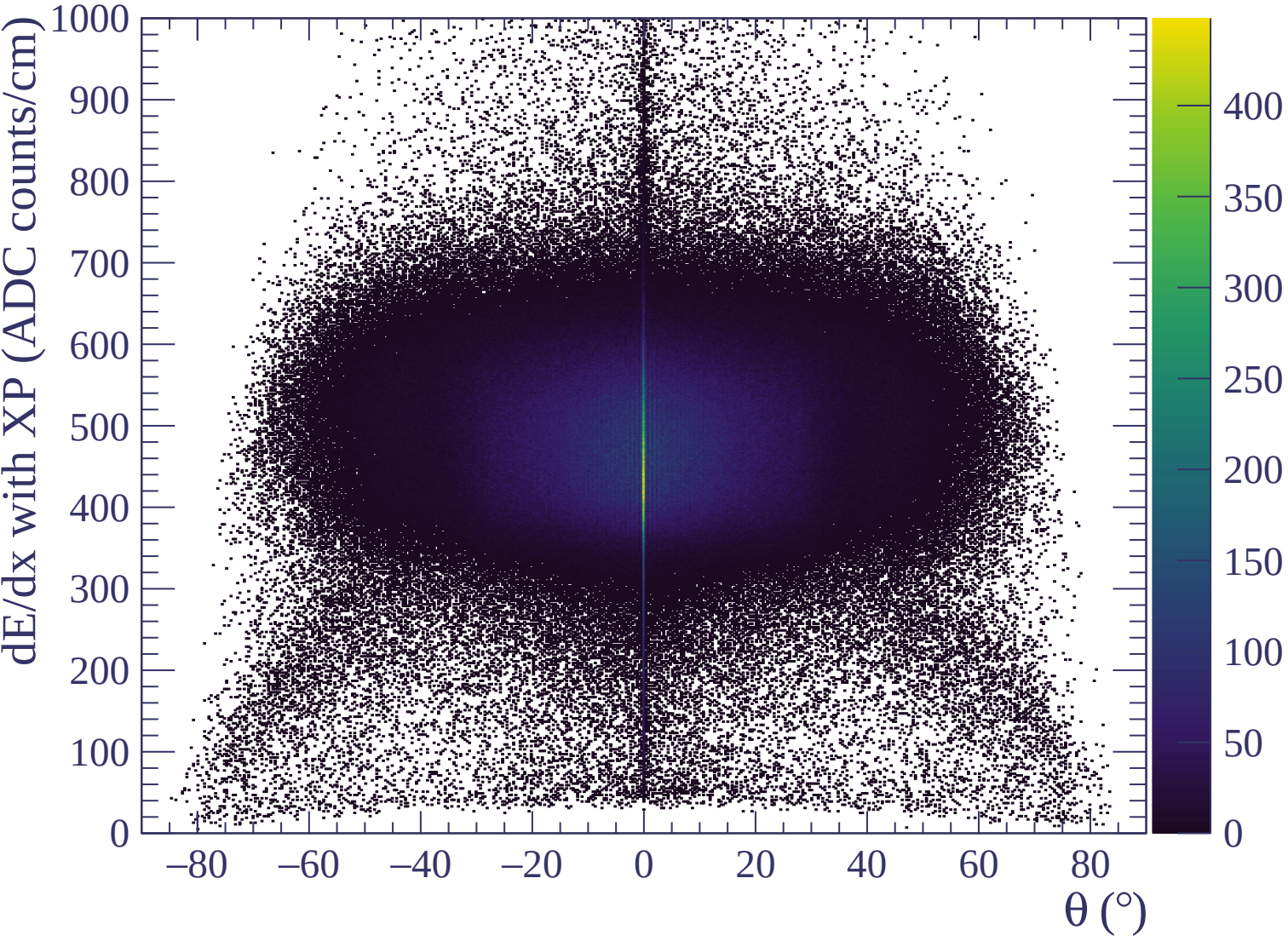


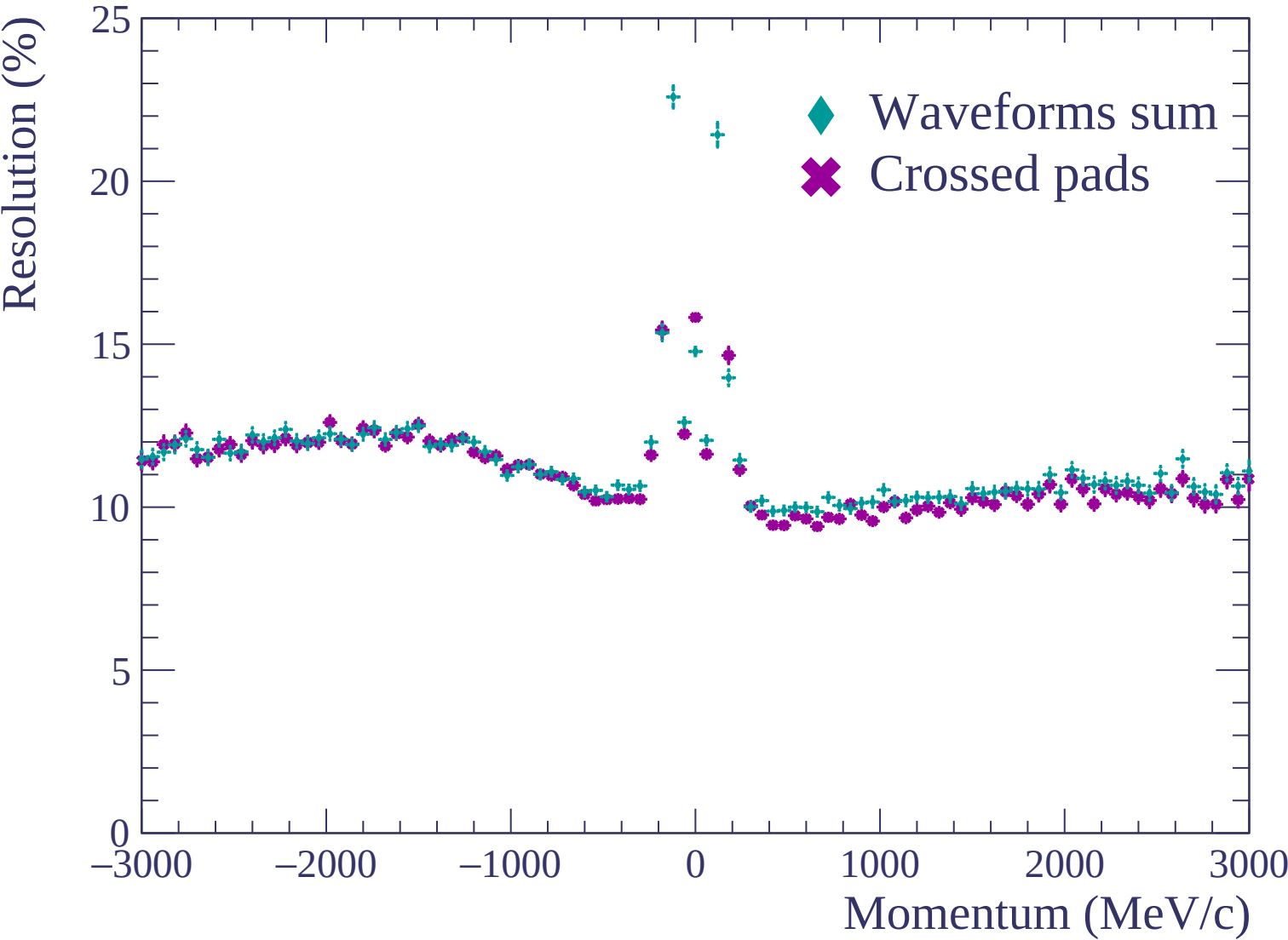


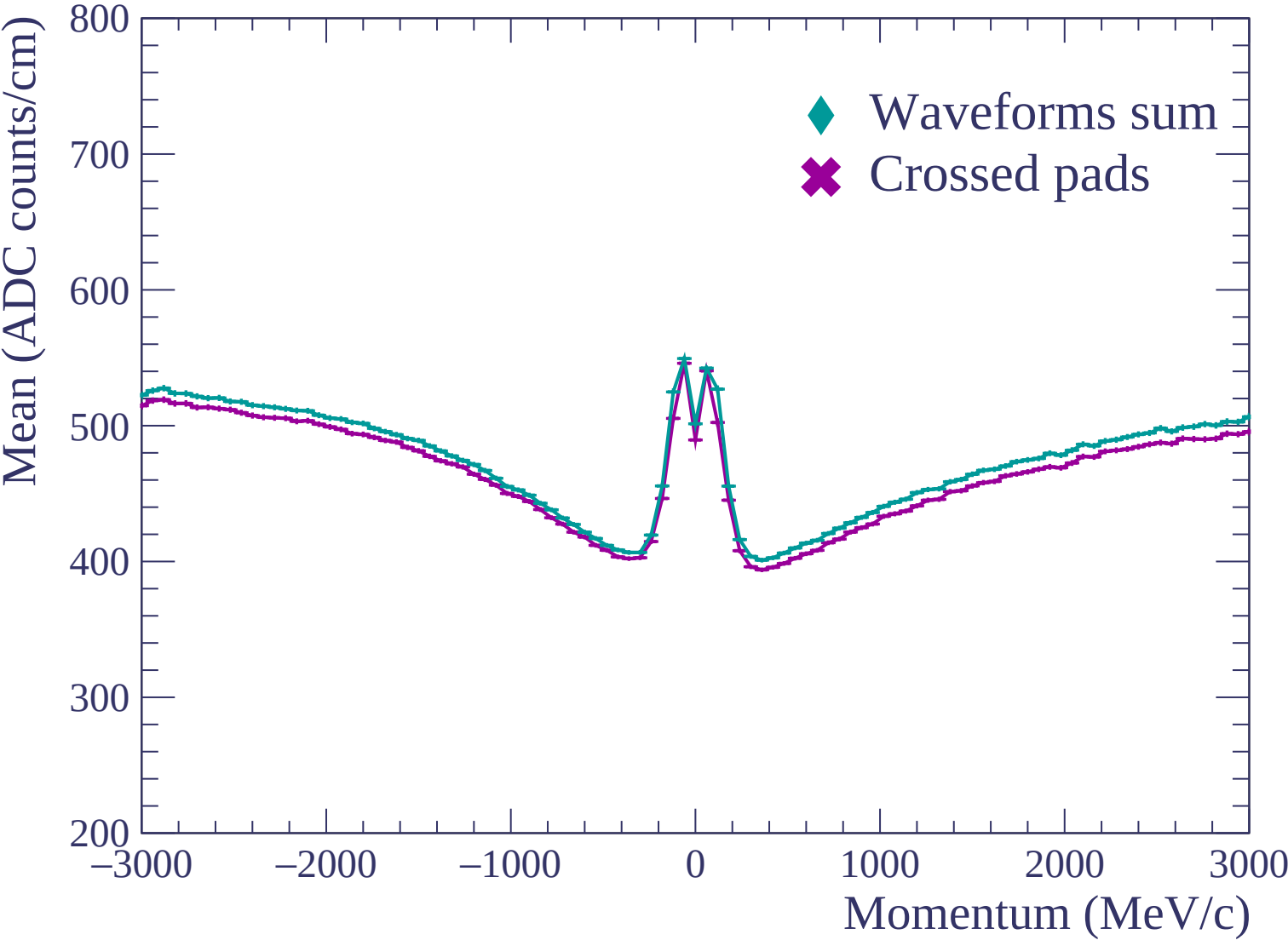


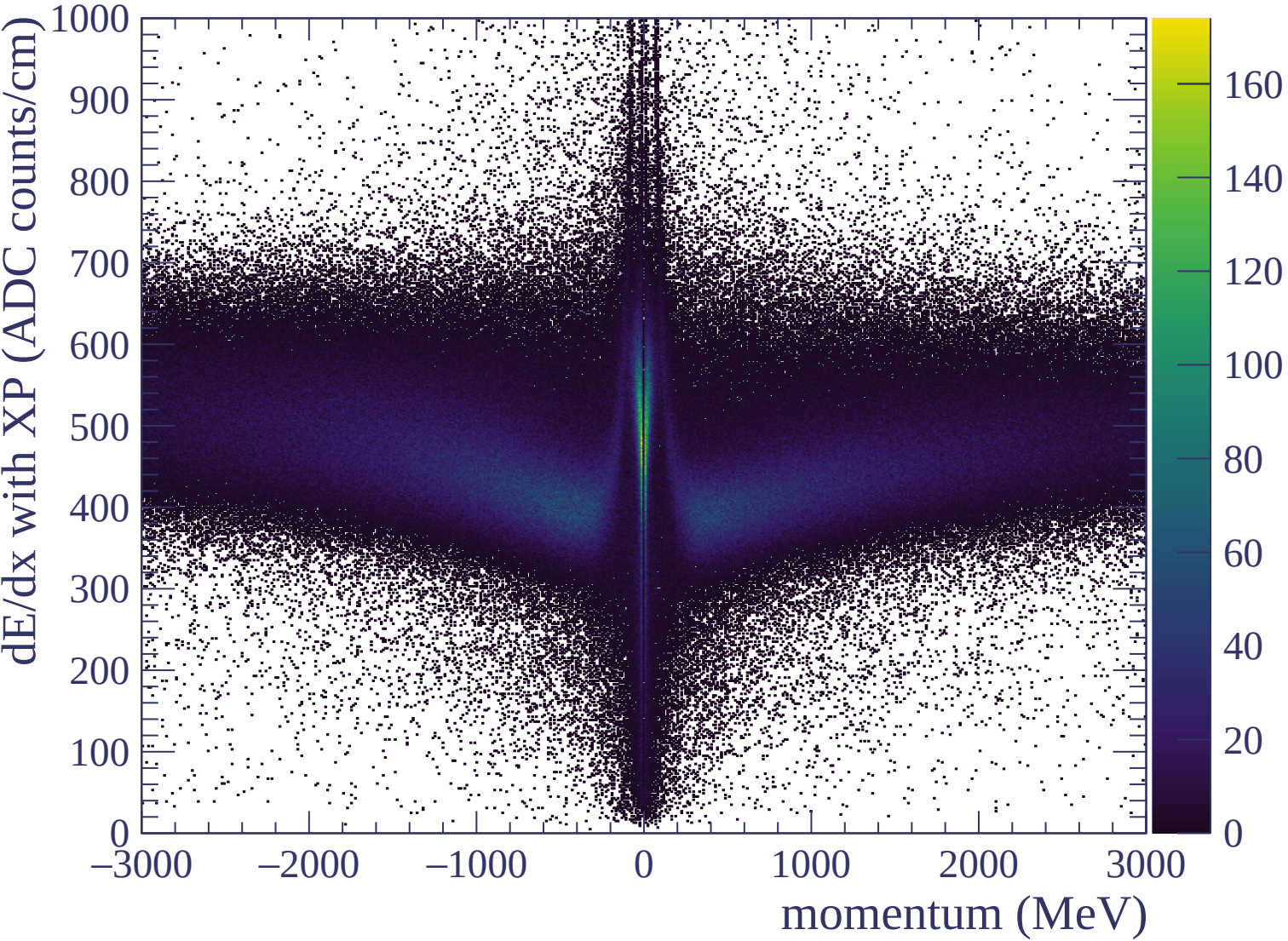


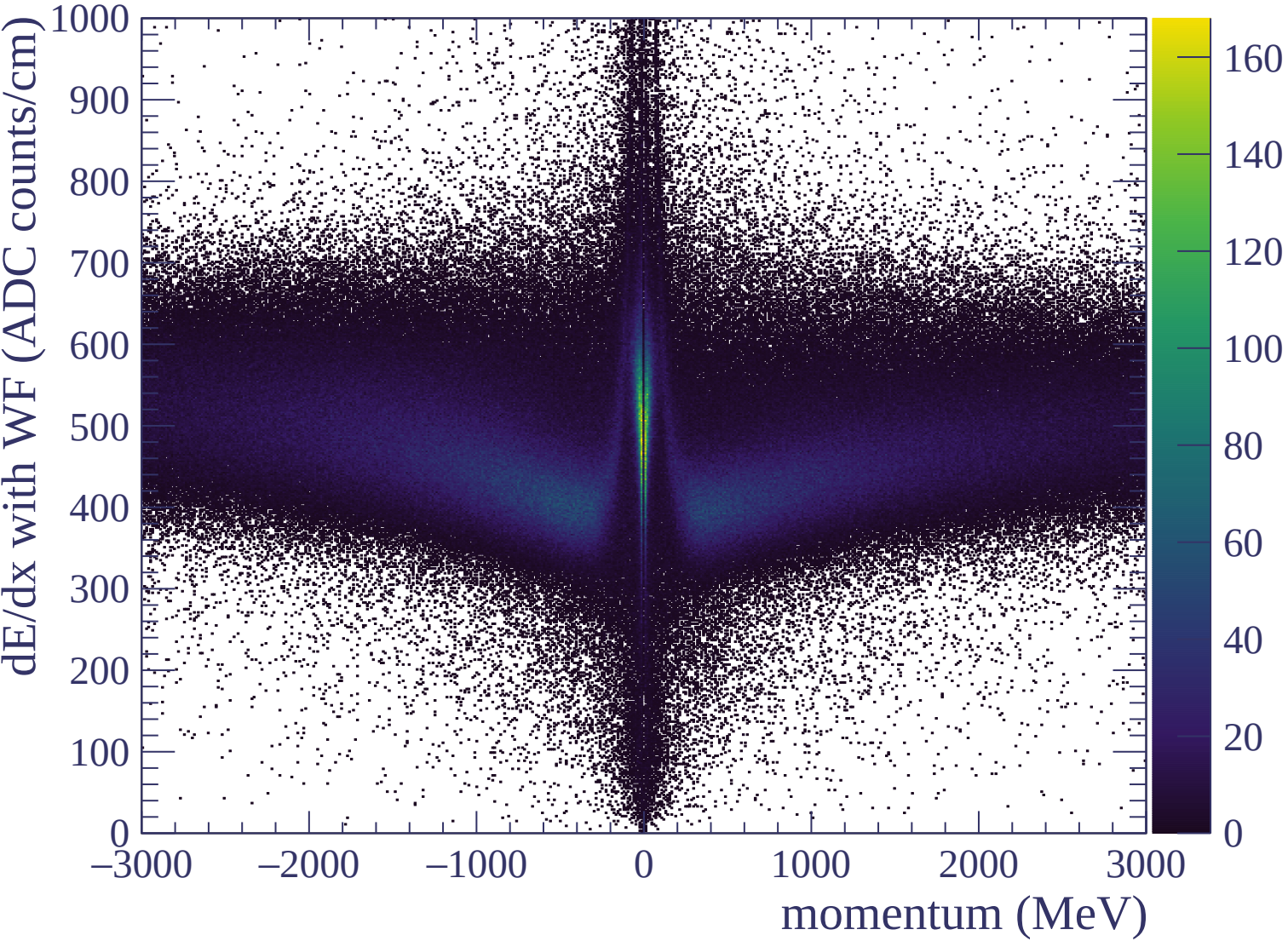


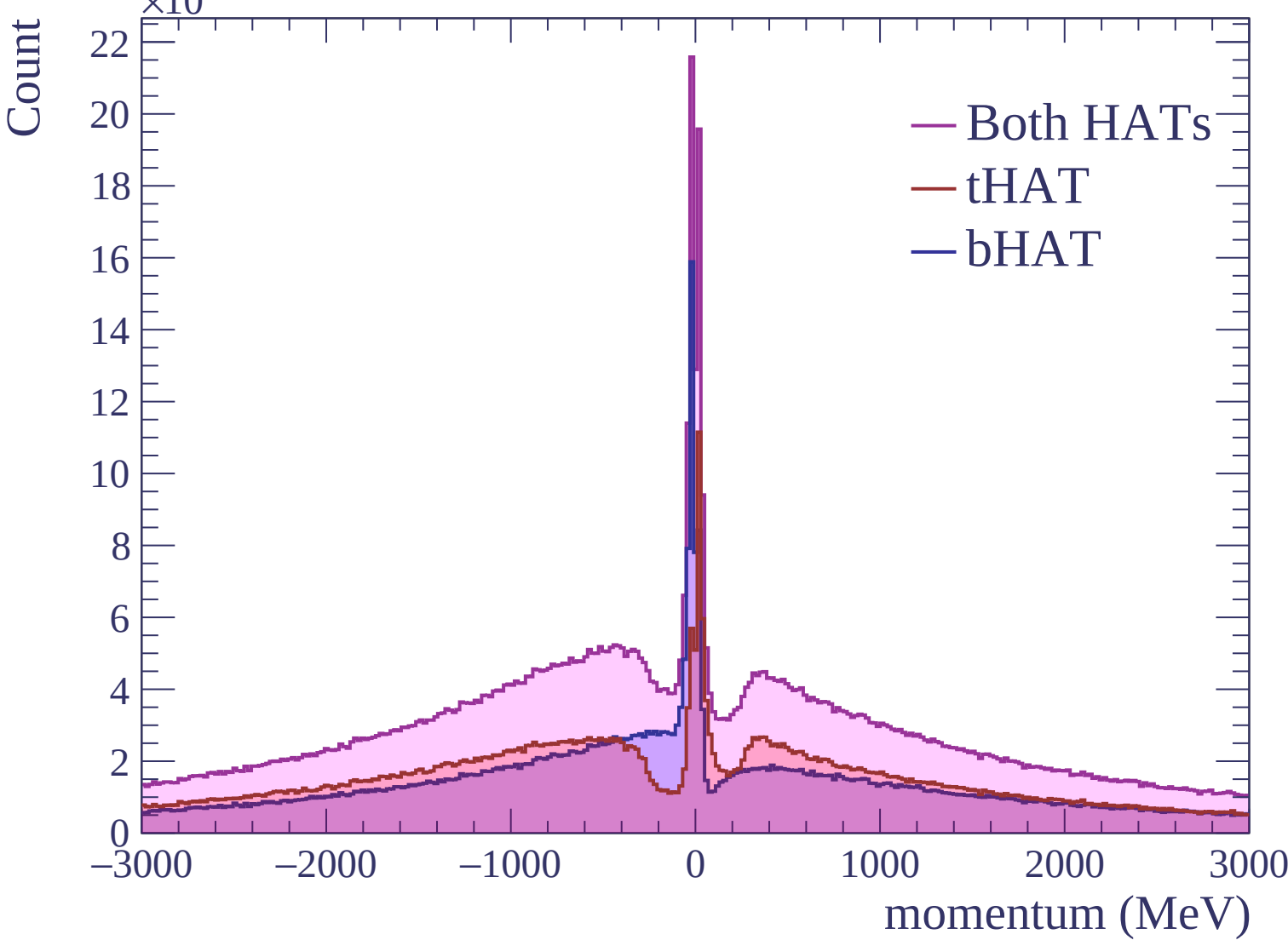


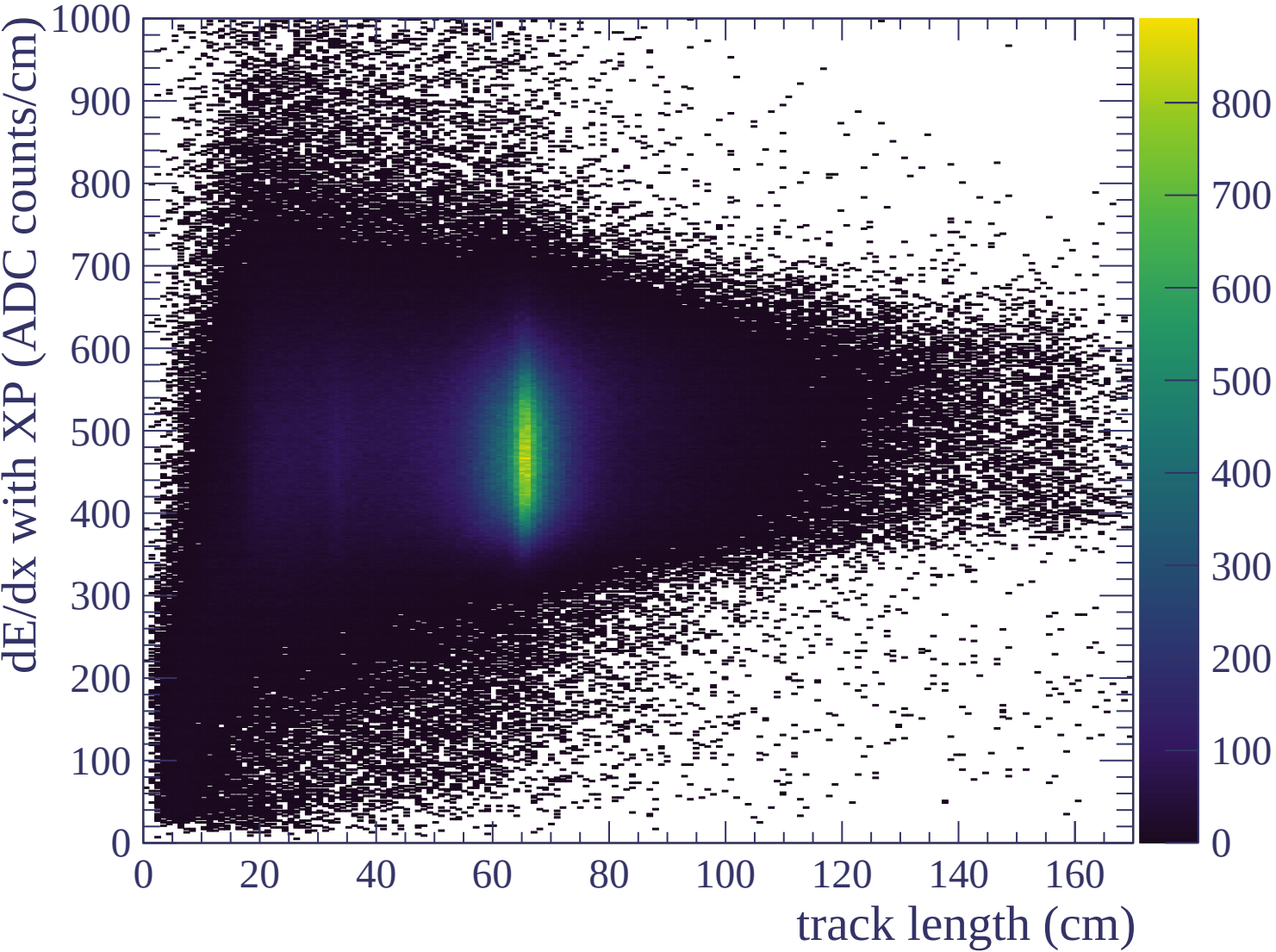


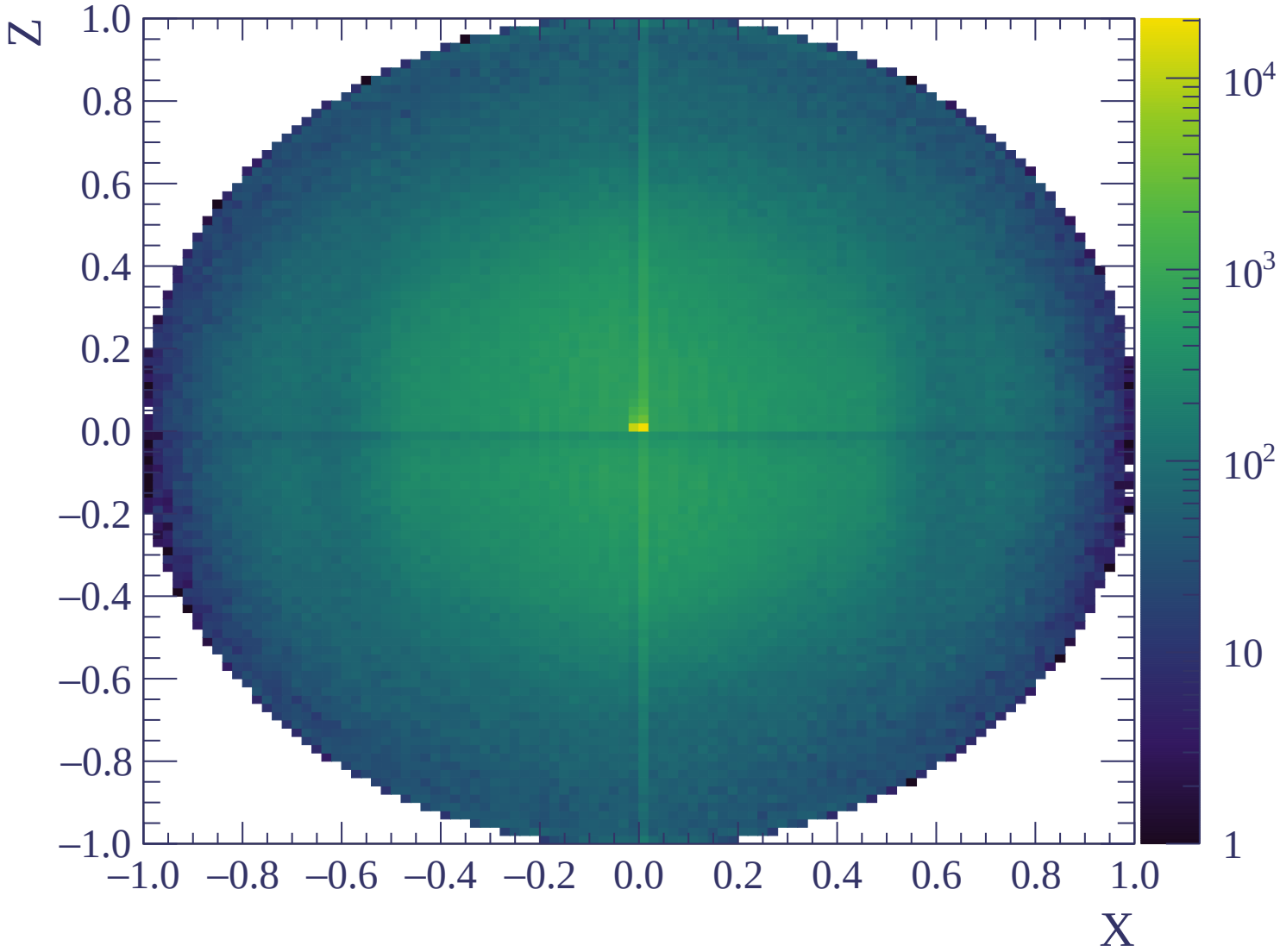


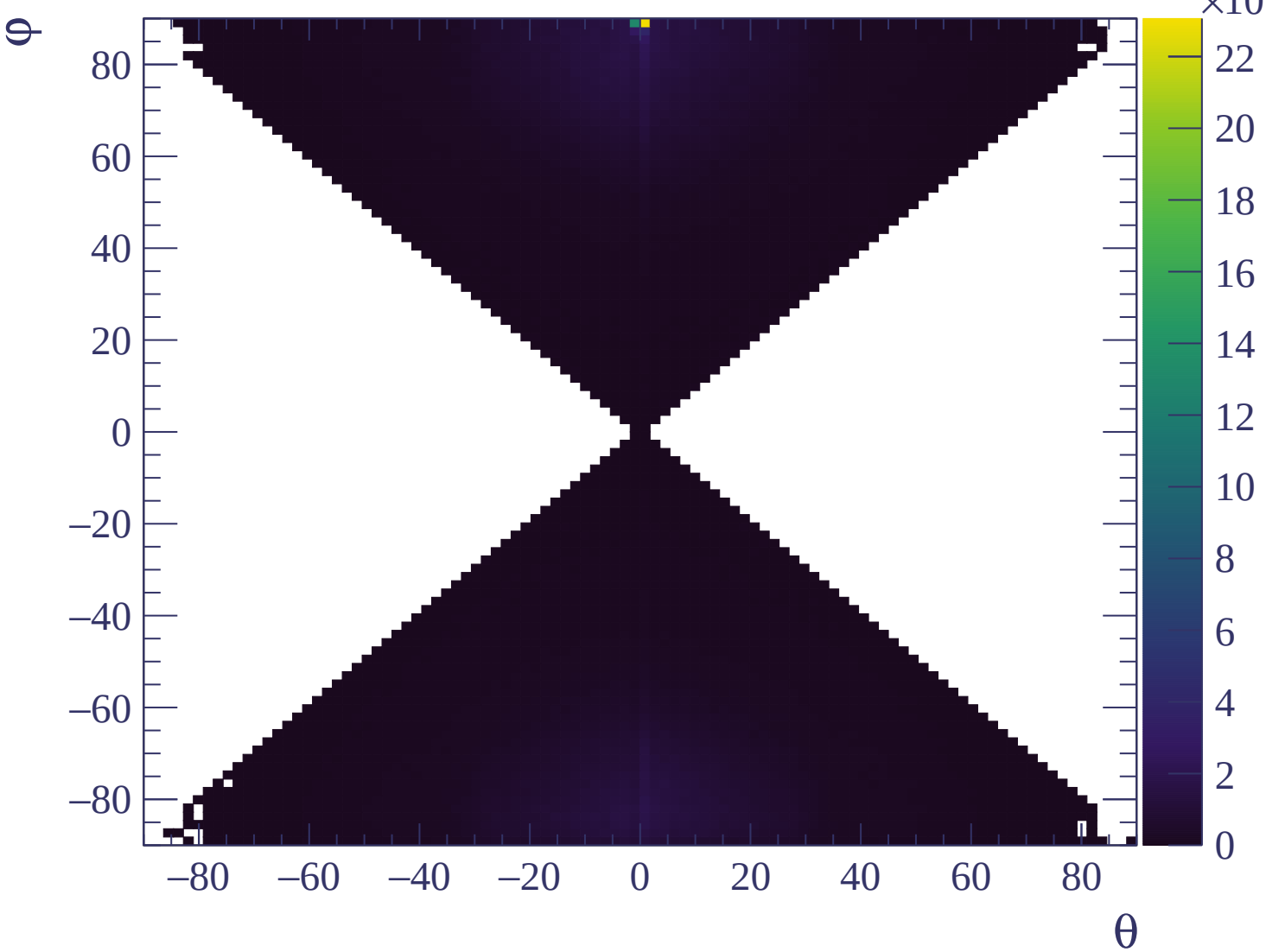


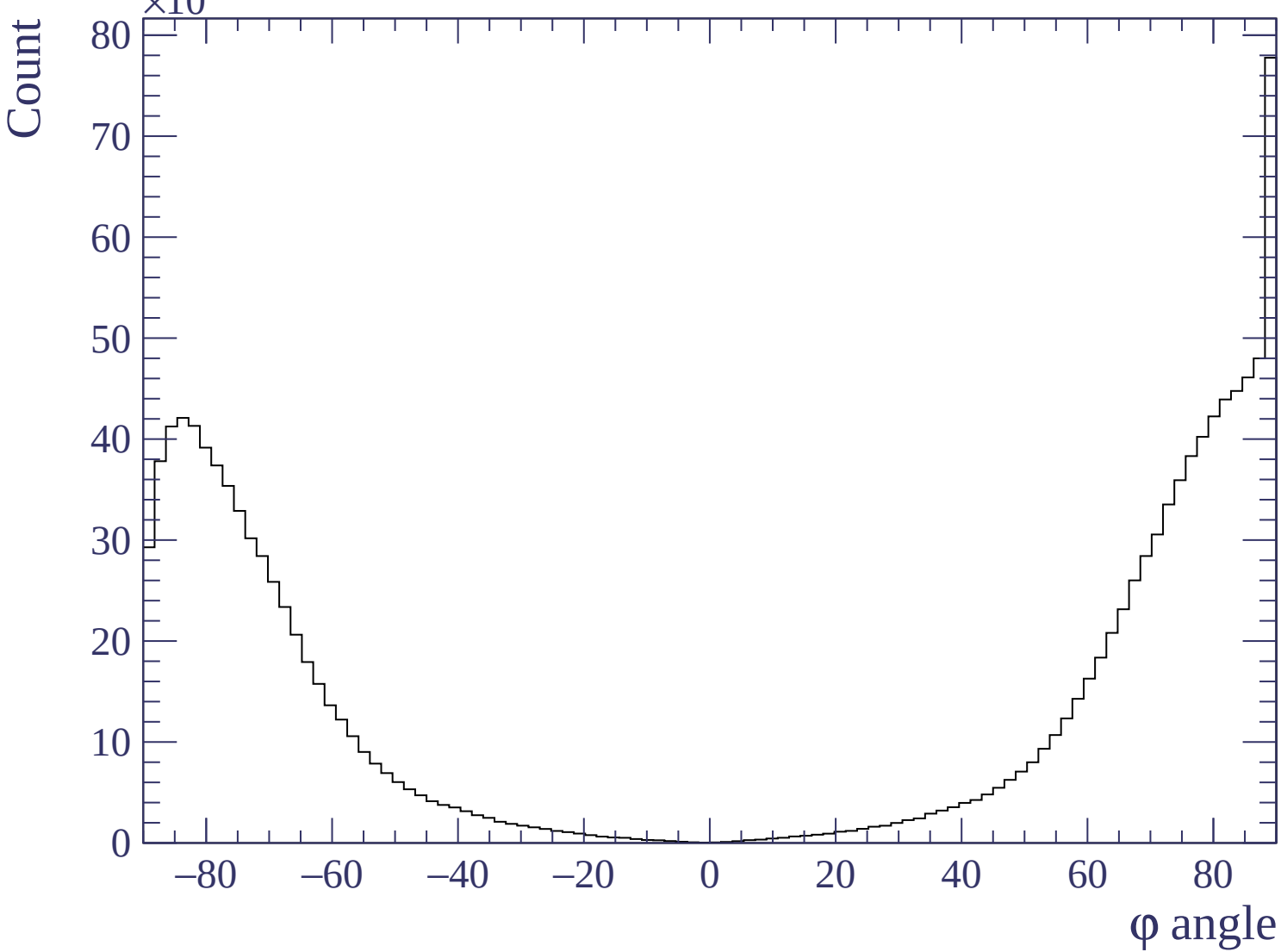


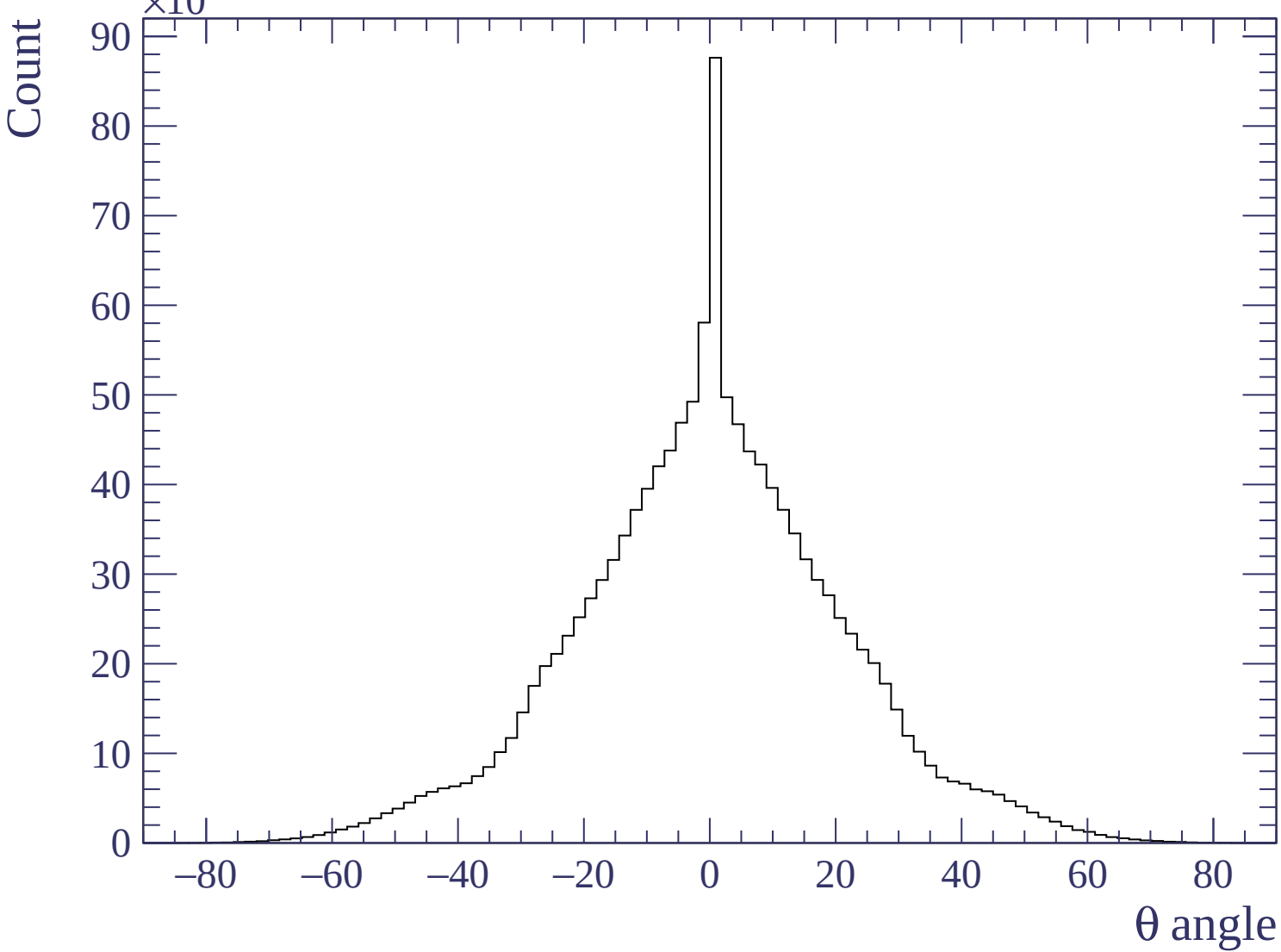


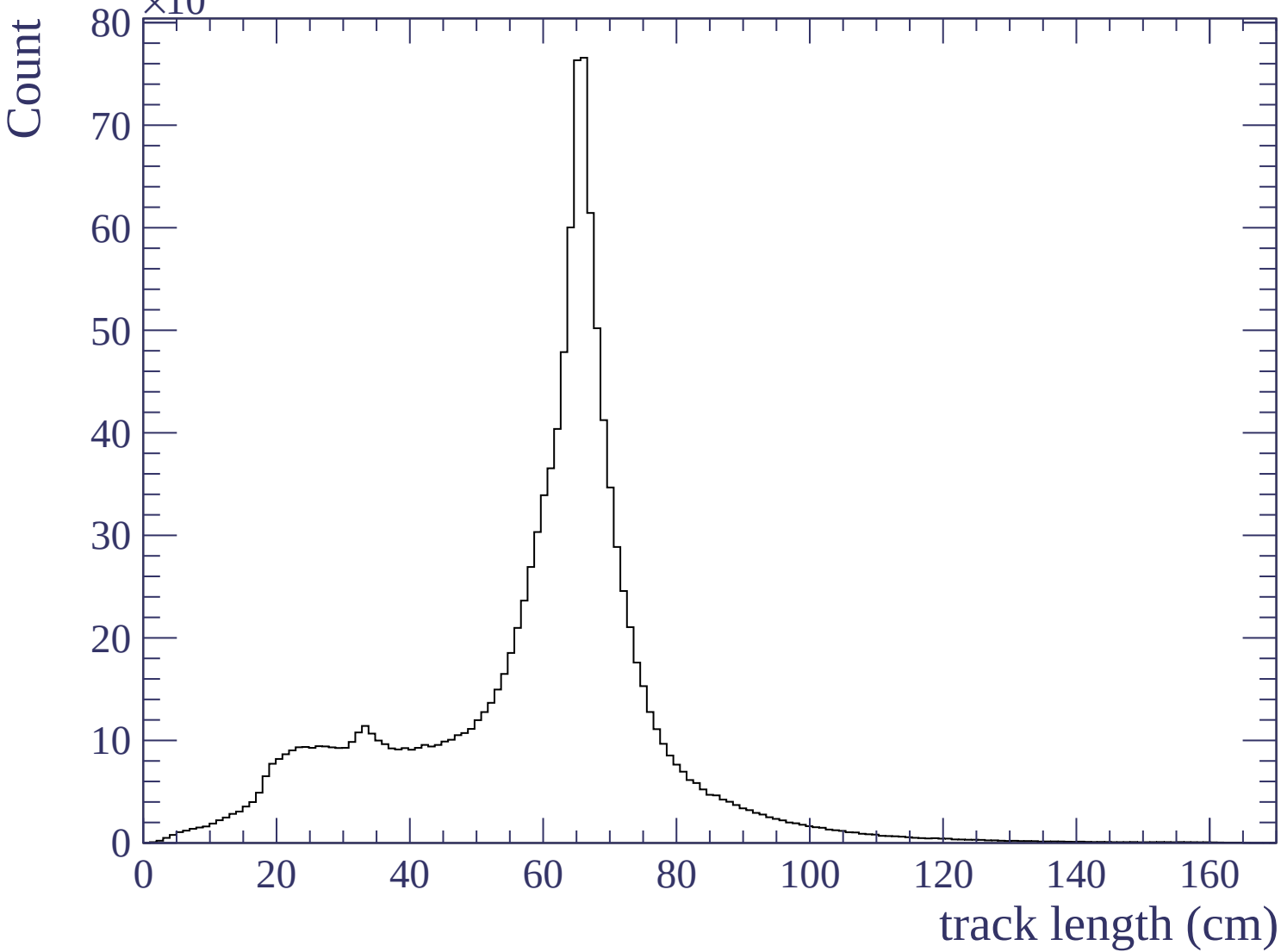


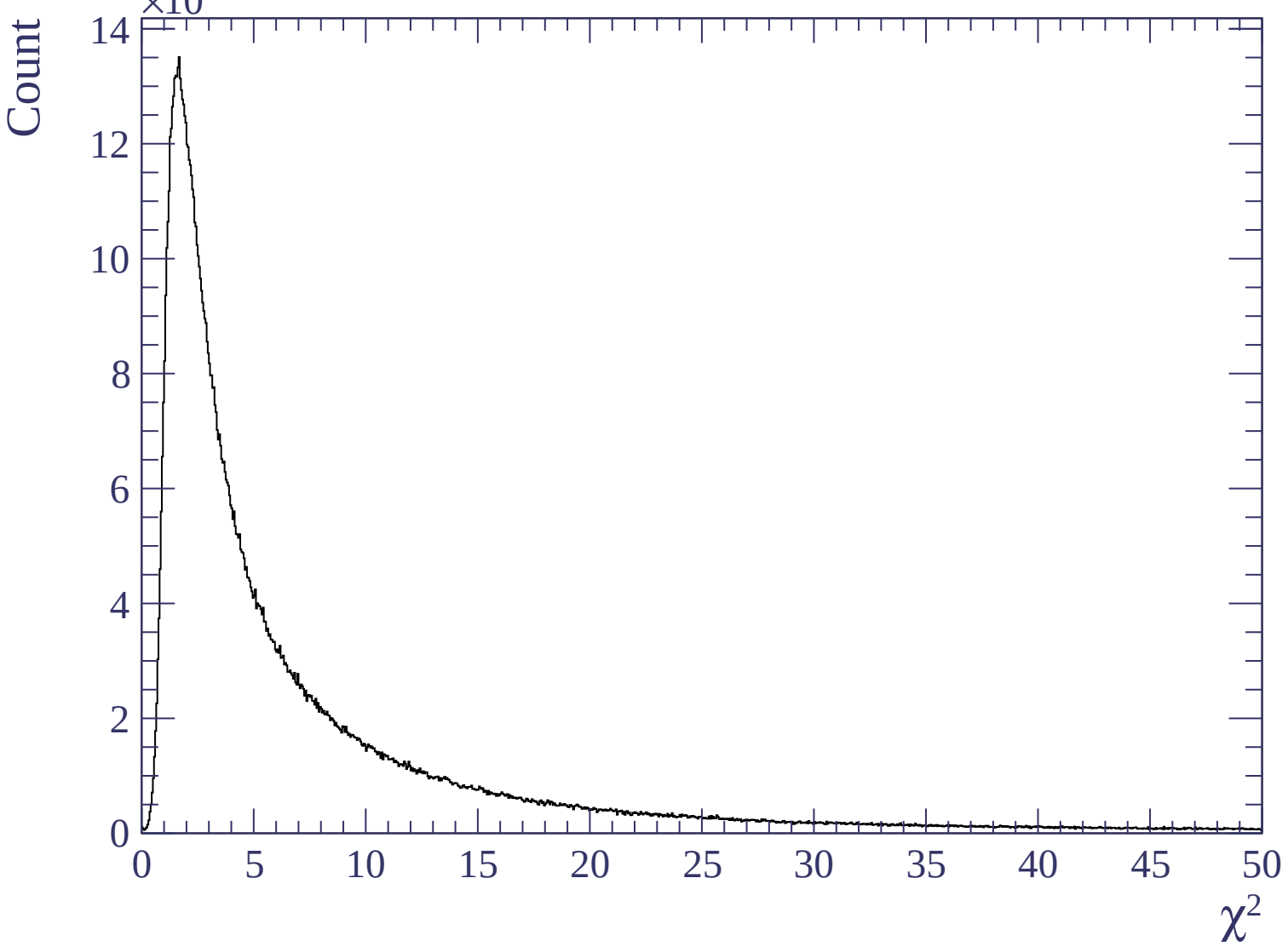


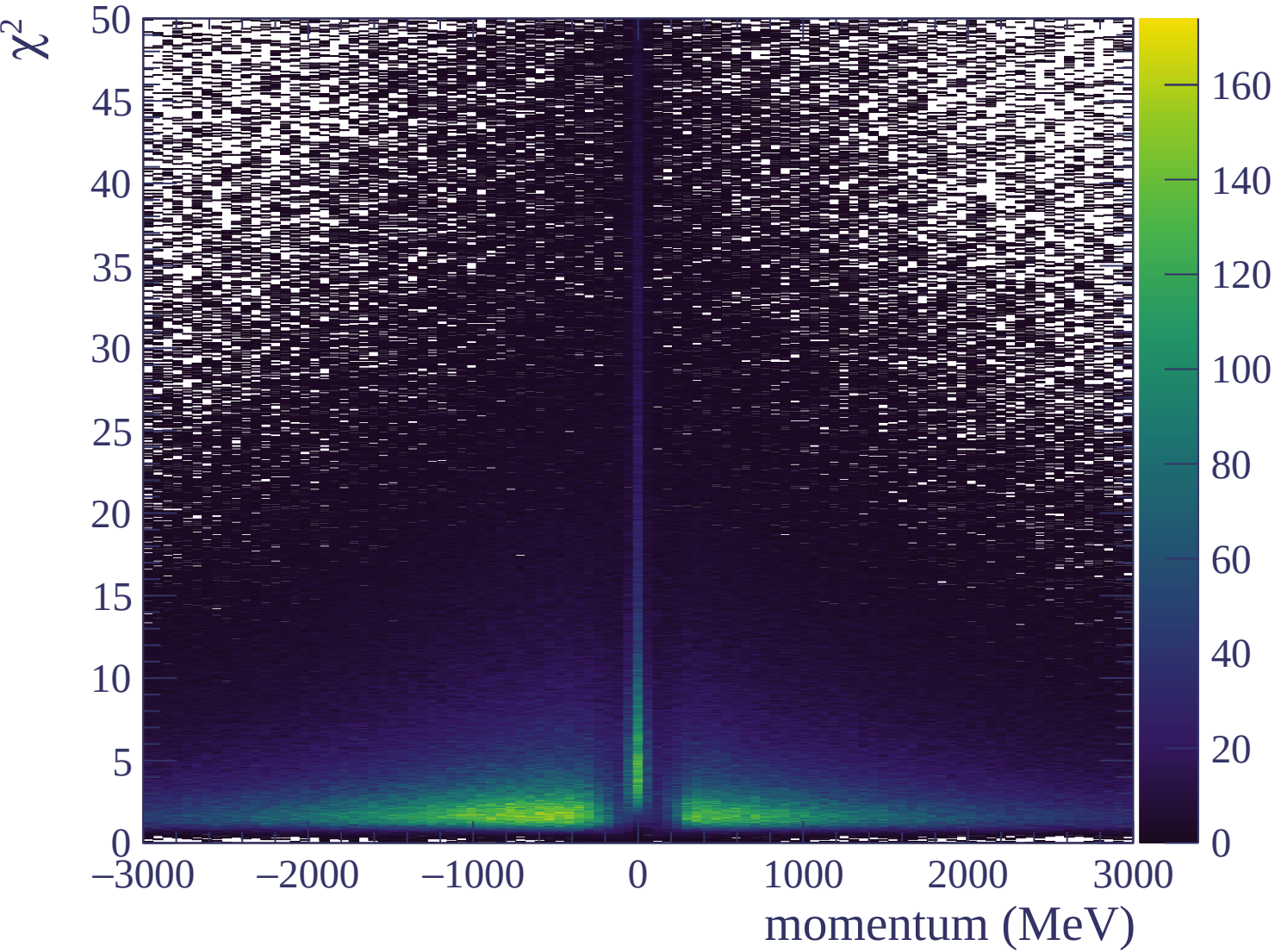


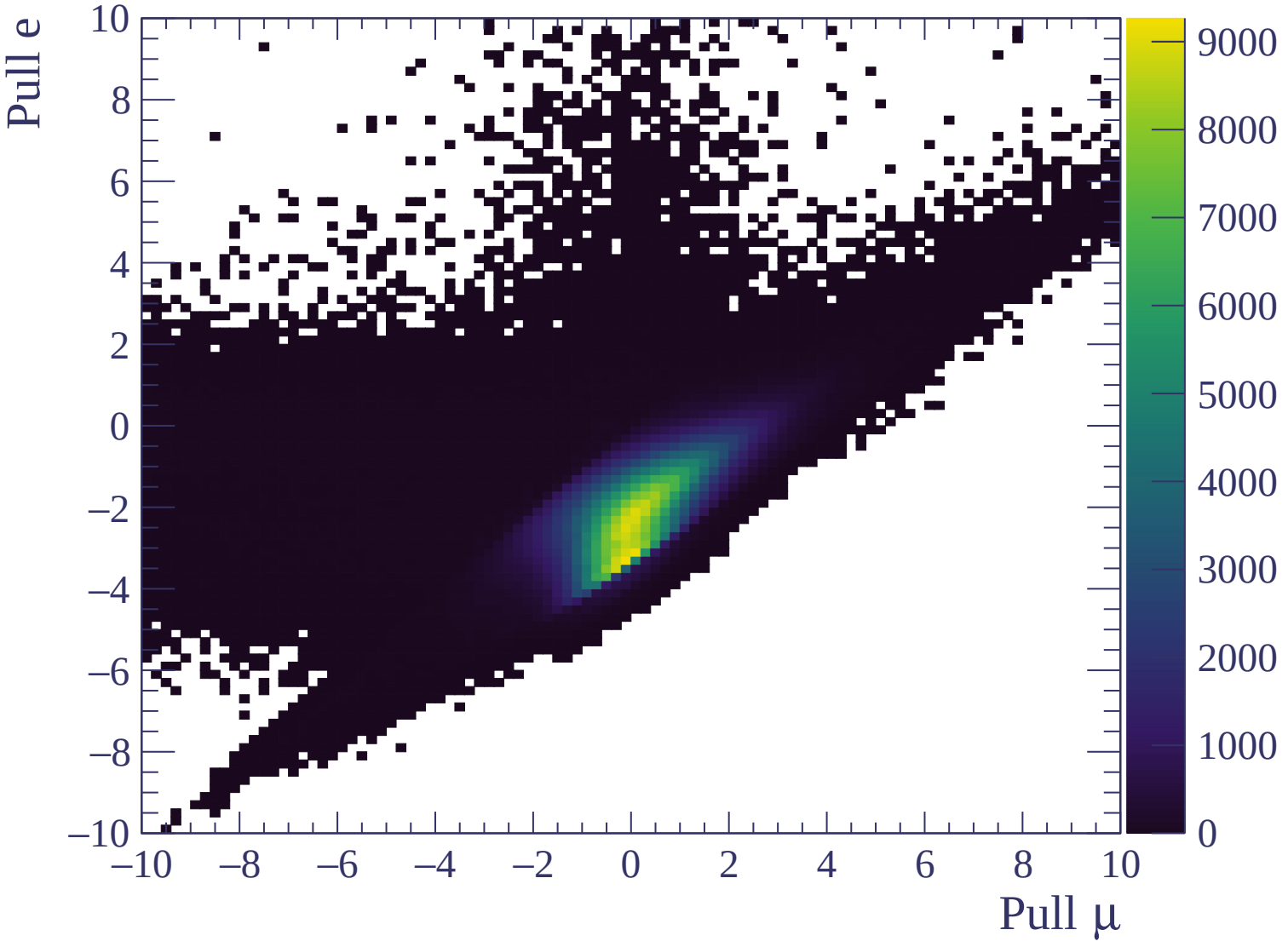


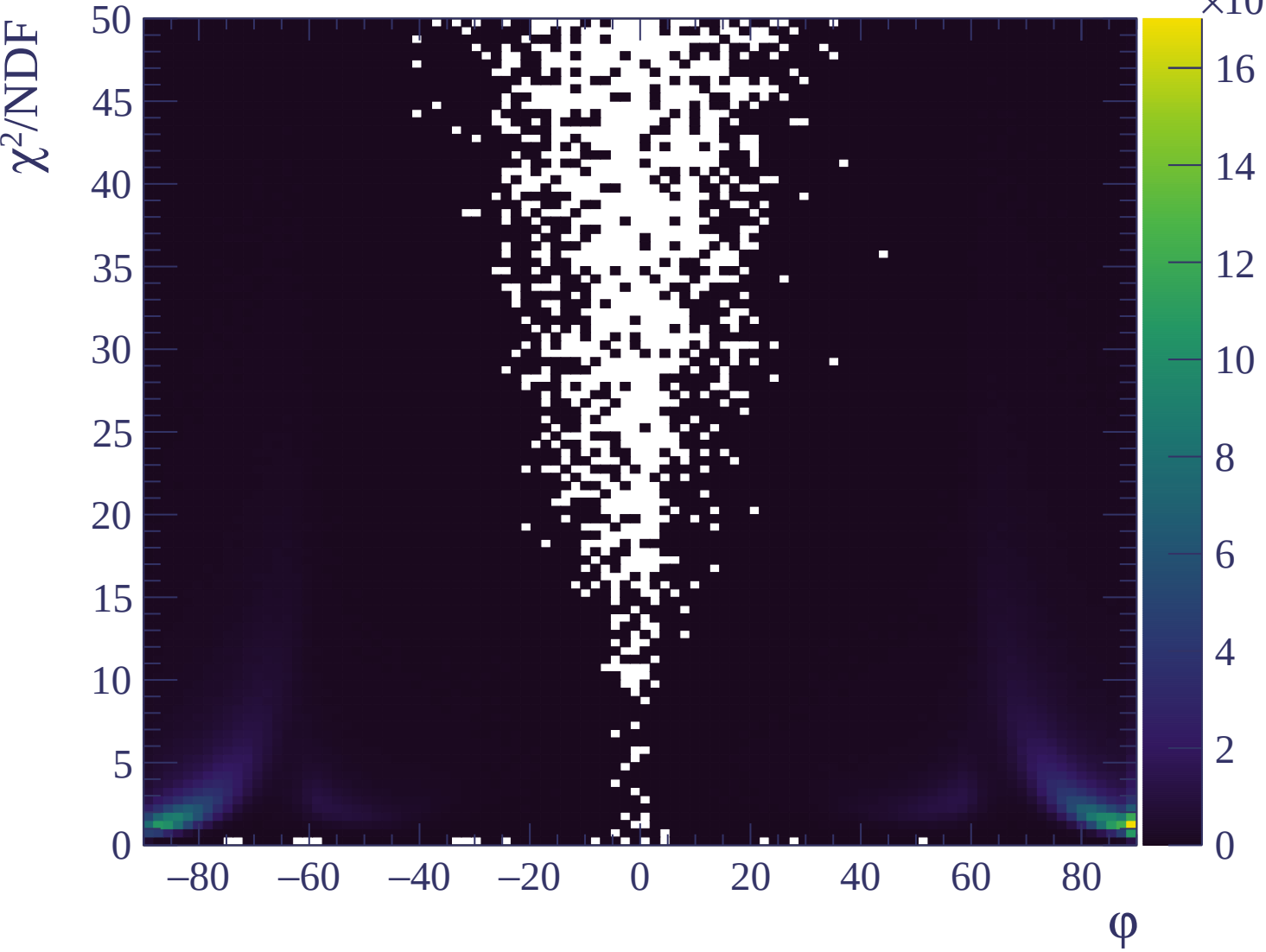


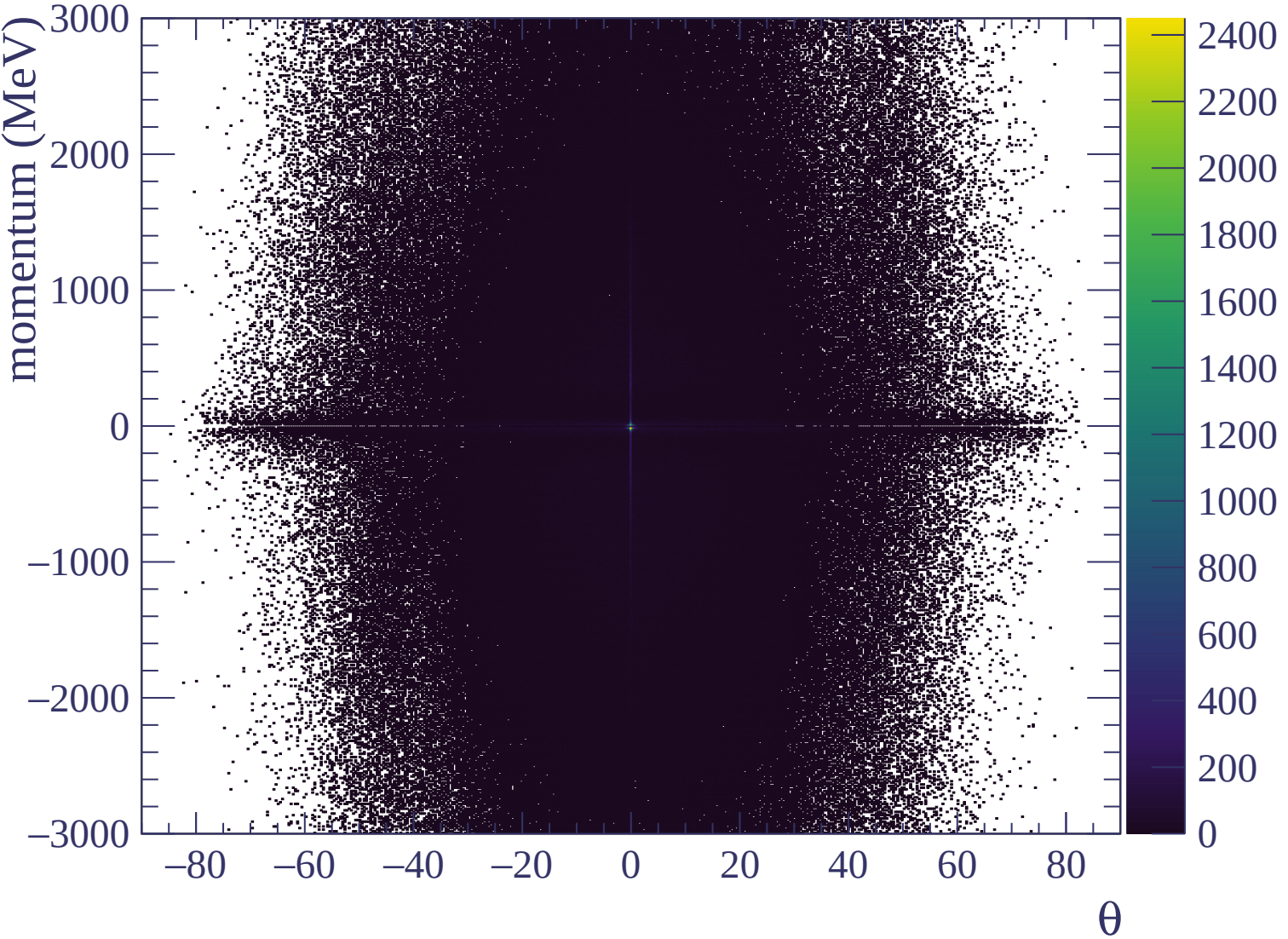


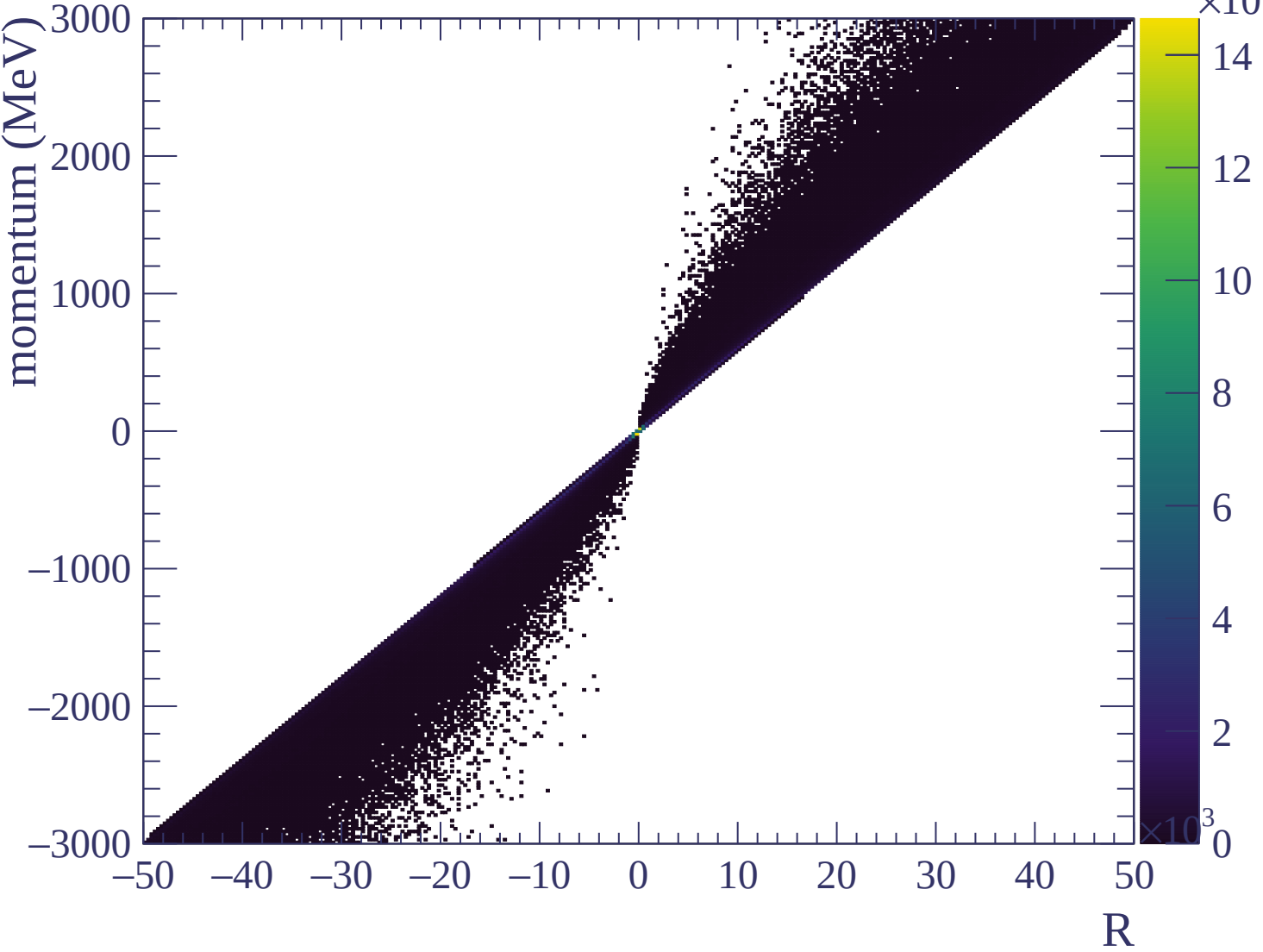


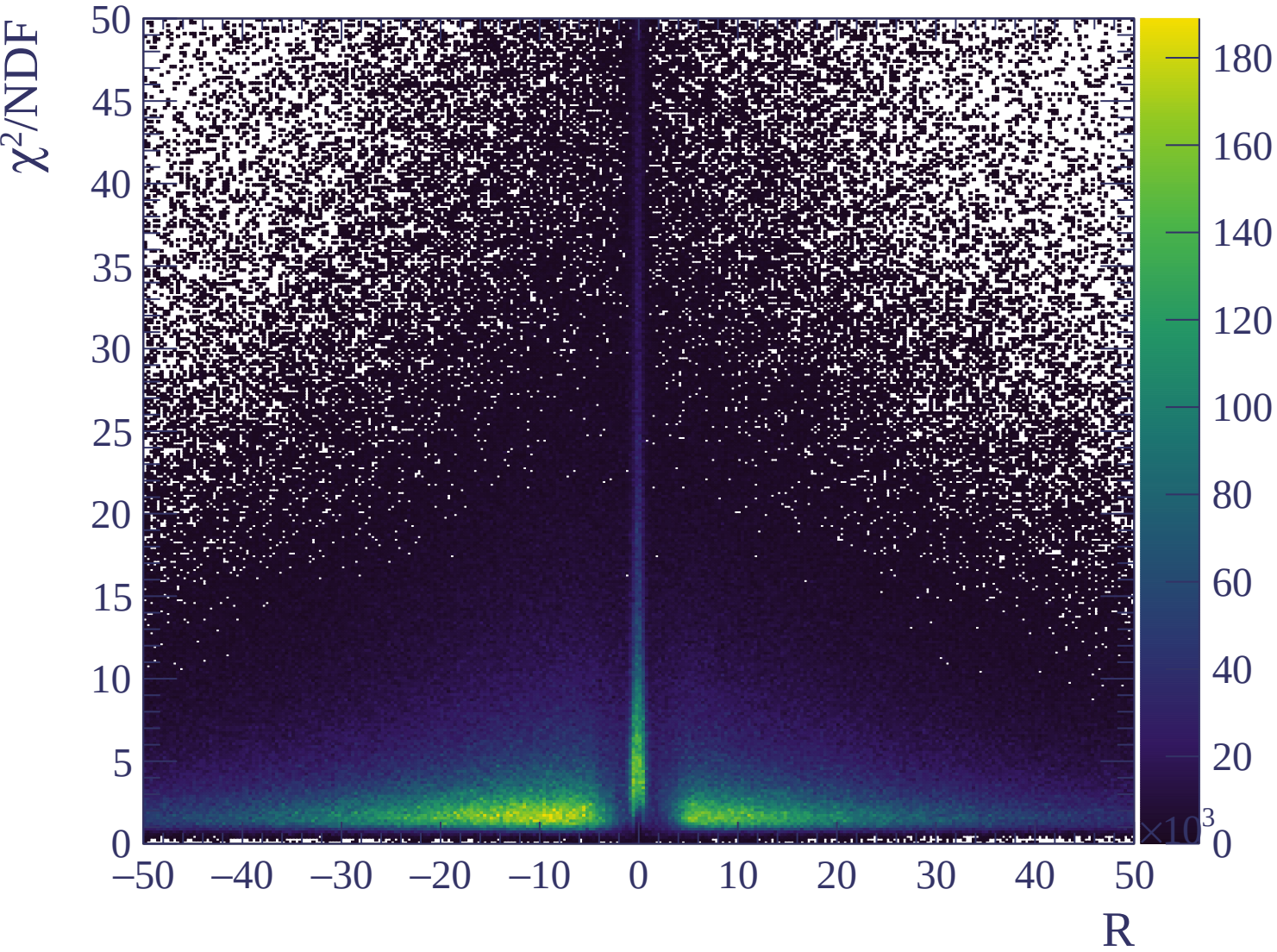


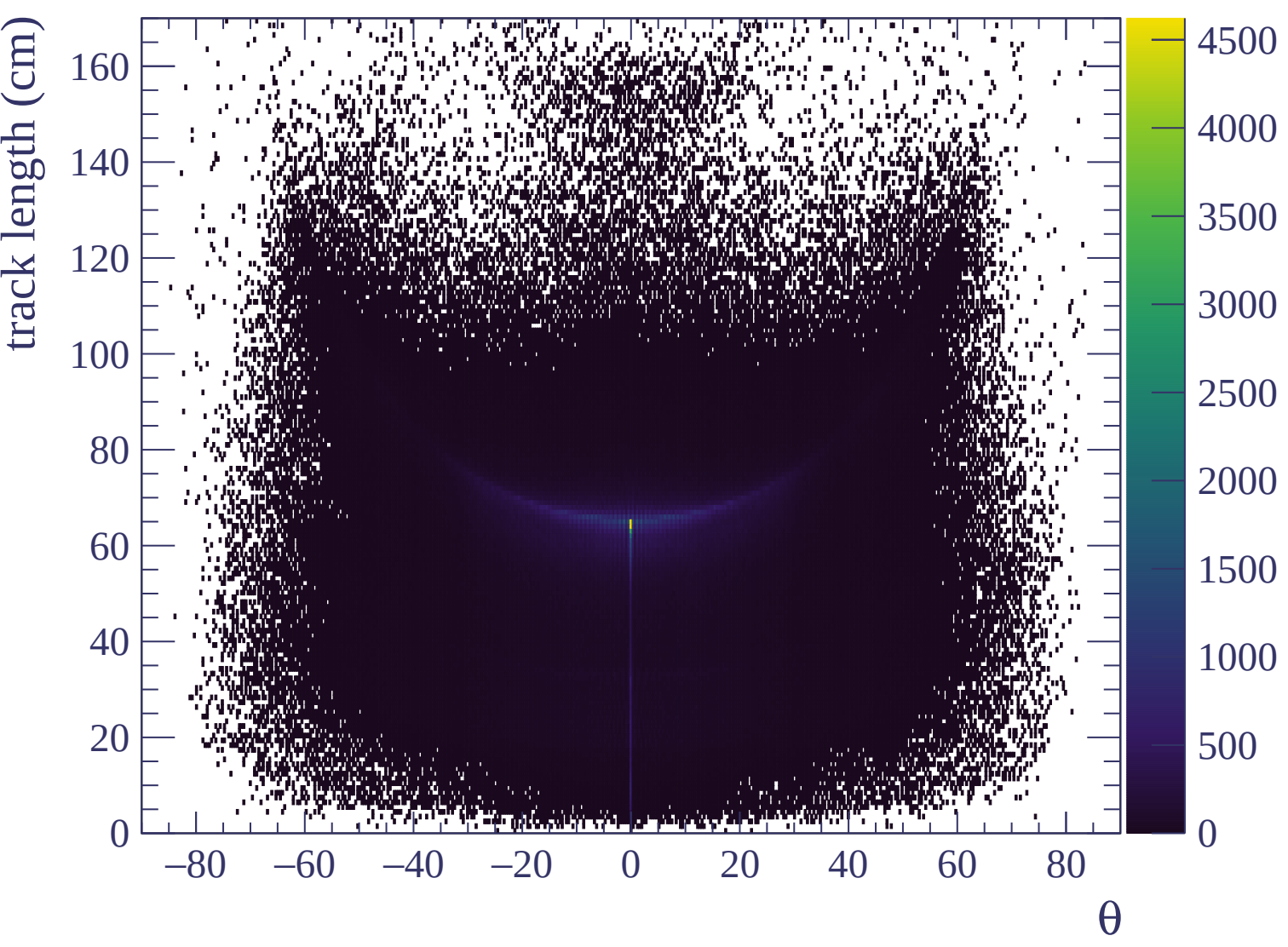


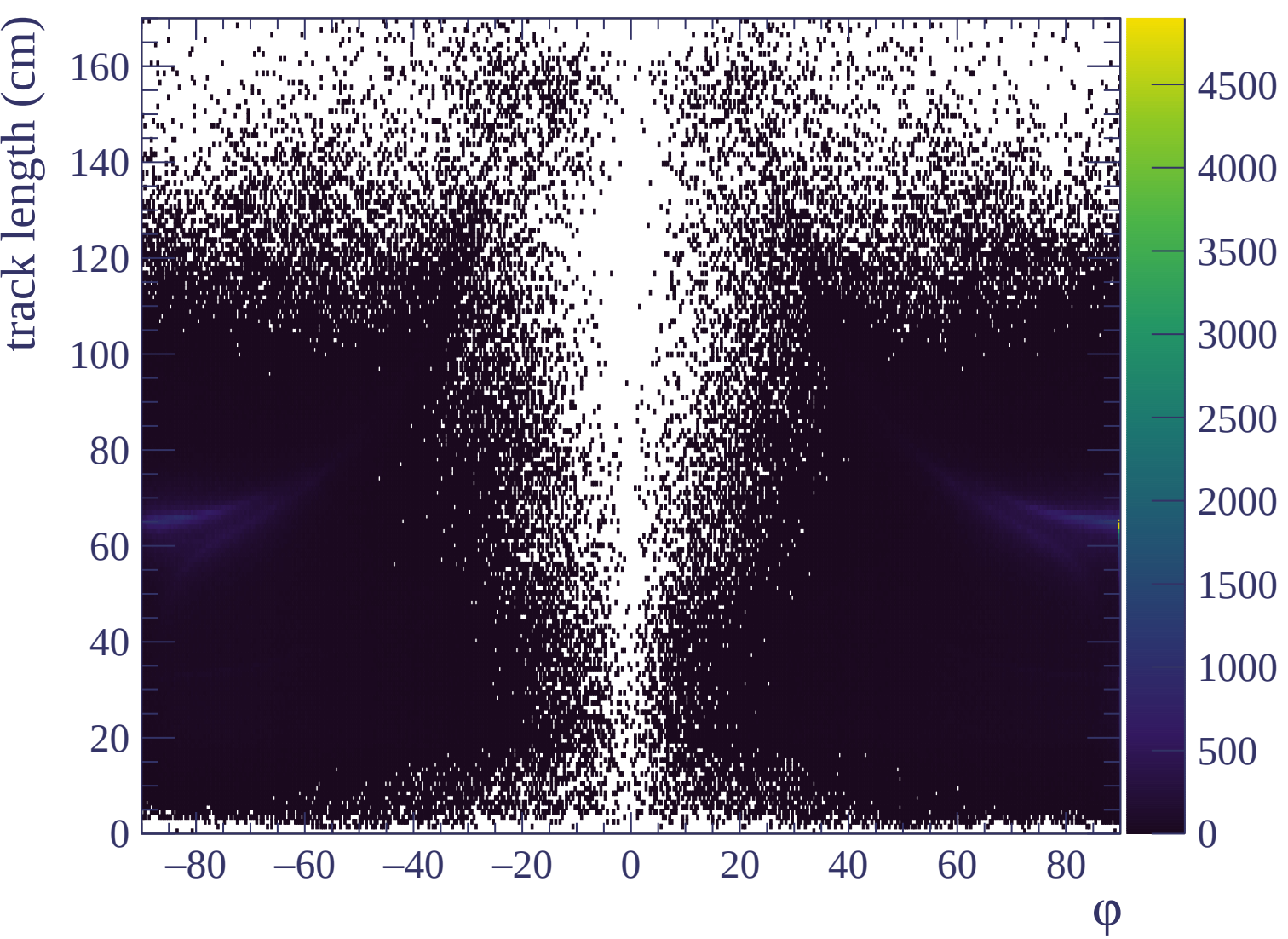






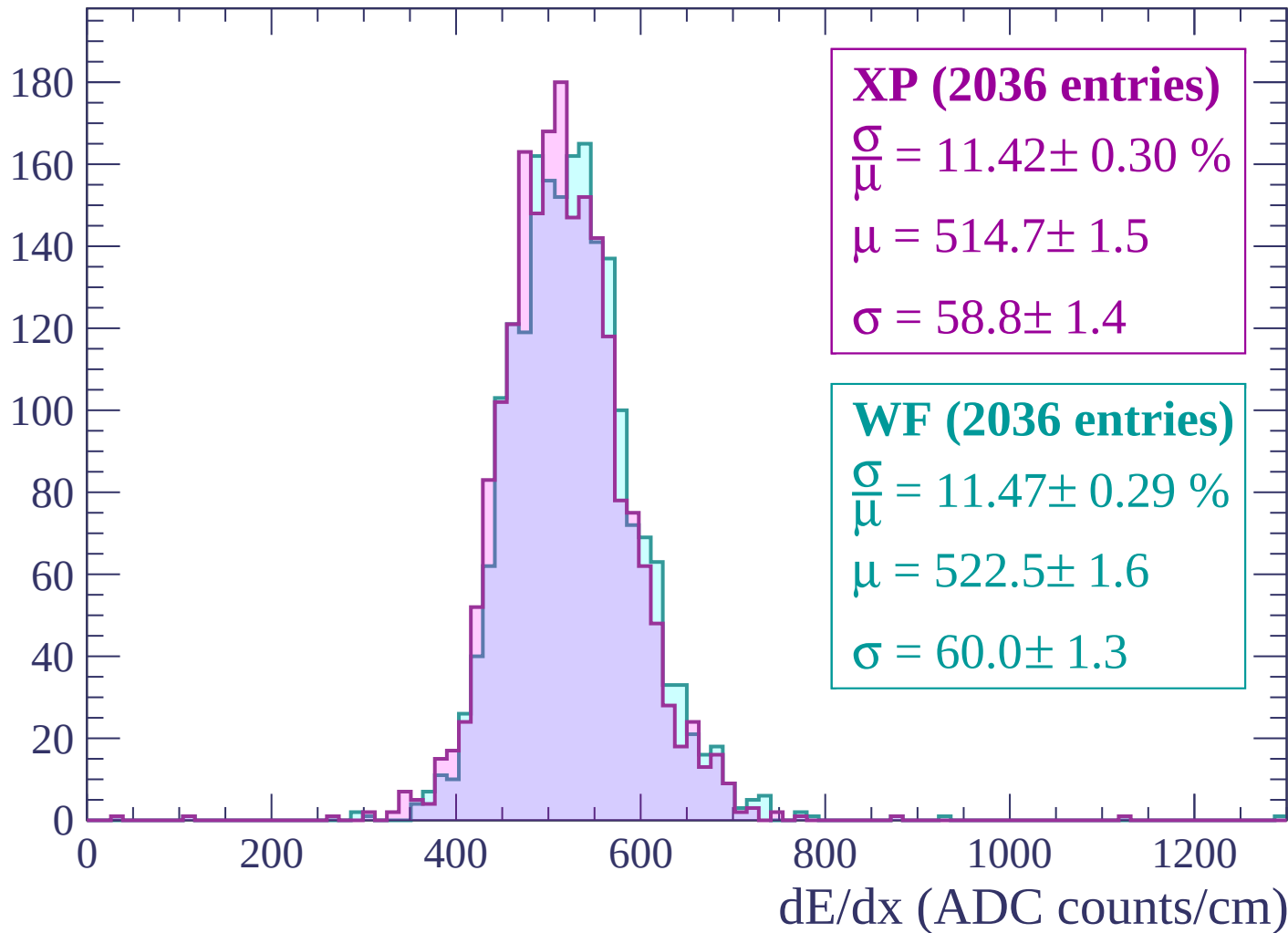






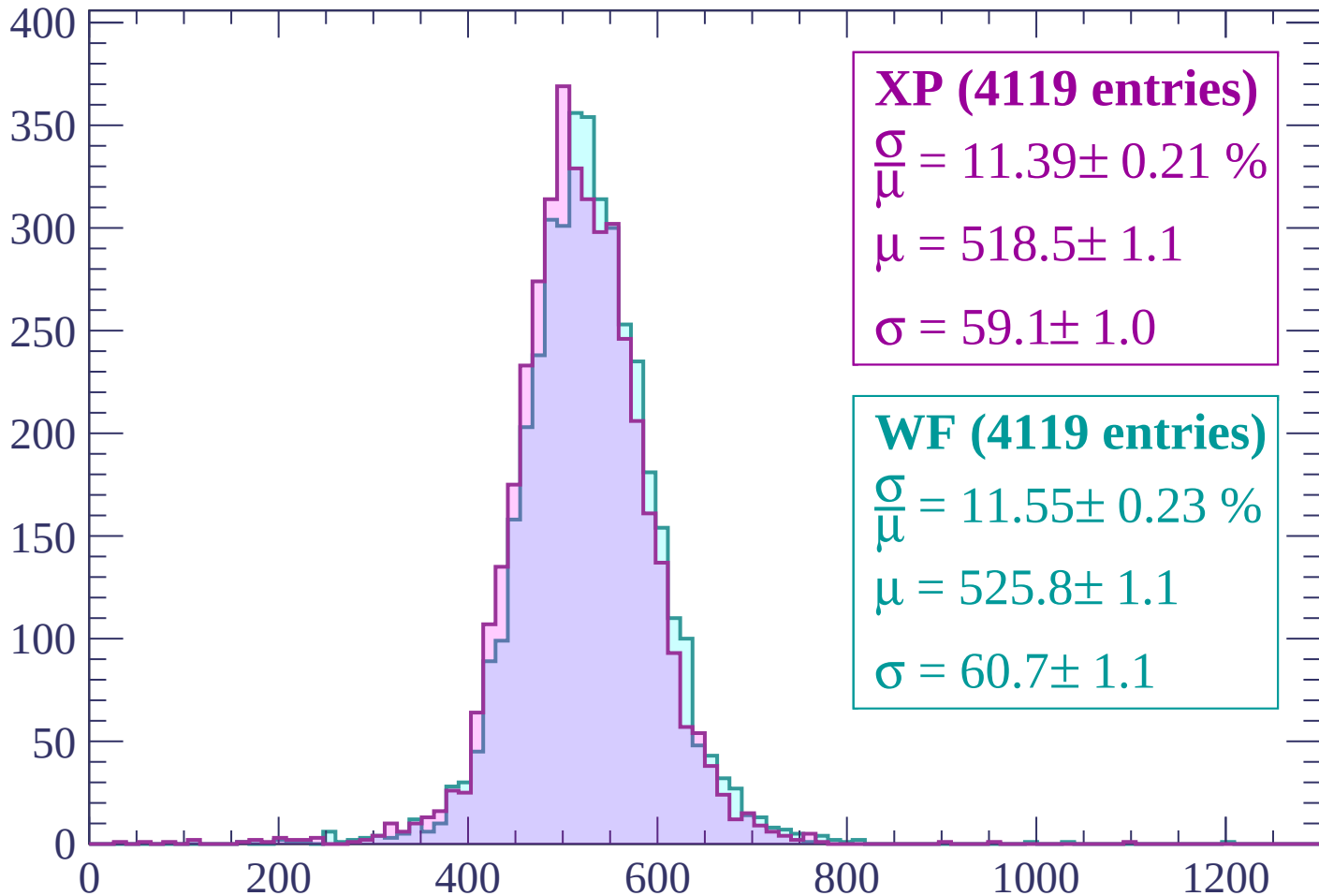
Energy loss | $-3000 < p < -2940$

Count



Energy loss | $-2940 < p < -2880$

Count



XP (4119 entries)

$$\frac{\sigma}{\mu} = 11.39 \pm 0.21 \%$$

$$\mu = 518.5 \pm 1.1$$

$$\sigma = 59.1 \pm 1.0$$

WF (4119 entries)

$$\frac{\sigma}{\mu} = 11.55 \pm 0.23 \%$$

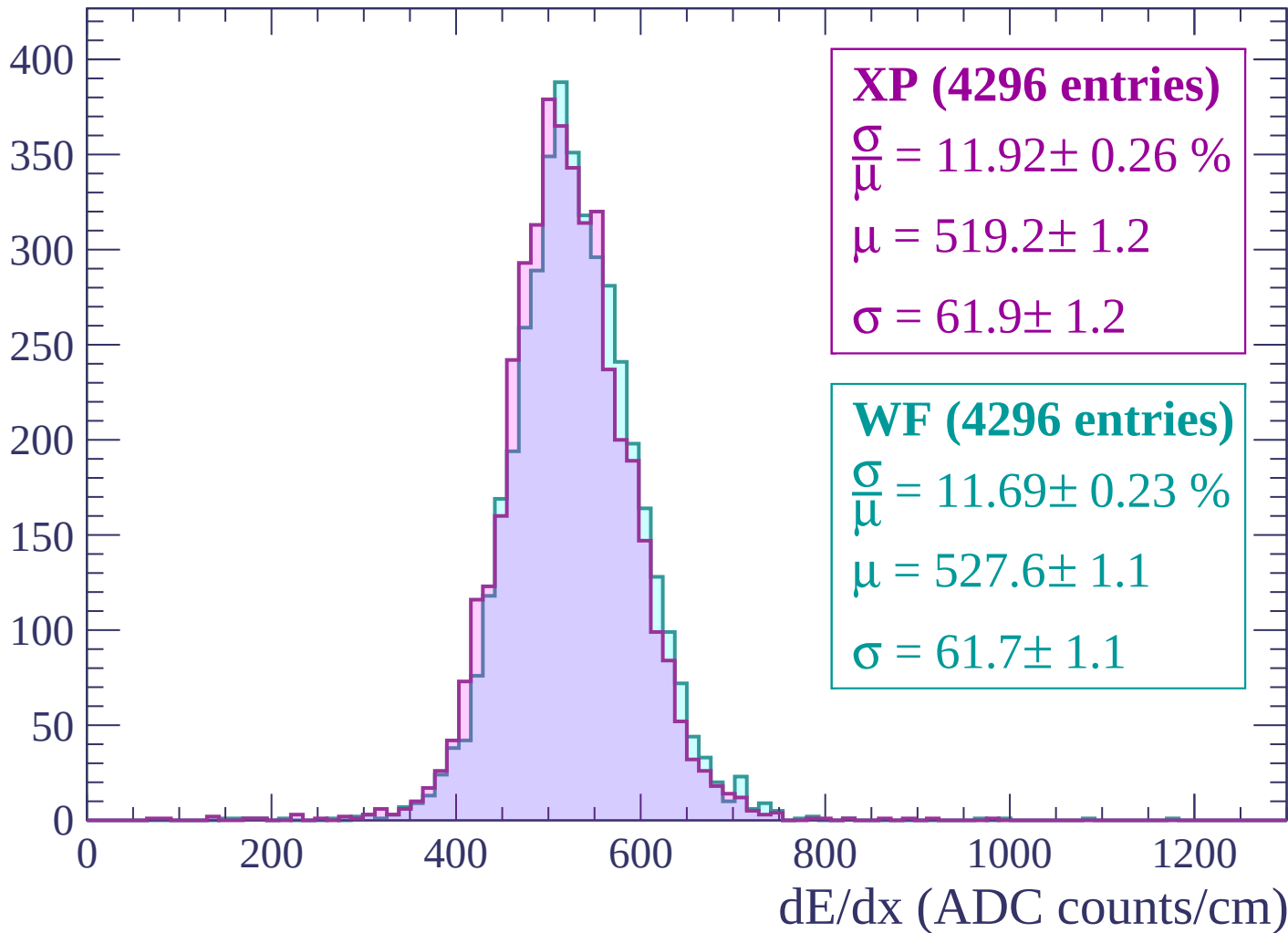
$$\mu = 525.8 \pm 1.1$$

$$\sigma = 60.7 \pm 1.1$$

dE/dx (ADC counts/cm)

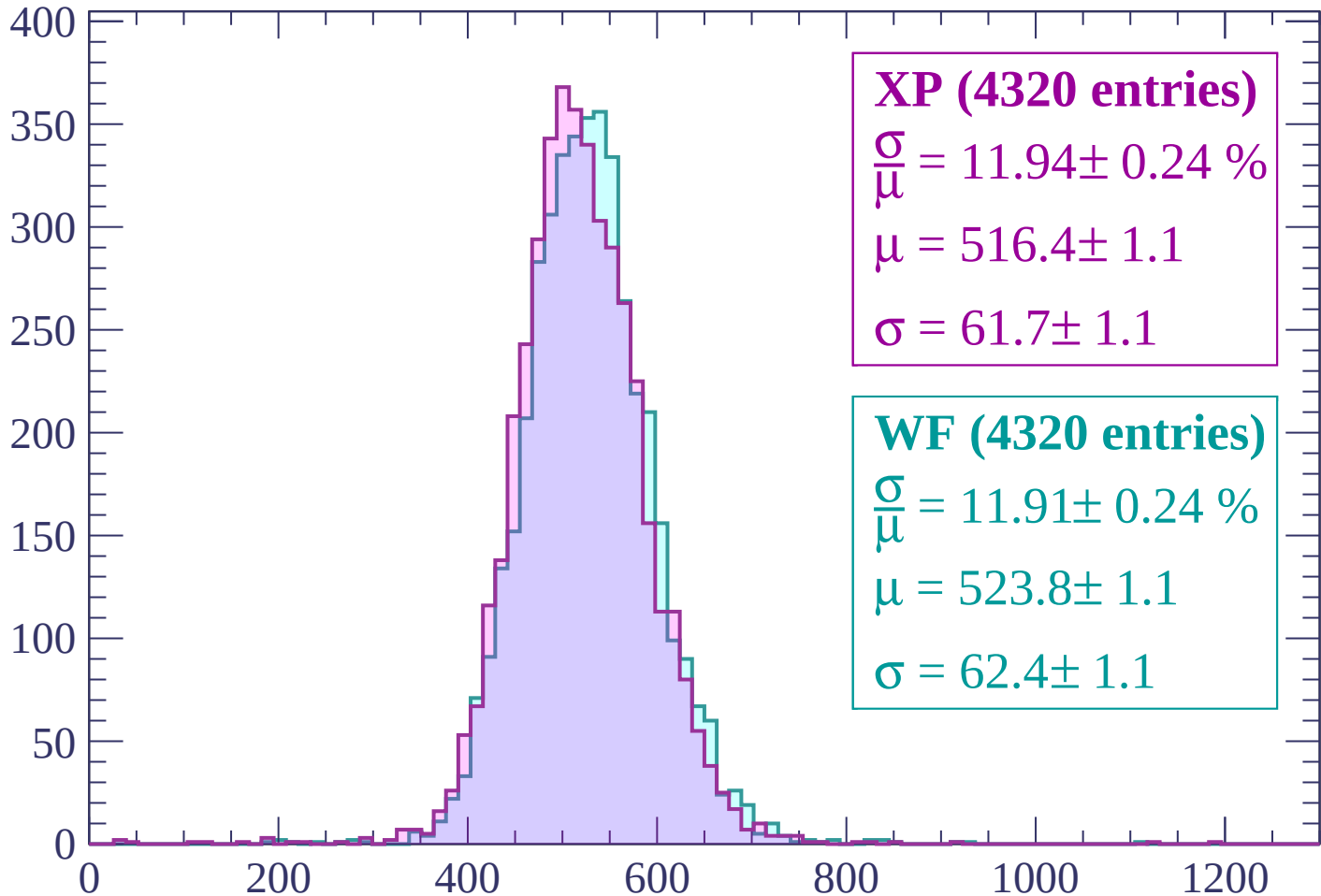
Energy loss | $-2880 < p < -2820$

Count



Energy loss | -2820 < p < -2760

Count



XP (4320 entries)

$$\frac{\sigma}{\mu} = 11.94 \pm 0.24 \%$$

$$\mu = 516.4 \pm 1.1$$

$$\sigma = 61.7 \pm 1.1$$

WF (4320 entries)

$$\frac{\sigma}{\mu} = 11.91 \pm 0.24 \%$$

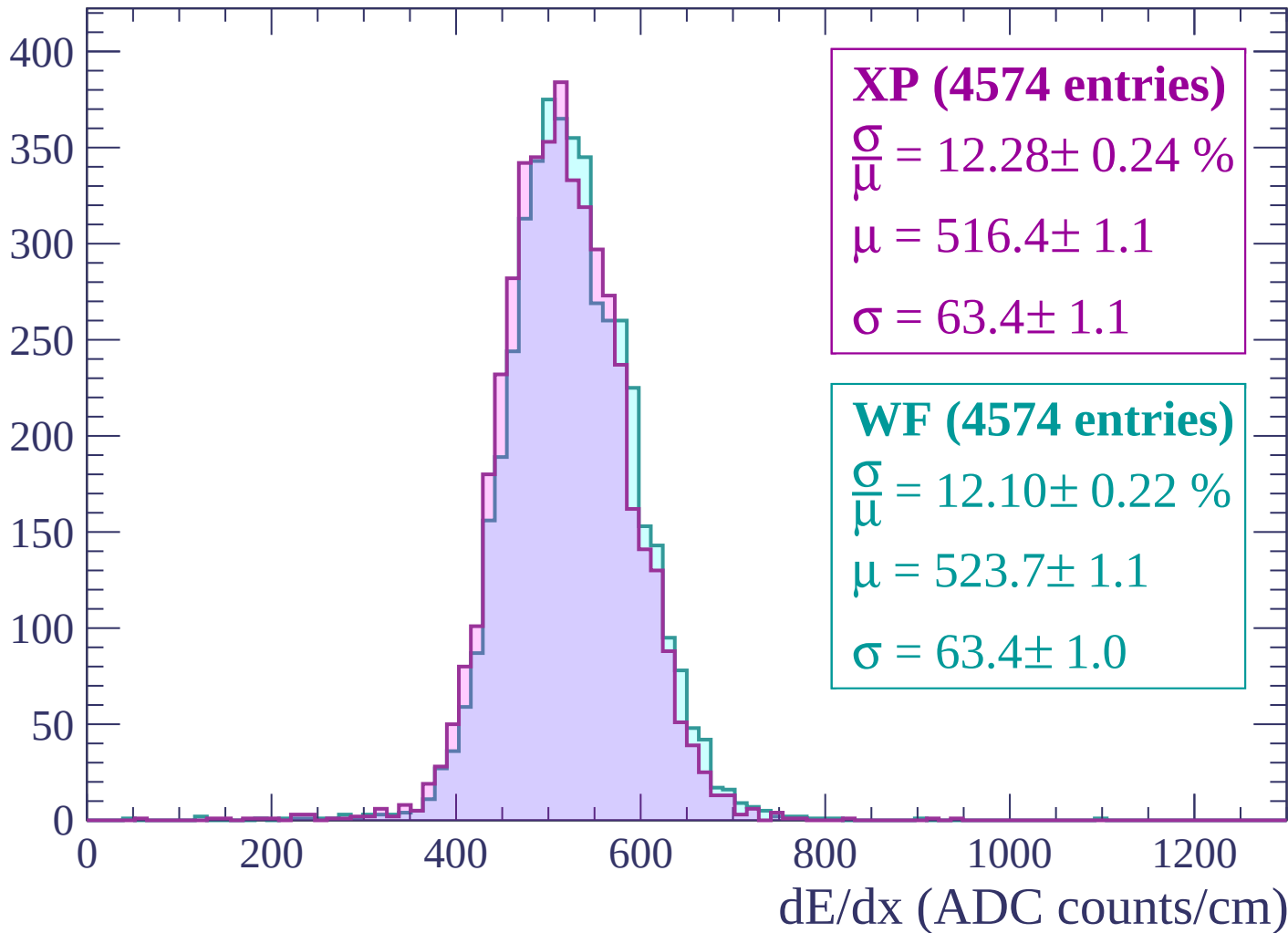
$$\mu = 523.8 \pm 1.1$$

$$\sigma = 62.4 \pm 1.1$$

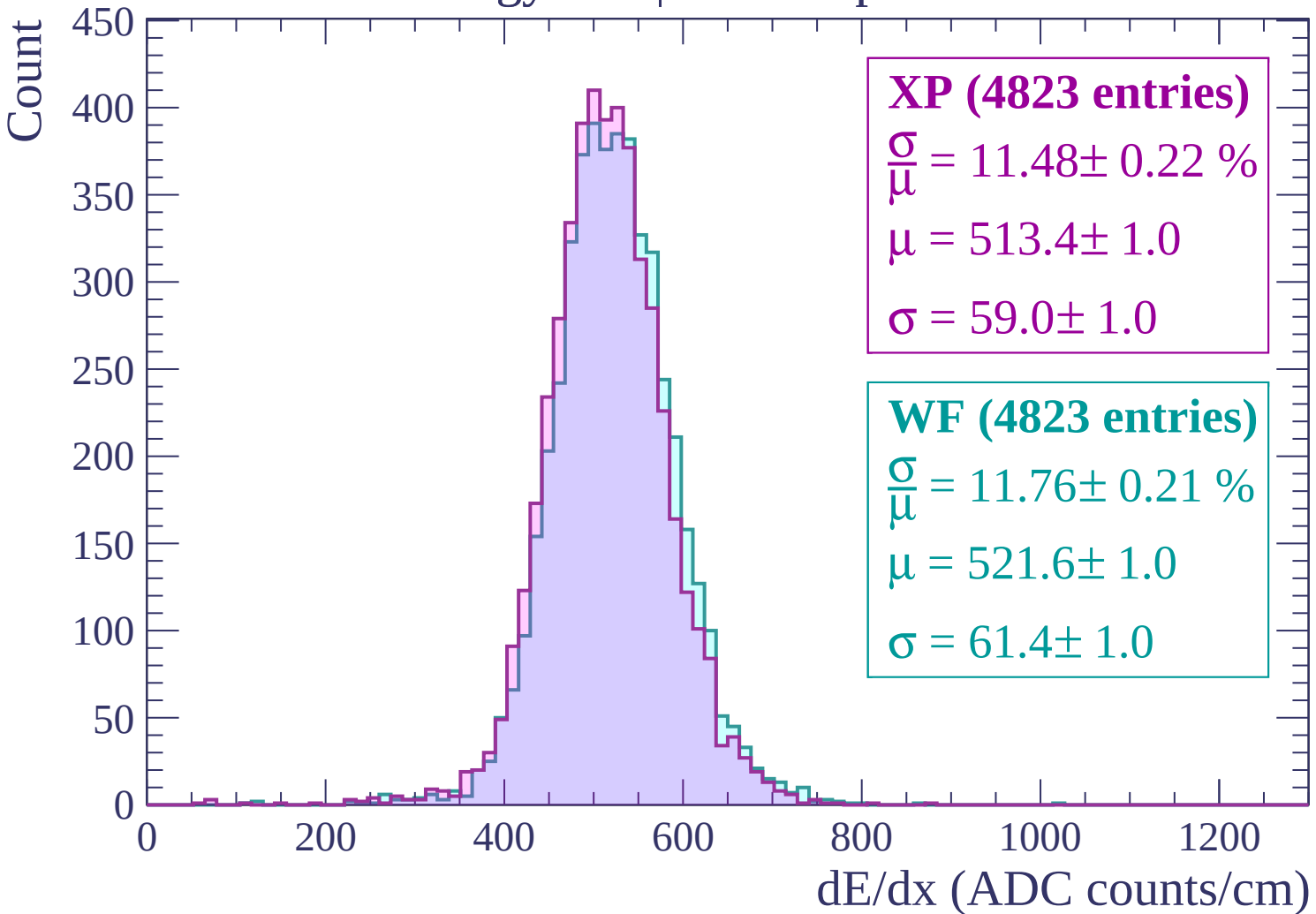
dE/dx (ADC counts/cm)

Energy loss | $-2760 < p < -2700$

Count

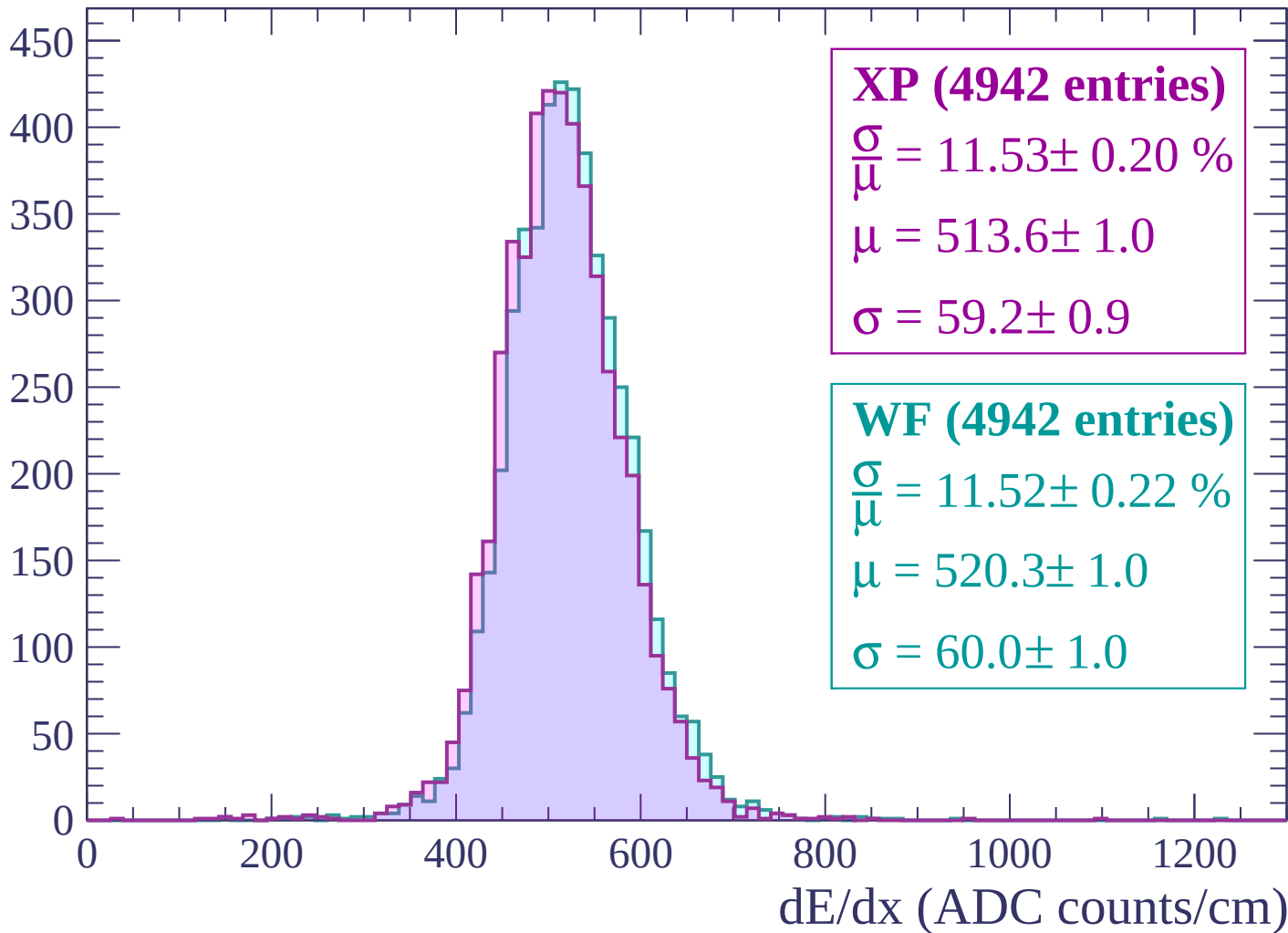


Energy loss | -2700 < p < -2640



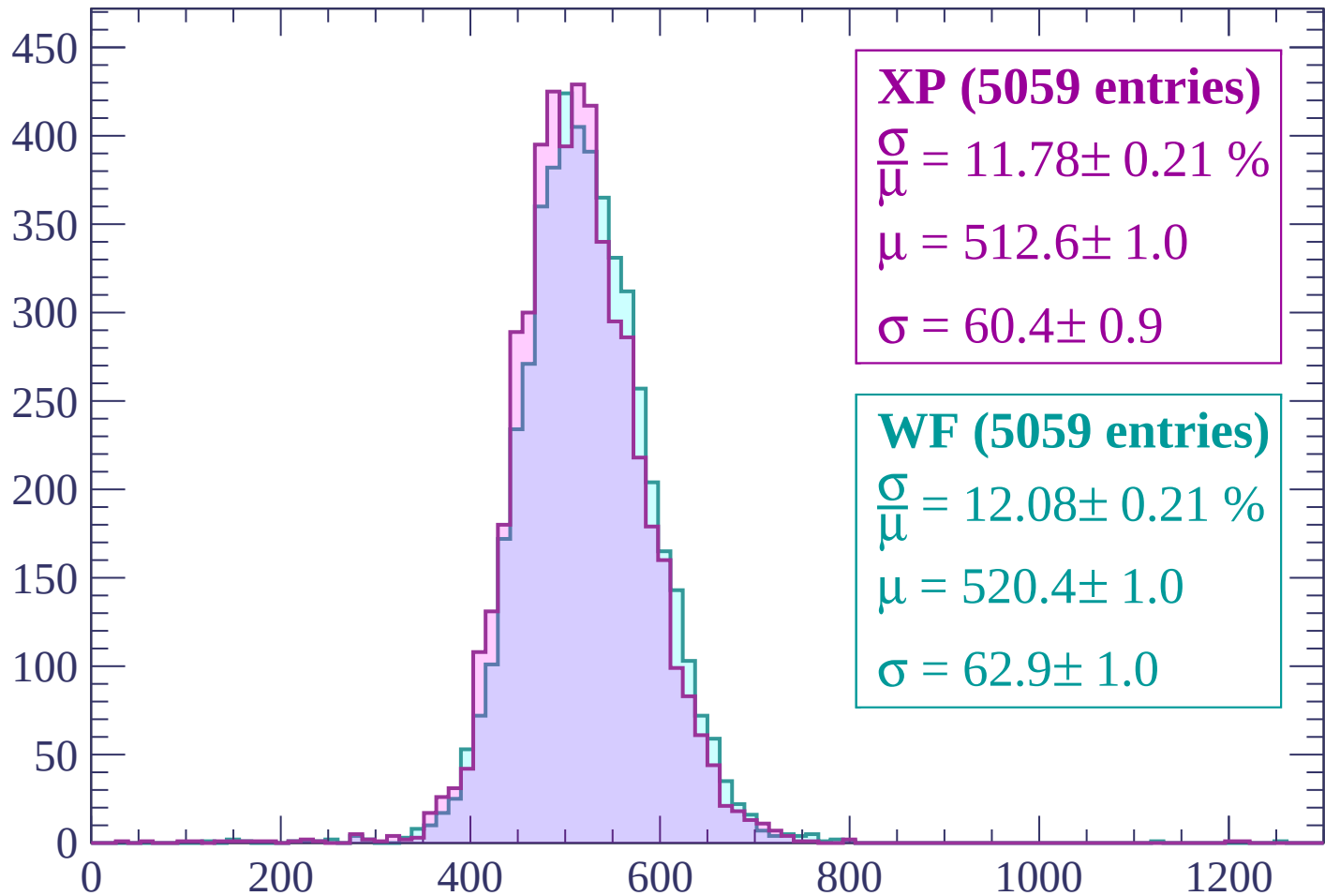
Energy loss | $-2640 < p < -2580$

Count



Energy loss | $-2580 < p < -2520$

Count



XP (5059 entries)

$$\frac{\sigma}{\mu} = 11.78 \pm 0.21 \%$$

$$\mu = 512.6 \pm 1.0$$

$$\sigma = 60.4 \pm 0.9$$

WF (5059 entries)

$$\frac{\sigma}{\mu} = 12.08 \pm 0.21 \%$$

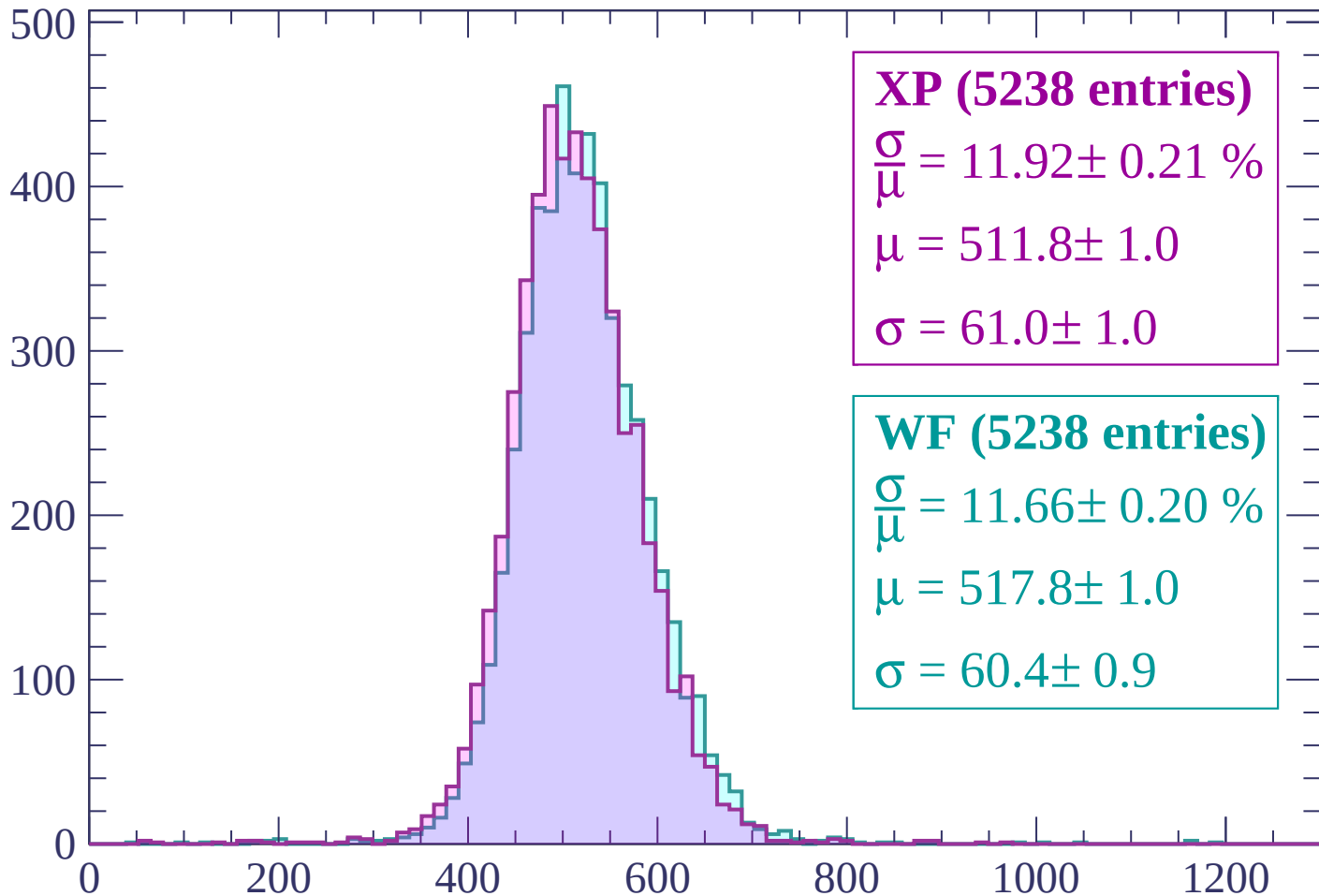
$$\mu = 520.4 \pm 1.0$$

$$\sigma = 62.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | -2520 < p < -2460

Count



XP (5238 entries)

$$\frac{\sigma}{\mu} = 11.92 \pm 0.21 \%$$

$$\mu = 511.8 \pm 1.0$$

$$\sigma = 61.0 \pm 1.0$$

WF (5238 entries)

$$\frac{\sigma}{\mu} = 11.66 \pm 0.20 \%$$

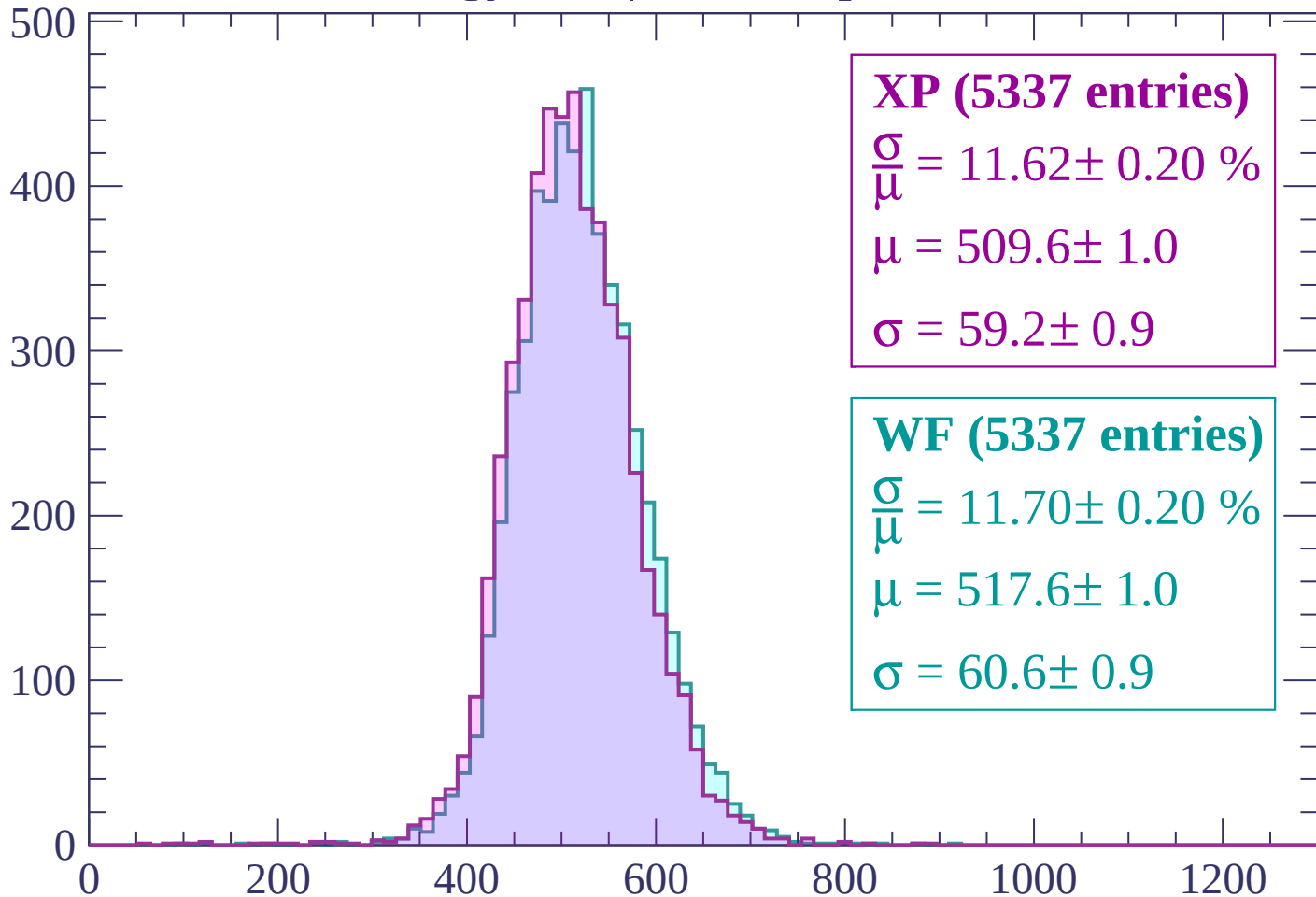
$$\mu = 517.8 \pm 1.0$$

$$\sigma = 60.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-2460 < p < -2400$

Count



XP (5337 entries)

$$\frac{\sigma}{\mu} = 11.62 \pm 0.20 \%$$

$$\mu = 509.6 \pm 1.0$$

$$\sigma = 59.2 \pm 0.9$$

WF (5337 entries)

$$\frac{\sigma}{\mu} = 11.70 \pm 0.20 \%$$

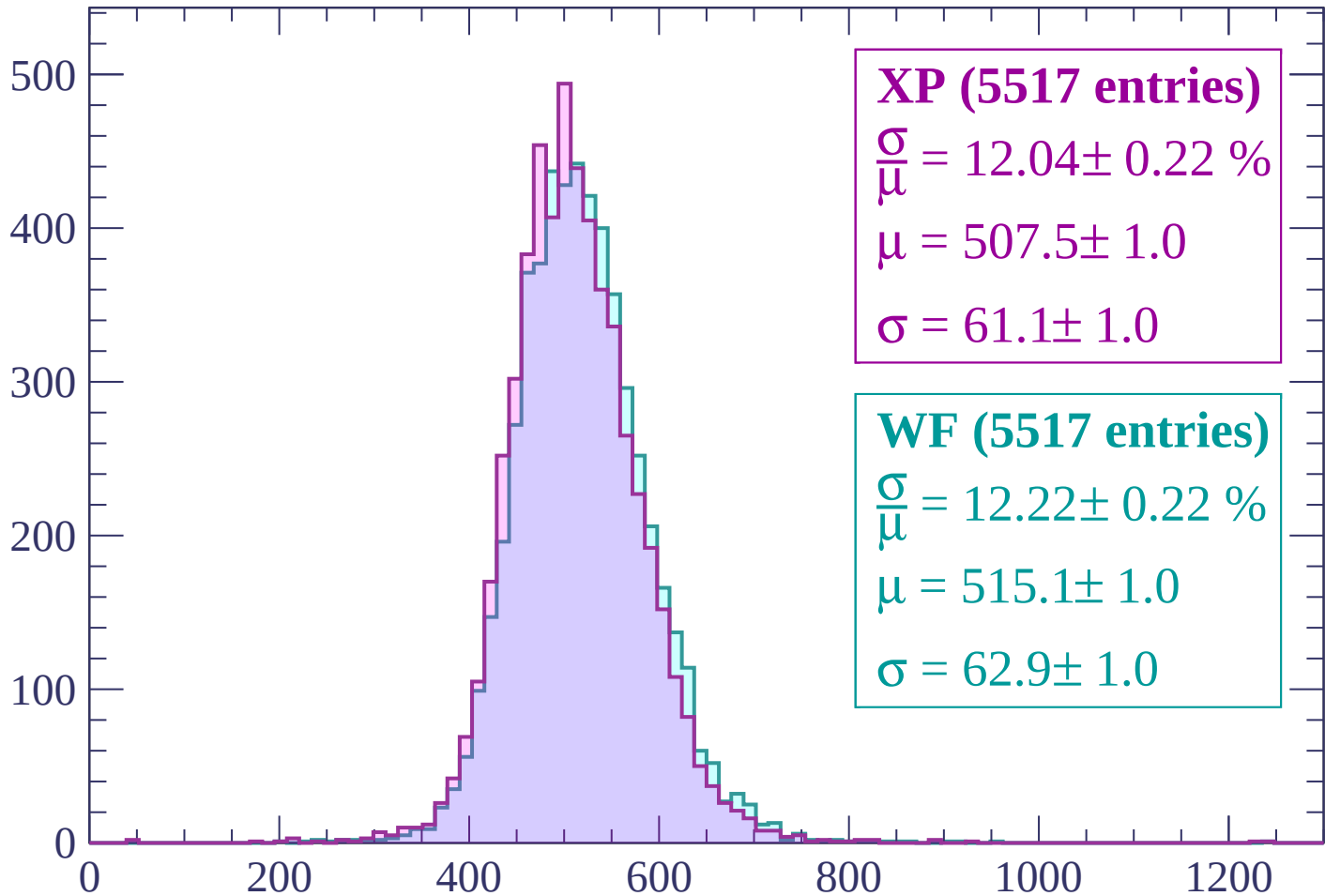
$$\mu = 517.6 \pm 1.0$$

$$\sigma = 60.6 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | -2400 < p < -2340

Count



XP (5517 entries)

$$\frac{\sigma}{\mu} = 12.04 \pm 0.22 \%$$

$$\mu = 507.5 \pm 1.0$$

$$\sigma = 61.1 \pm 1.0$$

WF (5517 entries)

$$\frac{\sigma}{\mu} = 12.22 \pm 0.22 \%$$

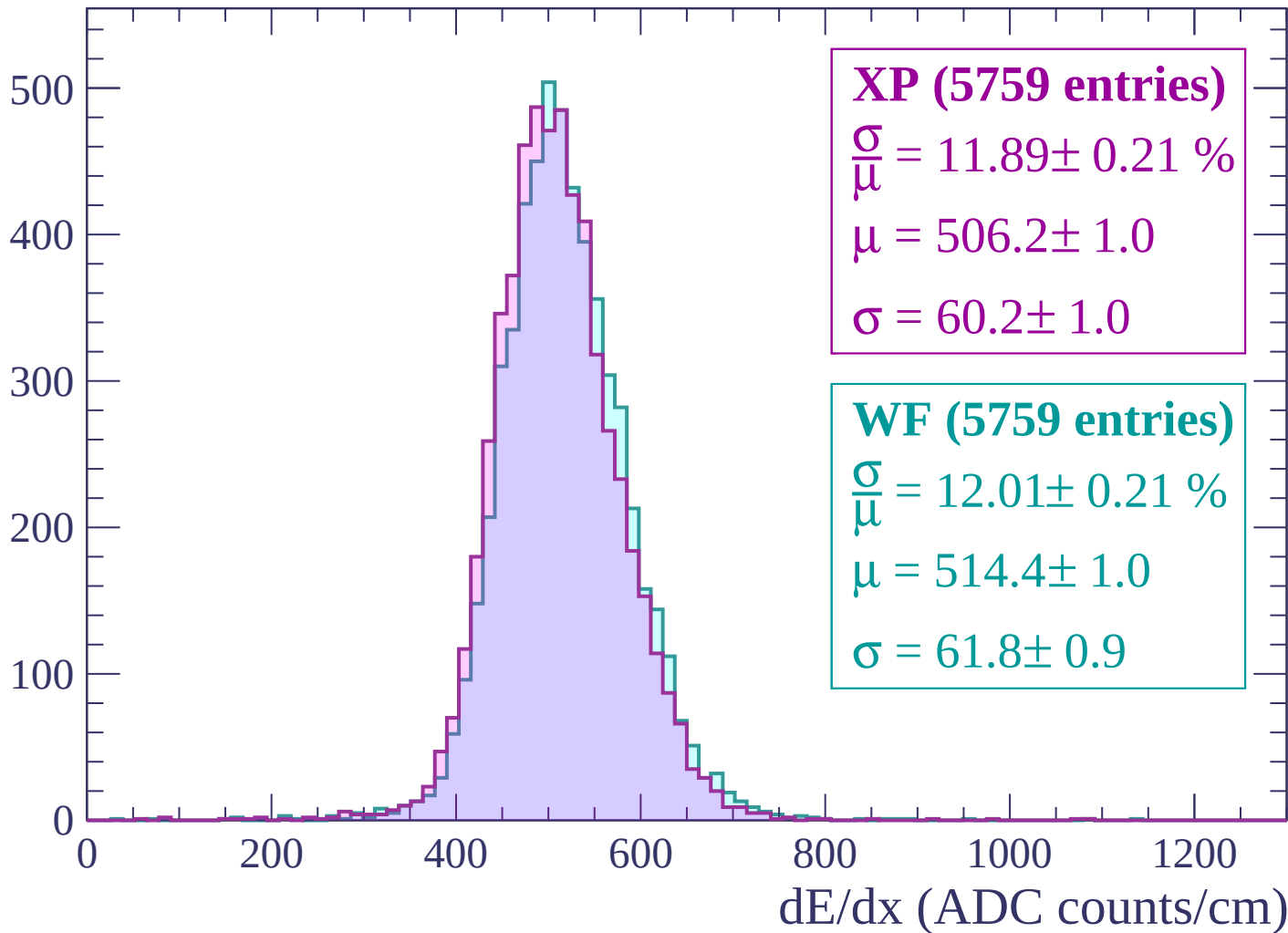
$$\mu = 515.1 \pm 1.0$$

$$\sigma = 62.9 \pm 1.0$$

dE/dx (ADC counts/cm)

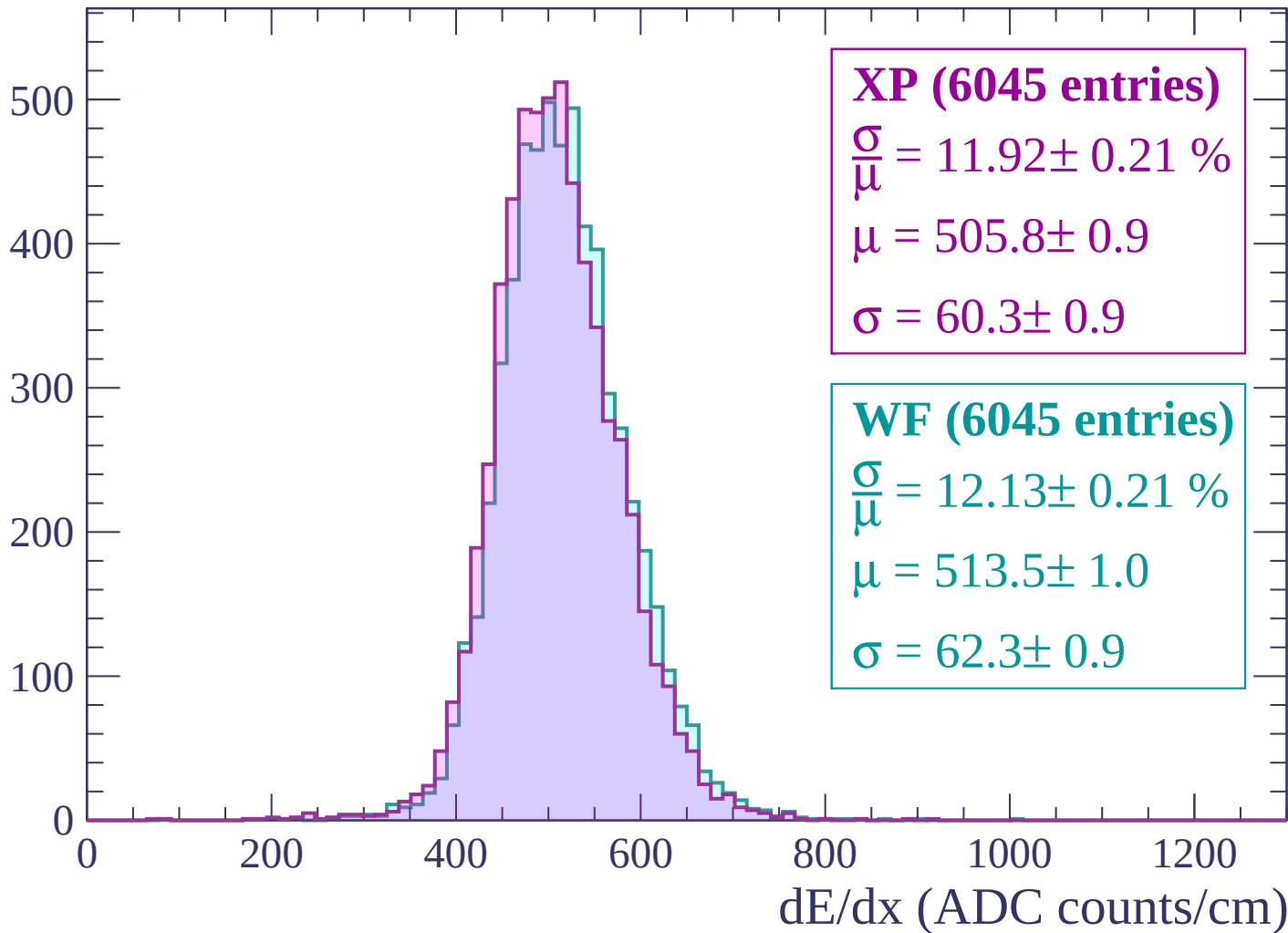
Energy loss | $-2340 < p < -2280$

Count



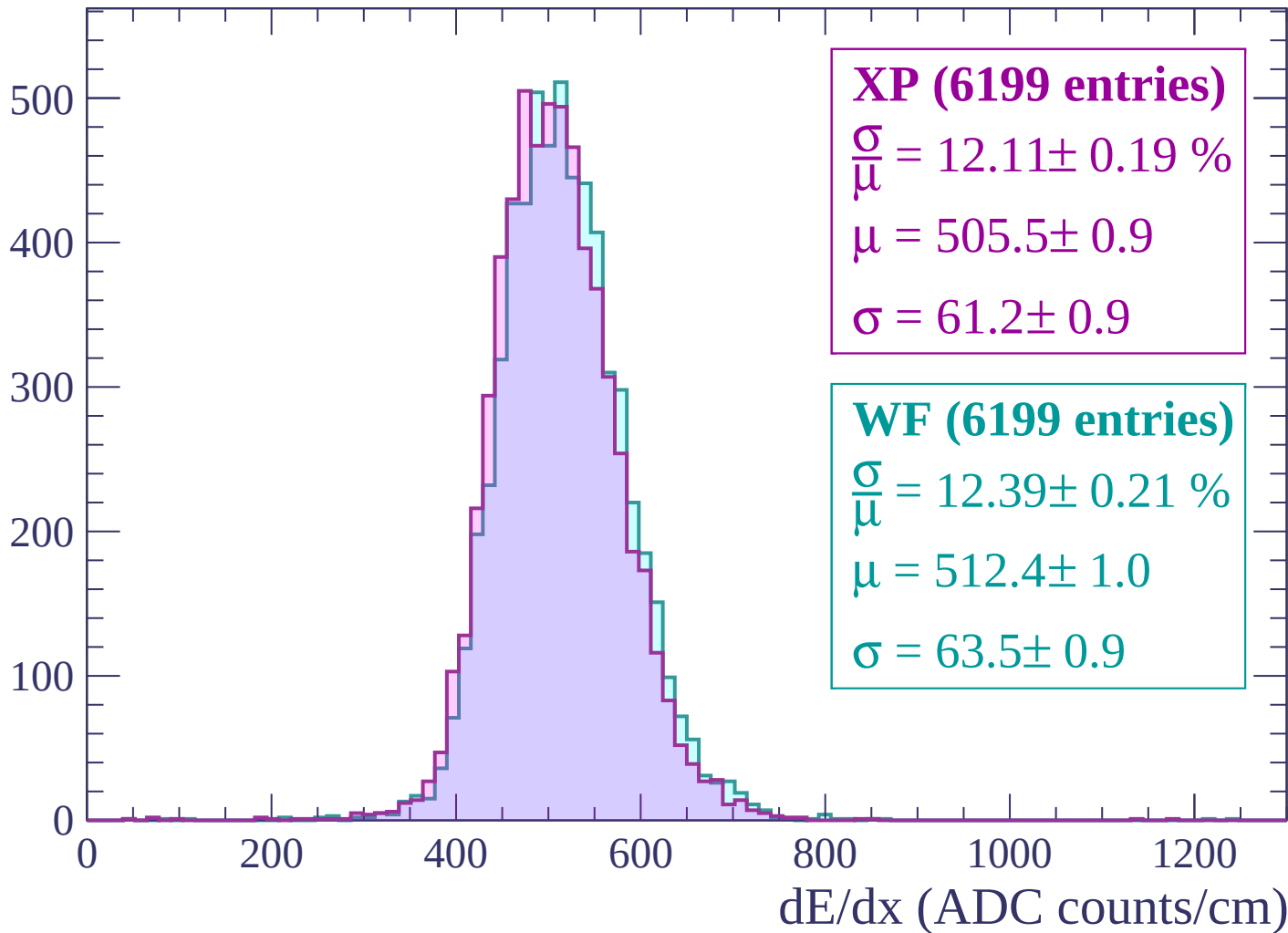
Energy loss | $-2280 < p < -2220$

Count



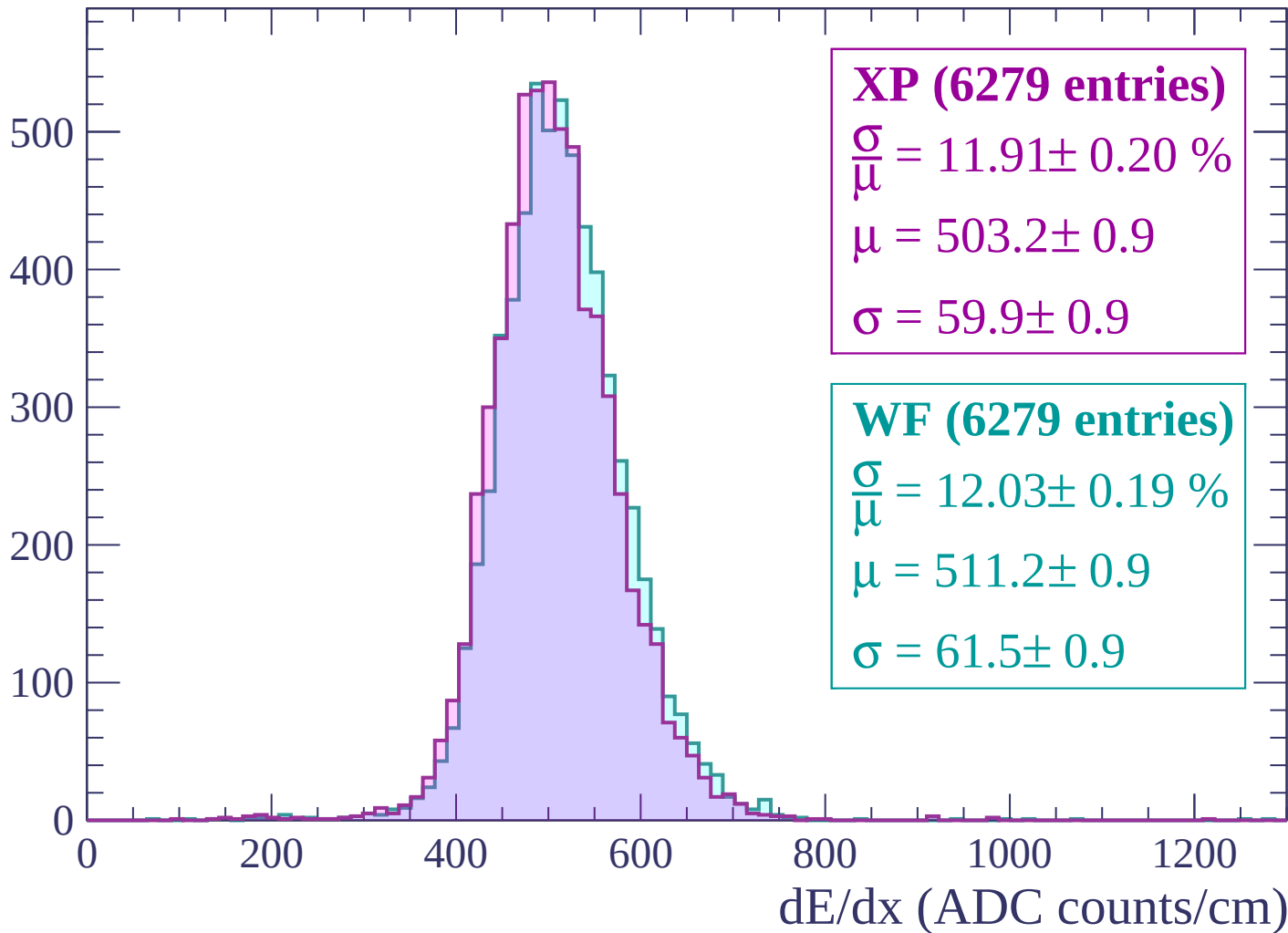
Energy loss | -2220 < p < -2160

Count



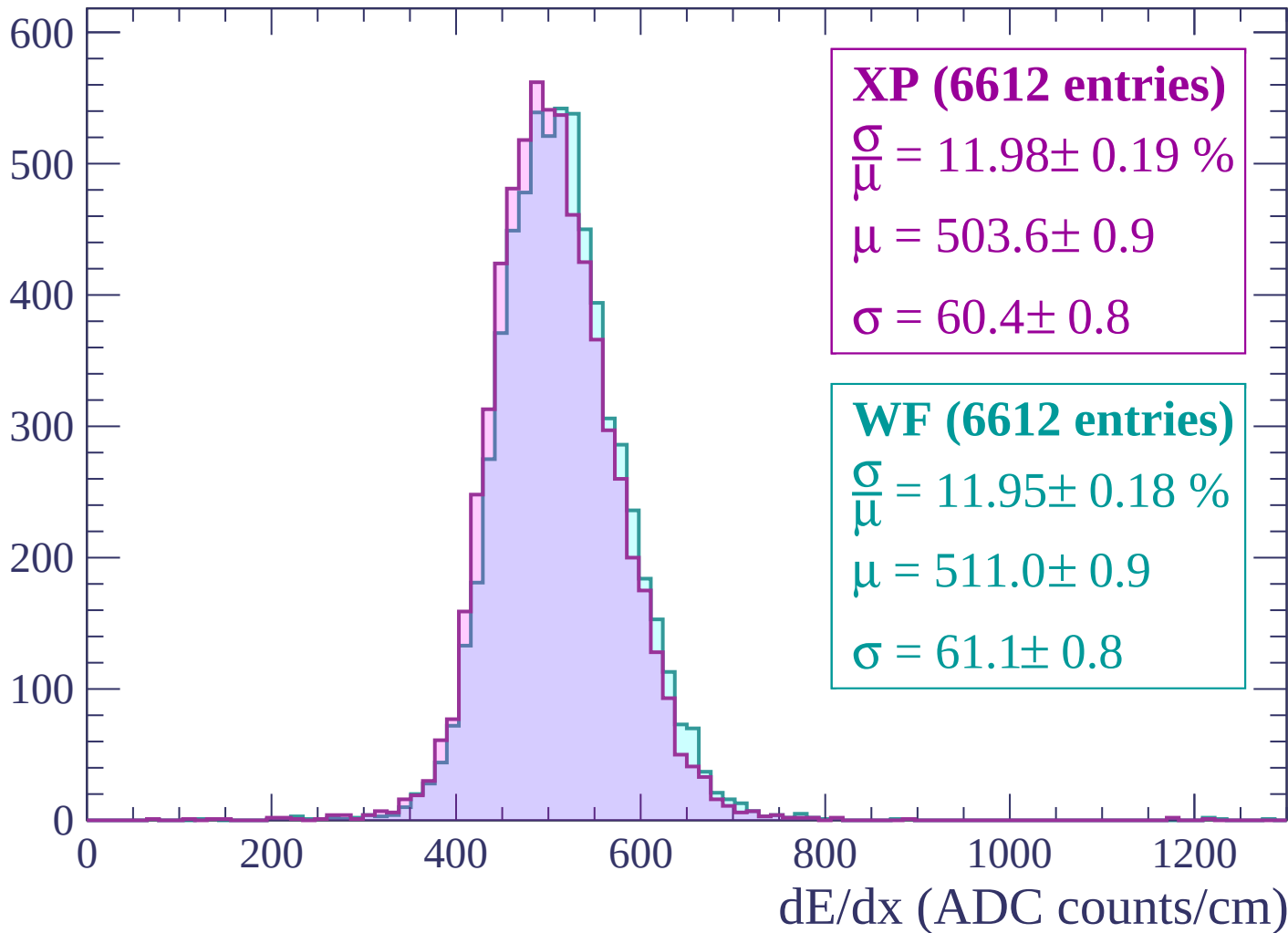
Energy loss | $-2160 < p < -2100$

Count



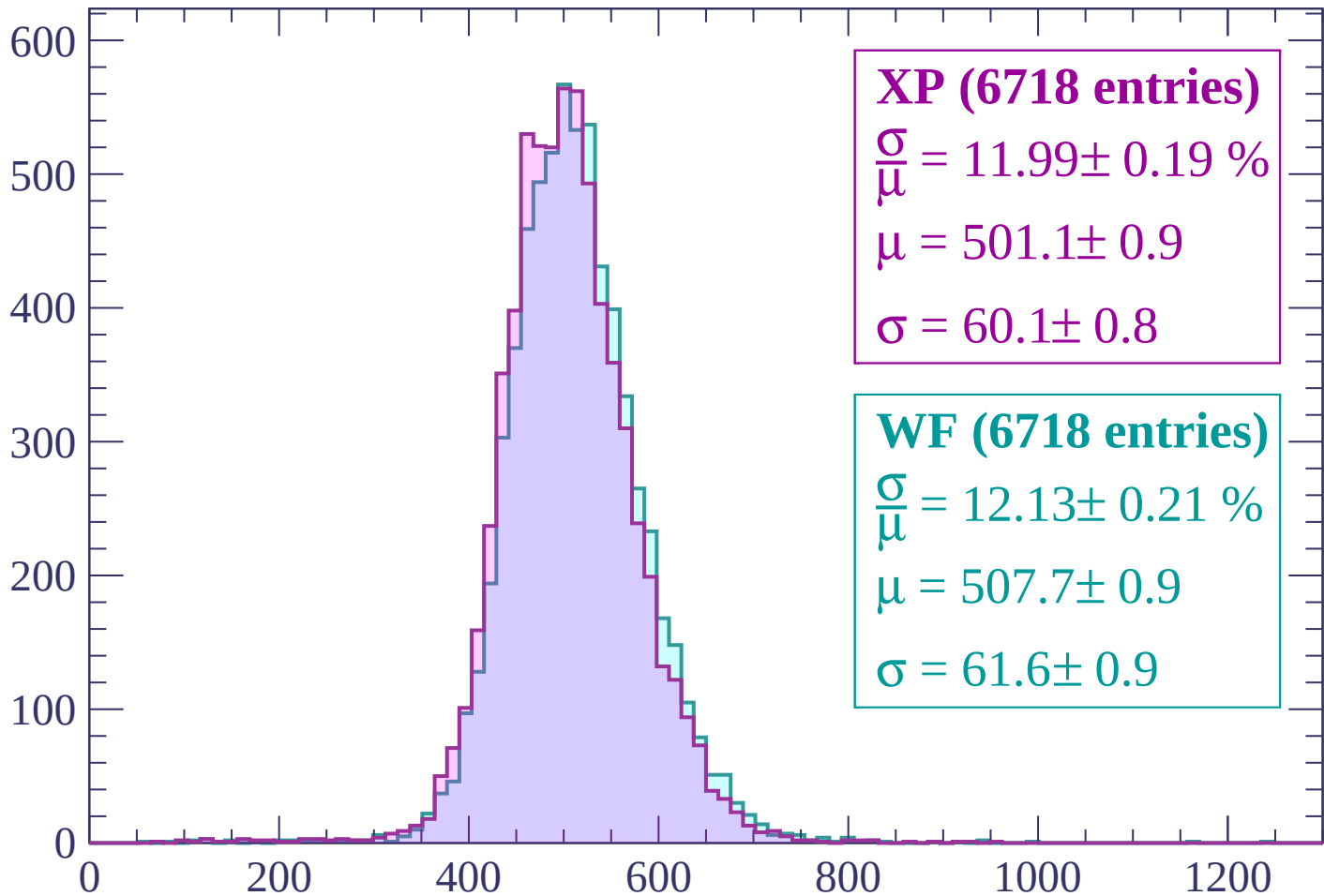
Energy loss | -2100 < p < -2040

Count



Energy loss | $-2040 < p < -1980$

Count



XP (6718 entries)

$$\frac{\sigma}{\mu} = 11.99 \pm 0.19 \%$$

$$\mu = 501.1 \pm 0.9$$

$$\sigma = 60.1 \pm 0.8$$

WF (6718 entries)

$$\frac{\sigma}{\mu} = 12.13 \pm 0.21 \%$$

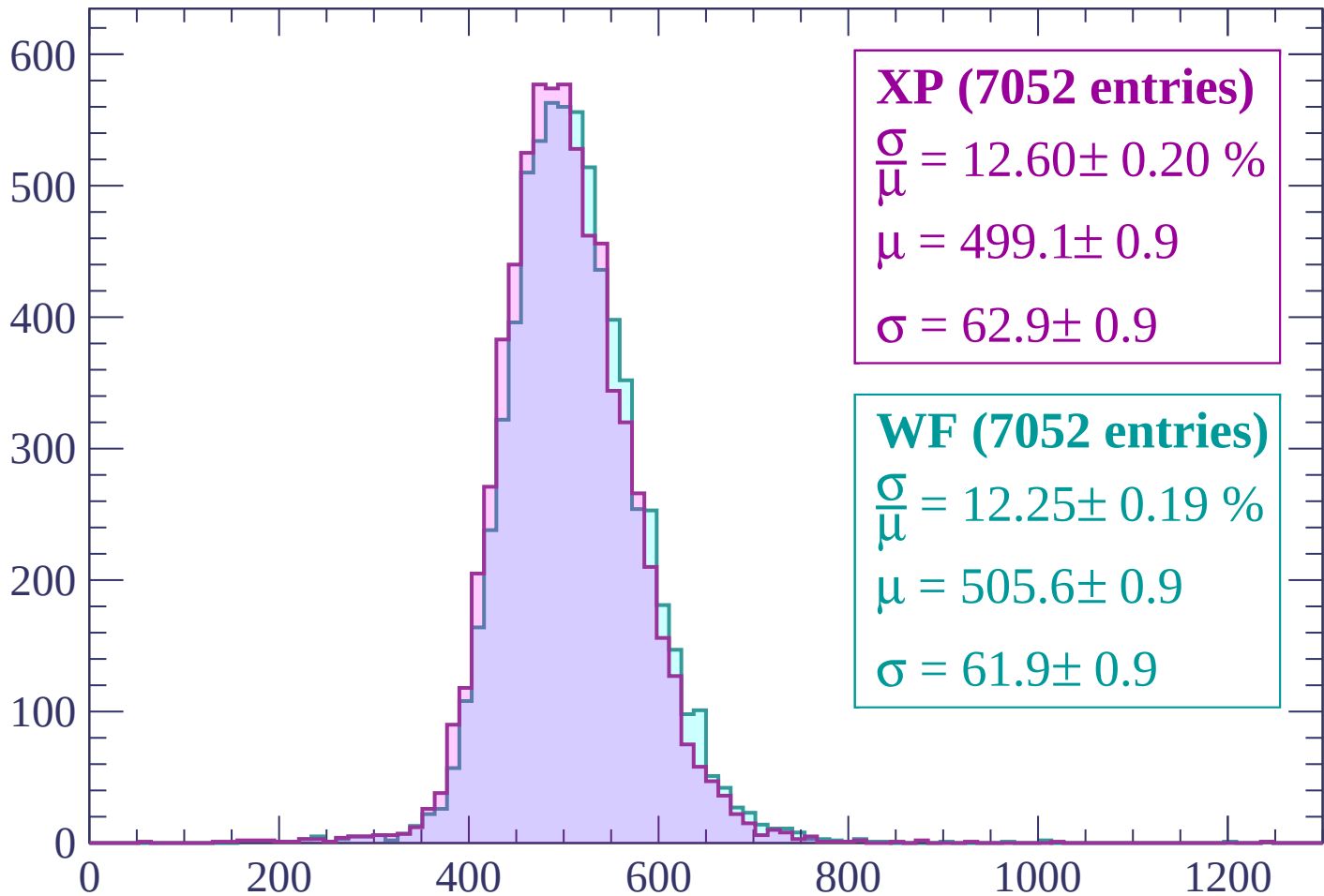
$$\mu = 507.7 \pm 0.9$$

$$\sigma = 61.6 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | -1980 < p < -1920

Count



XP (7052 entries)

$$\frac{\sigma}{\mu} = 12.60 \pm 0.20 \%$$

$$\mu = 499.1 \pm 0.9$$

$$\sigma = 62.9 \pm 0.9$$

WF (7052 entries)

$$\frac{\sigma}{\mu} = 12.25 \pm 0.19 \%$$

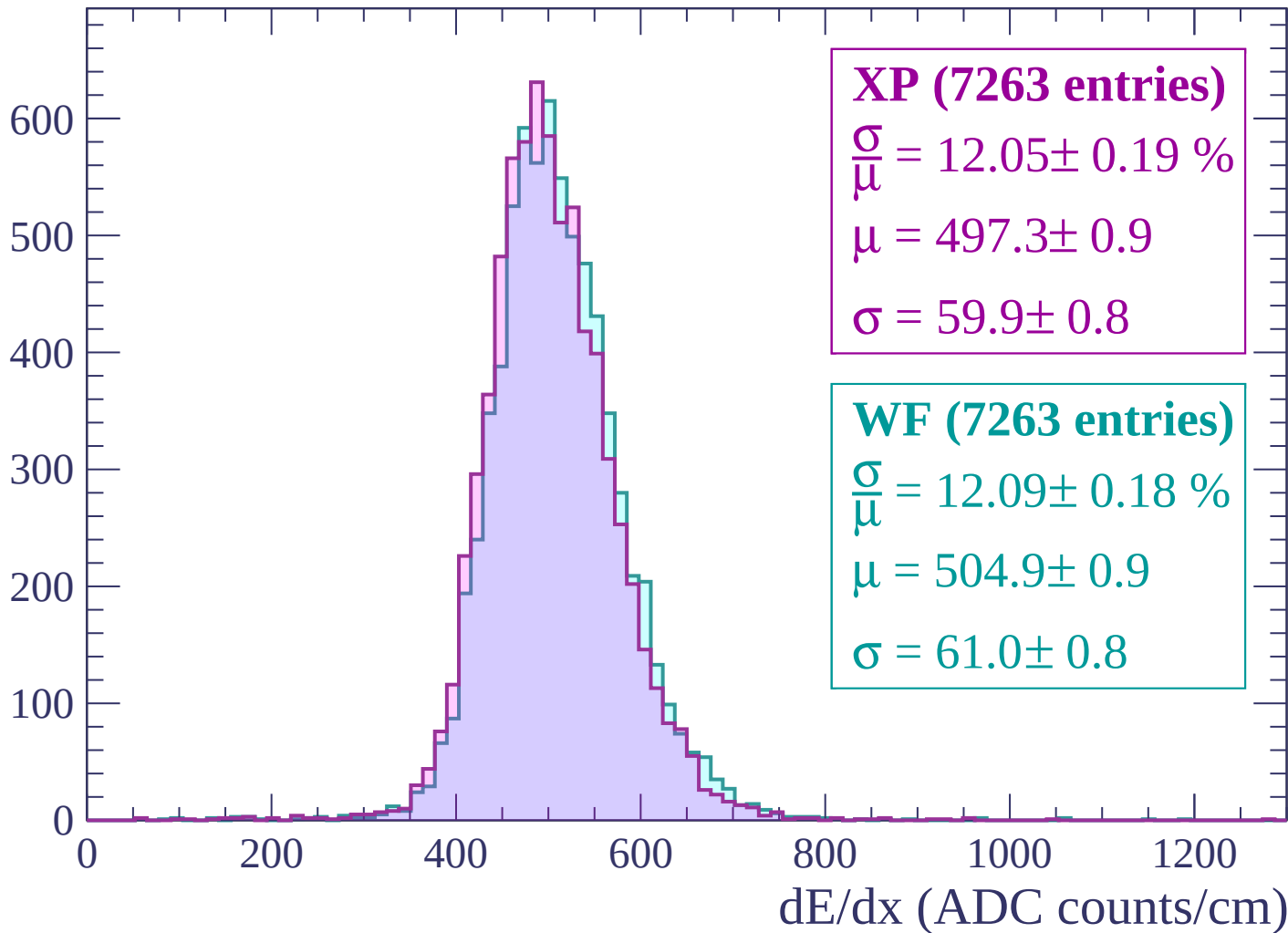
$$\mu = 505.6 \pm 0.9$$

$$\sigma = 61.9 \pm 0.9$$

dE/dx (ADC counts/cm)

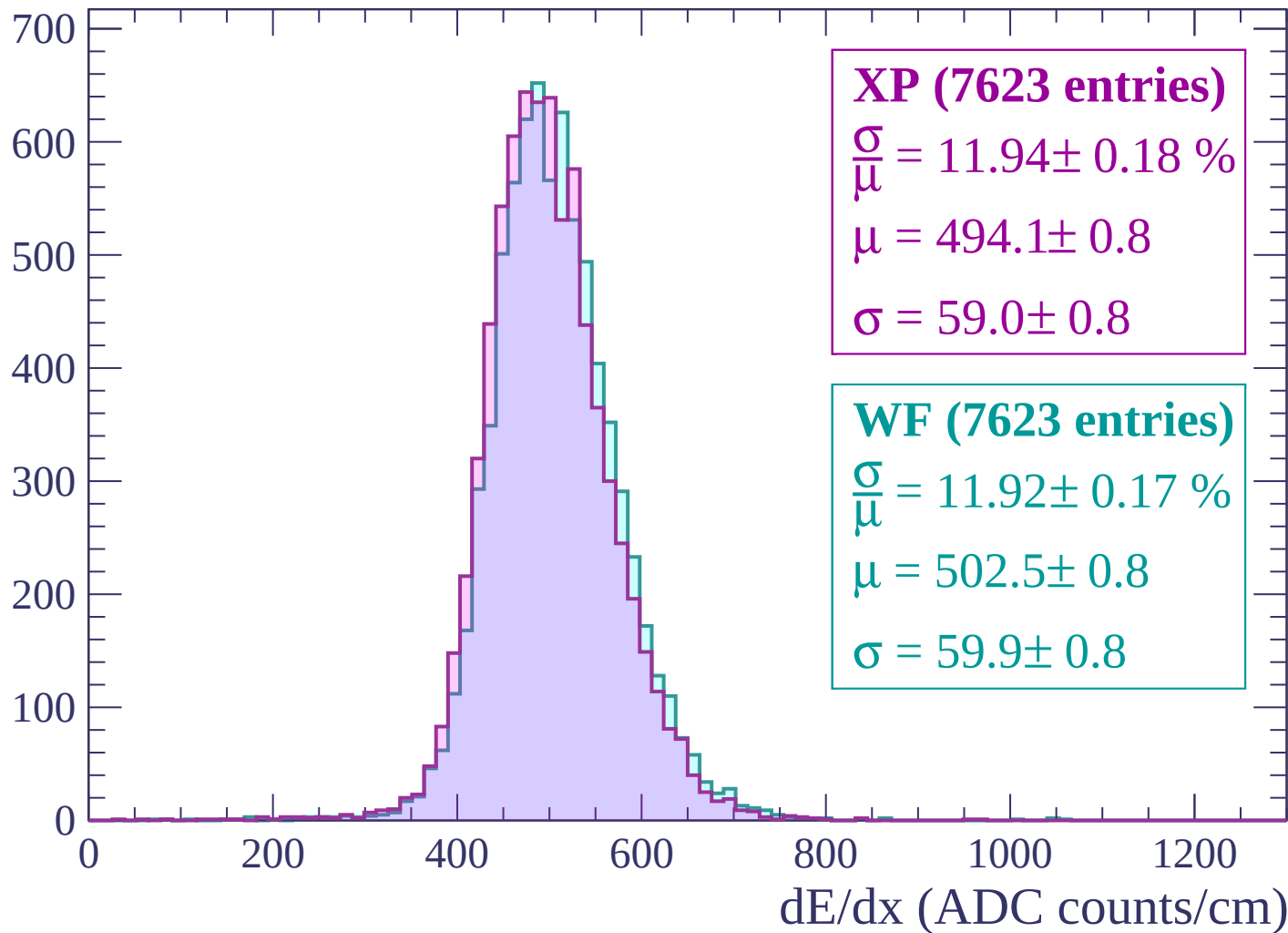
Energy loss | -1920 < p < -1860

Count



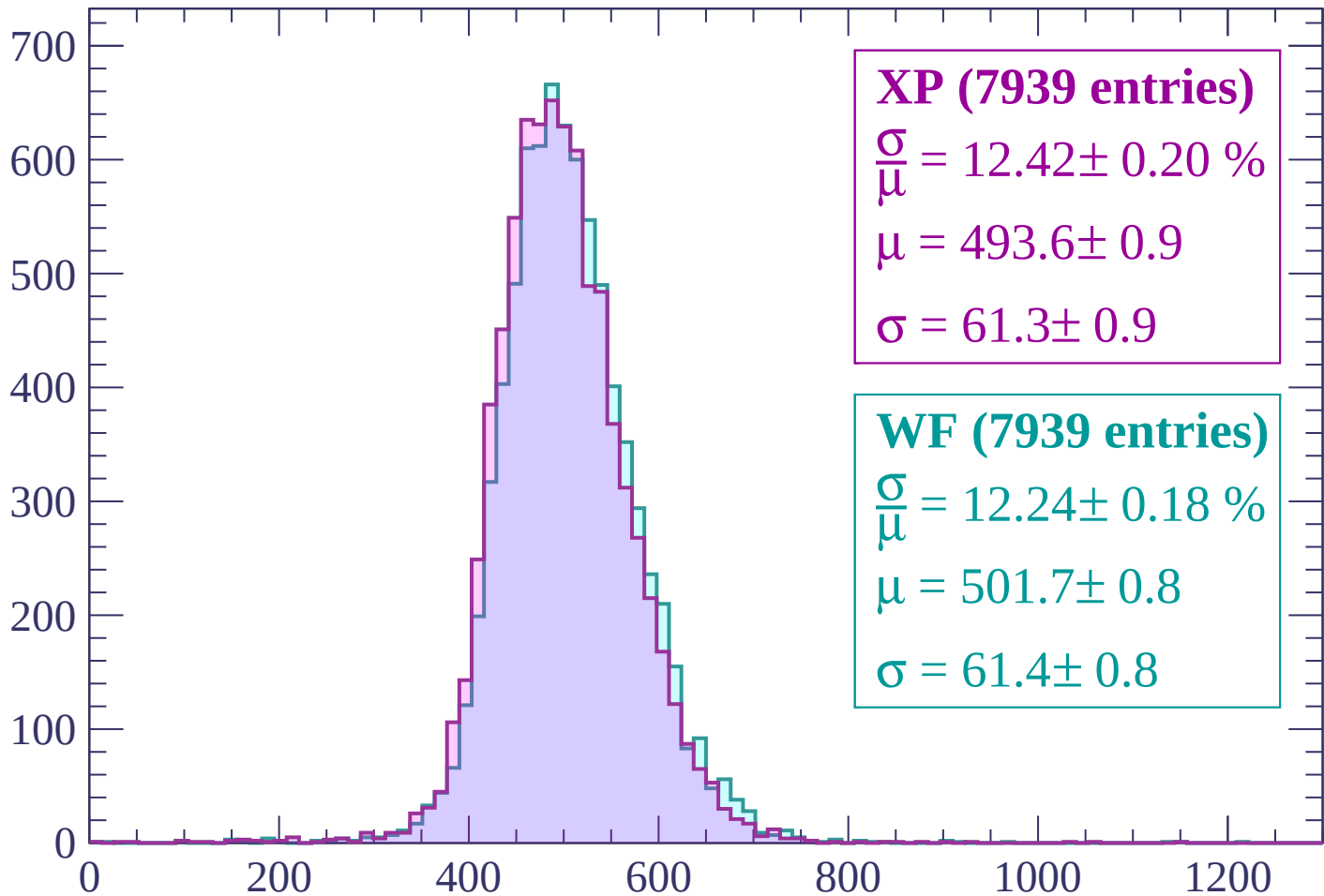
Energy loss | -1860 < p < -1800

Count



Energy loss | -1800 < p < -1740

Count



XP (7939 entries)

$$\frac{\sigma}{\mu} = 12.42 \pm 0.20 \%$$

$$\mu = 493.6 \pm 0.9$$

$$\sigma = 61.3 \pm 0.9$$

WF (7939 entries)

$$\frac{\sigma}{\mu} = 12.24 \pm 0.18 \%$$

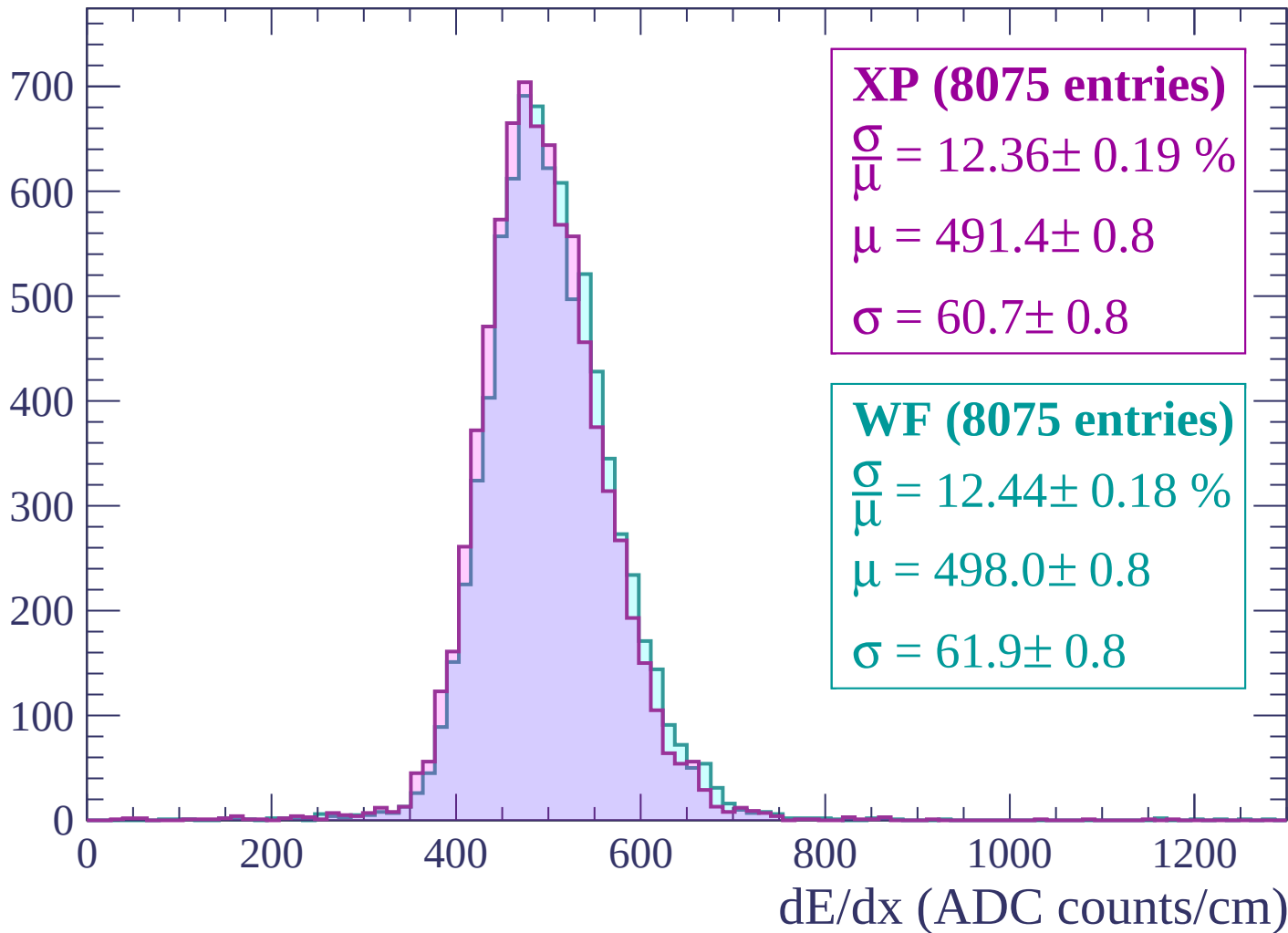
$$\mu = 501.7 \pm 0.8$$

$$\sigma = 61.4 \pm 0.8$$

dE/dx (ADC counts/cm)

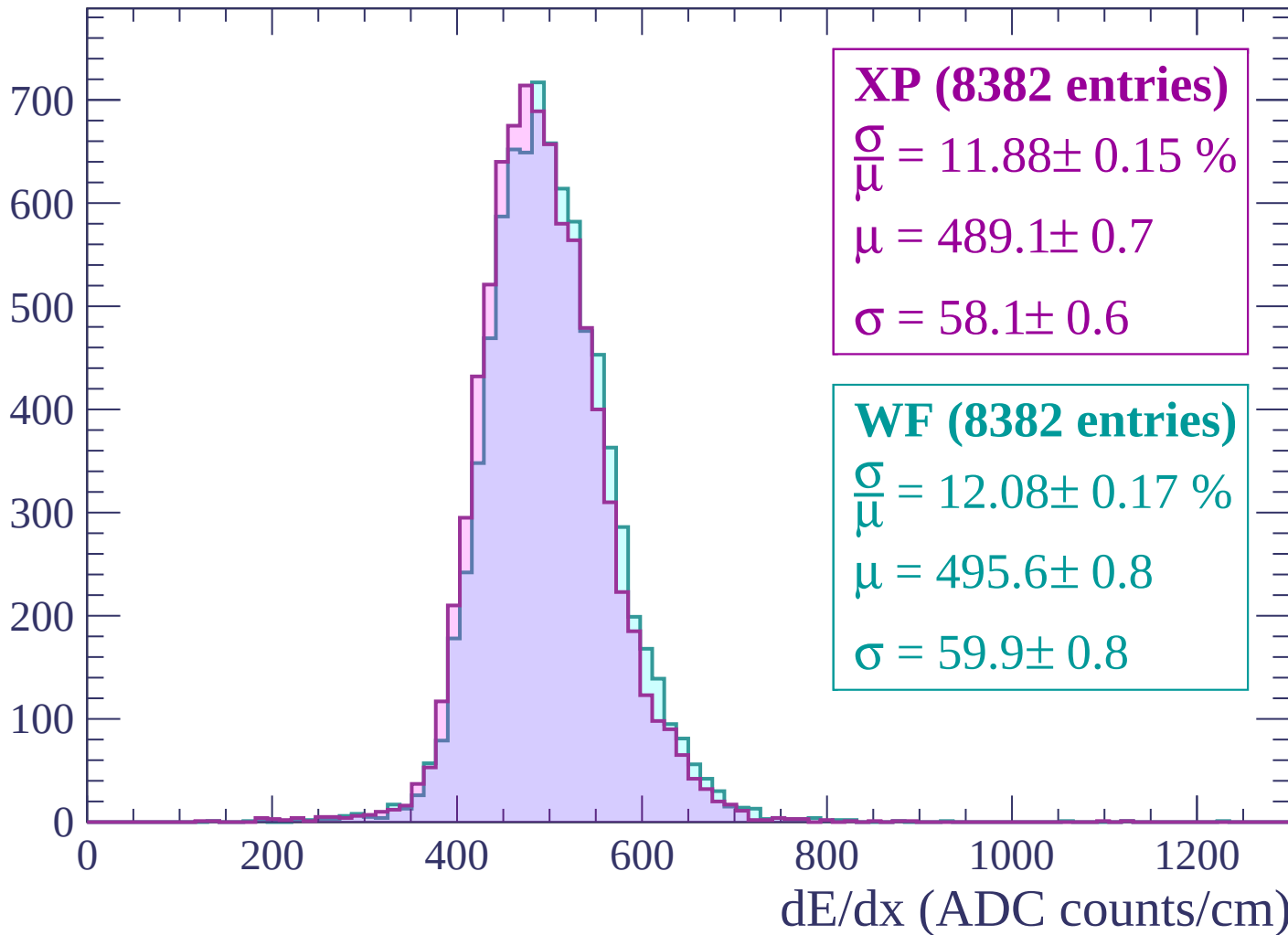
Energy loss | $-1740 < p < -1680$

Count



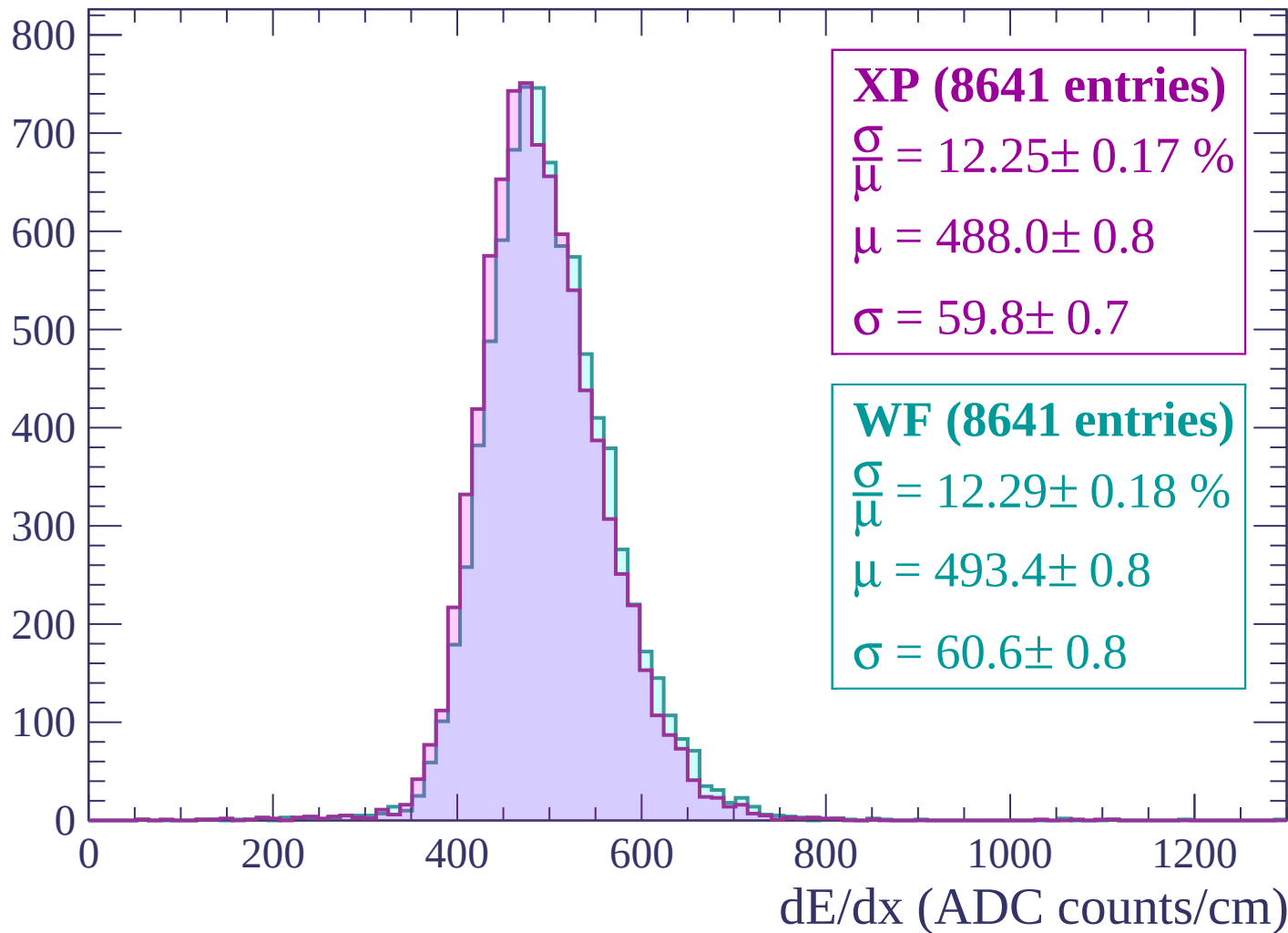
Energy loss | $-1680 < p < -1620$

Count



Energy loss | -1620 < p < -1560

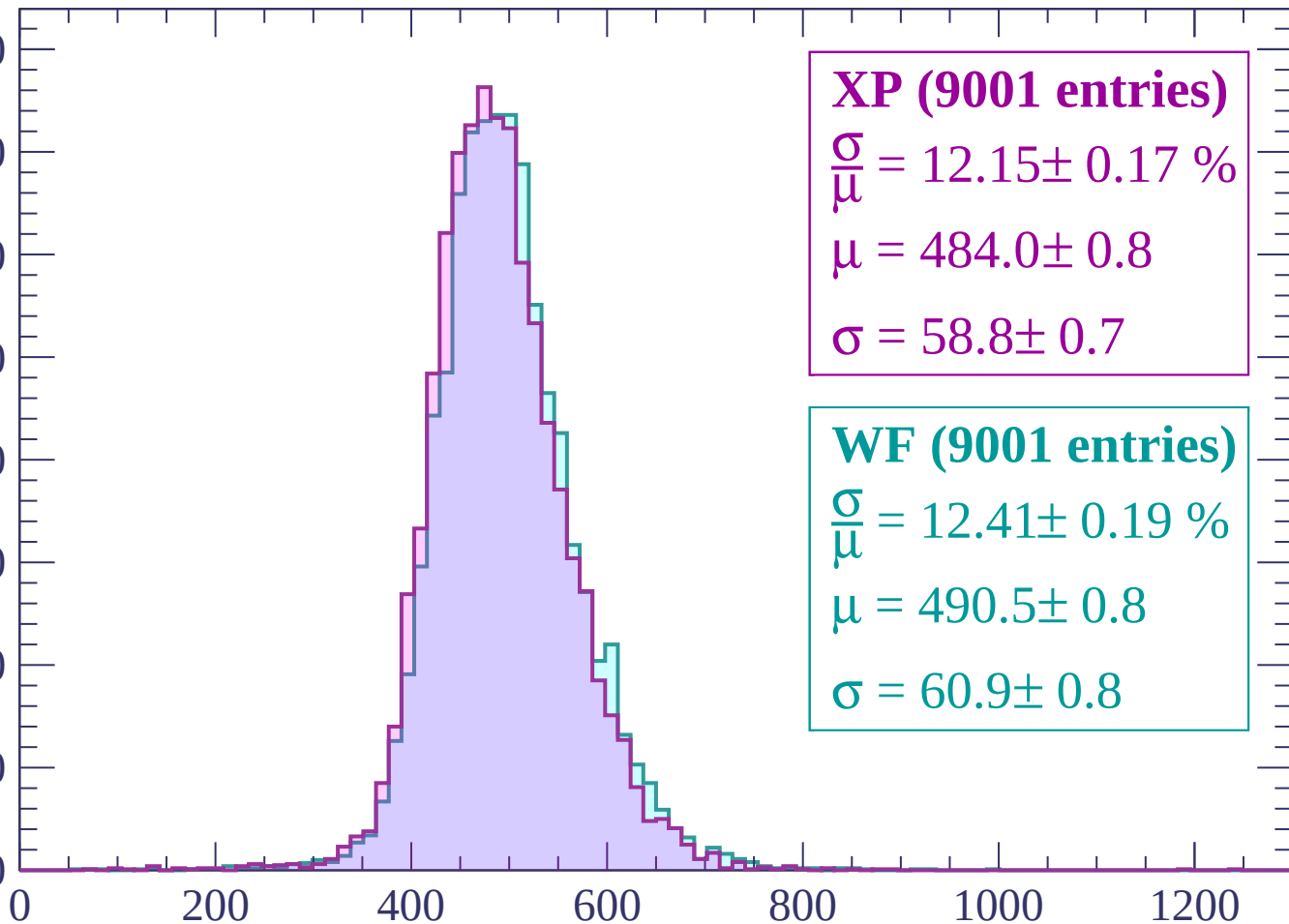
Count



Energy loss | -1560 < p < -1500

Count

800
700
600
500
400
300
200
100
0



XP (9001 entries)

$$\frac{\sigma}{\mu} = 12.15 \pm 0.17 \%$$

$$\mu = 484.0 \pm 0.8$$

$$\sigma = 58.8 \pm 0.7$$

WF (9001 entries)

$$\frac{\sigma}{\mu} = 12.41 \pm 0.19 \%$$

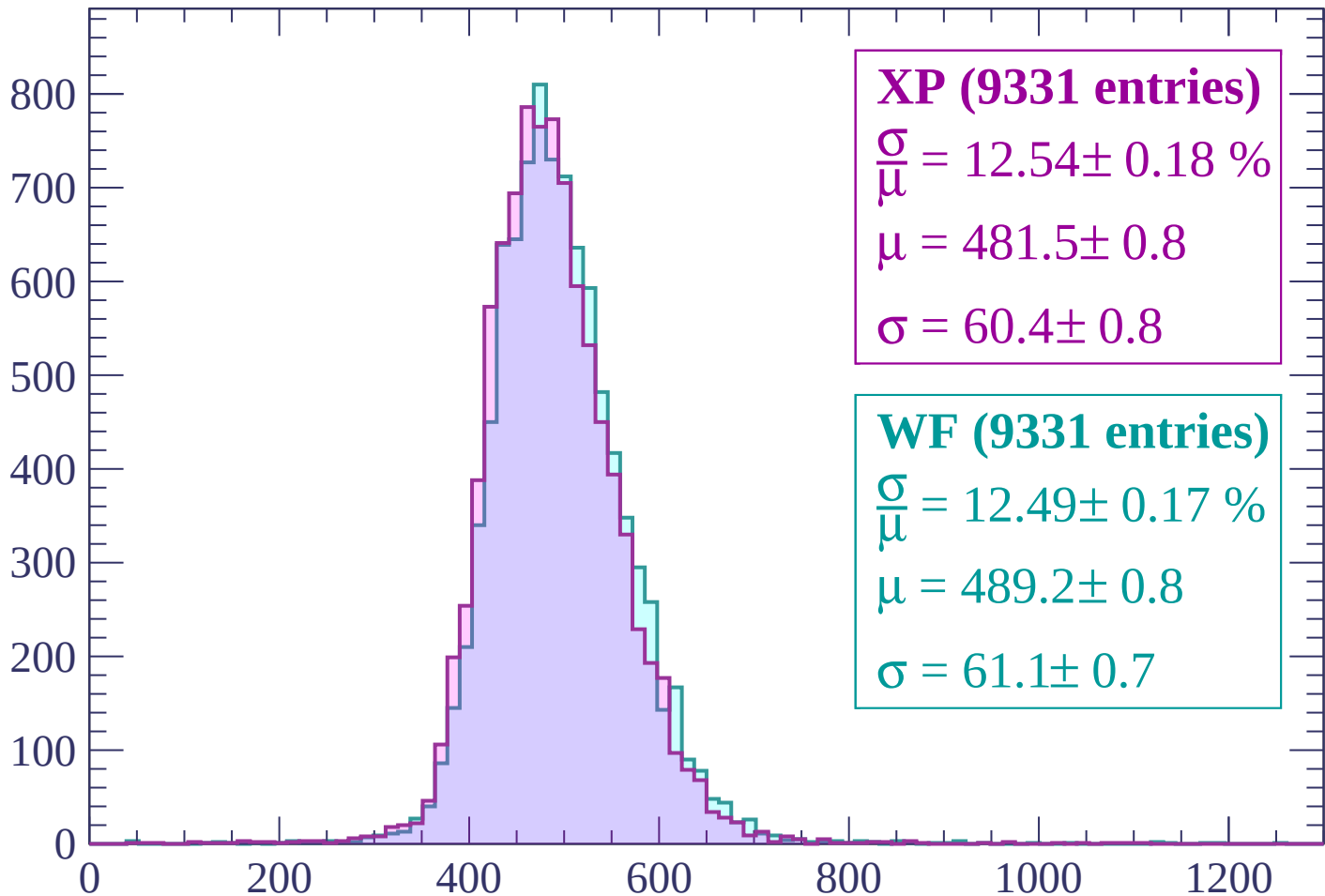
$$\mu = 490.5 \pm 0.8$$

$$\sigma = 60.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1500 < p < -1440$

Count



XP (9331 entries)

$$\frac{\sigma}{\mu} = 12.54 \pm 0.18 \%$$

$$\mu = 481.5 \pm 0.8$$

$$\sigma = 60.4 \pm 0.8$$

WF (9331 entries)

$$\frac{\sigma}{\mu} = 12.49 \pm 0.17 \%$$

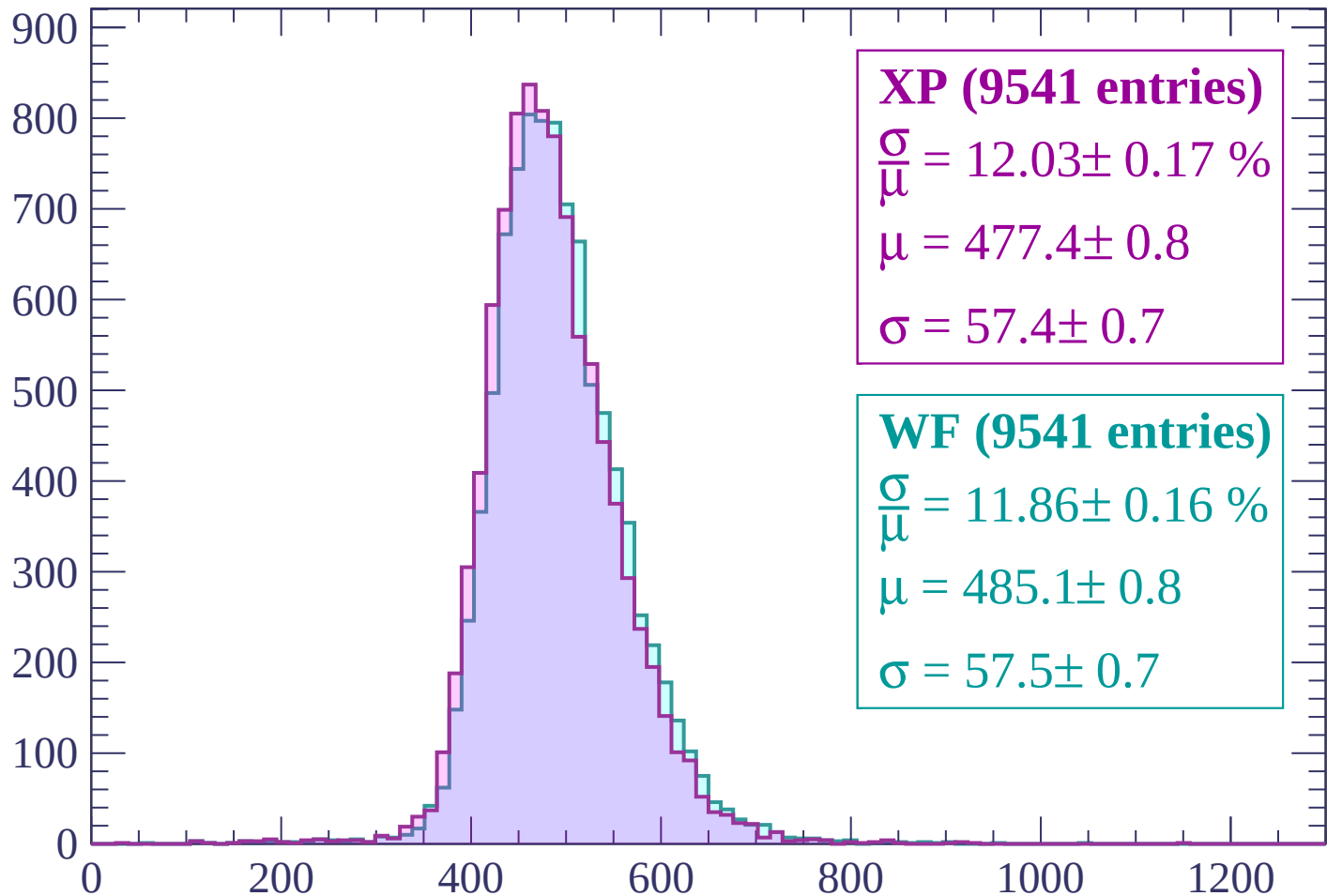
$$\mu = 489.2 \pm 0.8$$

$$\sigma = 61.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1440 < p < -1380$

Count



XP (9541 entries)

$$\frac{\sigma}{\mu} = 12.03 \pm 0.17 \%$$

$$\mu = 477.4 \pm 0.8$$

$$\sigma = 57.4 \pm 0.7$$

WF (9541 entries)

$$\frac{\sigma}{\mu} = 11.86 \pm 0.16 \%$$

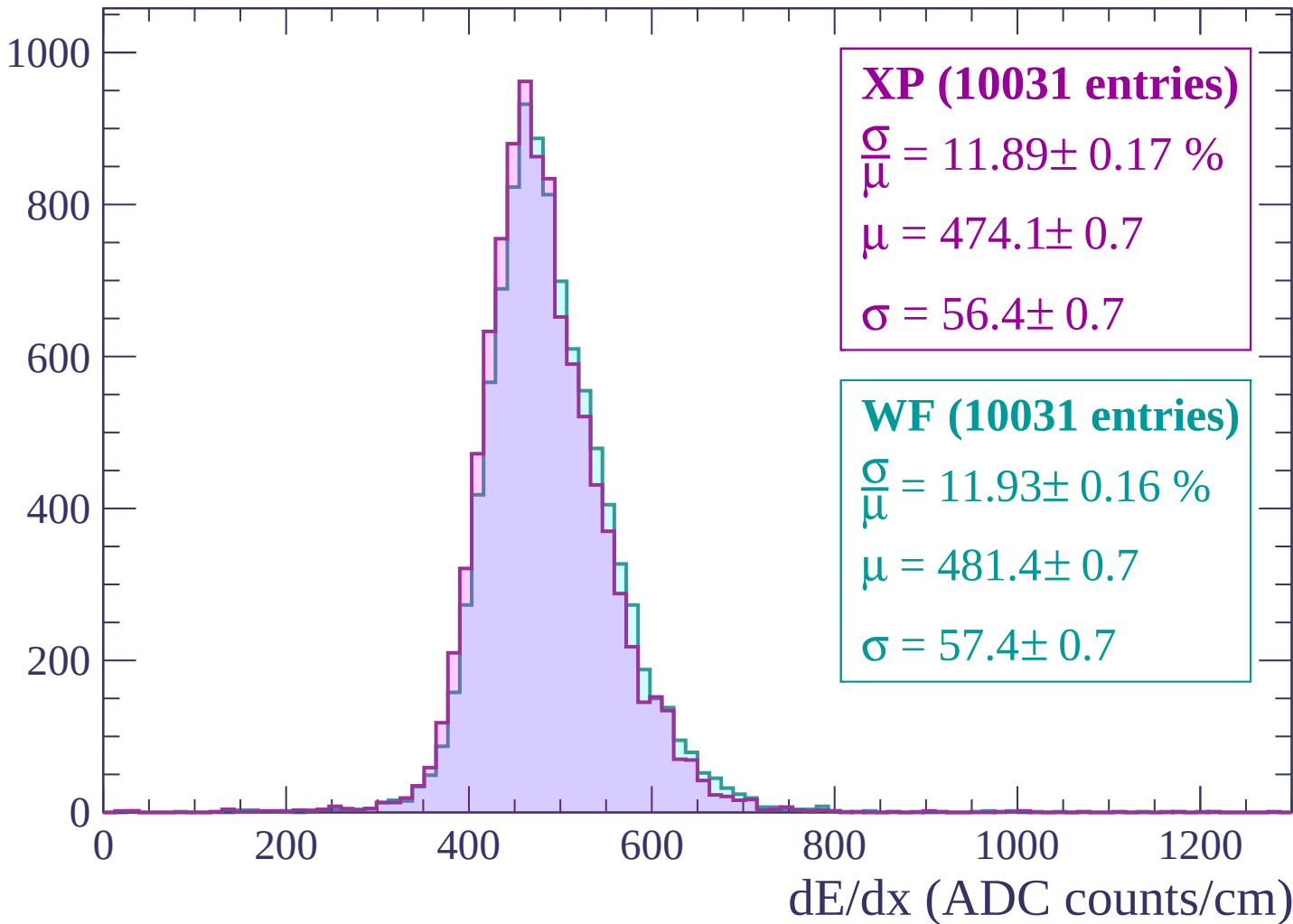
$$\mu = 485.1 \pm 0.8$$

$$\sigma = 57.5 \pm 0.7$$

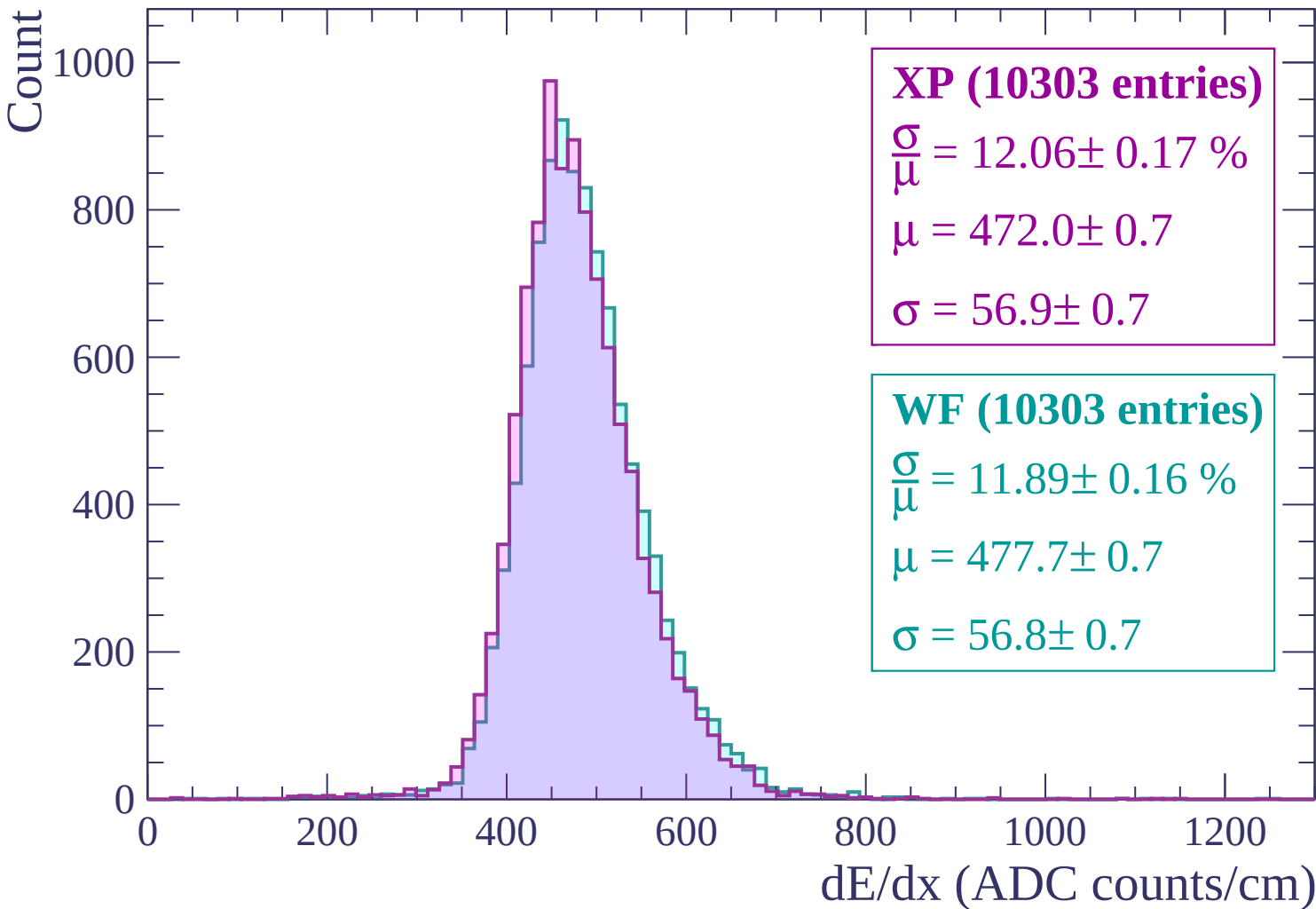
dE/dx (ADC counts/cm)

Energy loss | $-1380 < p < -1320$

Count



Energy loss | $-1320 < p < -1260$



Energy loss | $-1260 < p < -1200$

Count

1000

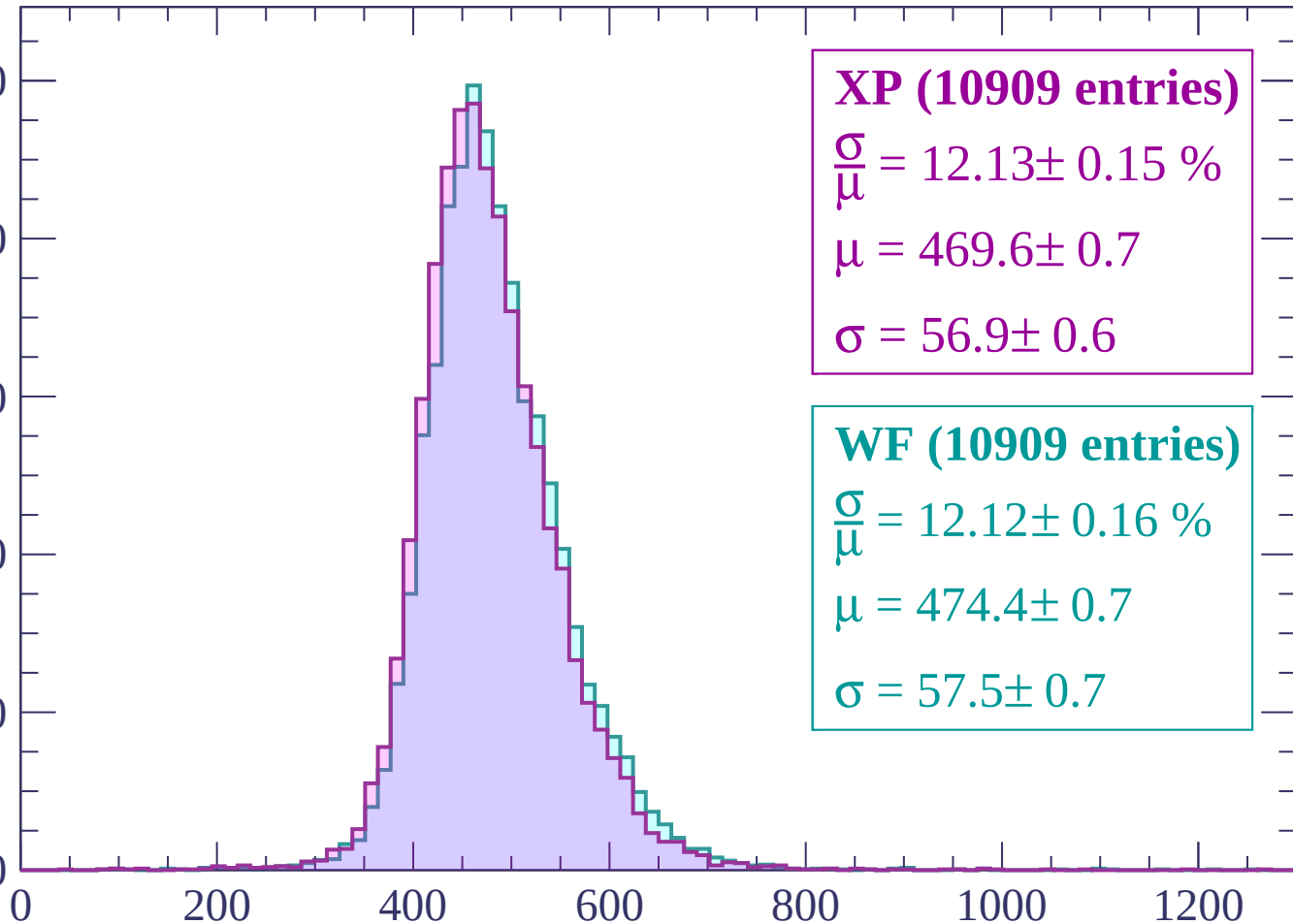
800

600

400

200

0



XP (10909 entries)

$\frac{\sigma}{\mu} = 12.13 \pm 0.15 \%$

$\mu = 469.6 \pm 0.7$

$\sigma = 56.9 \pm 0.6$

WF (10909 entries)

$\frac{\sigma}{\mu} = 12.12 \pm 0.16 \%$

$\mu = 474.4 \pm 0.7$

$\sigma = 57.5 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | -1200 < p < -1140

Count

1000

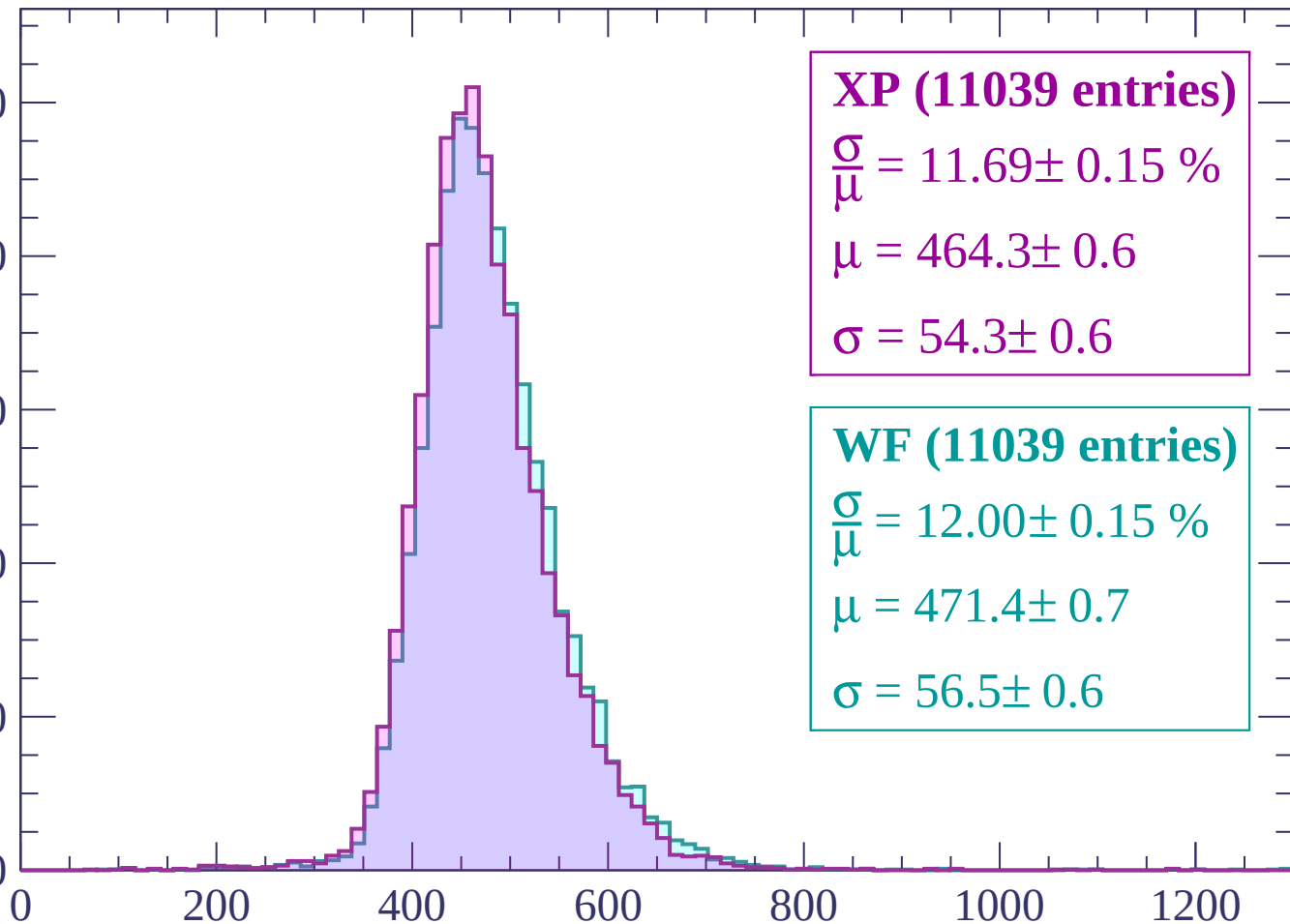
800

600

400

200

0



XP (11039 entries)

$\frac{\sigma}{\mu} = 11.69 \pm 0.15 \%$

$\mu = 464.3 \pm 0.6$

$\sigma = 54.3 \pm 0.6$

WF (11039 entries)

$\frac{\sigma}{\mu} = 12.00 \pm 0.15 \%$

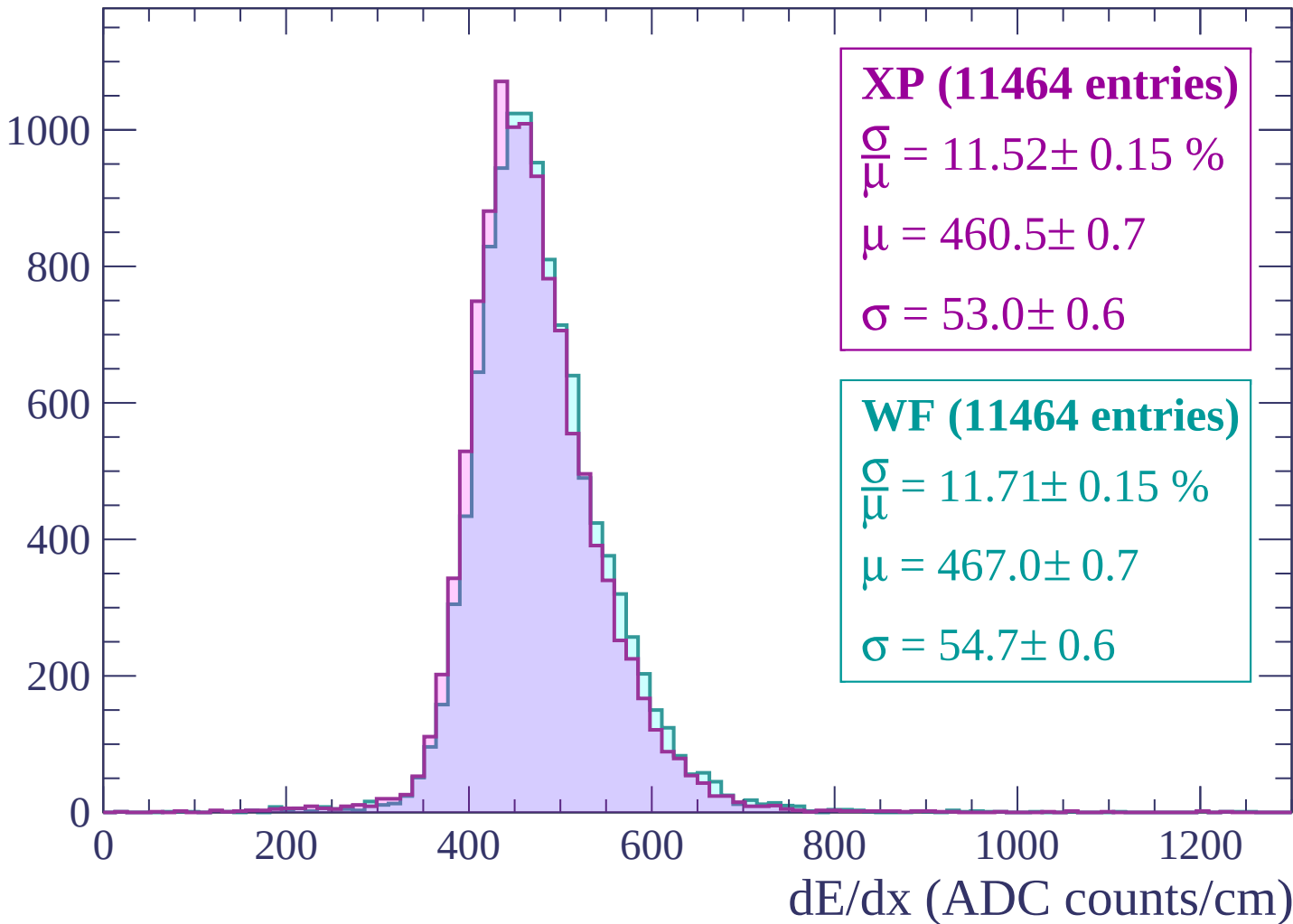
$\mu = 471.4 \pm 0.7$

$\sigma = 56.5 \pm 0.6$

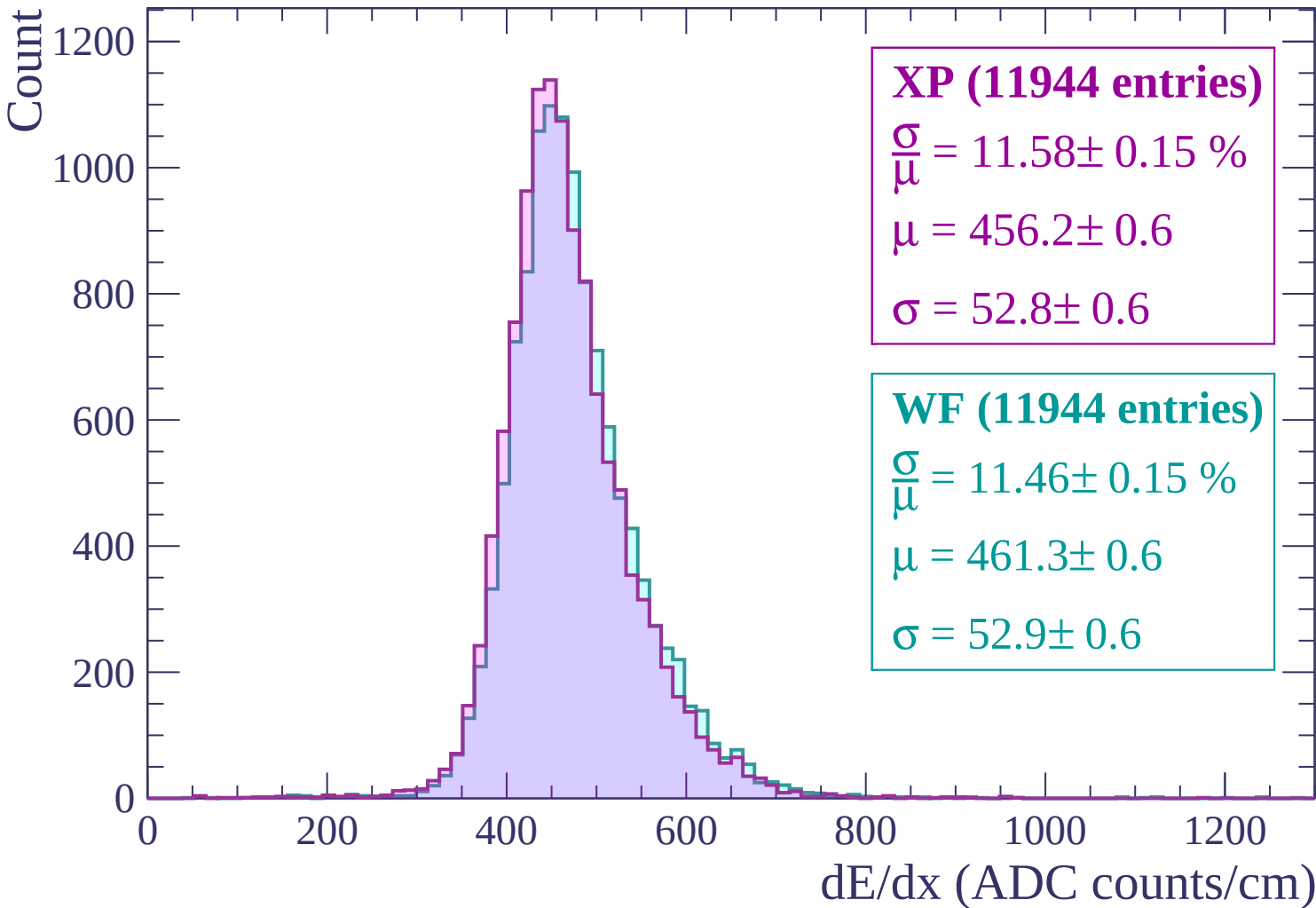
dE/dx (ADC counts/cm)

Energy loss | -1140 < p < -1080

Count



Energy loss | -1080 < p < -1020



Energy loss | $-1020 < p < -960$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (12436 entries)

$\frac{\sigma}{\mu} = 11.16 \pm 0.15 \%$

$\mu = 450.1 \pm 0.6$

$\sigma = 50.2 \pm 0.6$

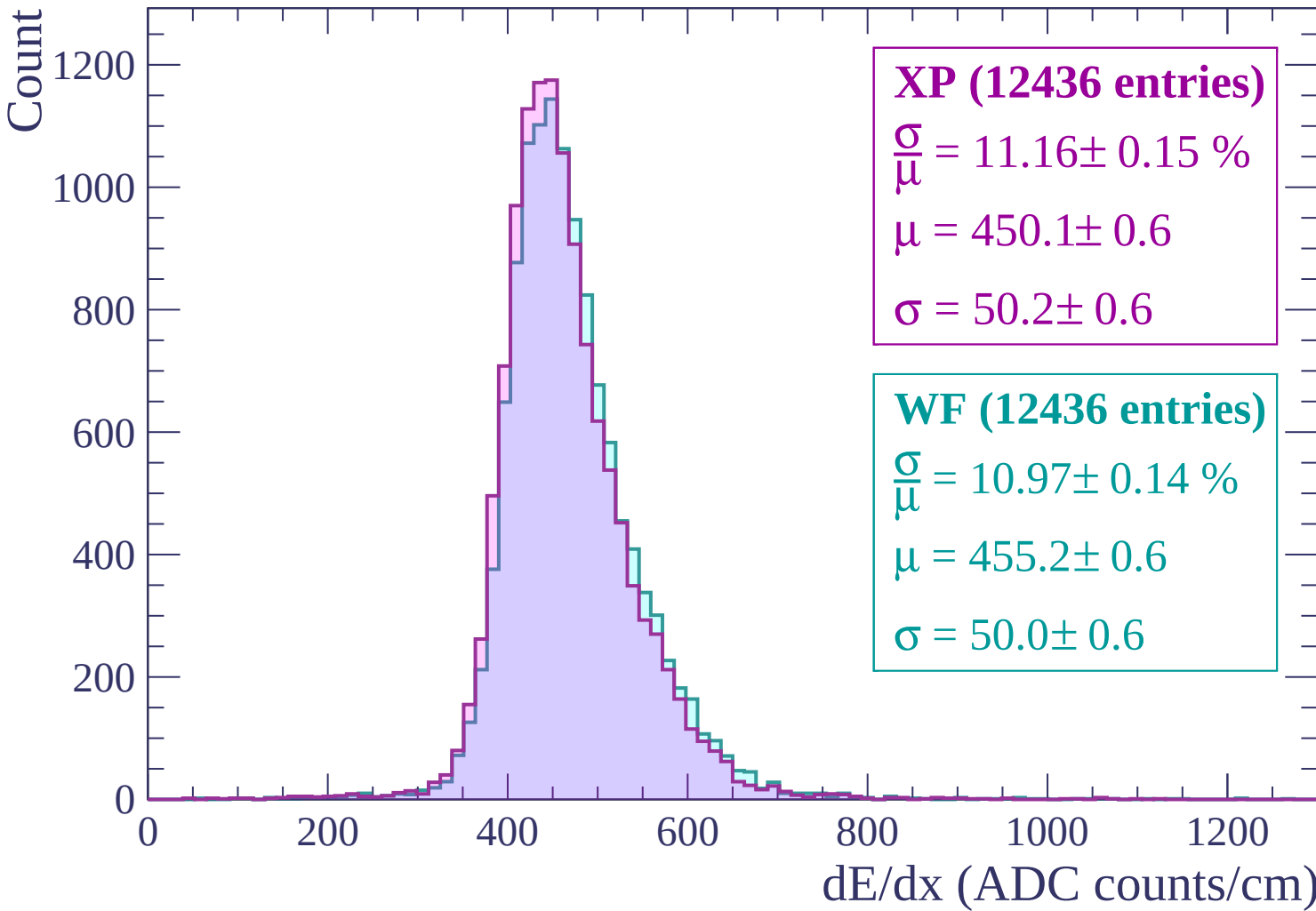
WF (12436 entries)

$\frac{\sigma}{\mu} = 10.97 \pm 0.14 \%$

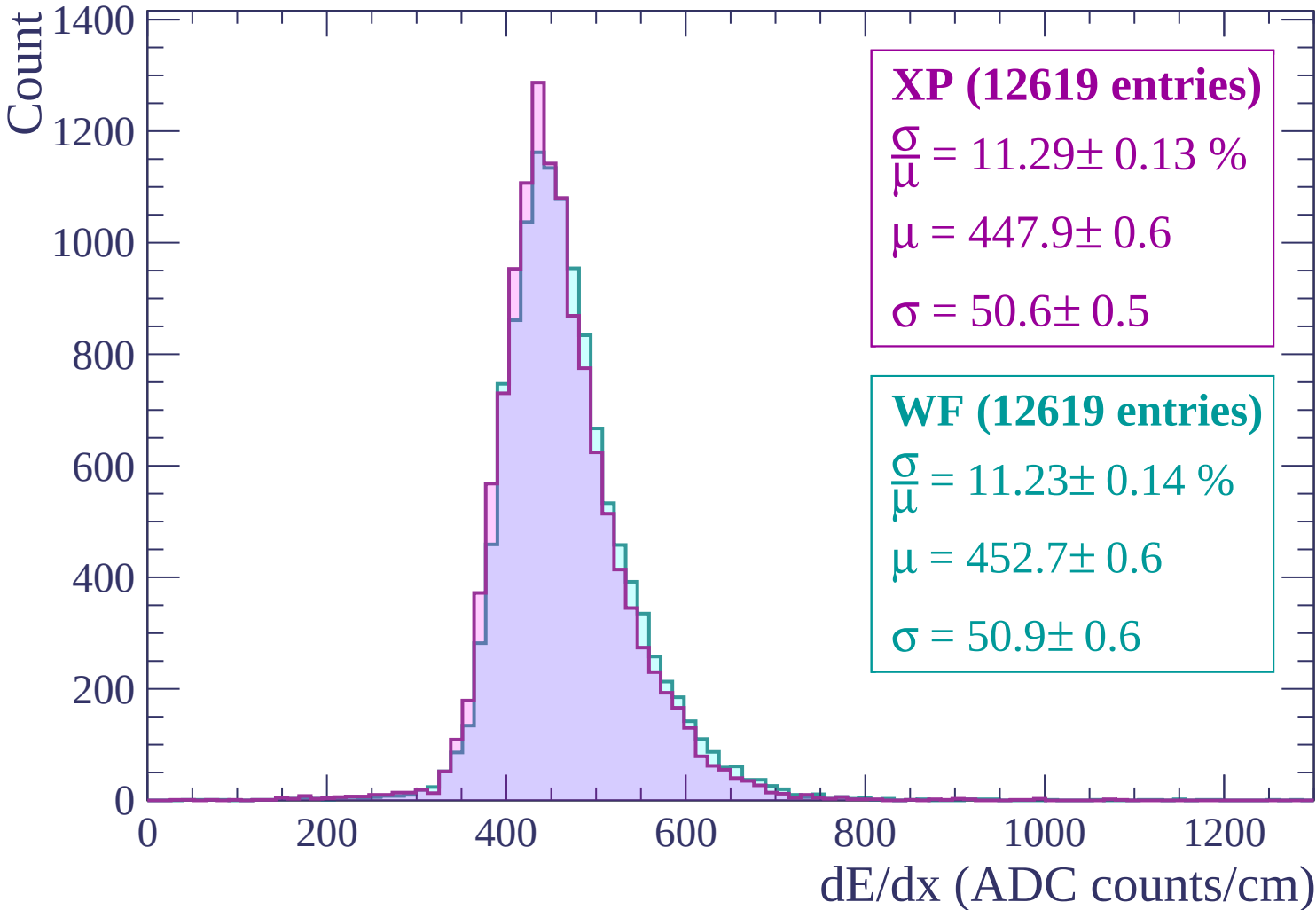
$\mu = 455.2 \pm 0.6$

$\sigma = 50.0 \pm 0.6$

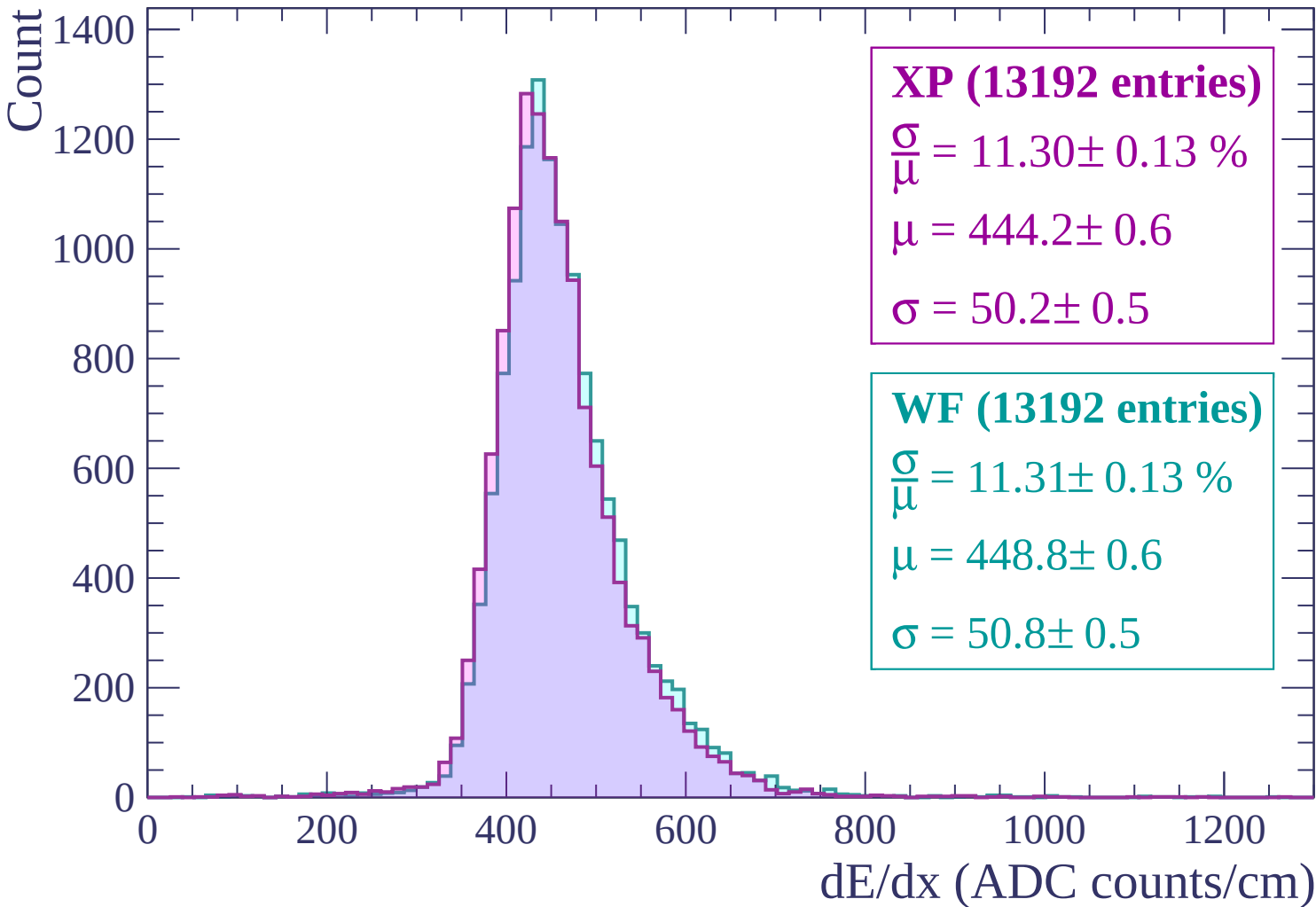
dE/dx (ADC counts/cm)



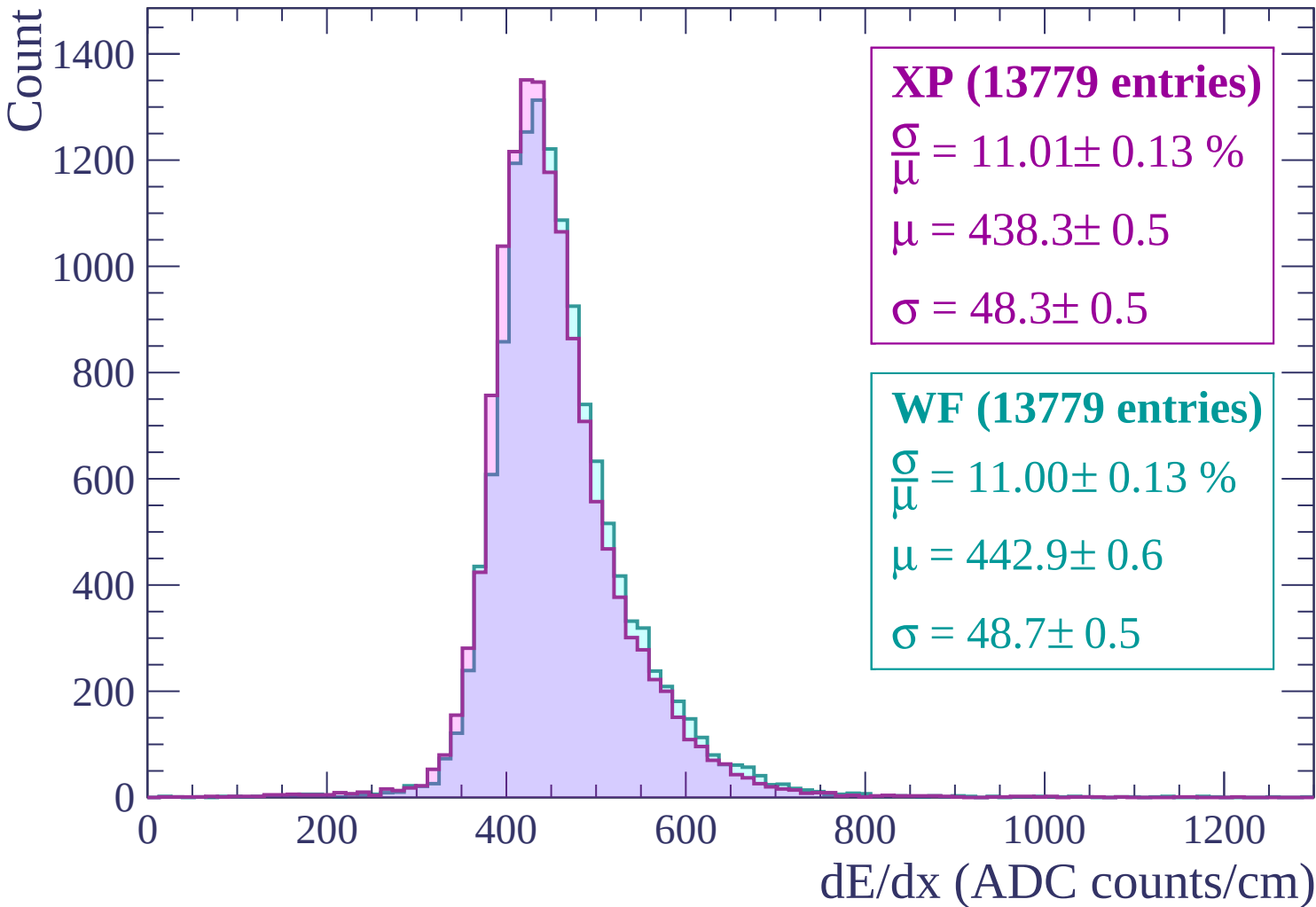
Energy loss | -960 < p < -900



Energy loss | $-900 < p < -840$



Energy loss | $-840 < p < -780$



Energy loss | $-780 < p < -720$

Count

1400
1200
1000
800
600
400
200
0

XP (13966 entries)

$$\frac{\sigma}{\mu} = 10.97 \pm 0.14 \%$$

$$\mu = 432.2 \pm 0.5$$

$$\sigma = 47.4 \pm 0.5$$

WF (13966 entries)

$$\frac{\sigma}{\mu} = 11.08 \pm 0.13 \%$$

$$\mu = 438.1 \pm 0.5$$

$$\sigma = 48.6 \pm 0.5$$

0

200

400

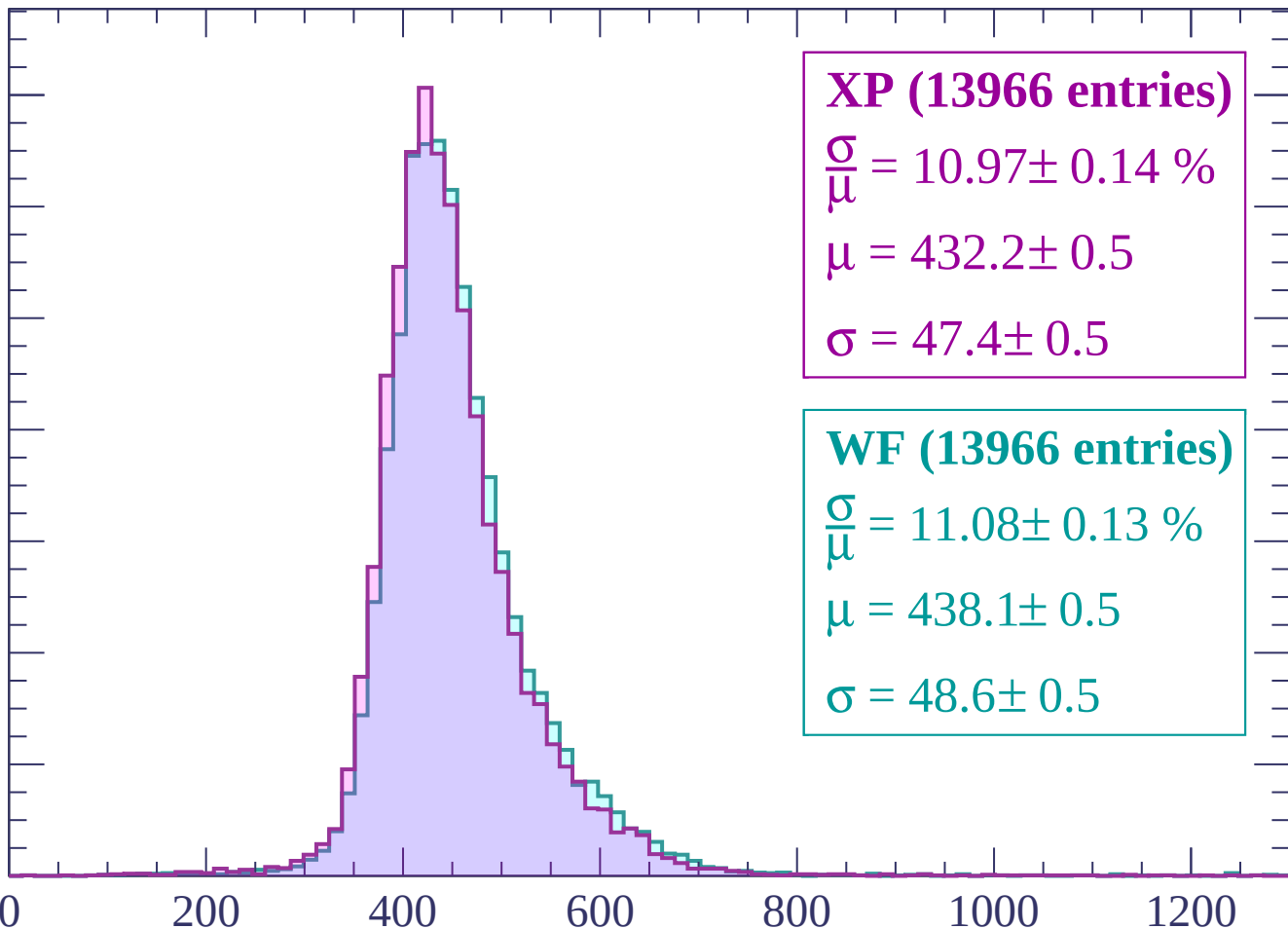
600

800

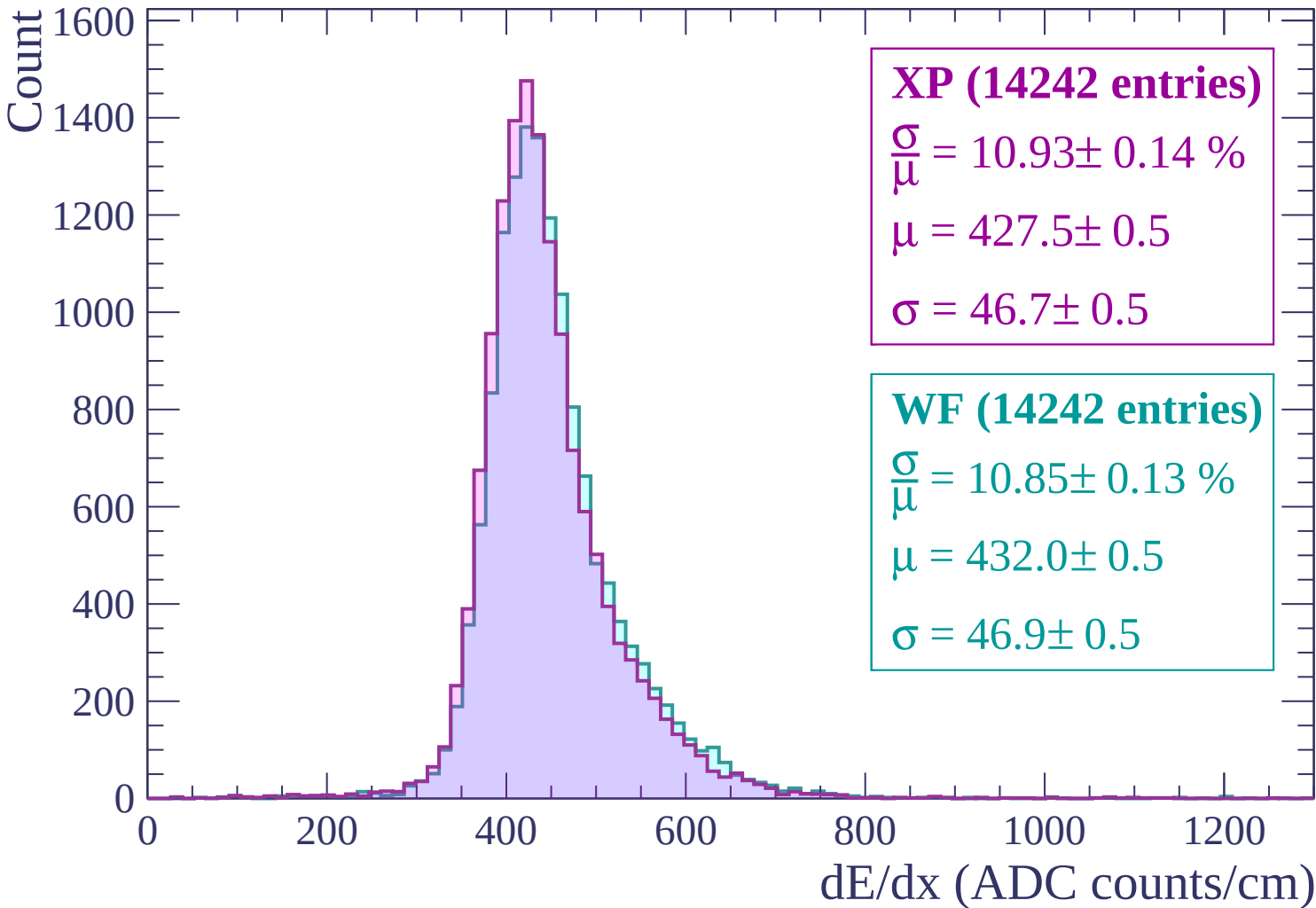
1000

1200

dE/dx (ADC counts/cm)



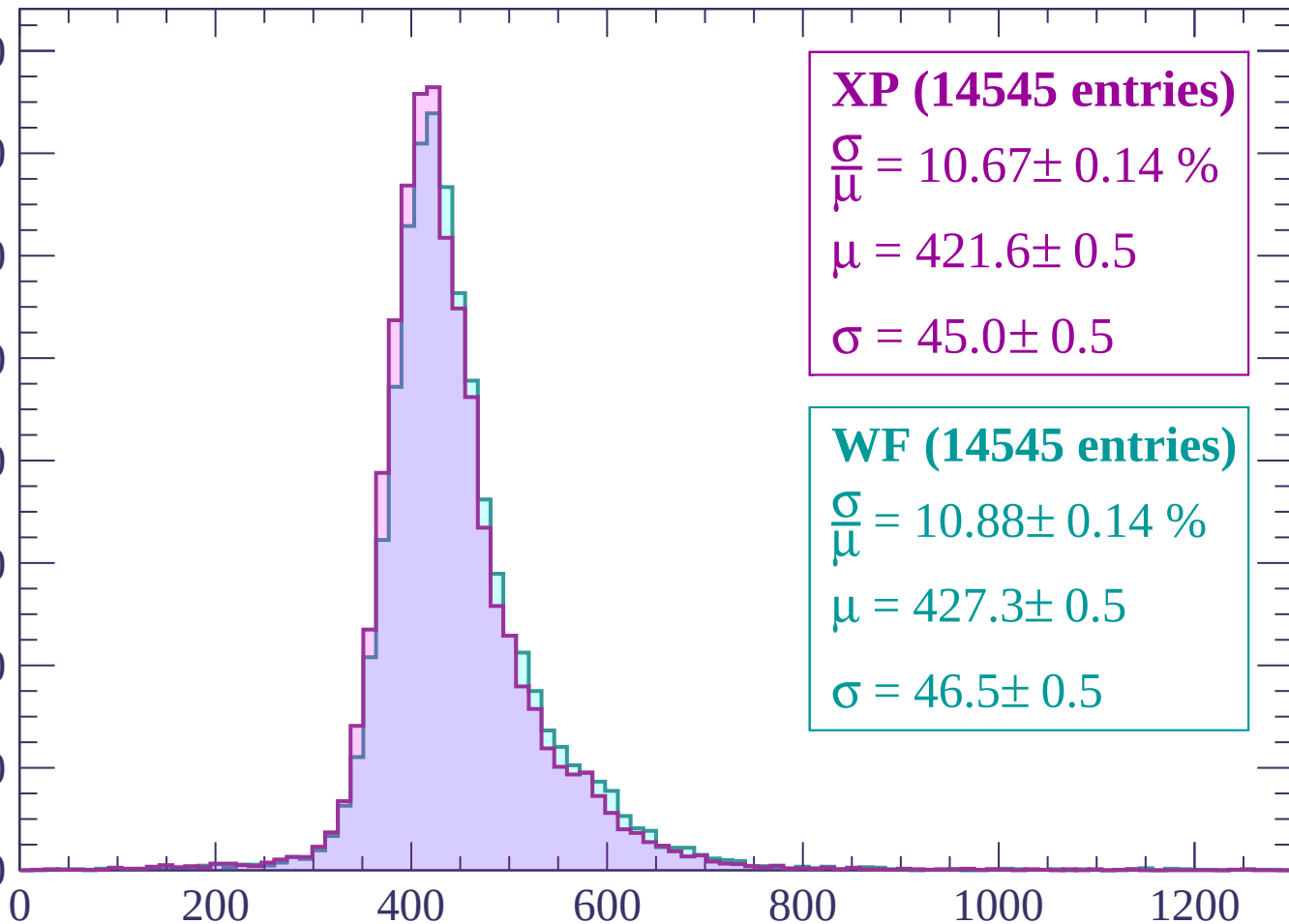
Energy loss | -720 < p < -660



Energy loss | $-660 < p < -600$

Count

1600
1400
1200
1000
800
600
400
200
0



XP (14545 entries)

$$\frac{\sigma}{\mu} = 10.67 \pm 0.14 \%$$

$$\mu = 421.6 \pm 0.5$$

$$\sigma = 45.0 \pm 0.5$$

WF (14545 entries)

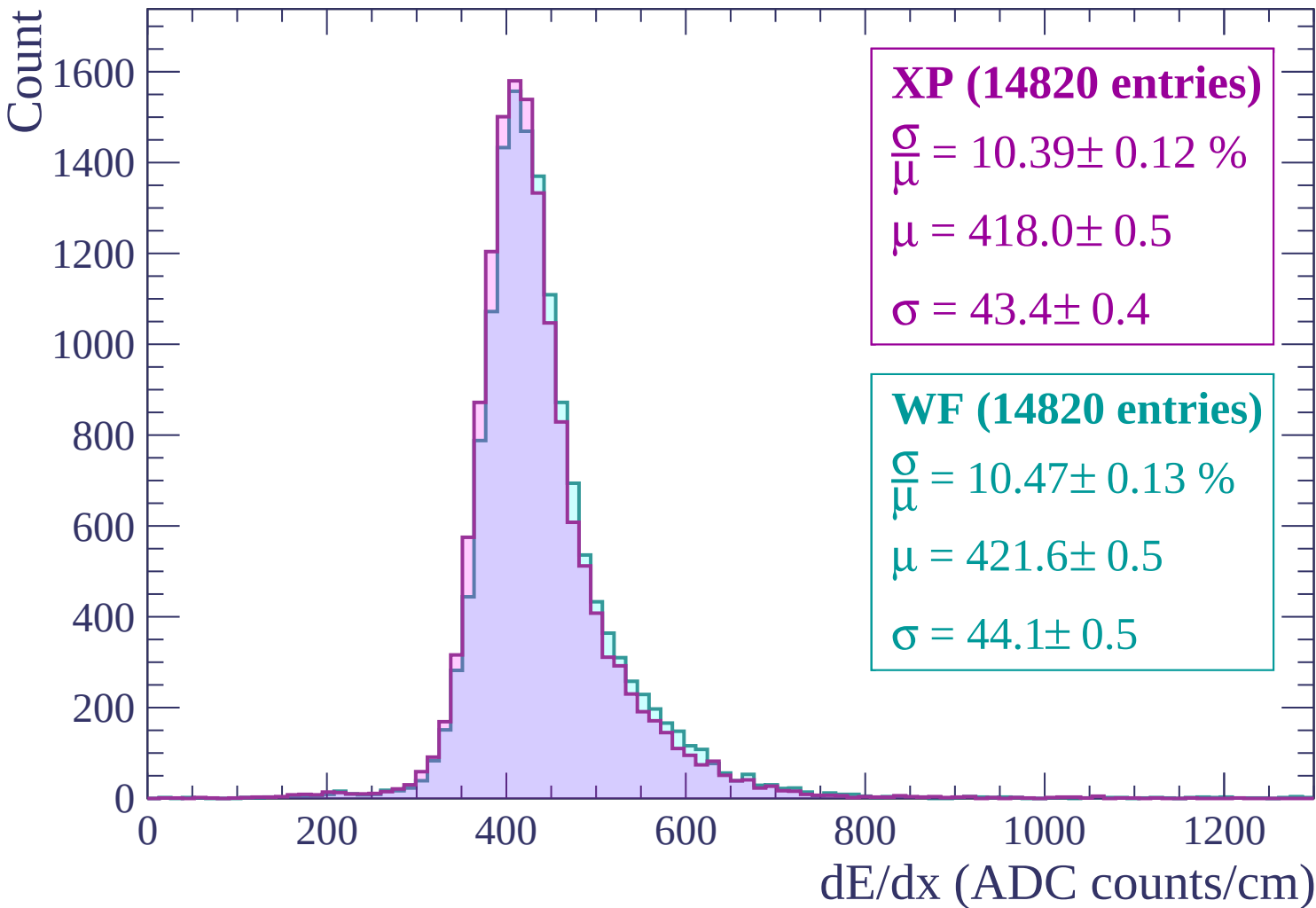
$$\frac{\sigma}{\mu} = 10.88 \pm 0.14 \%$$

$$\mu = 427.3 \pm 0.5$$

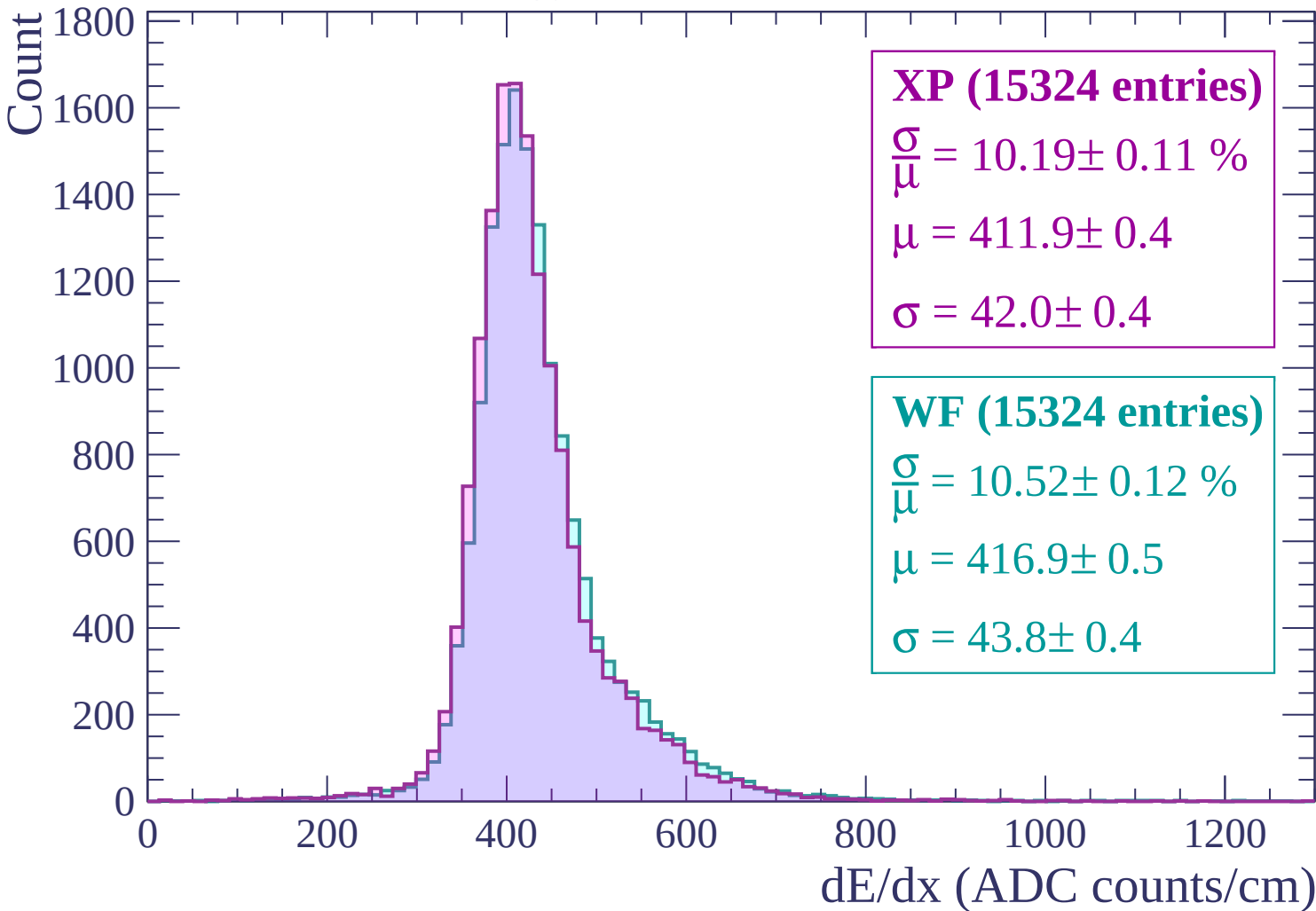
$$\sigma = 46.5 \pm 0.5$$

dE/dx (ADC counts/cm)

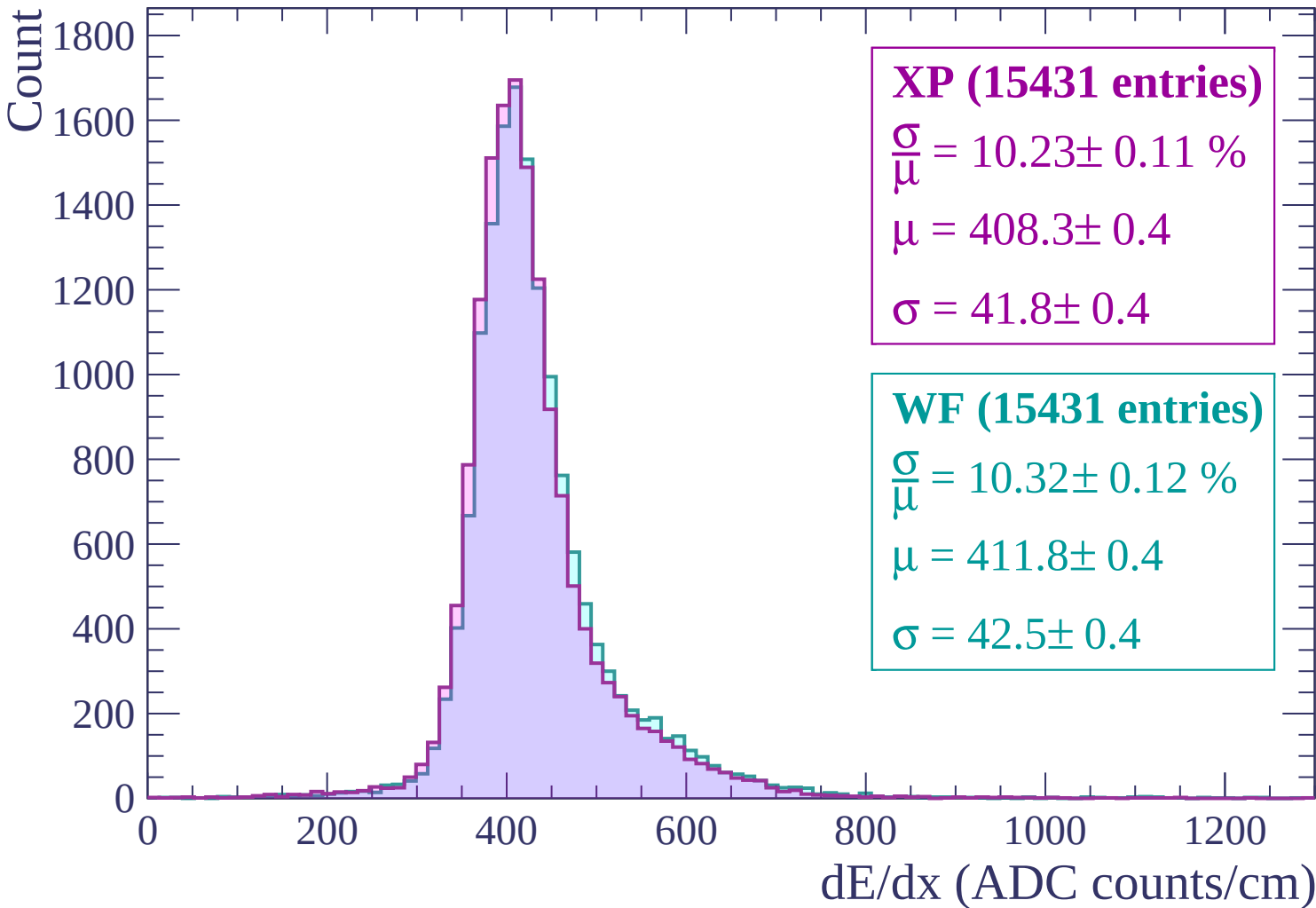
Energy loss | $-600 < p < -540$



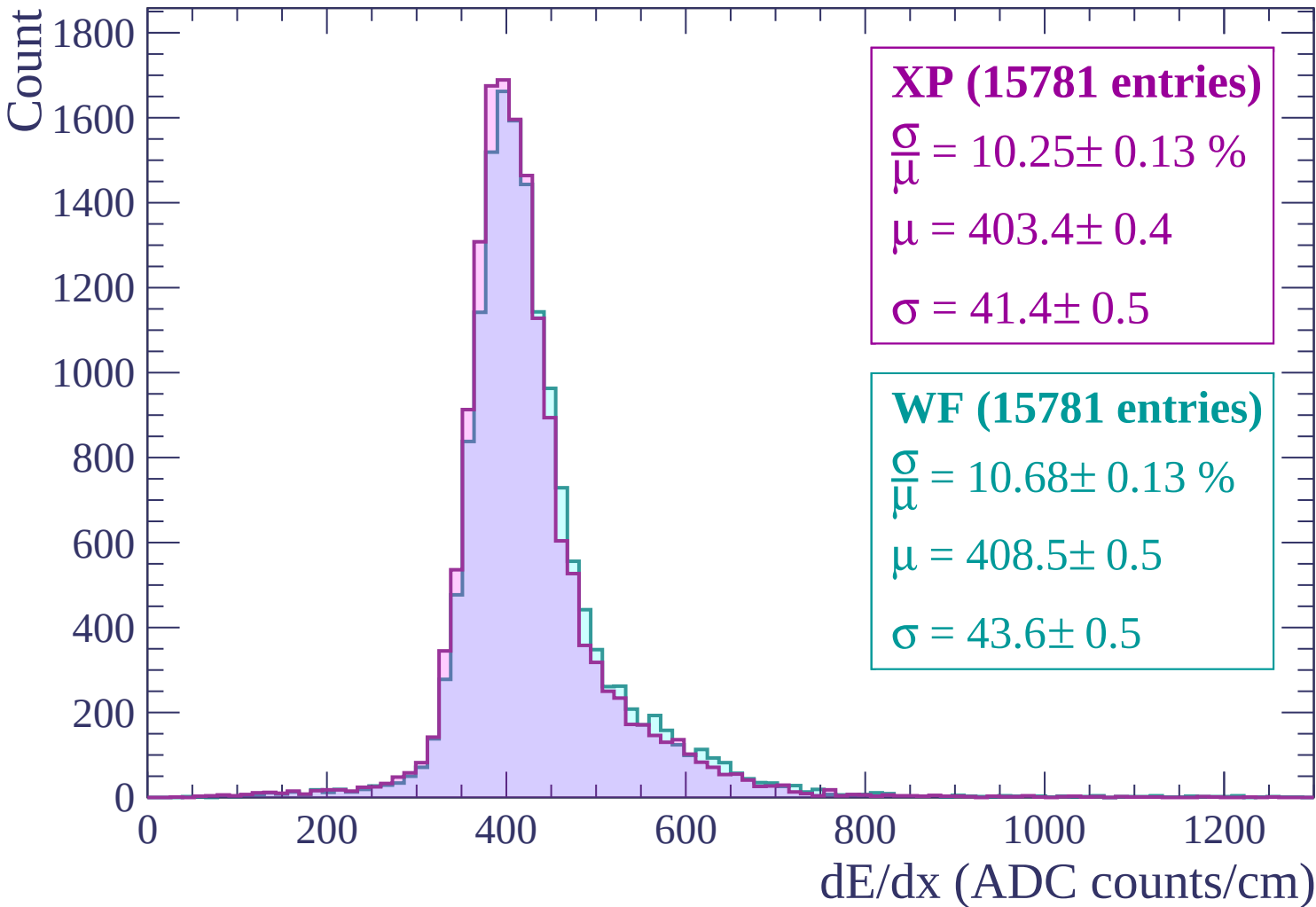
Energy loss | $-540 < p < -480$



Energy loss | $-480 < p < -420$



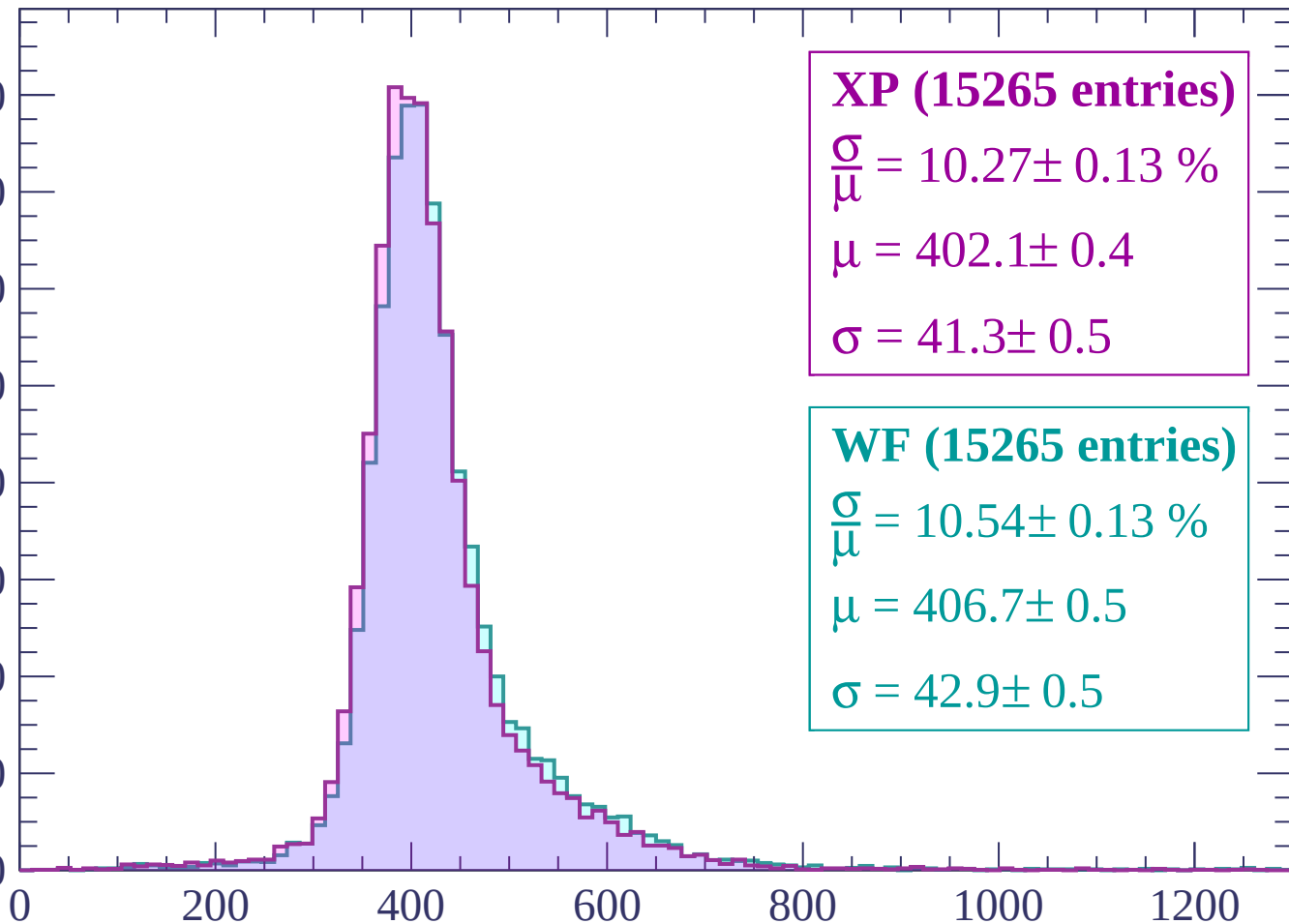
Energy loss | $-420 < p < -360$



Energy loss | $-360 < p < -300$

Count

1600
1400
1200
1000
800
600
400
200
0



XP (15265 entries)

$$\frac{\sigma}{\mu} = 10.27 \pm 0.13 \%$$

$$\mu = 402.1 \pm 0.4$$

$$\sigma = 41.3 \pm 0.5$$

WF (15265 entries)

$$\frac{\sigma}{\mu} = 10.54 \pm 0.13 \%$$

$$\mu = 406.7 \pm 0.5$$

$$\sigma = 42.9 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-300 < p < -240$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (14835 entries)

$$\frac{\sigma}{\mu} = 10.24 \pm 0.13 \%$$

$$\mu = 402.7 \pm 0.5$$

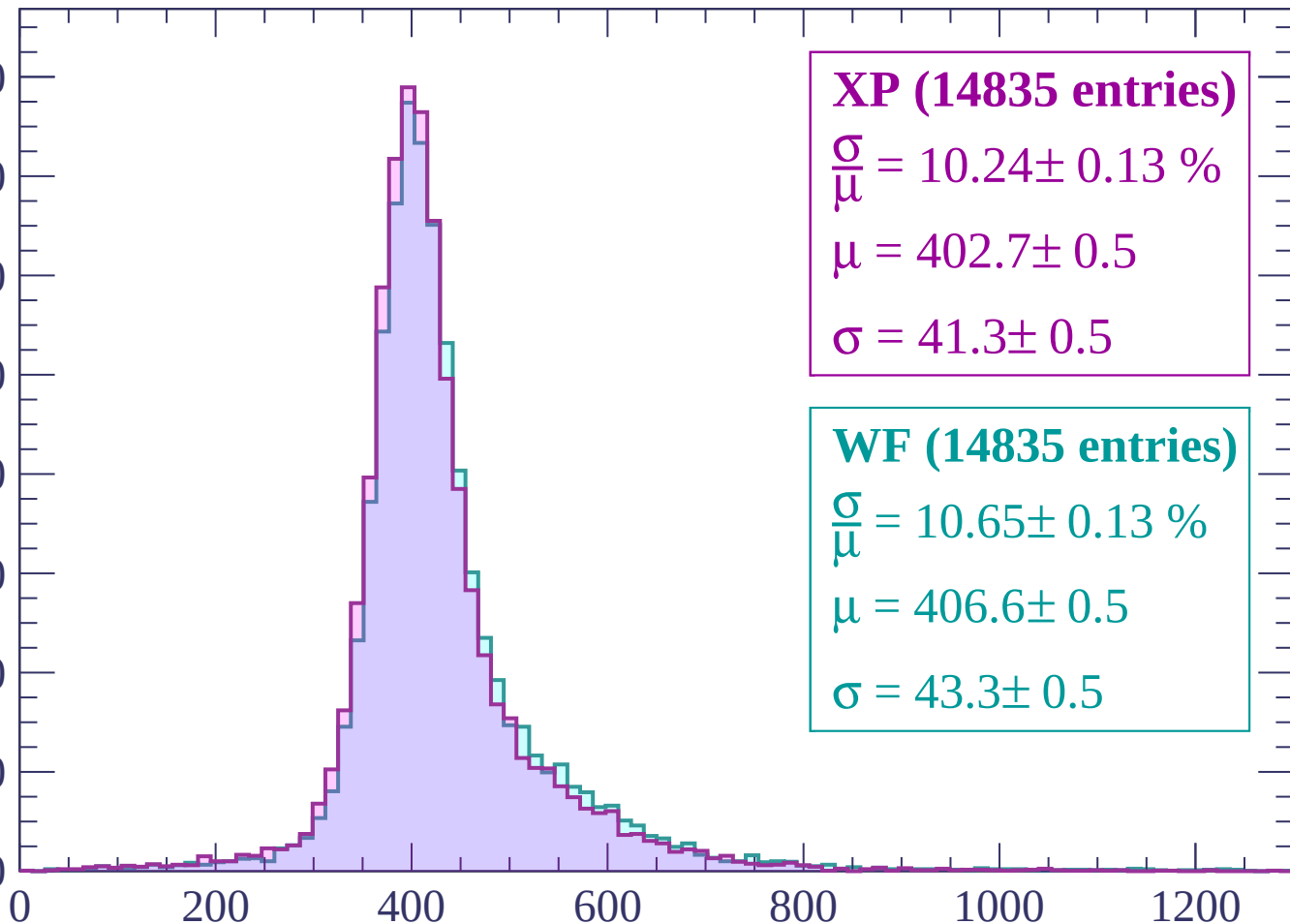
$$\sigma = 41.3 \pm 0.5$$

WF (14835 entries)

$$\frac{\sigma}{\mu} = 10.65 \pm 0.13 \%$$

$$\mu = 406.6 \pm 0.5$$

$$\sigma = 43.3 \pm 0.5$$



dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (13131 entries)

$$\frac{\sigma}{\mu} = 11.60 \pm 0.16 \%$$

$$\mu = 414.7 \pm 0.6$$

$$\sigma = 48.1 \pm 0.6$$

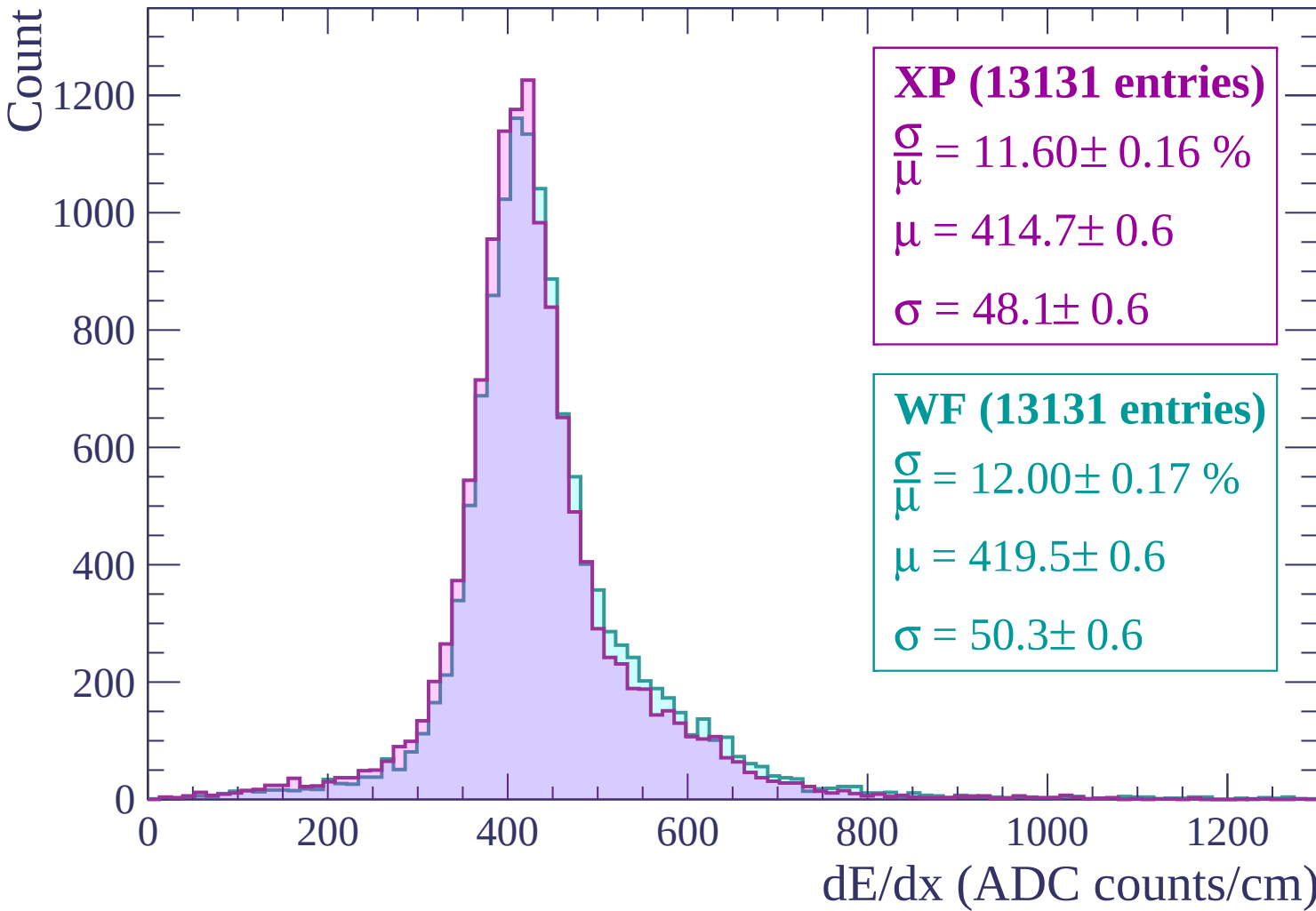
WF (13131 entries)

$$\frac{\sigma}{\mu} = 12.00 \pm 0.17 \%$$

$$\mu = 419.5 \pm 0.6$$

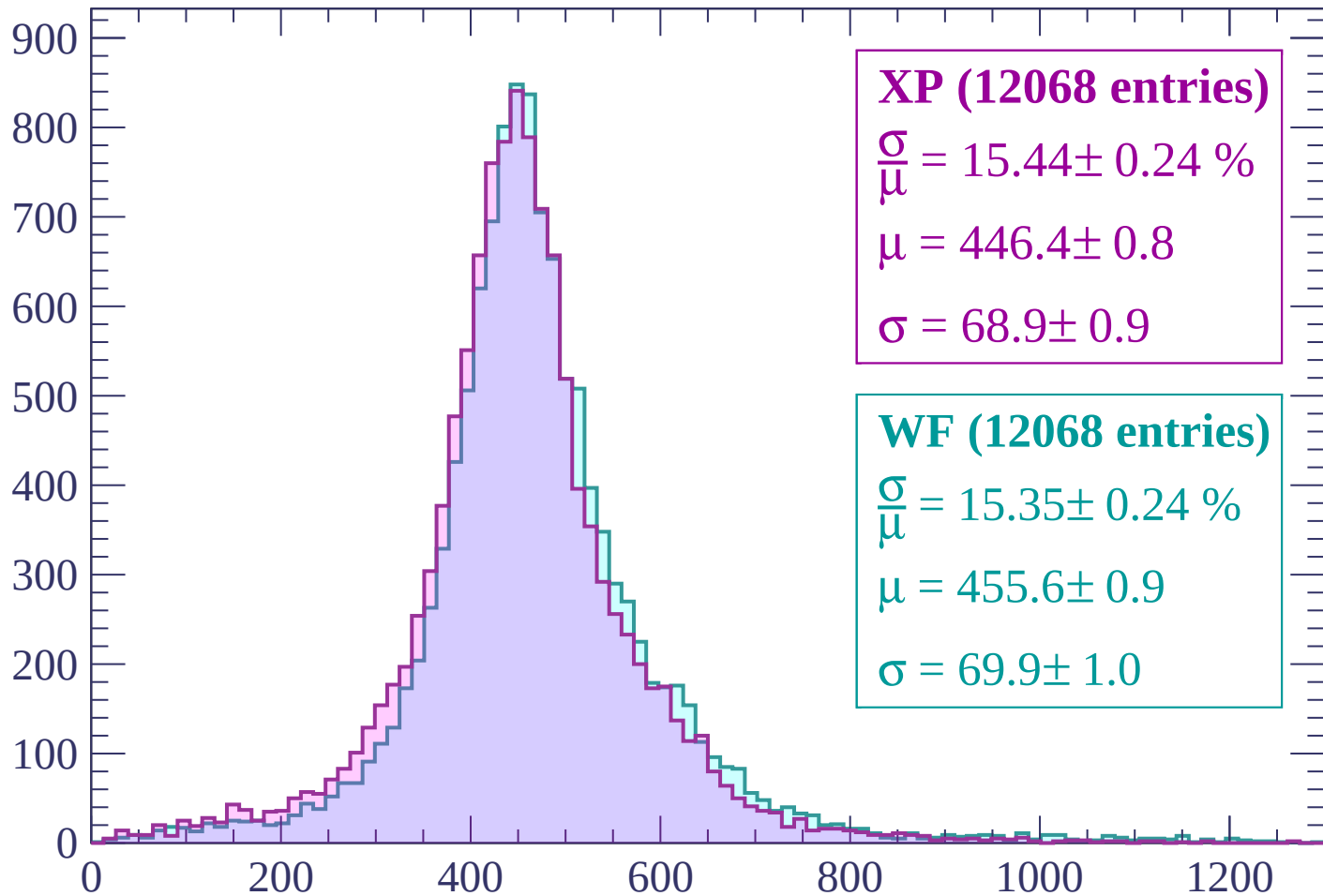
$$\sigma = 50.3 \pm 0.6$$

dE/dx (ADC counts/cm)



Energy loss | -180 < p < -120

Count



XP (12068 entries)

$$\frac{\sigma}{\mu} = 15.44 \pm 0.24 \%$$

$$\mu = 446.4 \pm 0.8$$

$$\sigma = 68.9 \pm 0.9$$

WF (12068 entries)

$$\frac{\sigma}{\mu} = 15.35 \pm 0.24 \%$$

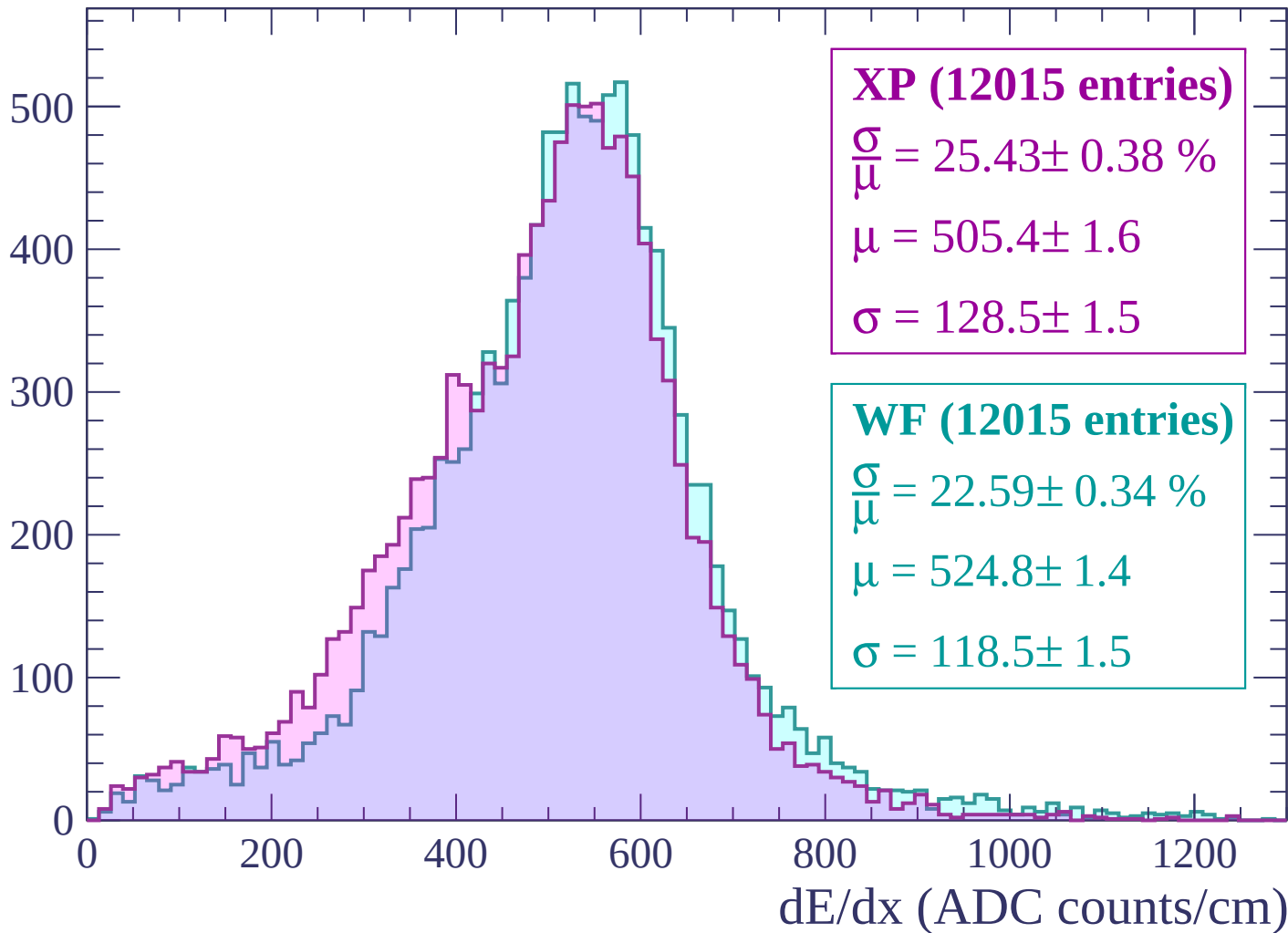
$$\mu = 455.6 \pm 0.9$$

$$\sigma = 69.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-120 < p < -60$

Count



Energy loss | $-60 < p < 0$

Count

1400
1200
1000
800
600
400
200
0

XP (22802 entries)

$\frac{\sigma}{\mu} = 12.24 \pm 0.14 \%$

$\mu = 546.0 \pm 0.6$

$\sigma = 66.8 \pm 0.7$

WF (22802 entries)

$\frac{\sigma}{\mu} = 12.61 \pm 0.13 \%$

$\mu = 549.4 \pm 0.6$

$\sigma = 69.3 \pm 0.6$

0

200

400

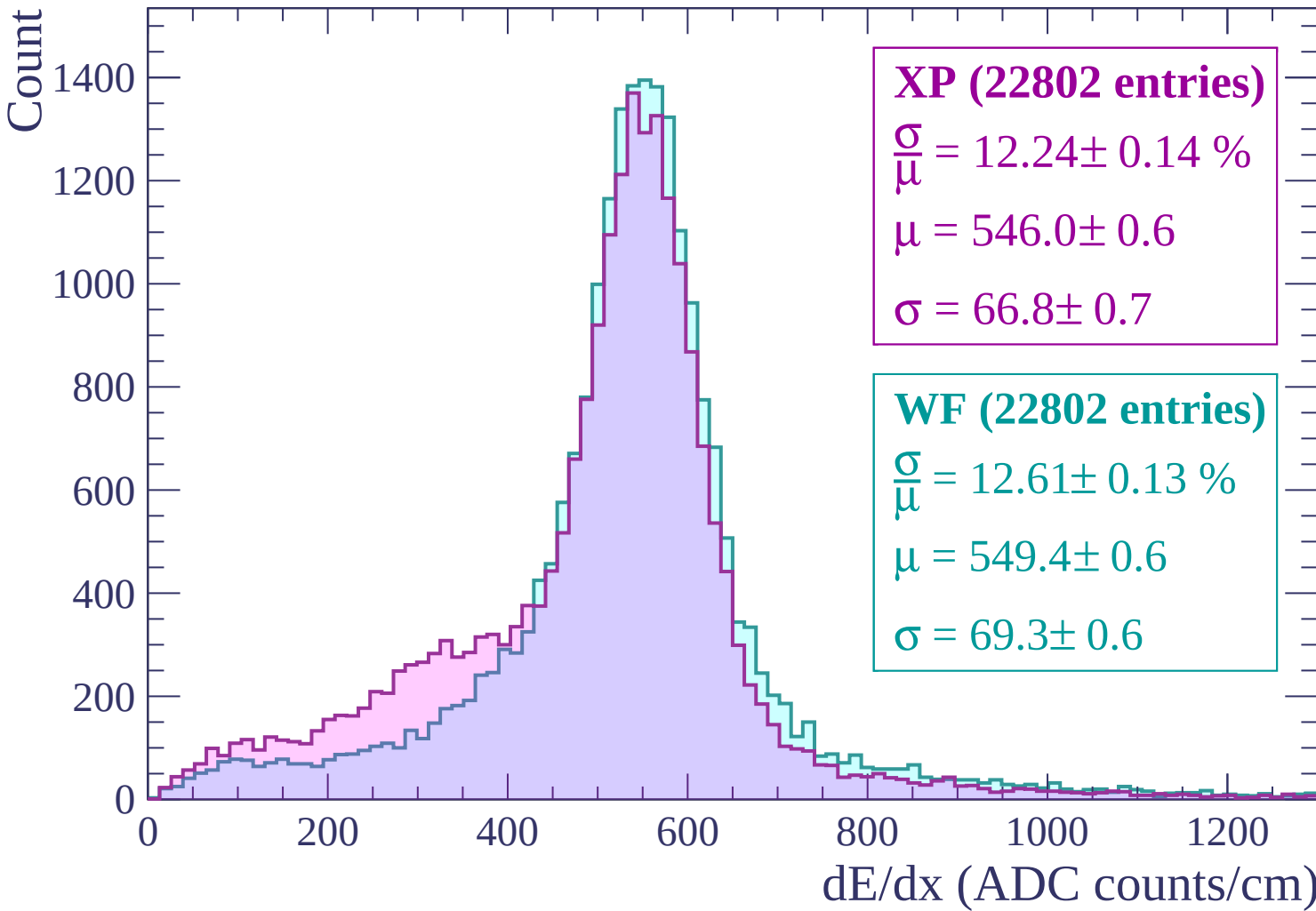
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $0 < p < 60$

Count

3500
3000
2500
2000
1500
1000
500
0

XP (54448 entries)

$\frac{\sigma}{\mu} = 15.82 \pm 0.11 \%$

$\mu = 489.5 \pm 0.4$

$\sigma = 77.4 \pm 0.5$

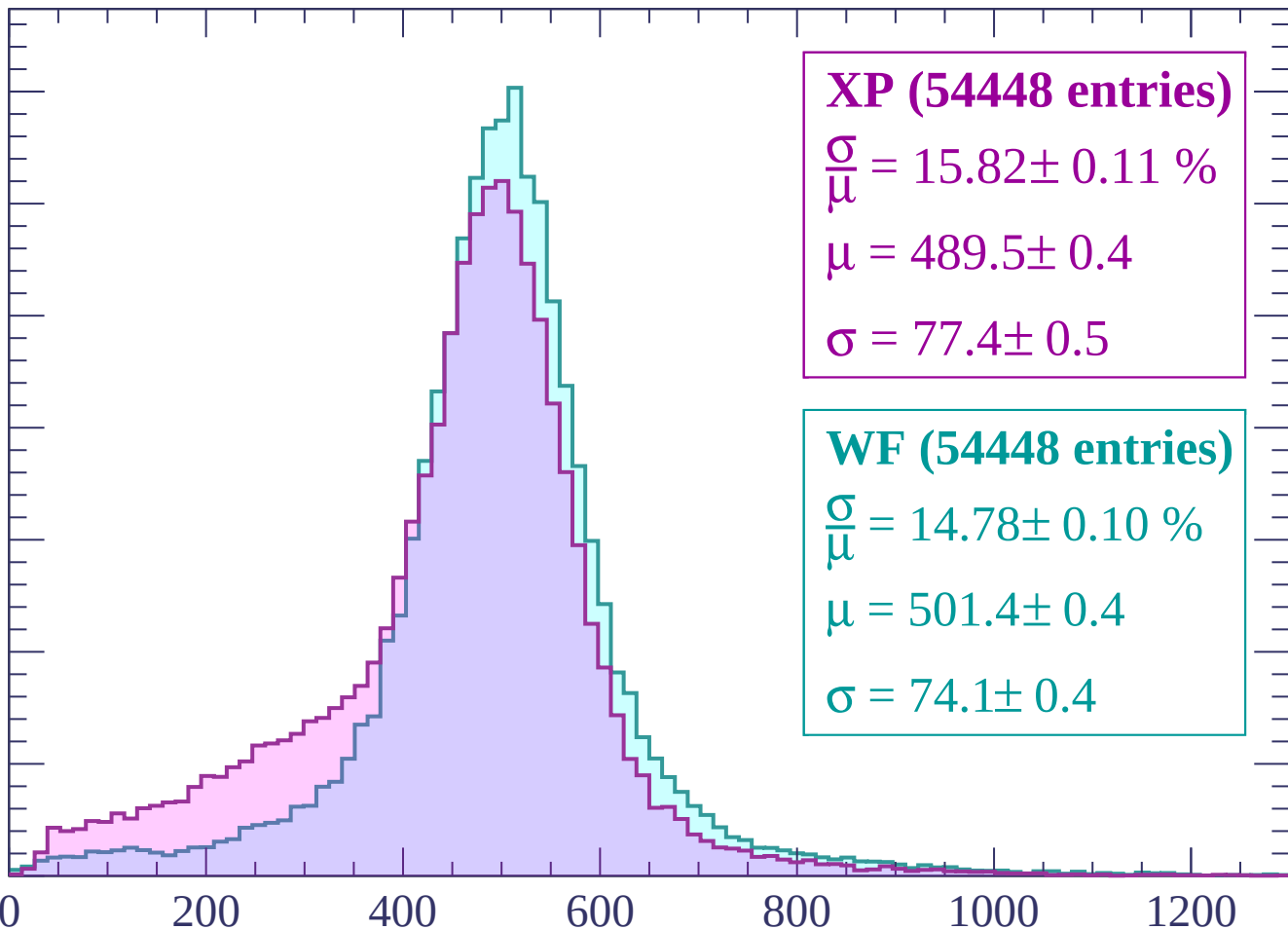
WF (54448 entries)

$\frac{\sigma}{\mu} = 14.78 \pm 0.10 \%$

$\mu = 501.4 \pm 0.4$

$\sigma = 74.1 \pm 0.4$

0 200 400 600 800 1000 1200
dE/dx (ADC counts/cm)



Energy loss | $60 < p < 120$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (18423 entries)

$\frac{\sigma}{\mu} = 11.63 \pm 0.14 \%$

$\mu = 540.4 \pm 0.7$

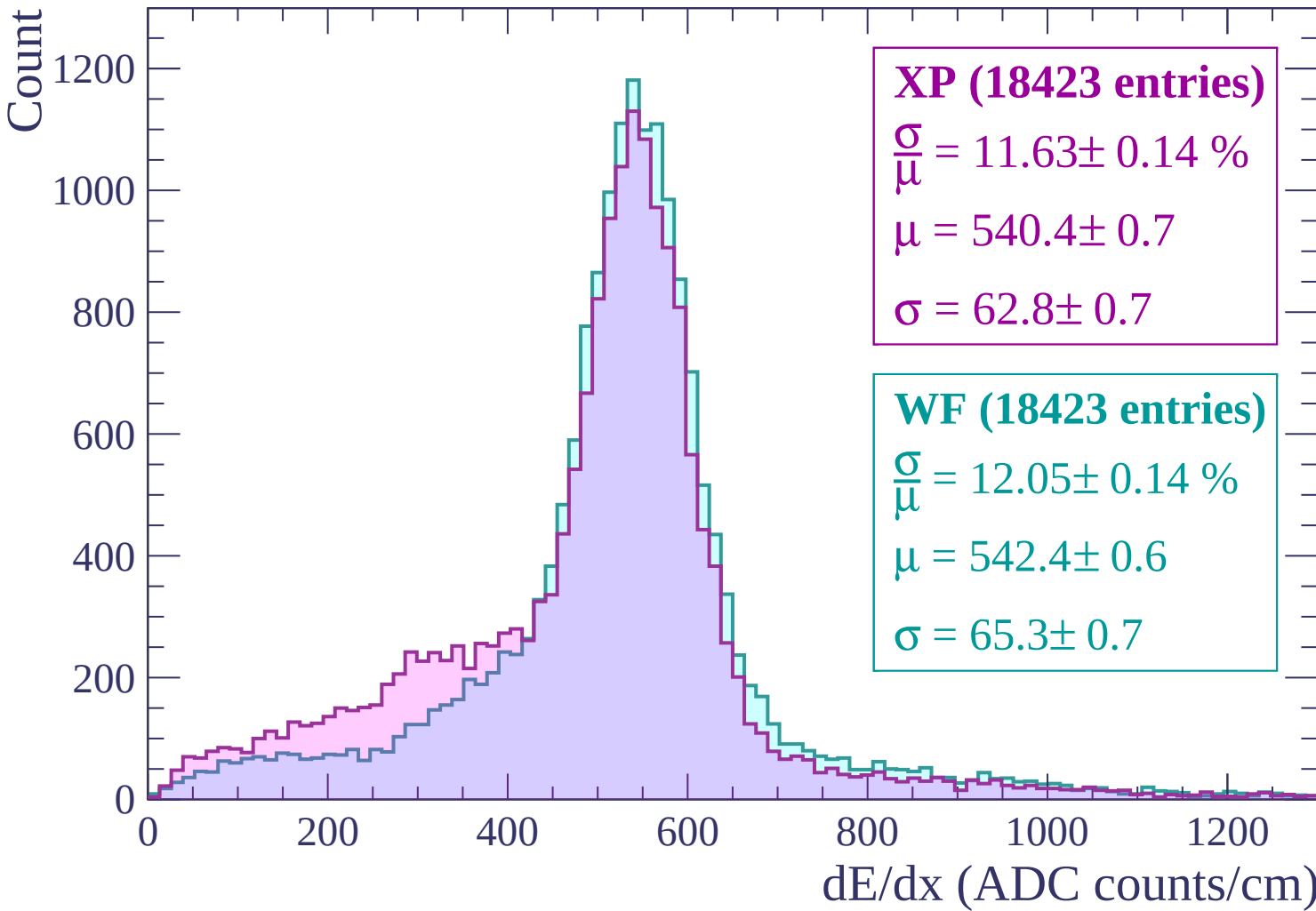
$\sigma = 62.8 \pm 0.7$

WF (18423 entries)

$\frac{\sigma}{\mu} = 12.05 \pm 0.14 \%$

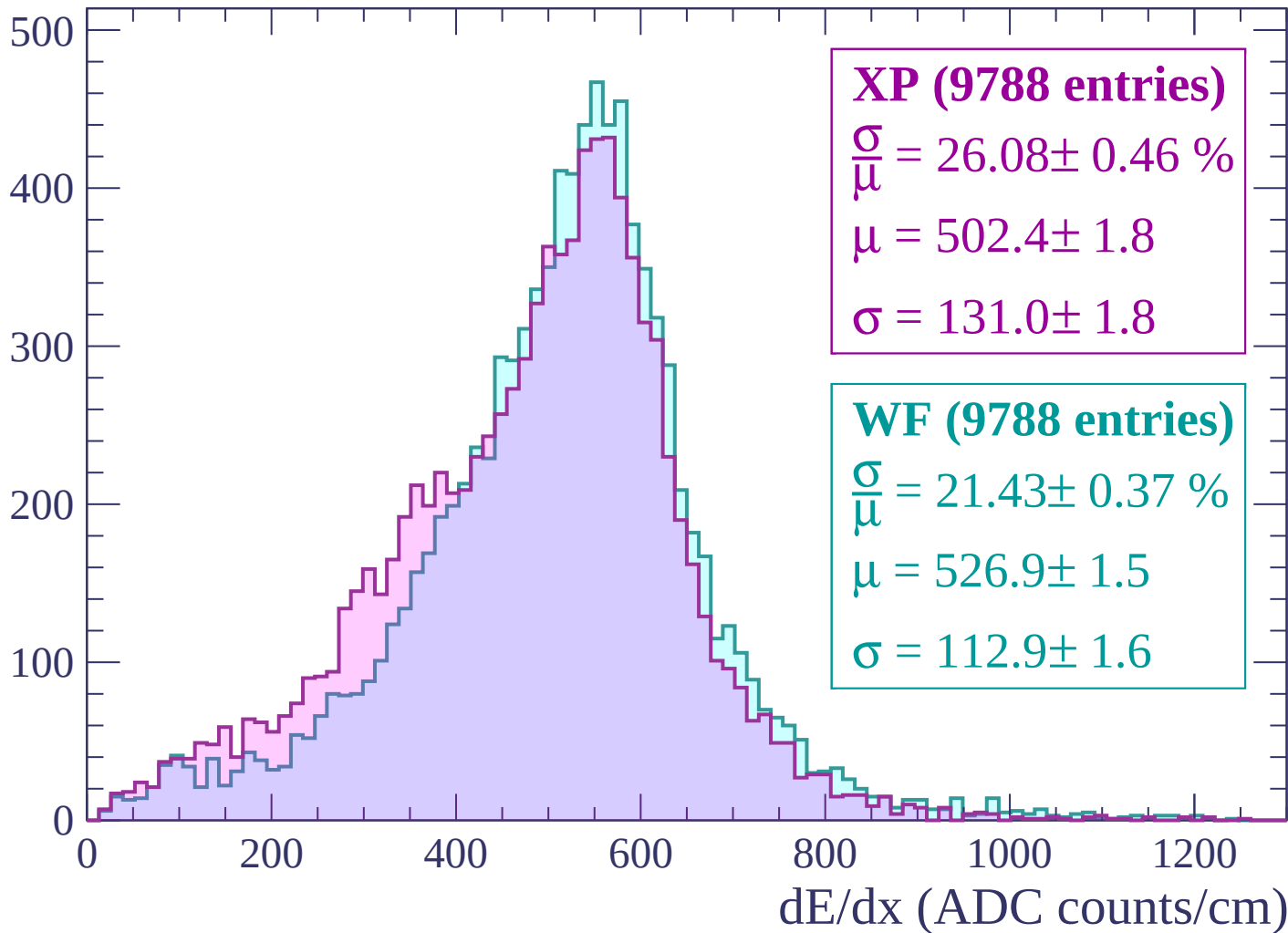
$\mu = 542.4 \pm 0.6$

$\sigma = 65.3 \pm 0.7$



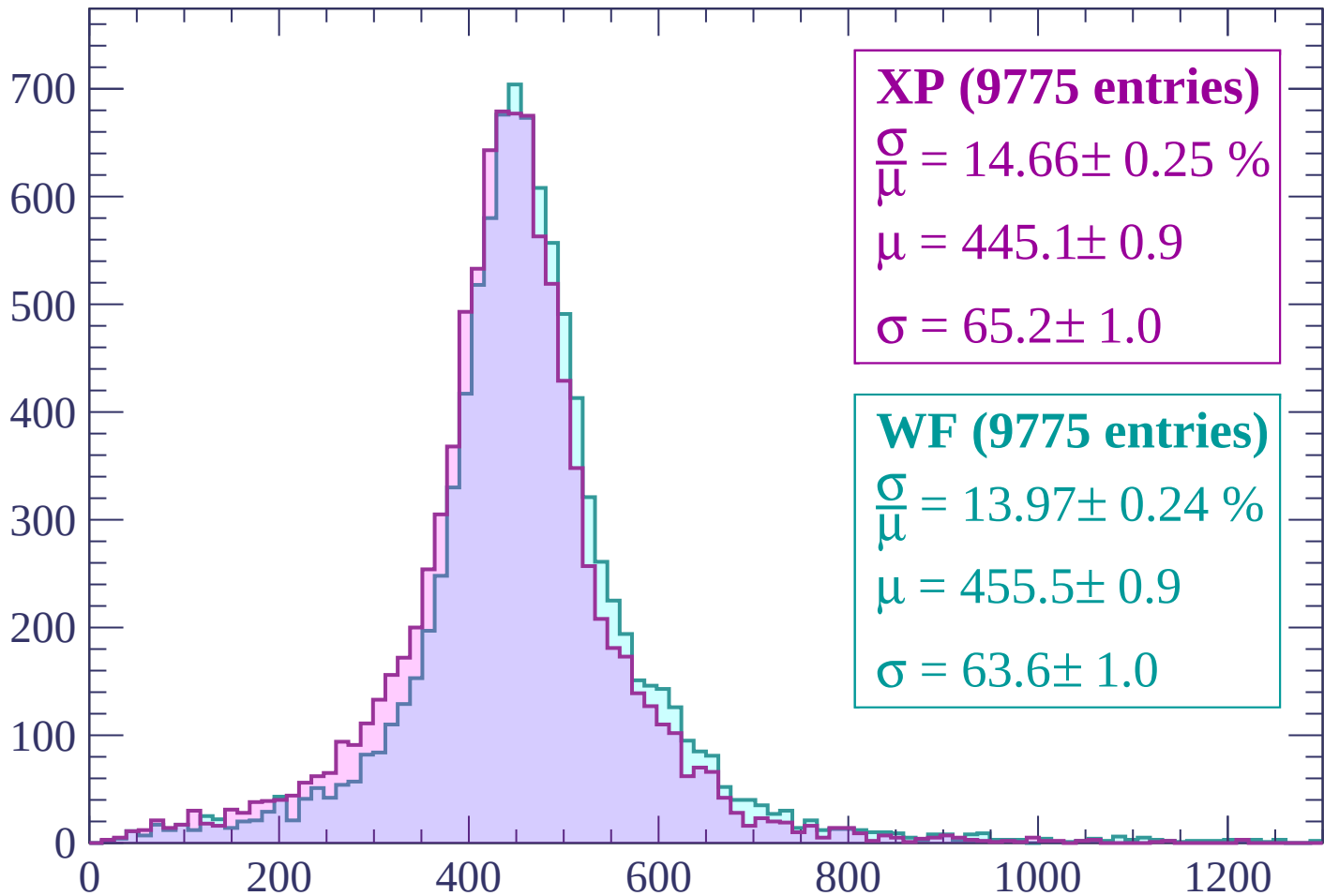
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



XP (9775 entries)

$$\frac{\sigma}{\mu} = 14.66 \pm 0.25 \%$$

$$\mu = 445.1 \pm 0.9$$

$$\sigma = 65.2 \pm 1.0$$

WF (9775 entries)

$$\frac{\sigma}{\mu} = 13.97 \pm 0.24 \%$$

$$\mu = 455.5 \pm 0.9$$

$$\sigma = 63.6 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (10883 entries)

$\frac{\sigma}{\mu} = 11.15 \pm 0.17 \%$

$\mu = 407.9 \pm 0.6$

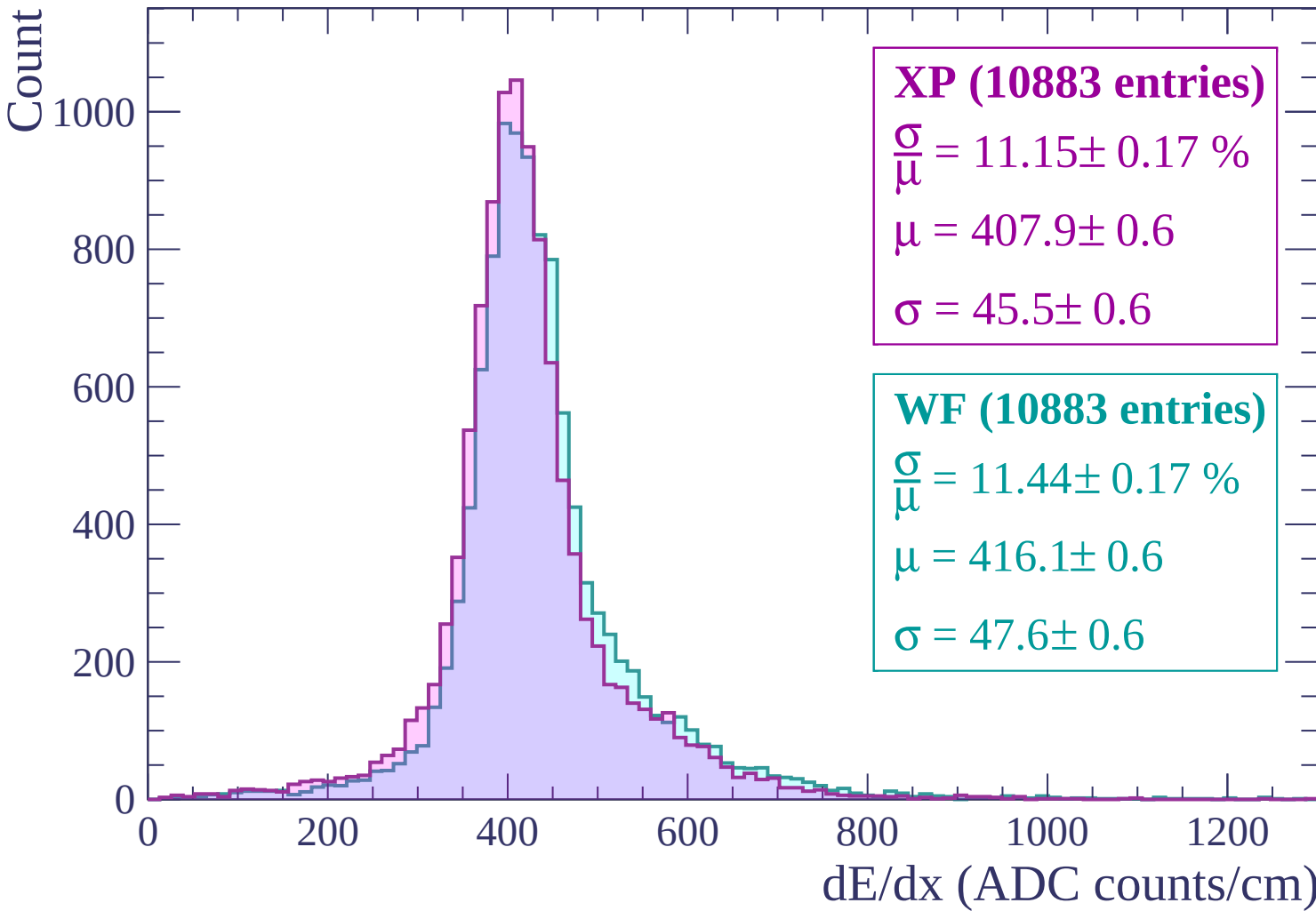
$\sigma = 45.5 \pm 0.6$

WF (10883 entries)

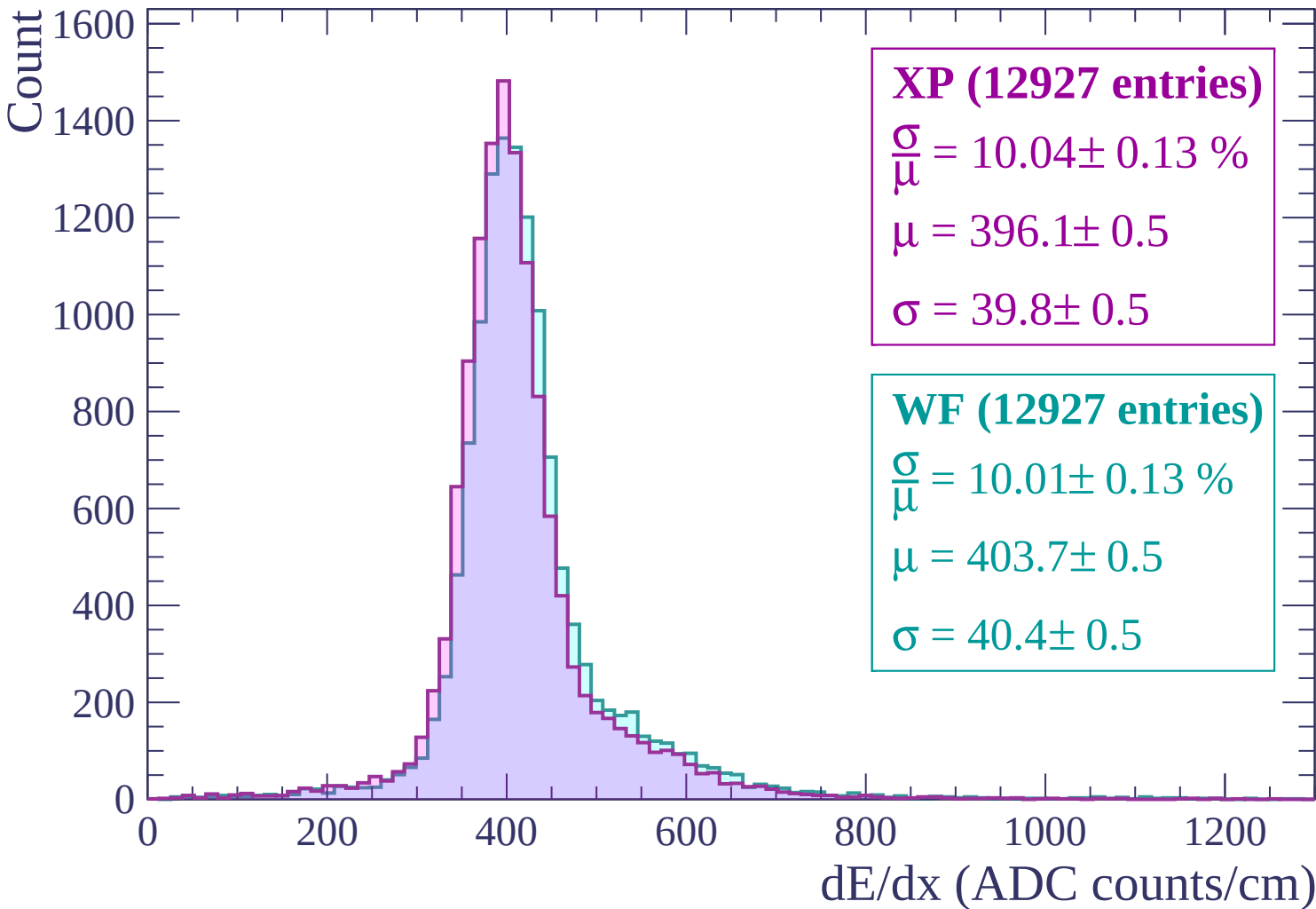
$\frac{\sigma}{\mu} = 11.44 \pm 0.17 \%$

$\mu = 416.1 \pm 0.6$

$\sigma = 47.6 \pm 0.6$



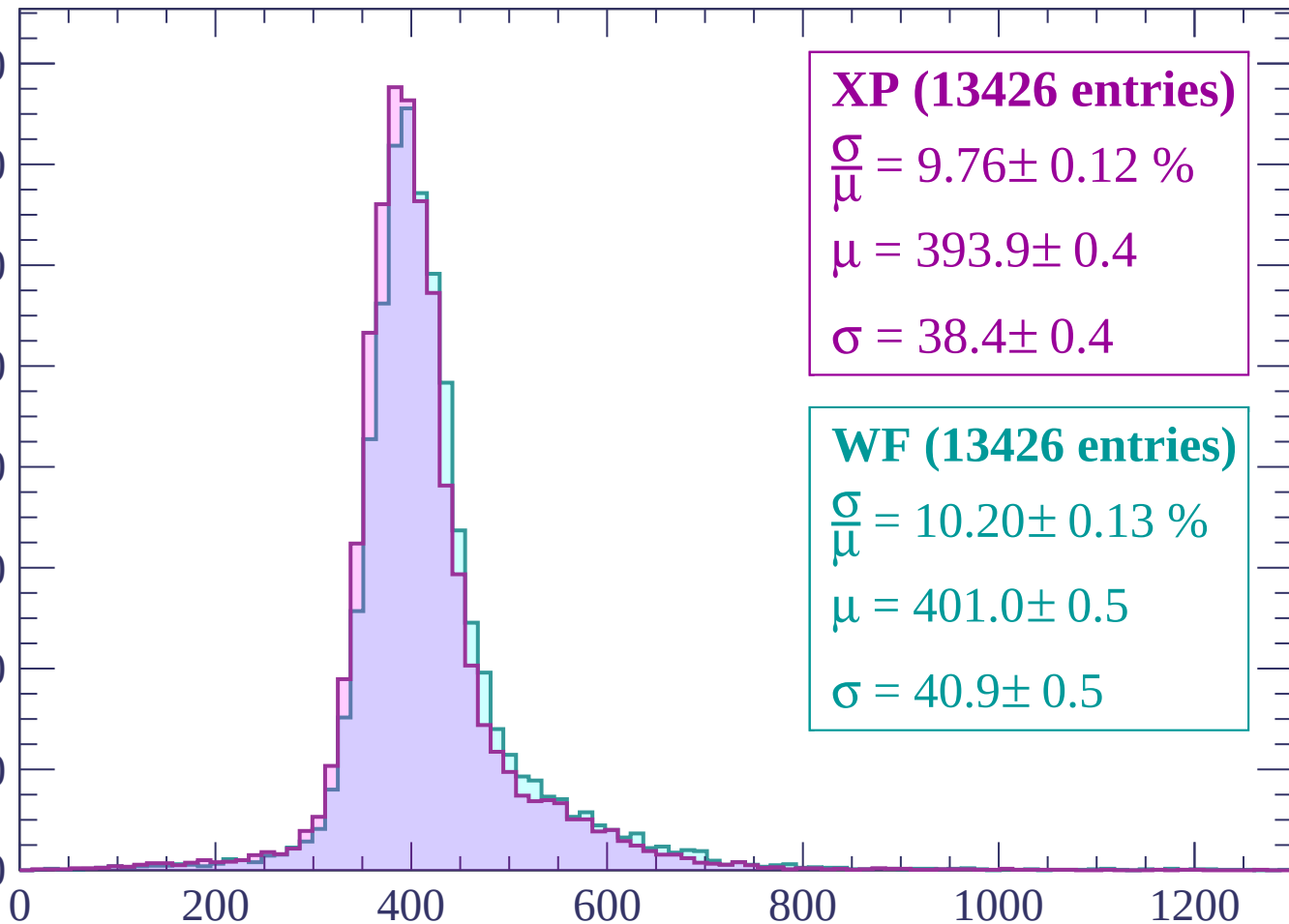
Energy loss | $300 < p < 360$



Energy loss | $360 < p < 420$

Count

1600
1400
1200
1000
800
600
400
200
0



XP (13426 entries)

$$\frac{\sigma}{\mu} = 9.76 \pm 0.12 \%$$

$$\mu = 393.9 \pm 0.4$$

$$\sigma = 38.4 \pm 0.4$$

WF (13426 entries)

$$\frac{\sigma}{\mu} = 10.20 \pm 0.13 \%$$

$$\mu = 401.0 \pm 0.5$$

$$\sigma = 40.9 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (13038 entries)

$$\frac{\sigma}{\mu} = 9.45 \pm 0.12 \%$$

$$\mu = 395.7 \pm 0.4$$

$$\sigma = 37.4 \pm 0.4$$

WF (13038 entries)

$$\frac{\sigma}{\mu} = 9.88 \pm 0.12 \%$$

$$\mu = 402.7 \pm 0.4$$

$$\sigma = 39.8 \pm 0.5$$

0

200

400

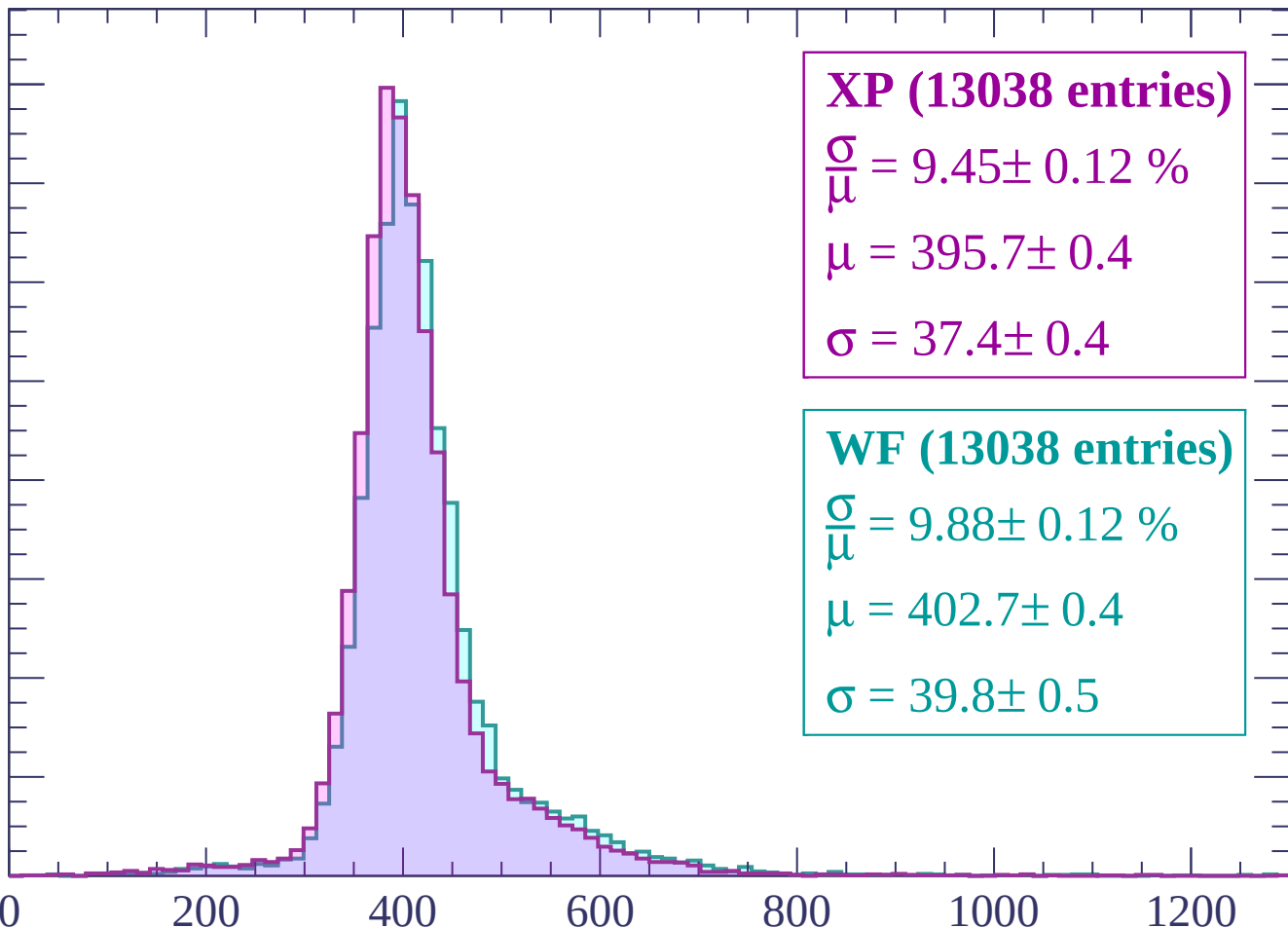
600

800

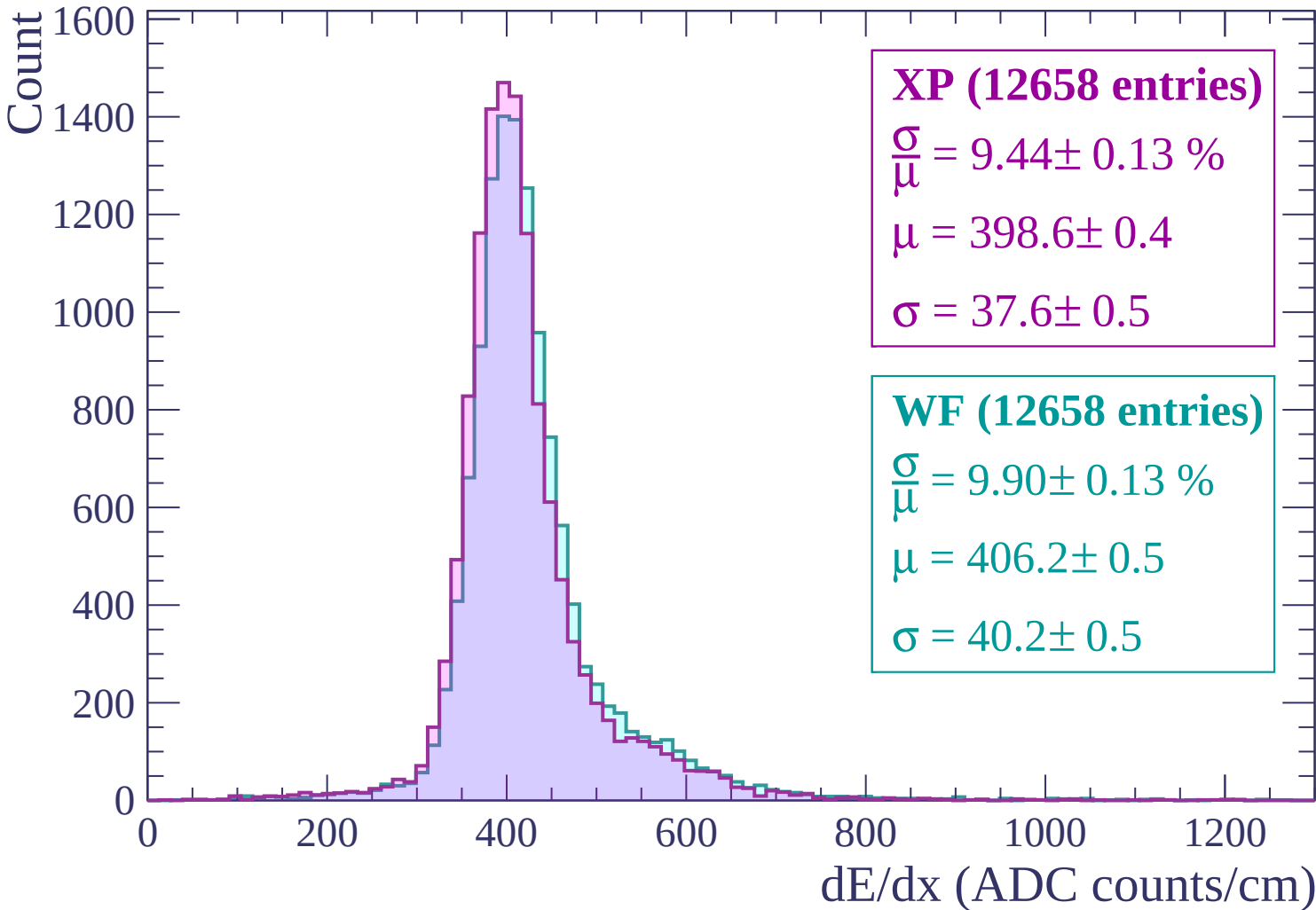
1000

1200

dE/dx (ADC counts/cm)



Energy loss | $480 < p < 540$



Energy loss | $540 < p < 600$

Count

1400
1200
1000
800
600
400
200
0

XP (12149 entries)

$$\frac{\sigma}{\mu} = 9.74 \pm 0.12 \%$$

$$\mu = 402.4 \pm 0.4$$

$$\sigma = 39.2 \pm 0.5$$

WF (12149 entries)

$$\frac{\sigma}{\mu} = 10.00 \pm 0.12 \%$$

$$\mu = 410.0 \pm 0.5$$

$$\sigma = 41.0 \pm 0.4$$

0

200

400

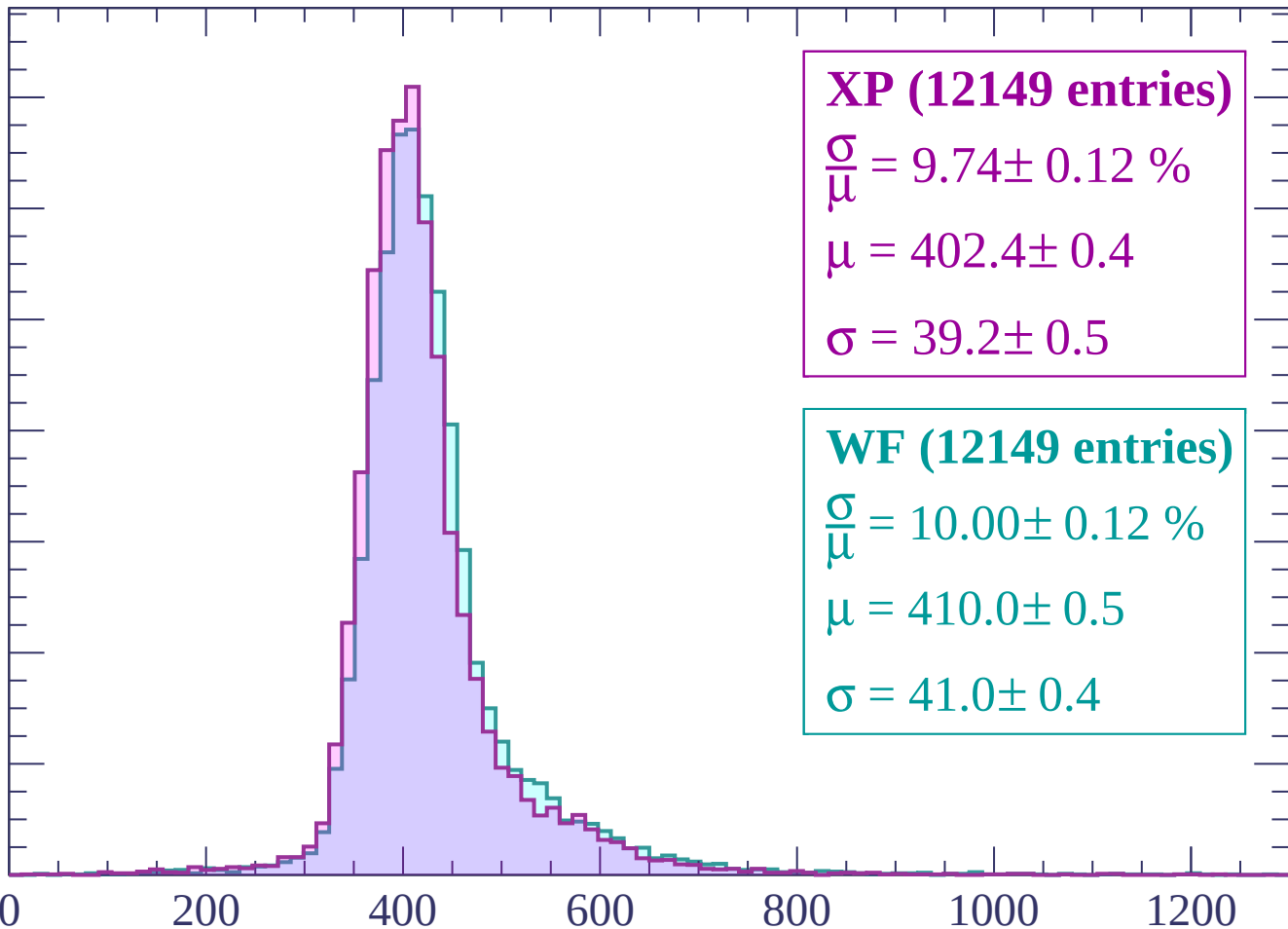
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $600 < p < 660$

Count

1400
1200
1000
800
600
400
200
0

XP (11583 entries)

$$\frac{\sigma}{\mu} = 9.64 \pm 0.12 \%$$

$$\mu = 405.7 \pm 0.5$$

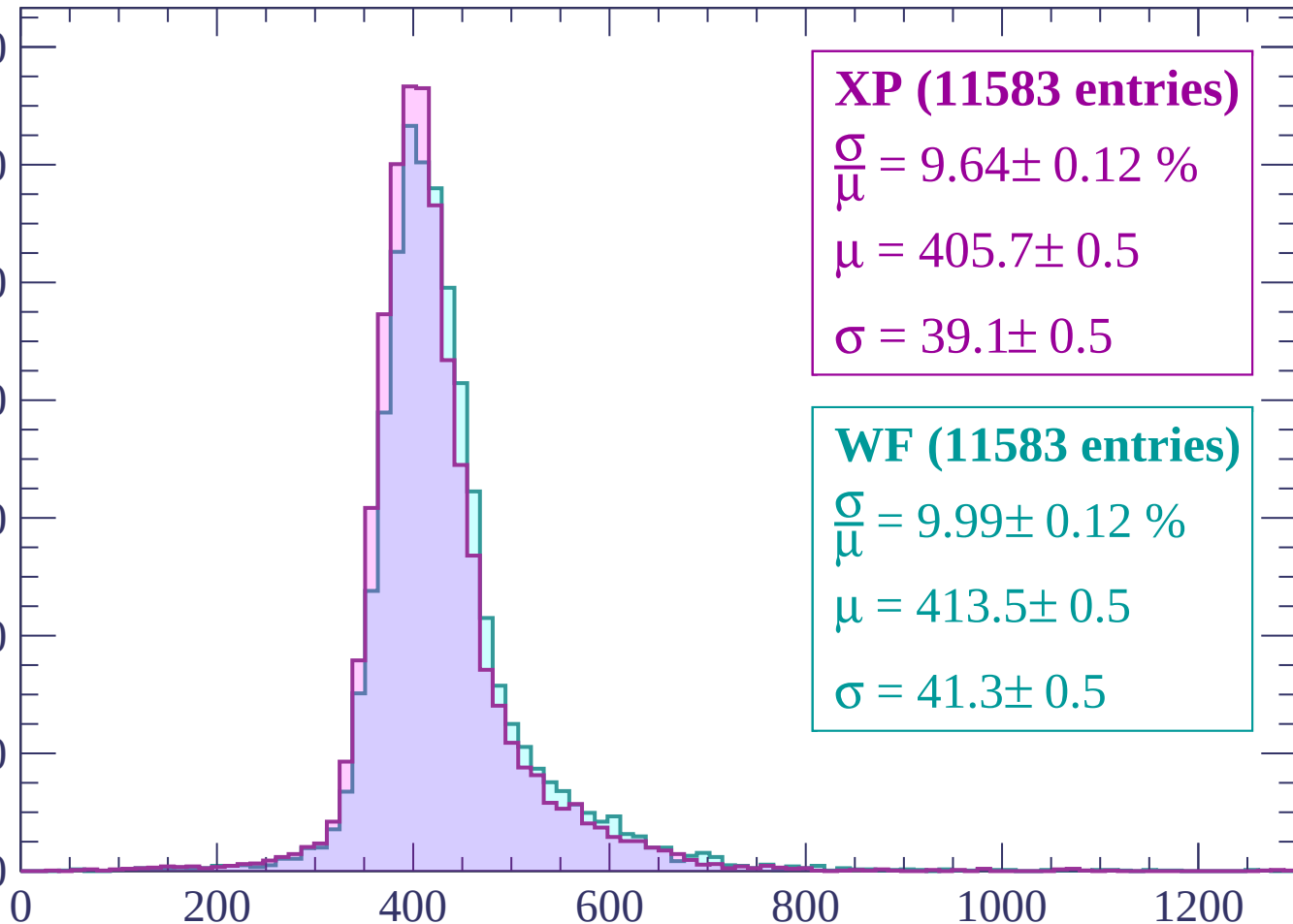
$$\sigma = 39.1 \pm 0.5$$

WF (11583 entries)

$$\frac{\sigma}{\mu} = 9.99 \pm 0.12 \%$$

$$\mu = 413.5 \pm 0.5$$

$$\sigma = 41.3 \pm 0.5$$



dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (11178 entries)

$$\frac{\sigma}{\mu} = 9.41 \pm 0.12 \%$$

$$\mu = 408.2 \pm 0.5$$

$$\sigma = 38.4 \pm 0.4$$

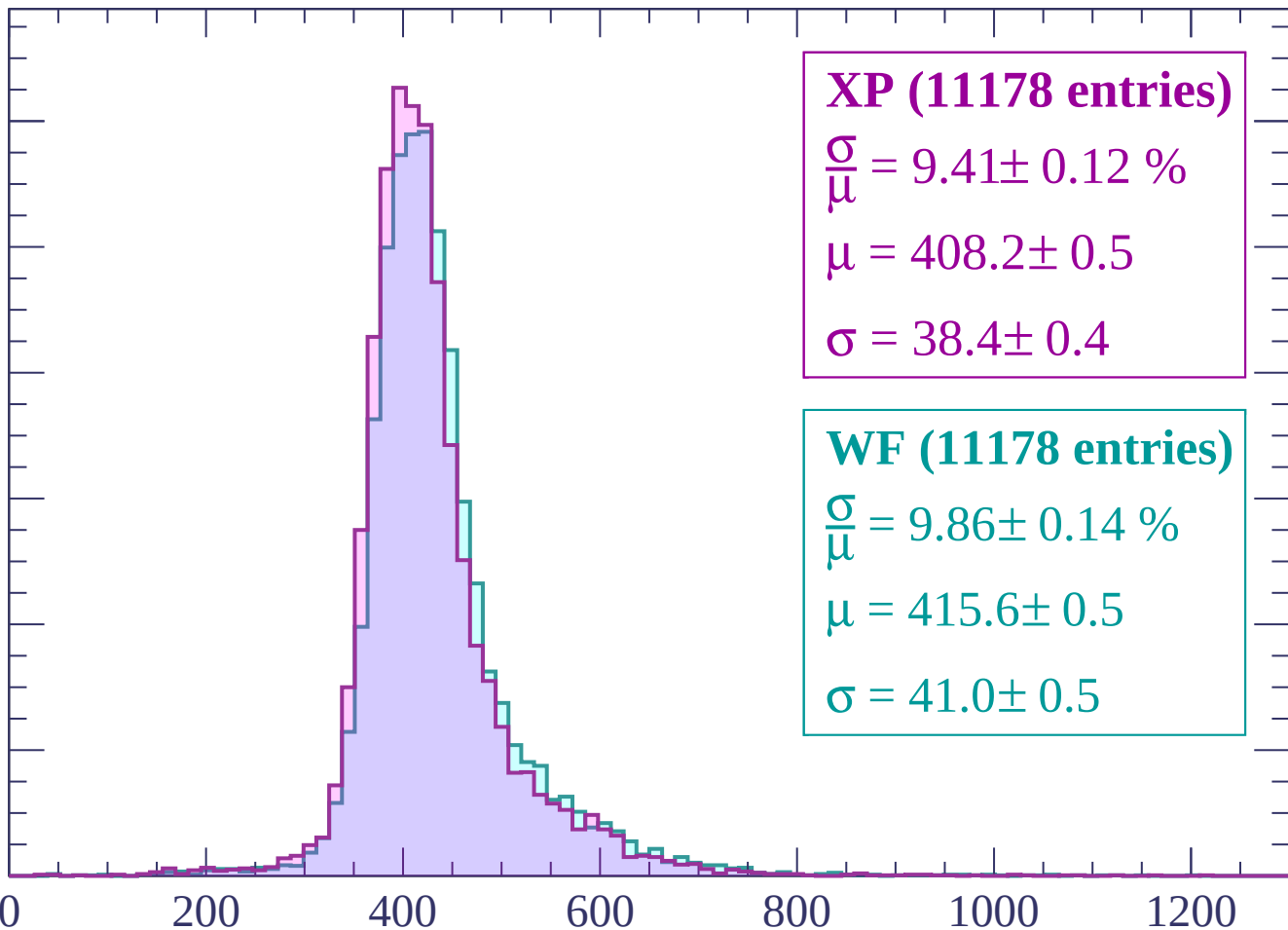
WF (11178 entries)

$$\frac{\sigma}{\mu} = 9.86 \pm 0.14 \%$$

$$\mu = 415.6 \pm 0.5$$

$$\sigma = 41.0 \pm 0.5$$

dE/dx (ADC counts/cm)



Energy loss | $720 < p < 780$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (10973 entries)

$\frac{\sigma}{\mu} = 9.69 \pm 0.12 \%$

$\mu = 413.8 \pm 0.5$

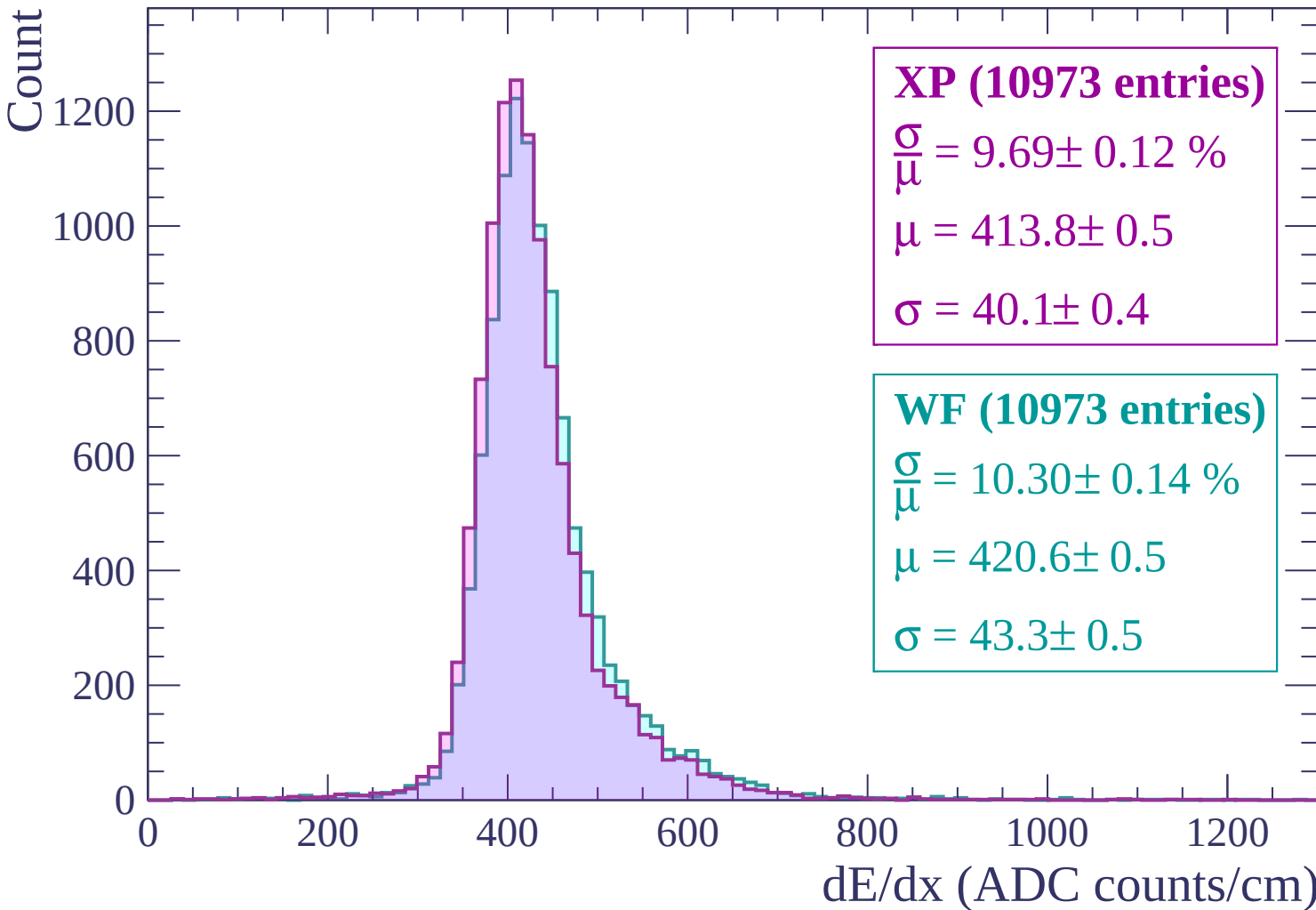
$\sigma = 40.1 \pm 0.4$

WF (10973 entries)

$\frac{\sigma}{\mu} = 10.30 \pm 0.14 \%$

$\mu = 420.6 \pm 0.5$

$\sigma = 43.3 \pm 0.5$



Energy loss | $780 < p < 840$

Count

1200

1000

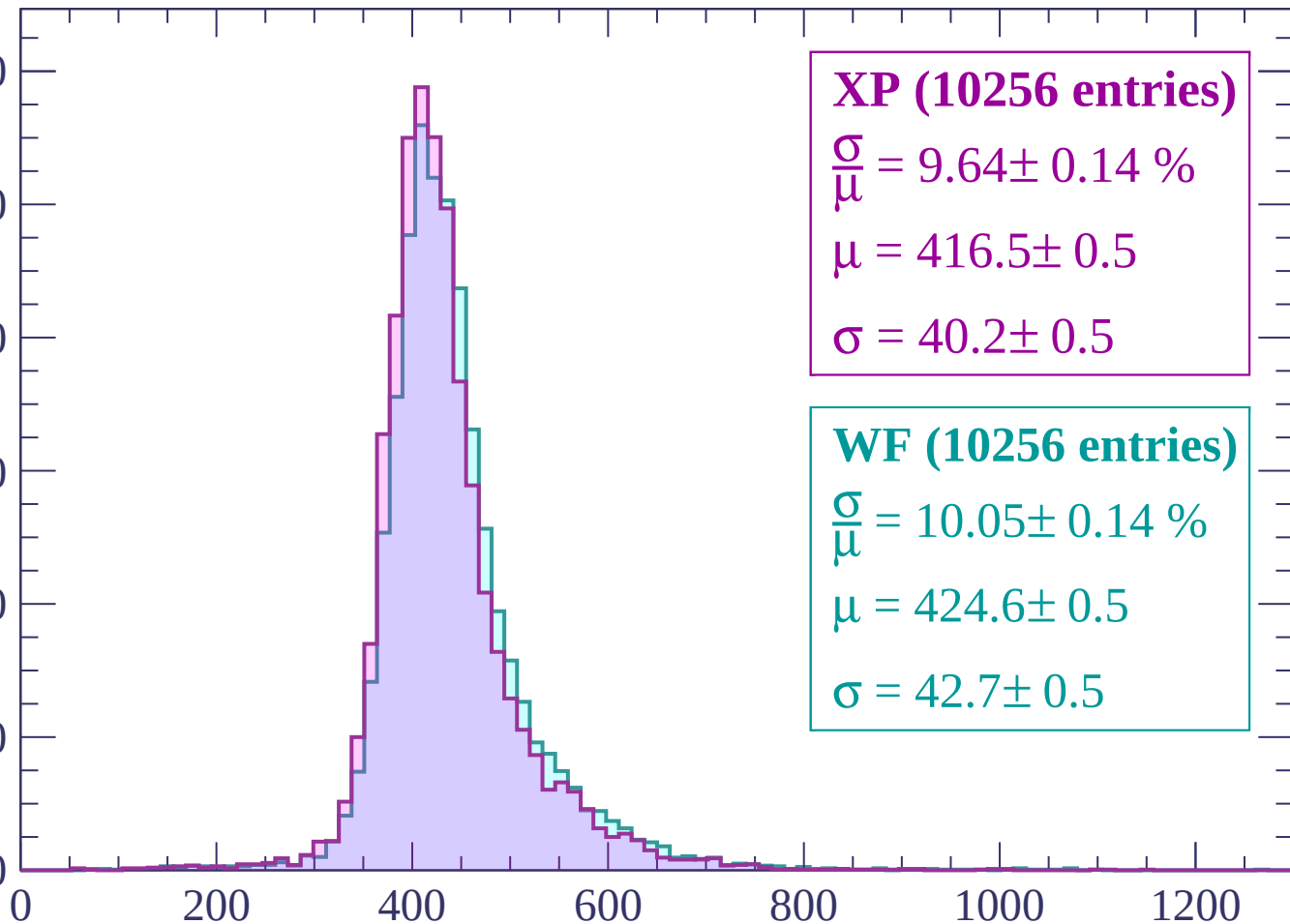
800

600

400

200

0



XP (10256 entries)

$\frac{\sigma}{\mu} = 9.64 \pm 0.14 \%$

$\mu = 416.5 \pm 0.5$

$\sigma = 40.2 \pm 0.5$

WF (10256 entries)

$\frac{\sigma}{\mu} = 10.05 \pm 0.14 \%$

$\mu = 424.6 \pm 0.5$

$\sigma = 42.7 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 900$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (9961 entries)

$\frac{\sigma}{\mu} = 10.10 \pm 0.14 \%$

$\mu = 421.6 \pm 0.5$

$\sigma = 42.6 \pm 0.5$

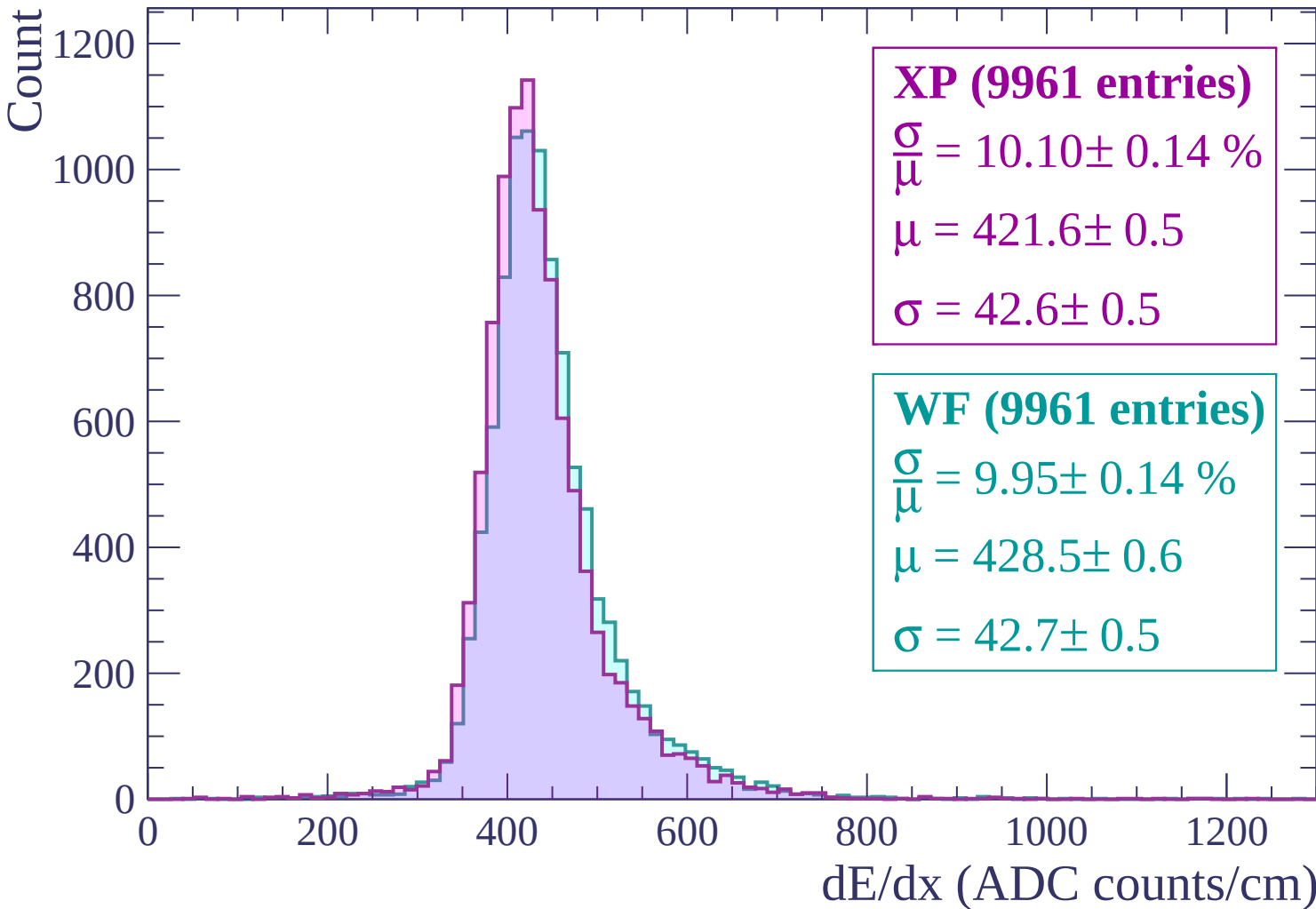
WF (9961 entries)

$\frac{\sigma}{\mu} = 9.95 \pm 0.14 \%$

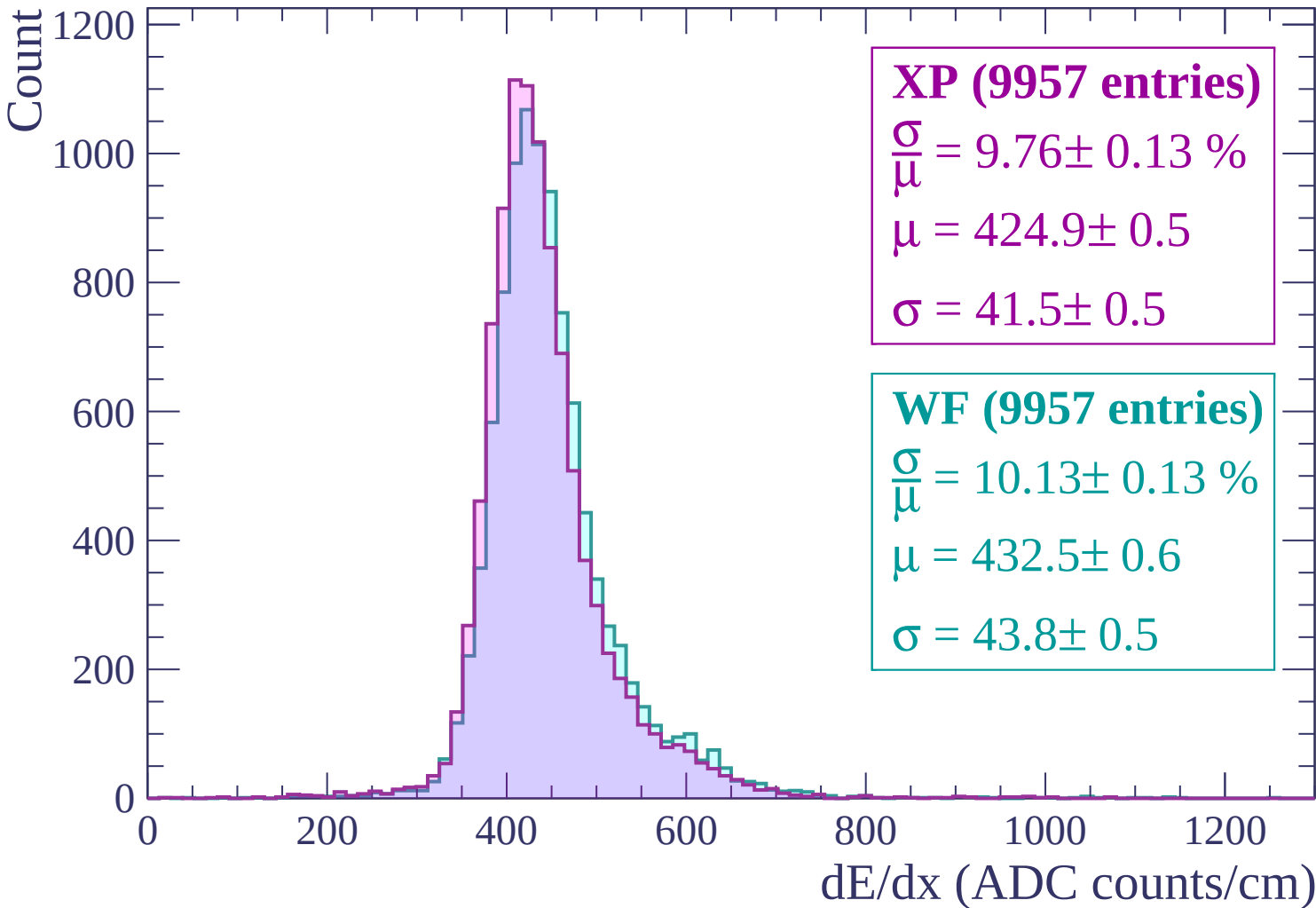
$\mu = 428.5 \pm 0.6$

$\sigma = 42.7 \pm 0.5$

dE/dx (ADC counts/cm)



Energy loss | $900 < p < 960$



Energy loss | $960 < p < 1020$

Count

1000

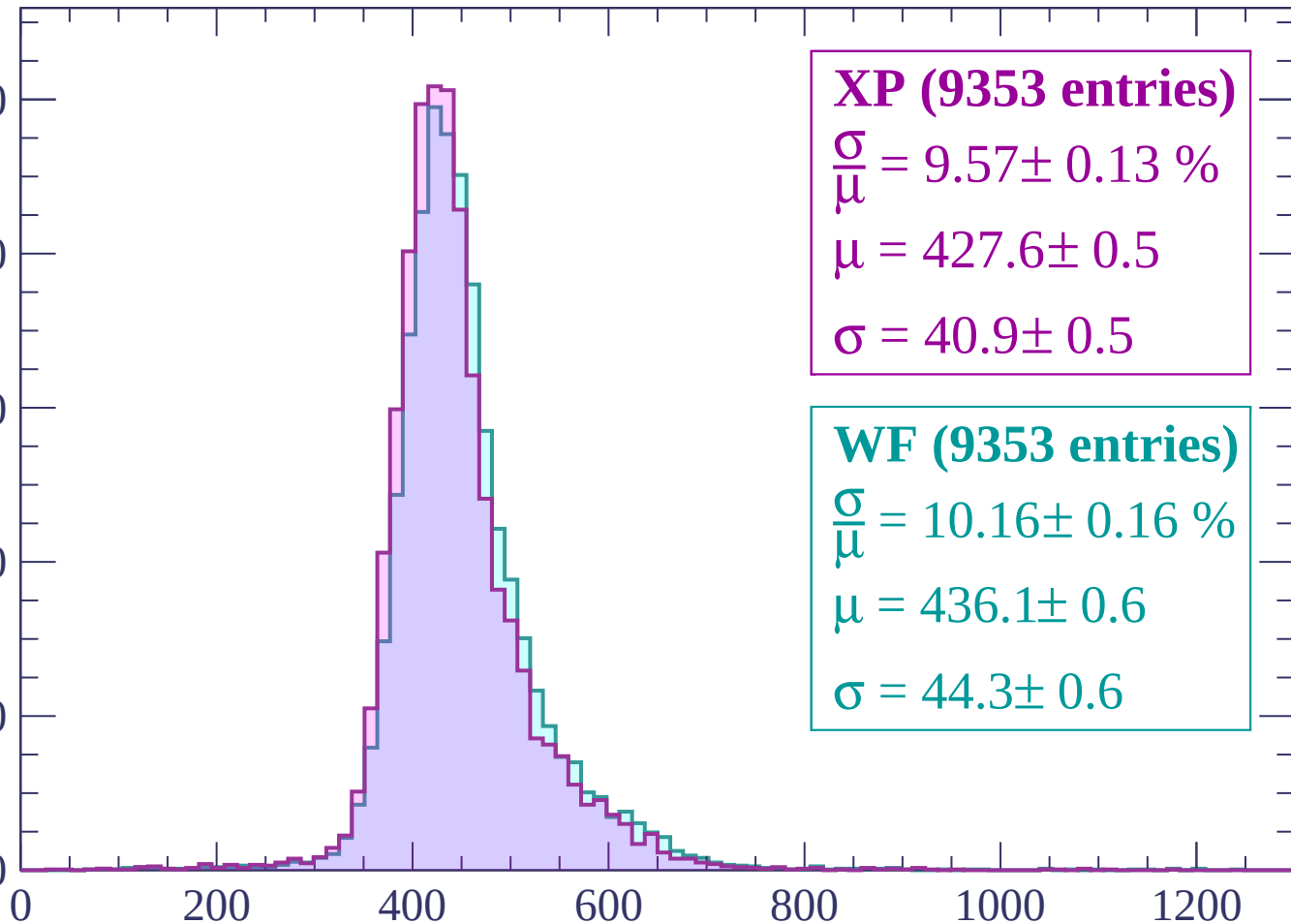
800

600

400

200

0



XP (9353 entries)

$\frac{\sigma}{\mu} = 9.57 \pm 0.13 \%$

$\mu = 427.6 \pm 0.5$

$\sigma = 40.9 \pm 0.5$

WF (9353 entries)

$\frac{\sigma}{\mu} = 10.16 \pm 0.16 \%$

$\mu = 436.1 \pm 0.6$

$\sigma = 44.3 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $1020 < p < 1080$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (9094 entries)

$\frac{\sigma}{\mu} = 10.00 \pm 0.14 \%$

$\mu = 433.3 \pm 0.6$

$\sigma = 43.3 \pm 0.5$

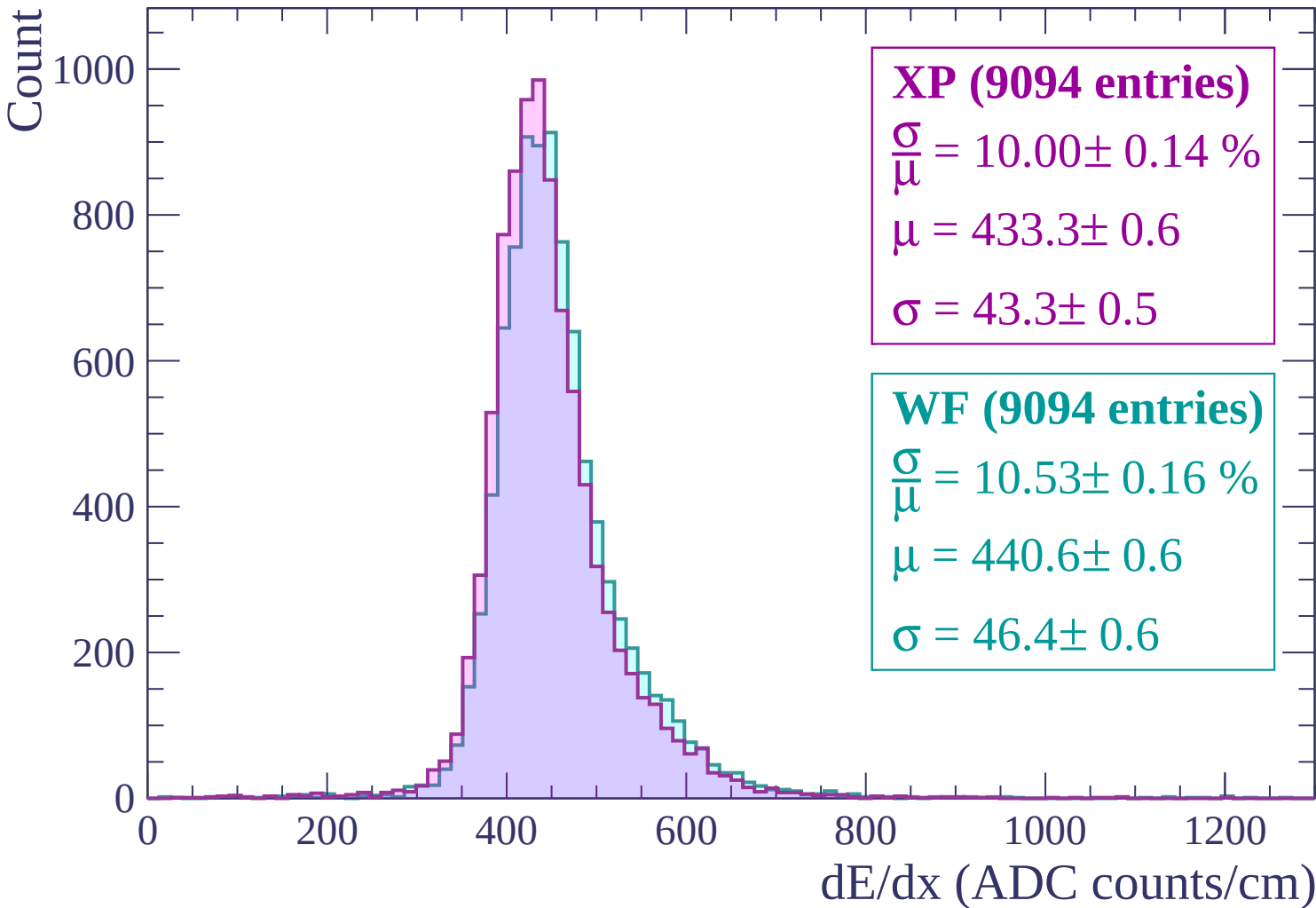
WF (9094 entries)

$\frac{\sigma}{\mu} = 10.53 \pm 0.16 \%$

$\mu = 440.6 \pm 0.6$

$\sigma = 46.4 \pm 0.6$

dE/dx (ADC counts/cm)



Energy loss | $1080 < p < 1140$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (8770 entries)

$\frac{\sigma}{\mu} = 10.17 \pm 0.16 \%$

$\mu = 435.1 \pm 0.6$

$\sigma = 44.3 \pm 0.6$

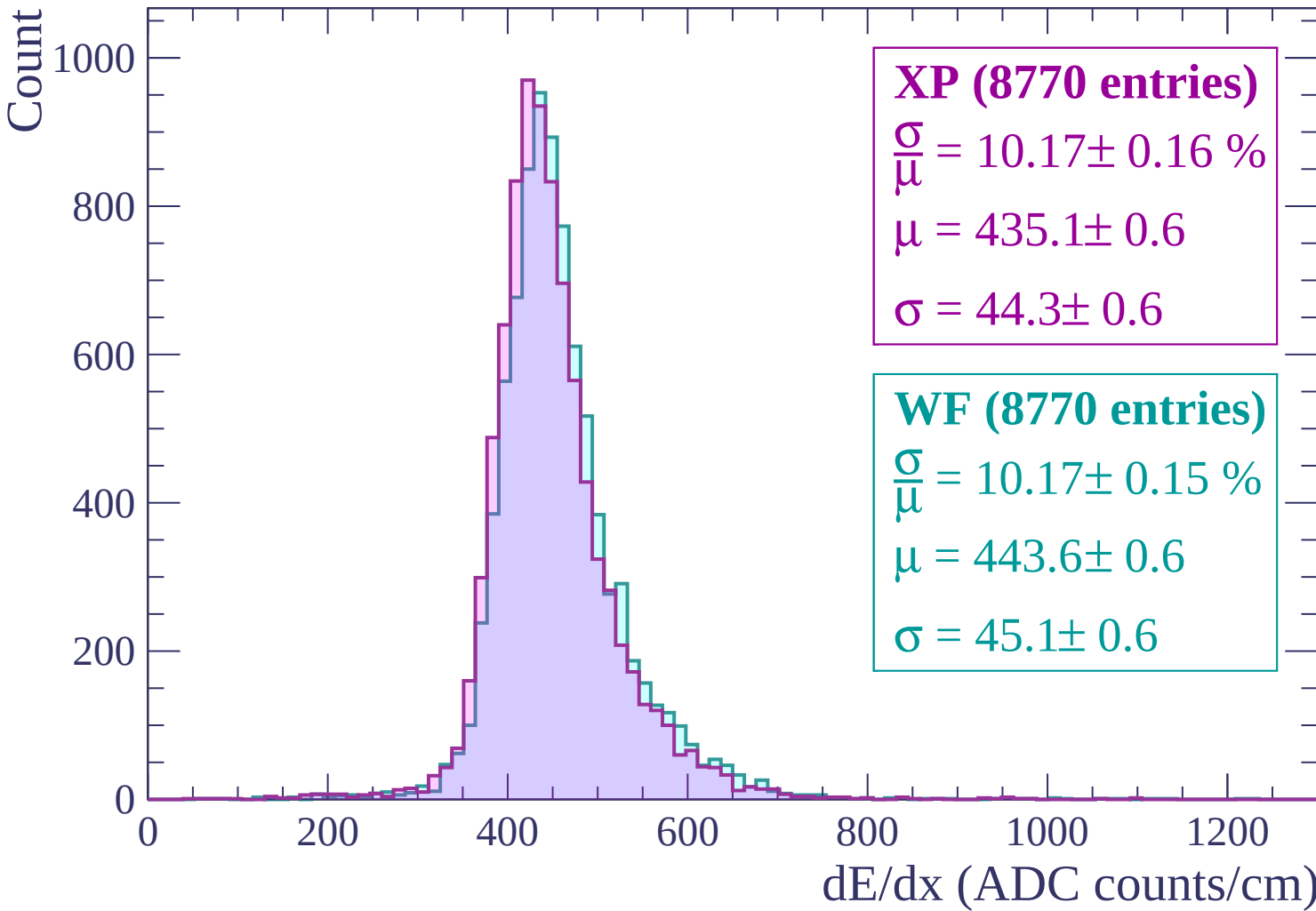
WF (8770 entries)

$\frac{\sigma}{\mu} = 10.17 \pm 0.15 \%$

$\mu = 443.6 \pm 0.6$

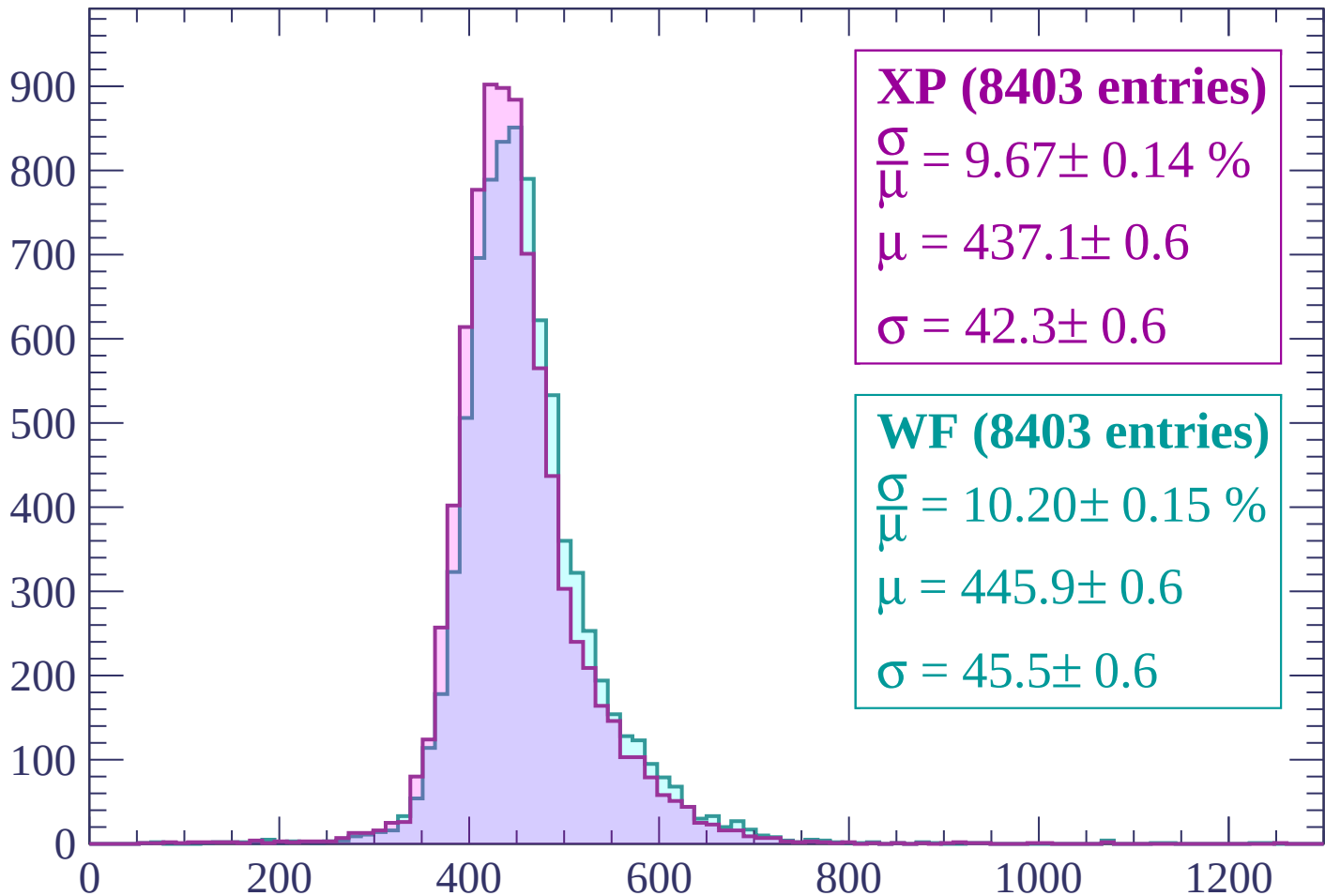
$\sigma = 45.1 \pm 0.6$

dE/dx (ADC counts/cm)



Energy loss | $1140 < p < 1200$

Count



XP (8403 entries)

$\frac{\sigma}{\mu} = 9.67 \pm 0.14 \%$

$\mu = 437.1 \pm 0.6$

$\sigma = 42.3 \pm 0.6$

WF (8403 entries)

$\frac{\sigma}{\mu} = 10.20 \pm 0.15 \%$

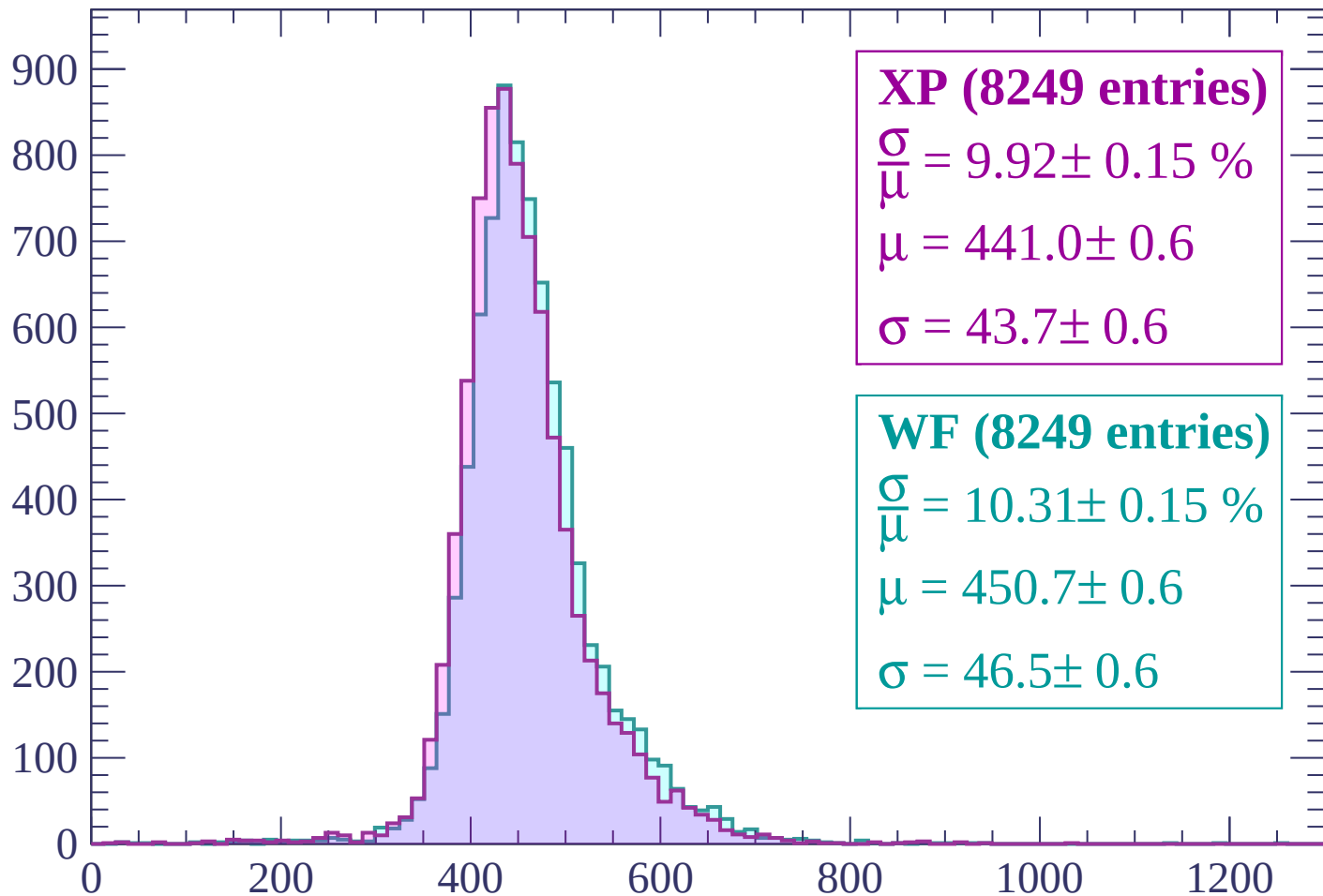
$\mu = 445.9 \pm 0.6$

$\sigma = 45.5 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1260$

Count



XP (8249 entries)

$$\frac{\sigma}{\mu} = 9.92 \pm 0.15 \%$$

$$\mu = 441.0 \pm 0.6$$

$$\sigma = 43.7 \pm 0.6$$

WF (8249 entries)

$$\frac{\sigma}{\mu} = 10.31 \pm 0.15 \%$$

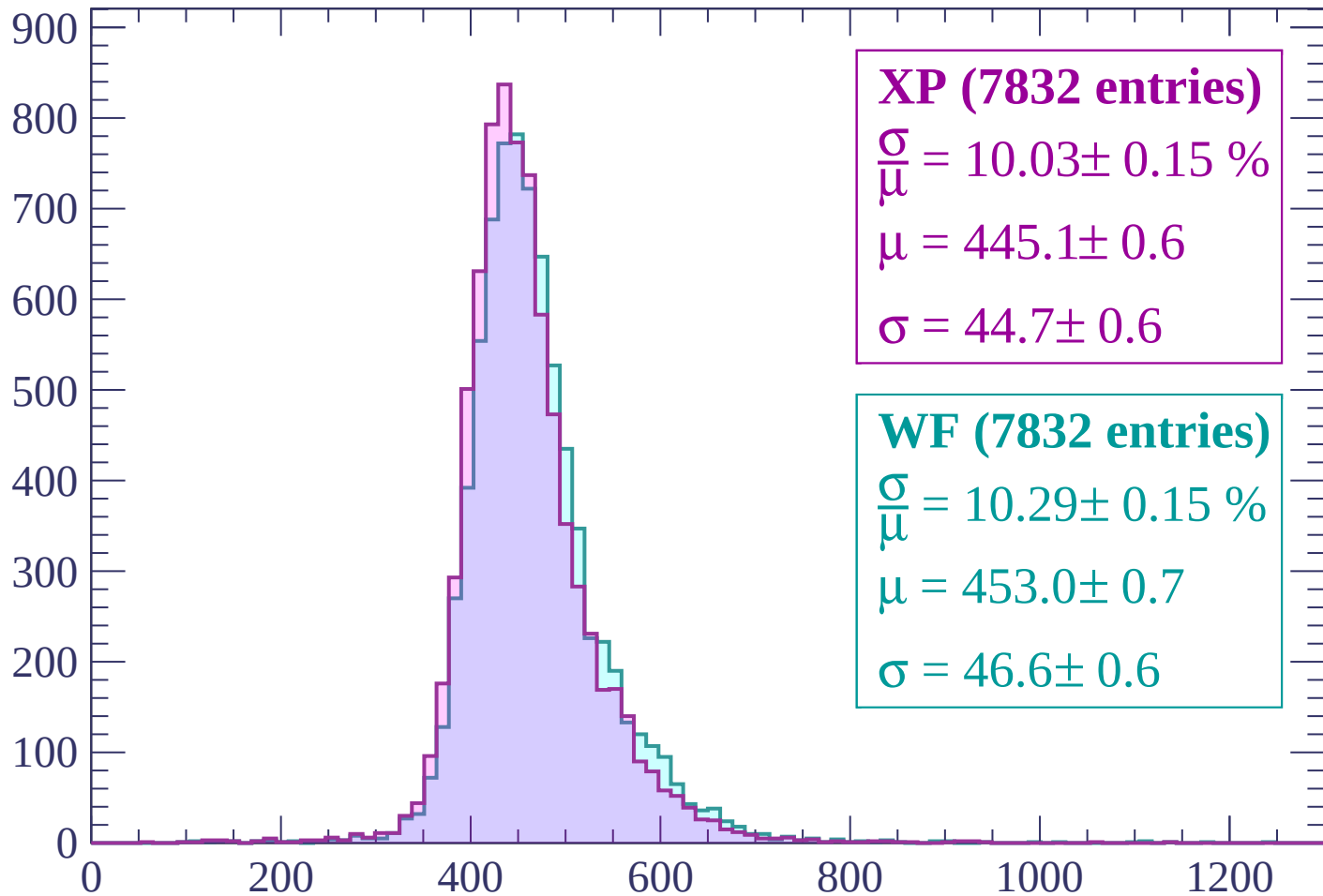
$$\mu = 450.7 \pm 0.6$$

$$\sigma = 46.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1260 < p < 1320$

Count



XP (7832 entries)

$\frac{\sigma}{\mu} = 10.03 \pm 0.15 \%$

$\mu = 445.1 \pm 0.6$

$\sigma = 44.7 \pm 0.6$

WF (7832 entries)

$\frac{\sigma}{\mu} = 10.29 \pm 0.15 \%$

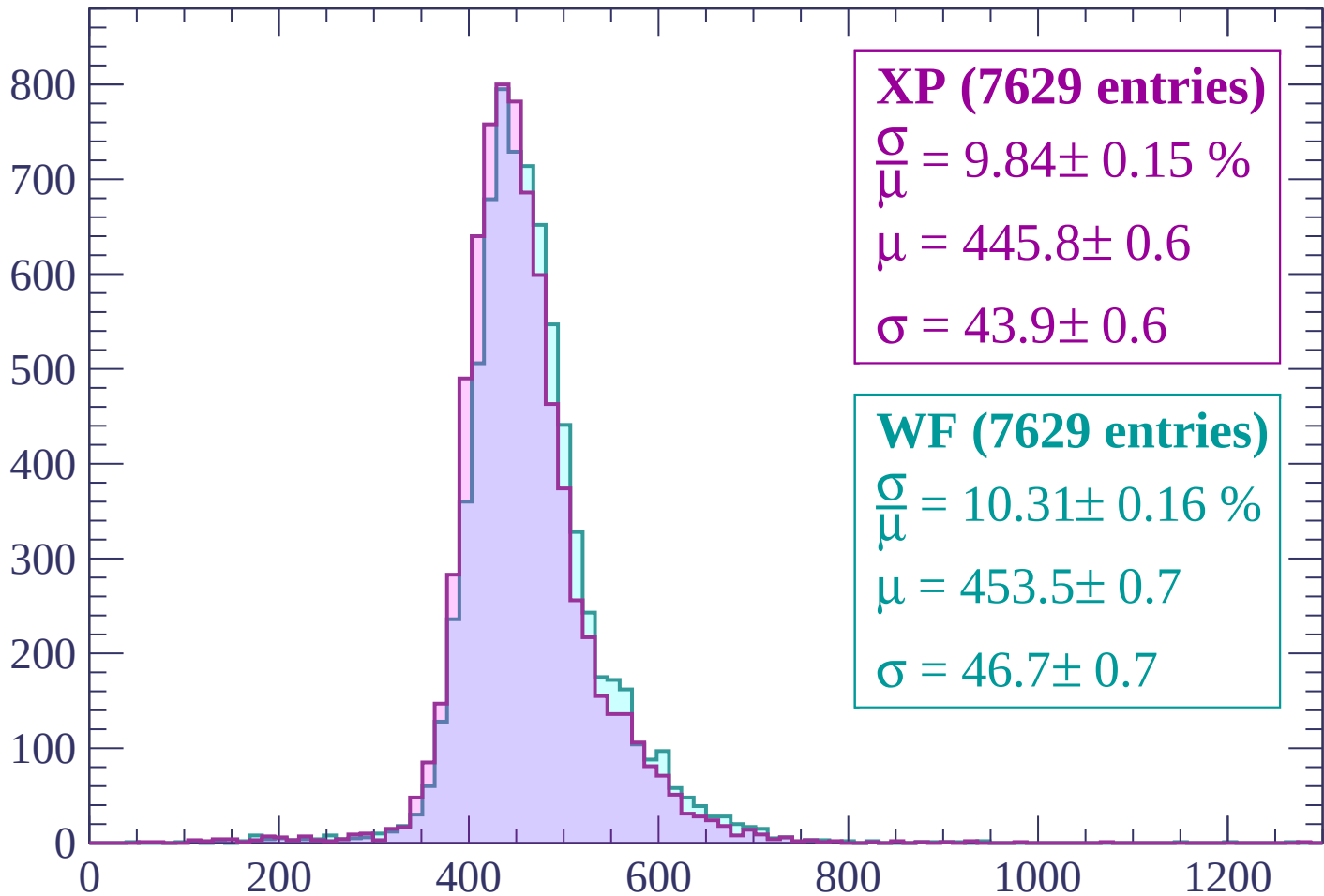
$\mu = 453.0 \pm 0.7$

$\sigma = 46.6 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1380$

Count



XP (7629 entries)

$$\frac{\sigma}{\mu} = 9.84 \pm 0.15 \%$$

$$\mu = 445.8 \pm 0.6$$

$$\sigma = 43.9 \pm 0.6$$

WF (7629 entries)

$$\frac{\sigma}{\mu} = 10.31 \pm 0.16 \%$$

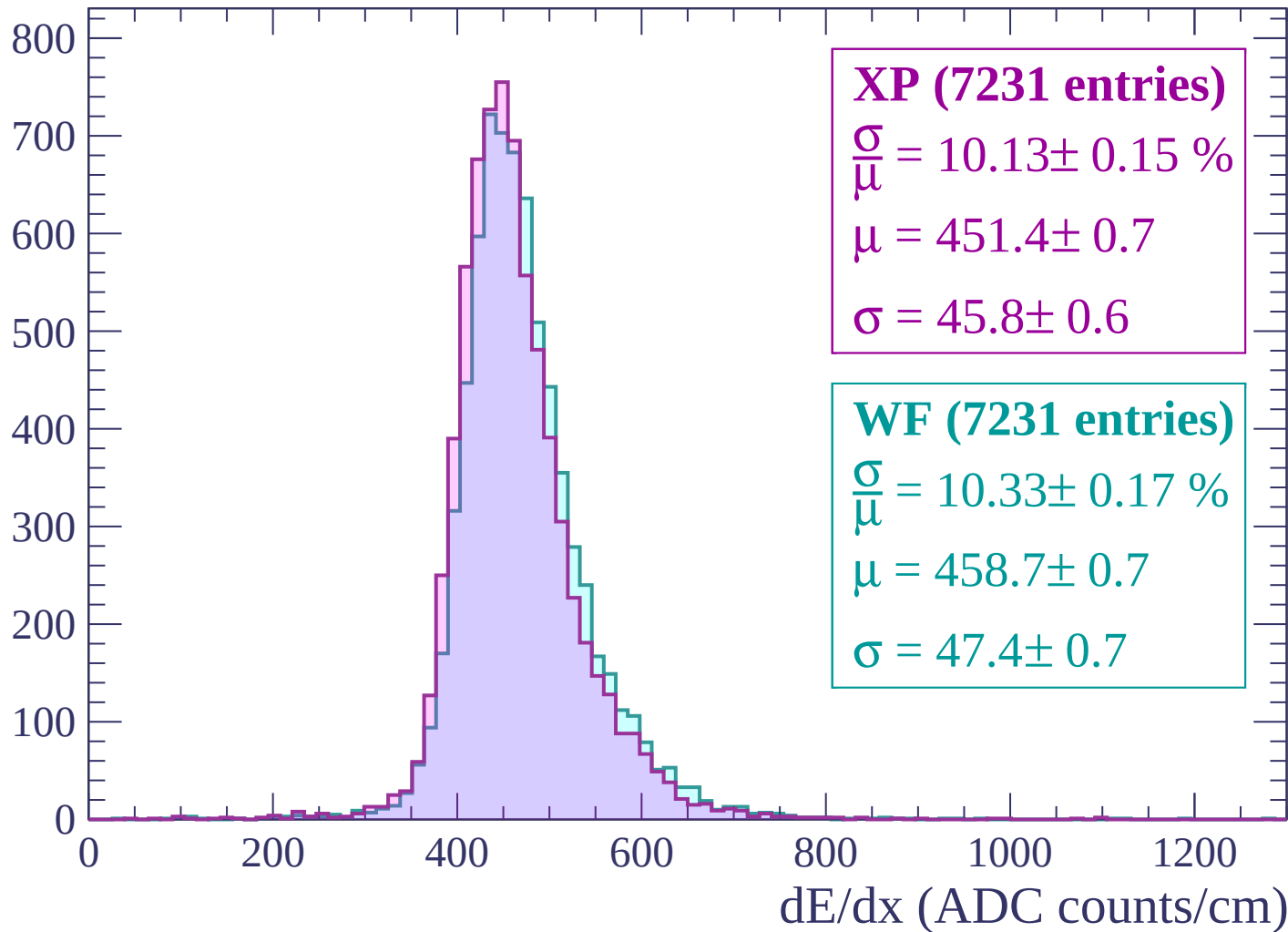
$$\mu = 453.5 \pm 0.7$$

$$\sigma = 46.7 \pm 0.7$$

dE/dx (ADC counts/cm)

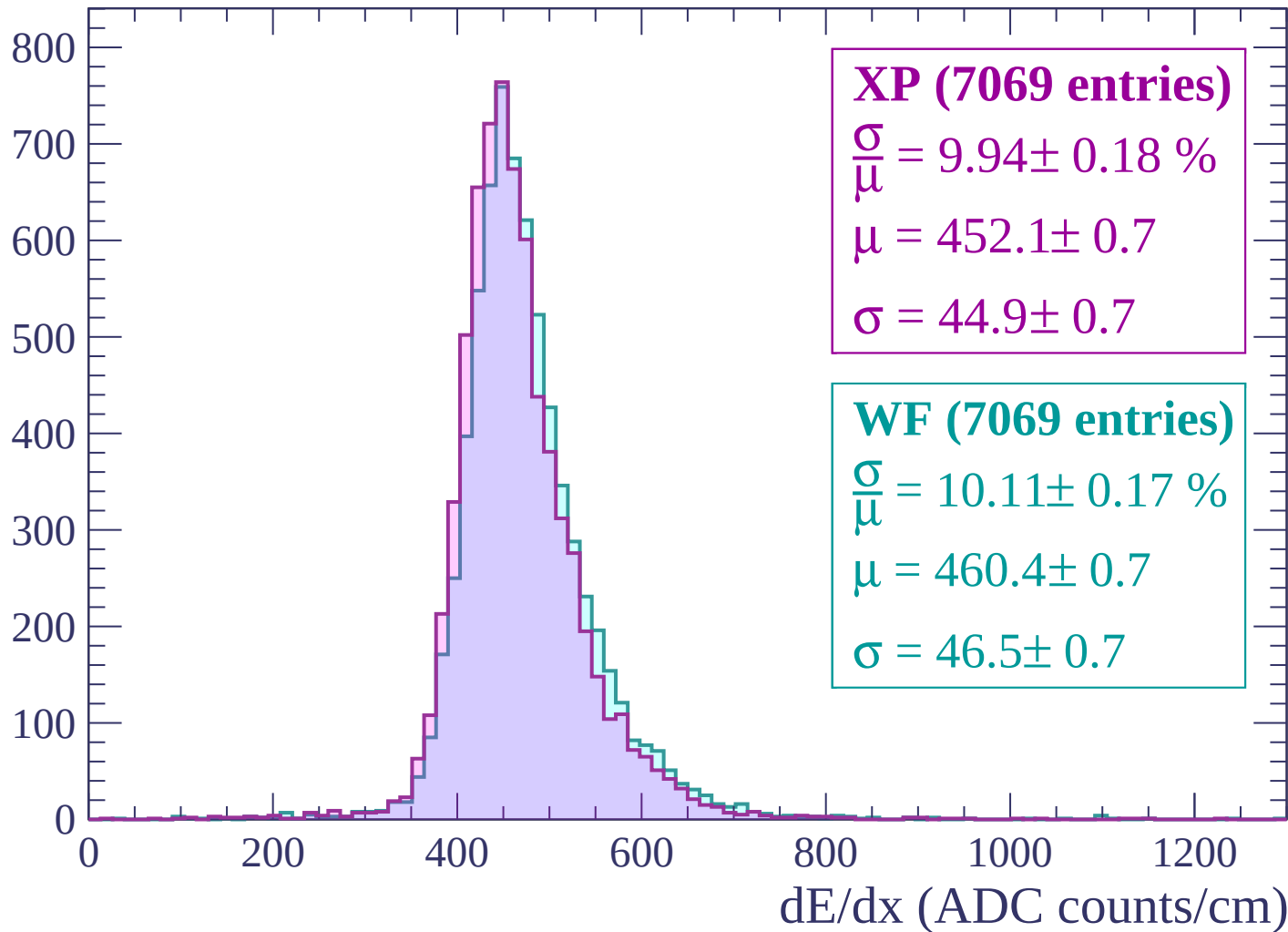
Energy loss | $1380 < p < 1440$

Count



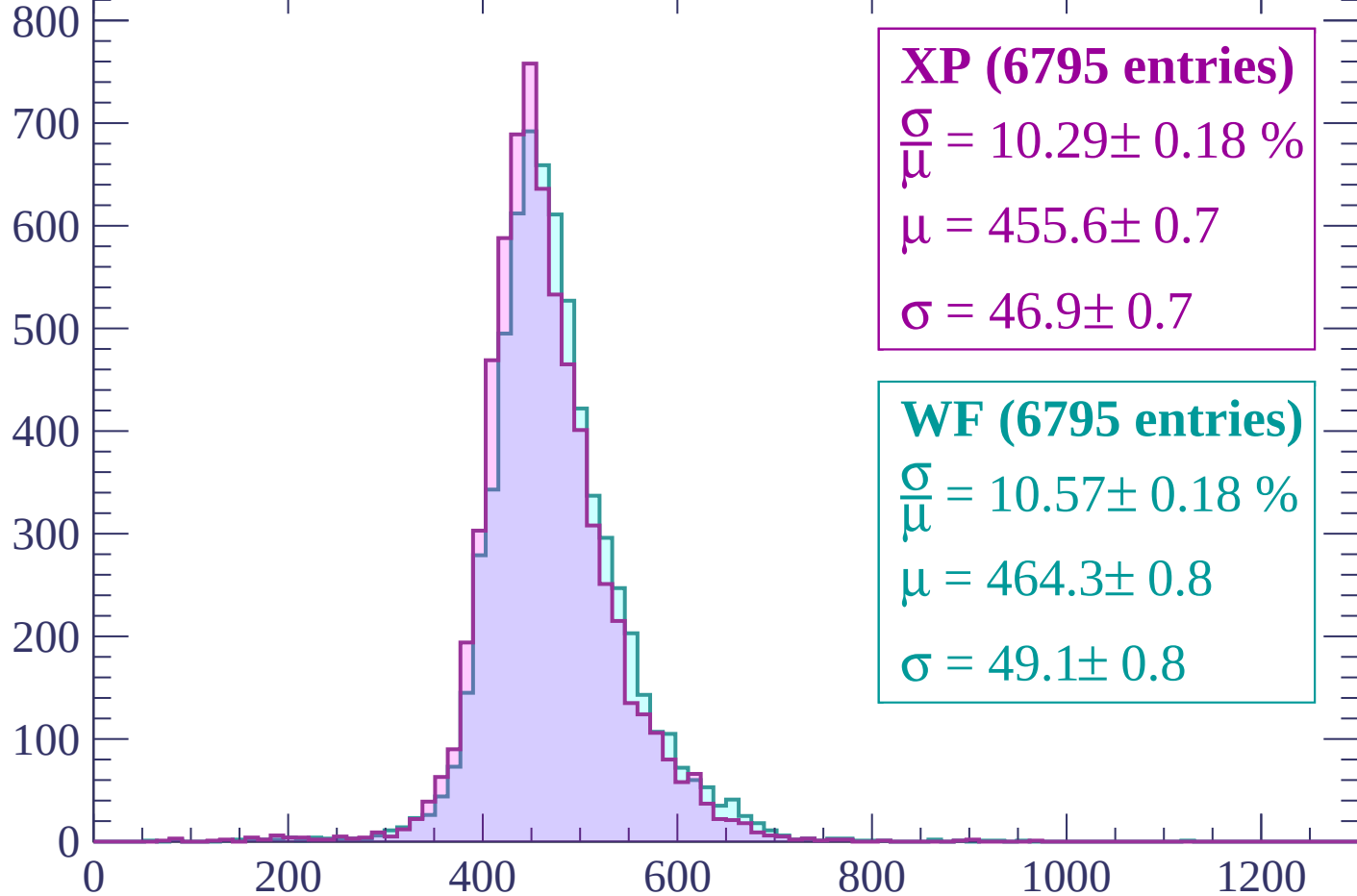
Energy loss | $1440 < p < 1500$

Count



Energy loss | $1500 < p < 1560$

Count



XP (6795 entries)

$\frac{\sigma}{\mu} = 10.29 \pm 0.18 \%$

$\mu = 455.6 \pm 0.7$

$\sigma = 46.9 \pm 0.7$

WF (6795 entries)

$\frac{\sigma}{\mu} = 10.57 \pm 0.18 \%$

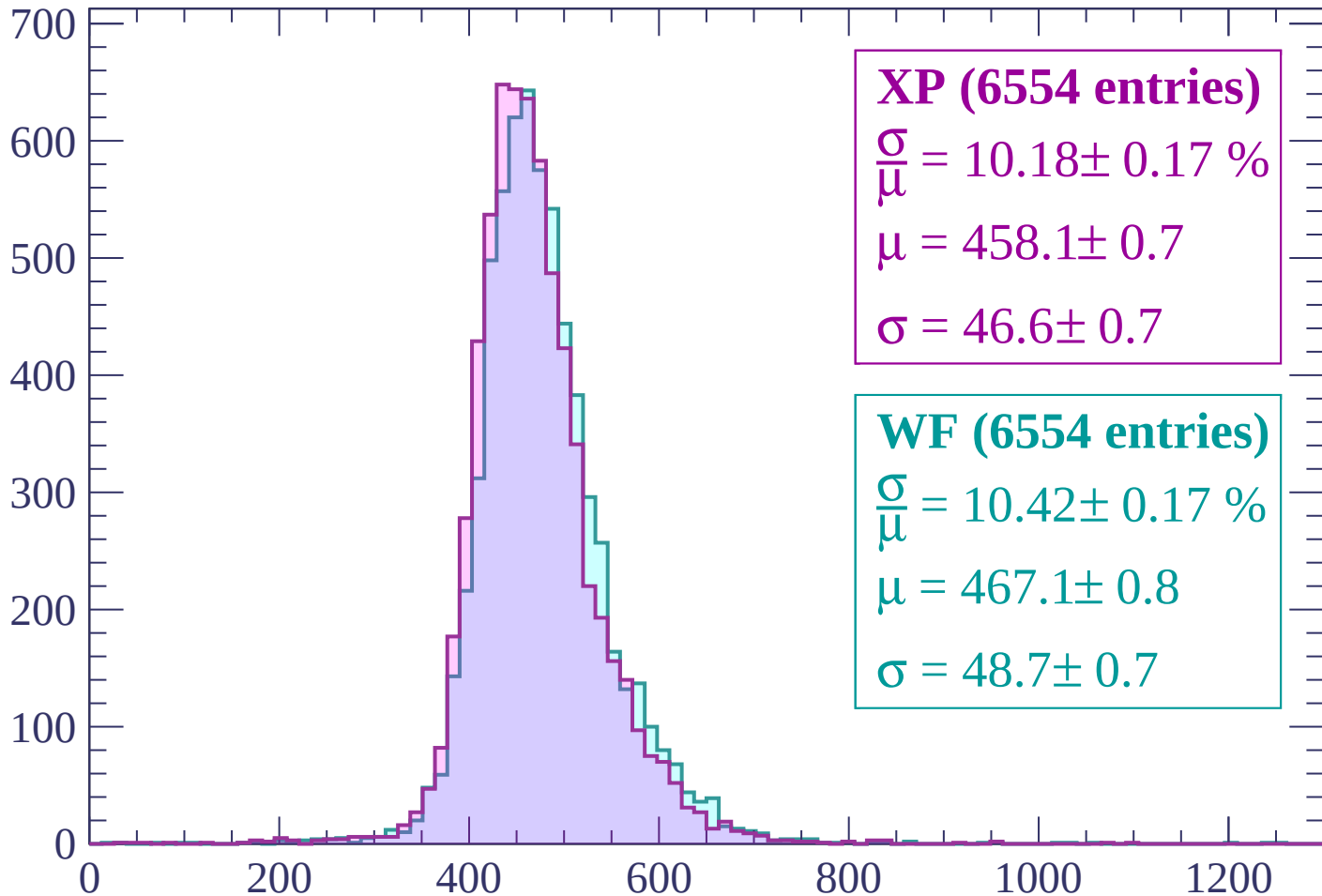
$\mu = 464.3 \pm 0.8$

$\sigma = 49.1 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $1560 < p < 1620$

Count



XP (6554 entries)

$\frac{\sigma}{\mu} = 10.18 \pm 0.17 \%$

$\mu = 458.1 \pm 0.7$

$\sigma = 46.6 \pm 0.7$

WF (6554 entries)

$\frac{\sigma}{\mu} = 10.42 \pm 0.17 \%$

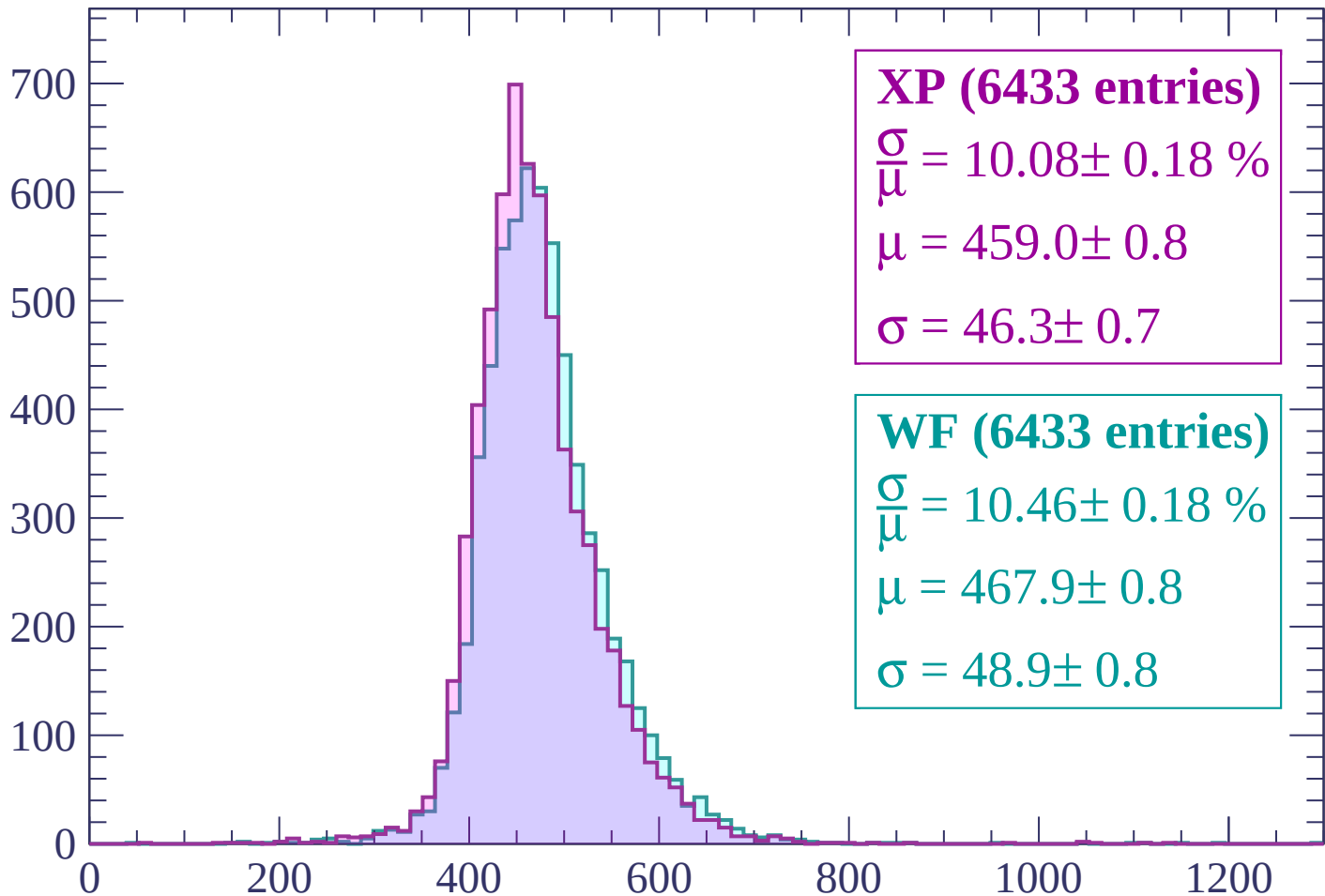
$\mu = 467.1 \pm 0.8$

$\sigma = 48.7 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $1620 < p < 1680$

Count



XP (6433 entries)

$\frac{\sigma}{\mu} = 10.08 \pm 0.18 \%$

$\mu = 459.0 \pm 0.8$

$\sigma = 46.3 \pm 0.7$

WF (6433 entries)

$\frac{\sigma}{\mu} = 10.46 \pm 0.18 \%$

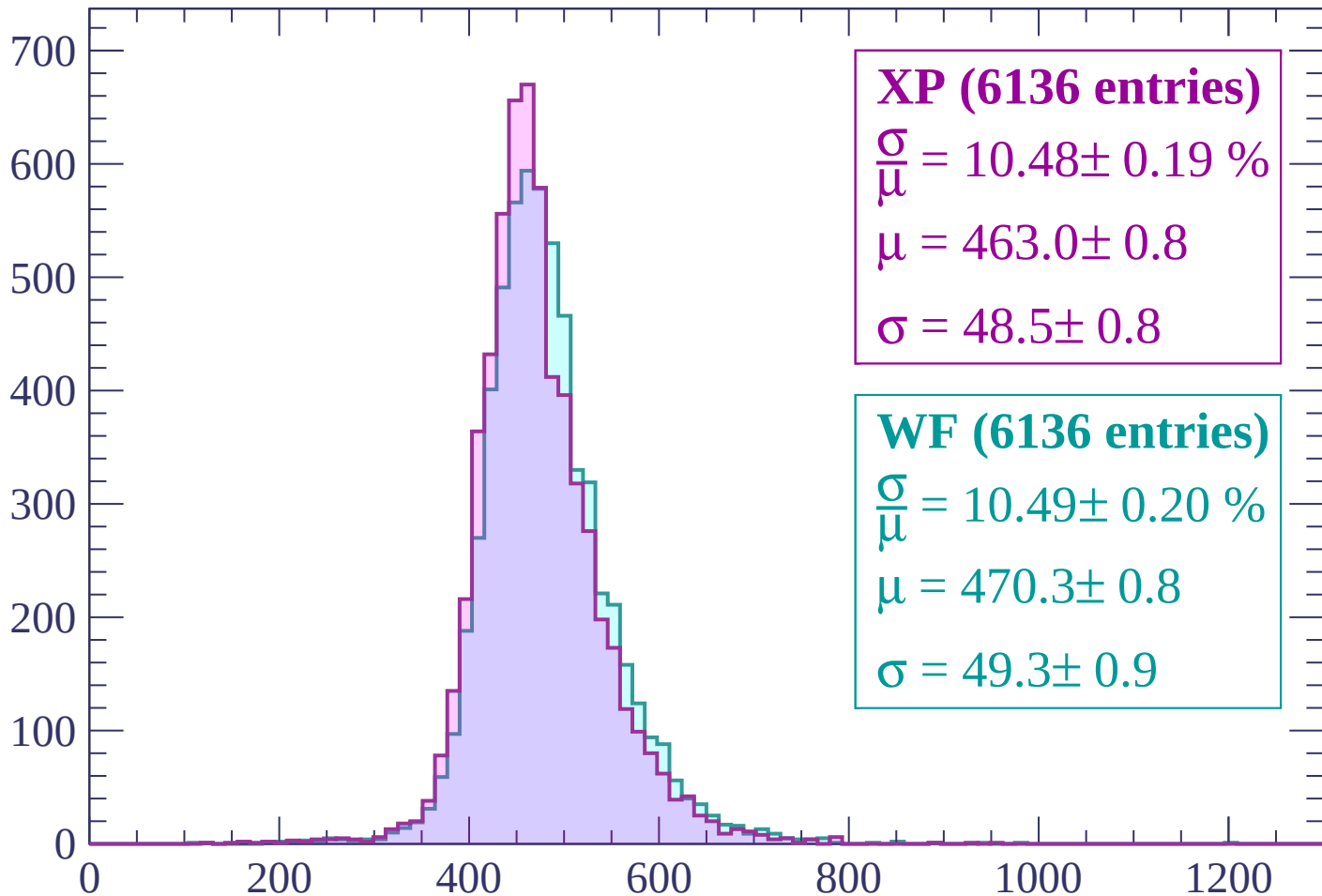
$\mu = 467.9 \pm 0.8$

$\sigma = 48.9 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1740$

Count



XP (6136 entries)

$\frac{\sigma}{\mu} = 10.48 \pm 0.19 \%$

$\mu = 463.0 \pm 0.8$

$\sigma = 48.5 \pm 0.8$

WF (6136 entries)

$\frac{\sigma}{\mu} = 10.49 \pm 0.20 \%$

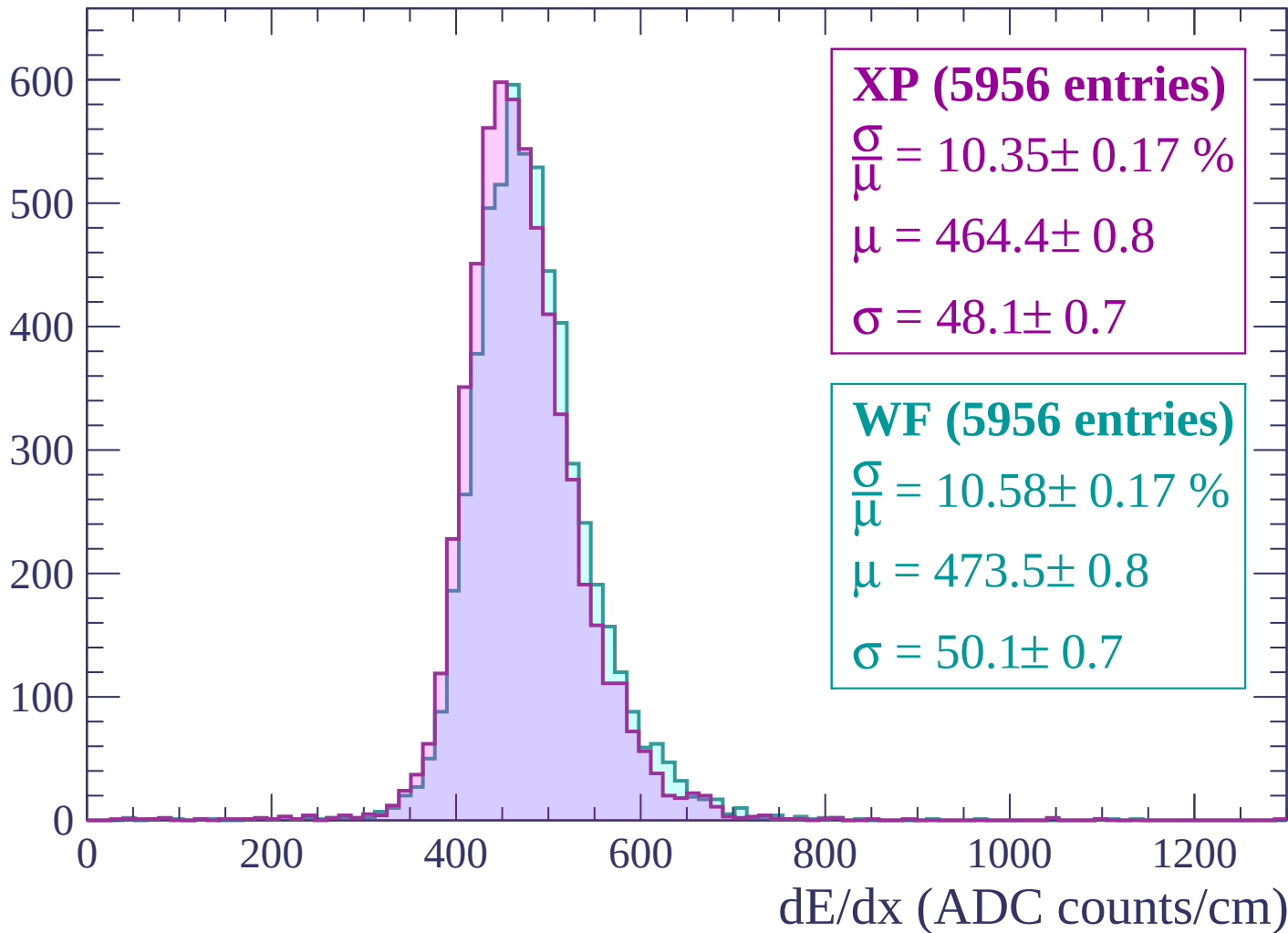
$\mu = 470.3 \pm 0.8$

$\sigma = 49.3 \pm 0.9$

dE/dx (ADC counts/cm)

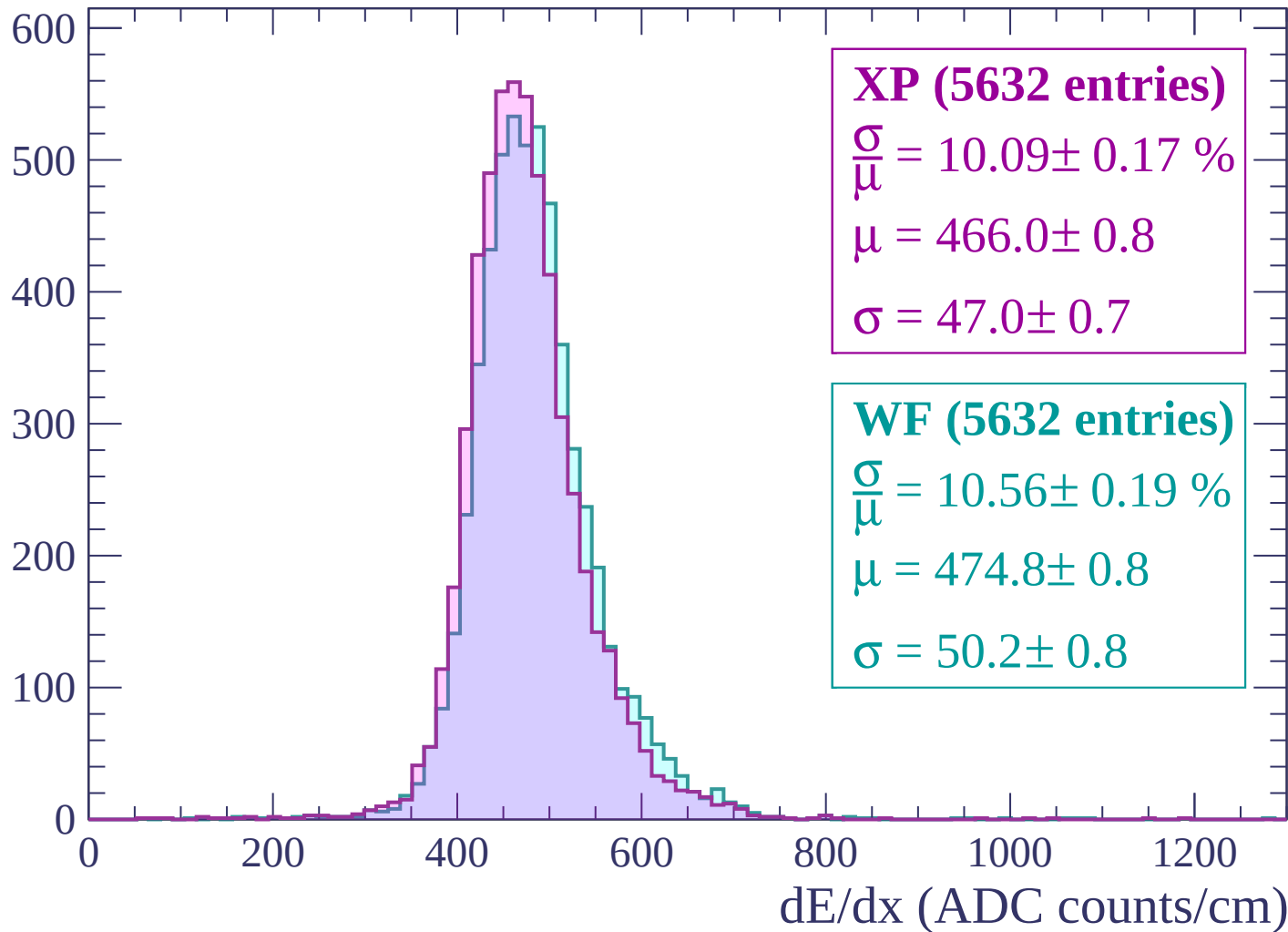
Energy loss | $1740 < p < 1800$

Count



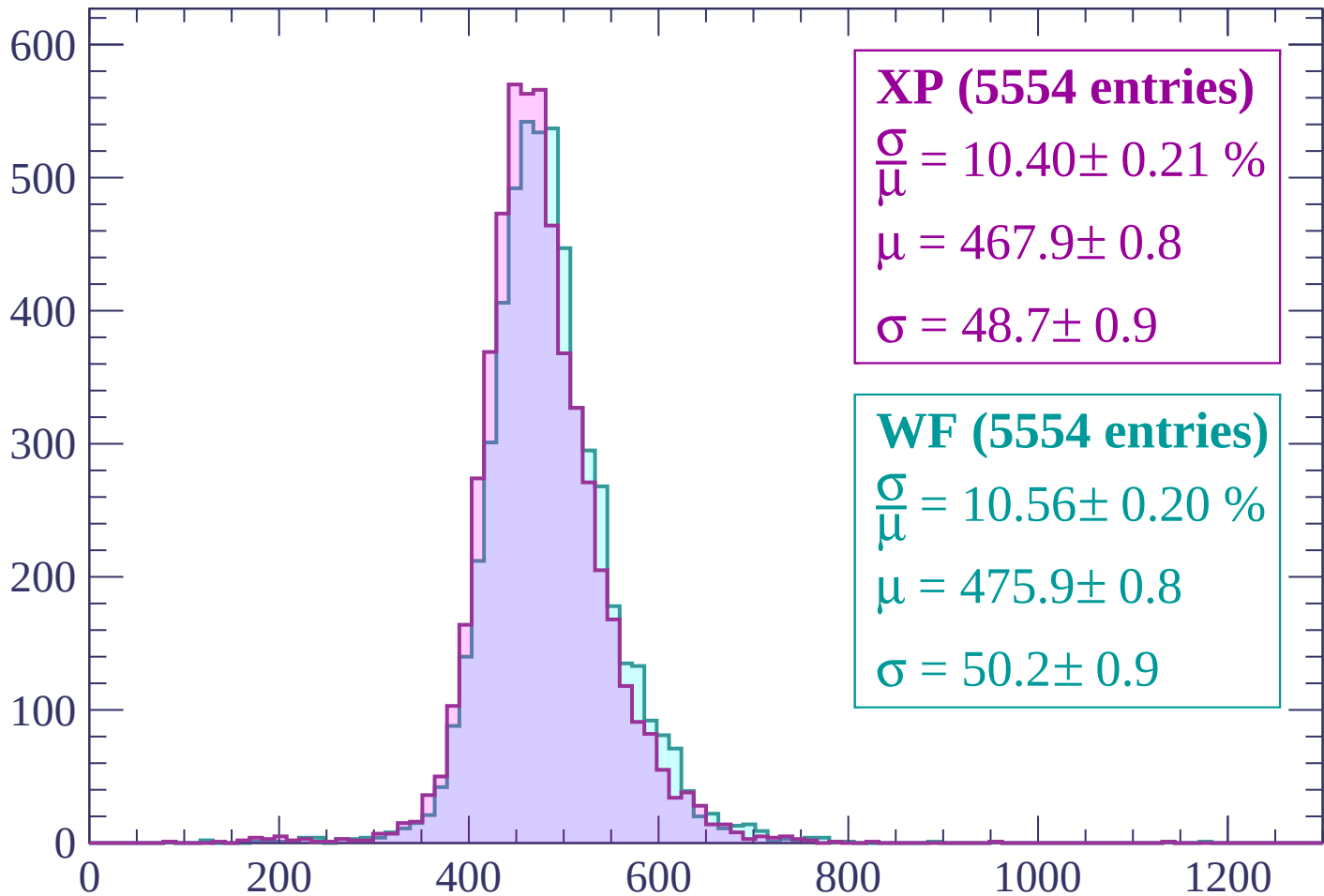
Energy loss | $1800 < p < 1860$

Count



Energy loss | $1860 < p < 1920$

Count



XP (5554 entries)

$$\frac{\sigma}{\mu} = 10.40 \pm 0.21 \%$$

$$\mu = 467.9 \pm 0.8$$

$$\sigma = 48.7 \pm 0.9$$

WF (5554 entries)

$$\frac{\sigma}{\mu} = 10.56 \pm 0.20 \%$$

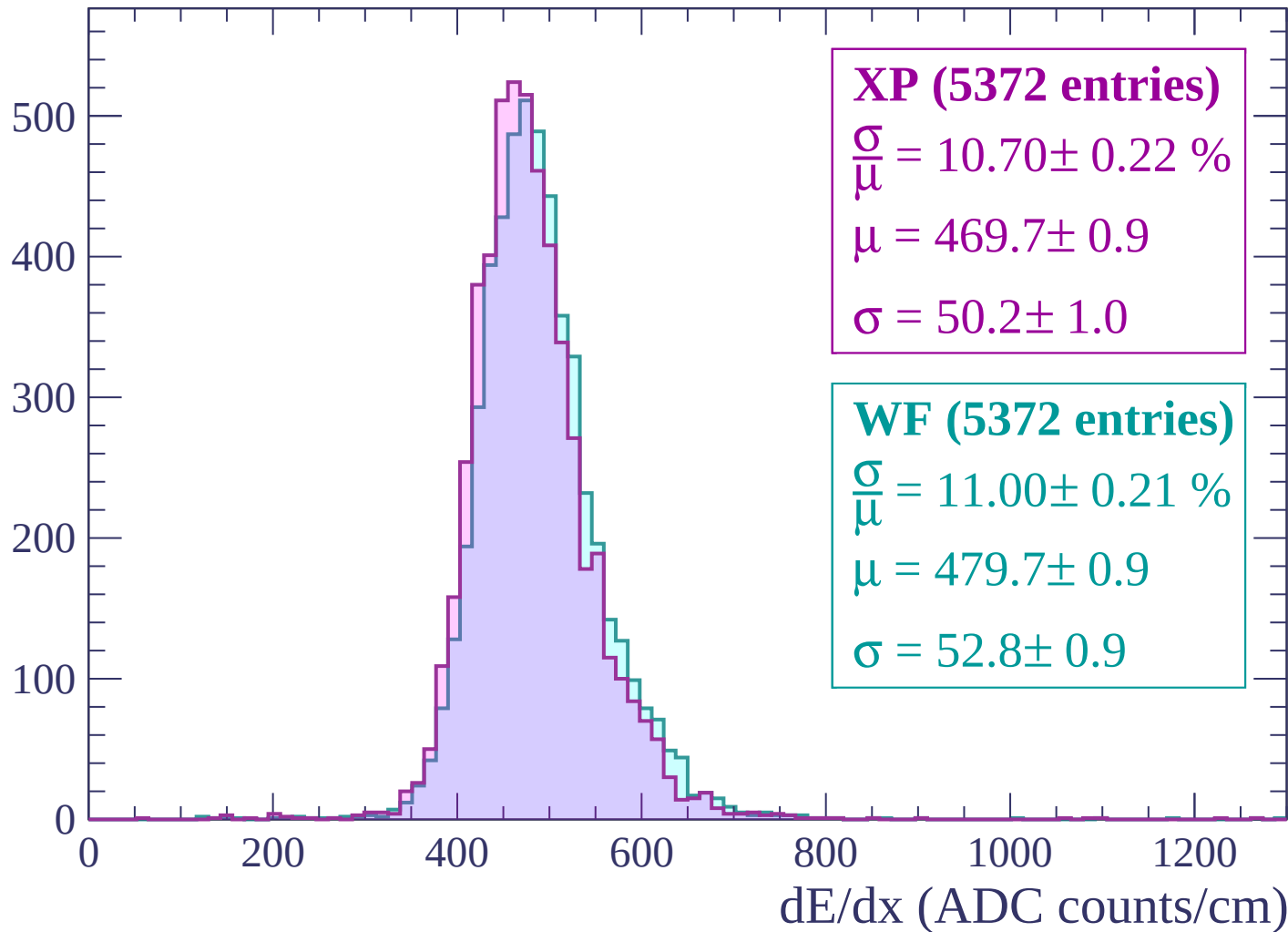
$$\mu = 475.9 \pm 0.8$$

$$\sigma = 50.2 \pm 0.9$$

dE/dx (ADC counts/cm)

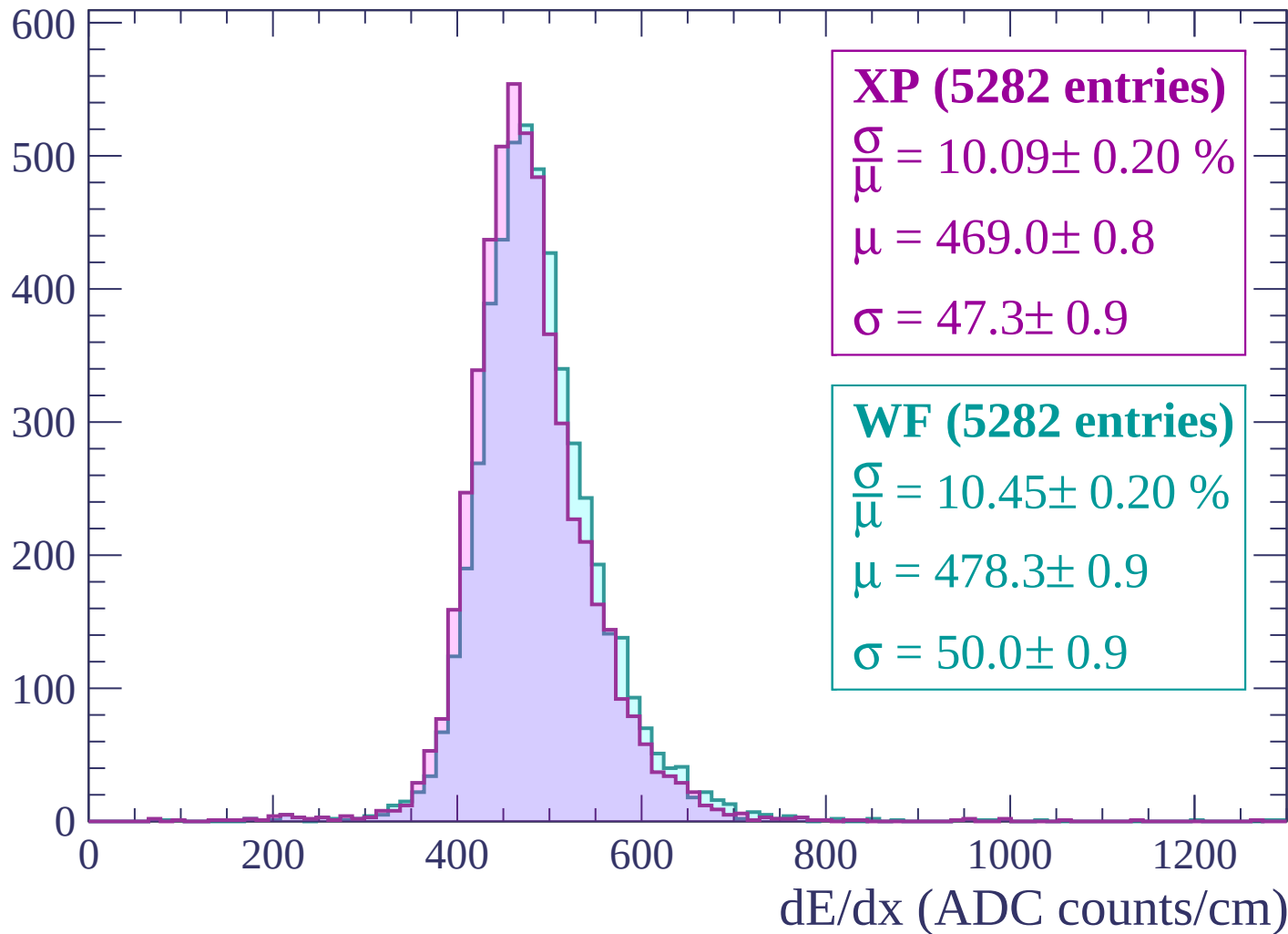
Energy loss | $1920 < p < 1980$

Count



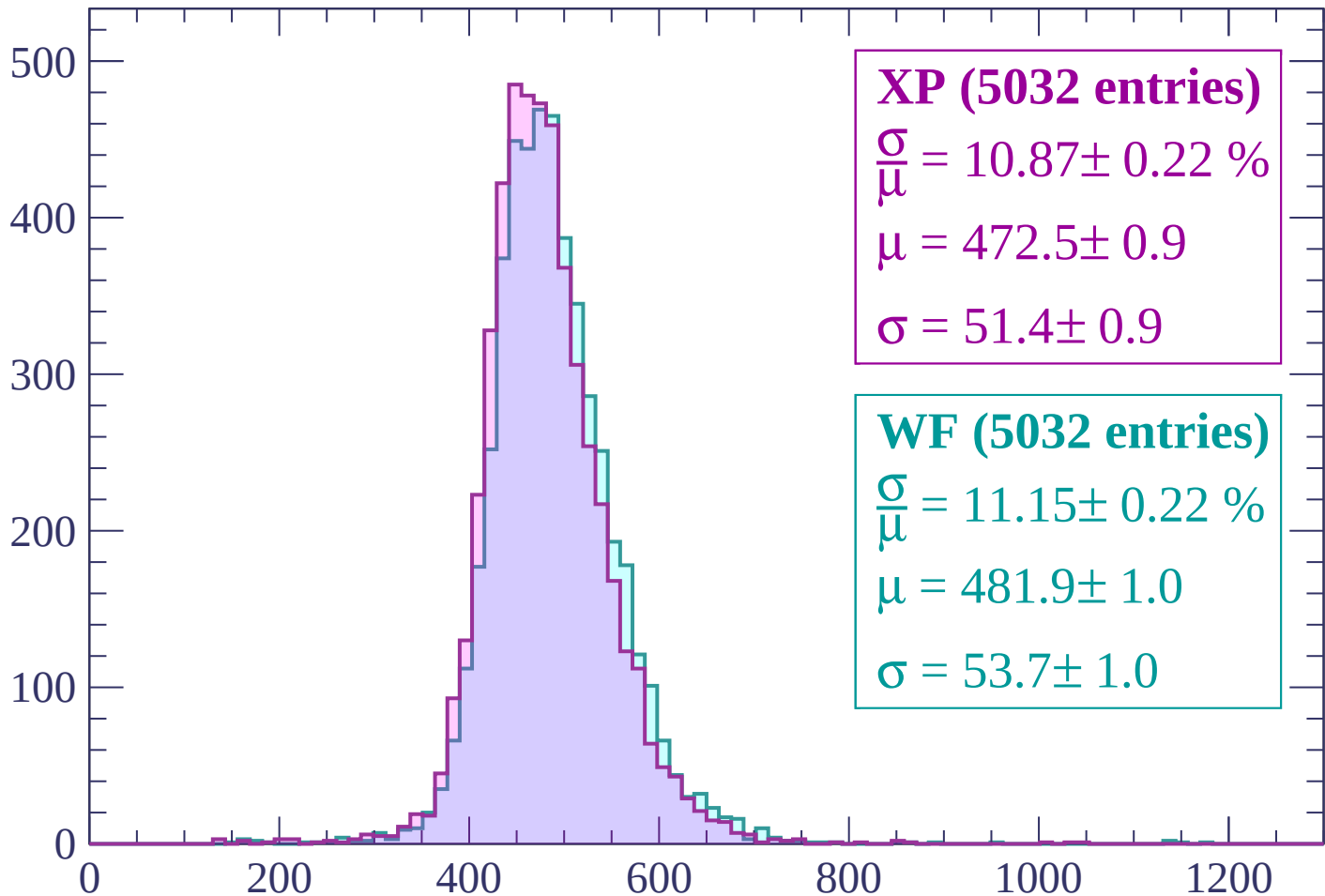
Energy loss | $1980 < p < 2040$

Count



Energy loss | $2040 < p < 2100$

Count



XP (5032 entries)

$$\frac{\sigma}{\mu} = 10.87 \pm 0.22 \%$$

$$\mu = 472.5 \pm 0.9$$

$$\sigma = 51.4 \pm 0.9$$

WF (5032 entries)

$$\frac{\sigma}{\mu} = 11.15 \pm 0.22 \%$$

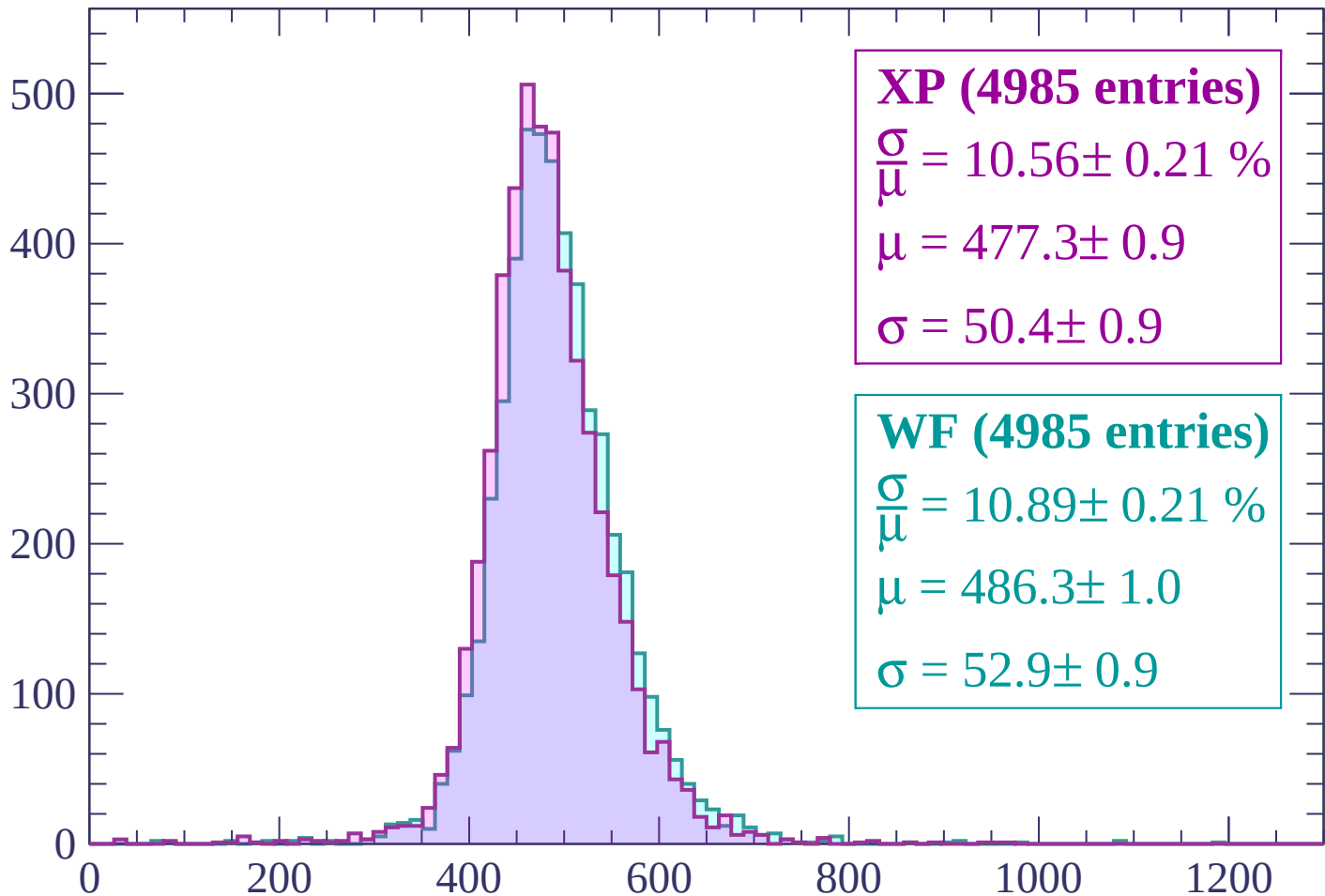
$$\mu = 481.9 \pm 1.0$$

$$\sigma = 53.7 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $2100 < p < 2160$

Count



XP (4985 entries)

$$\frac{\sigma}{\mu} = 10.56 \pm 0.21 \%$$

$$\mu = 477.3 \pm 0.9$$

$$\sigma = 50.4 \pm 0.9$$

WF (4985 entries)

$$\frac{\sigma}{\mu} = 10.89 \pm 0.21 \%$$

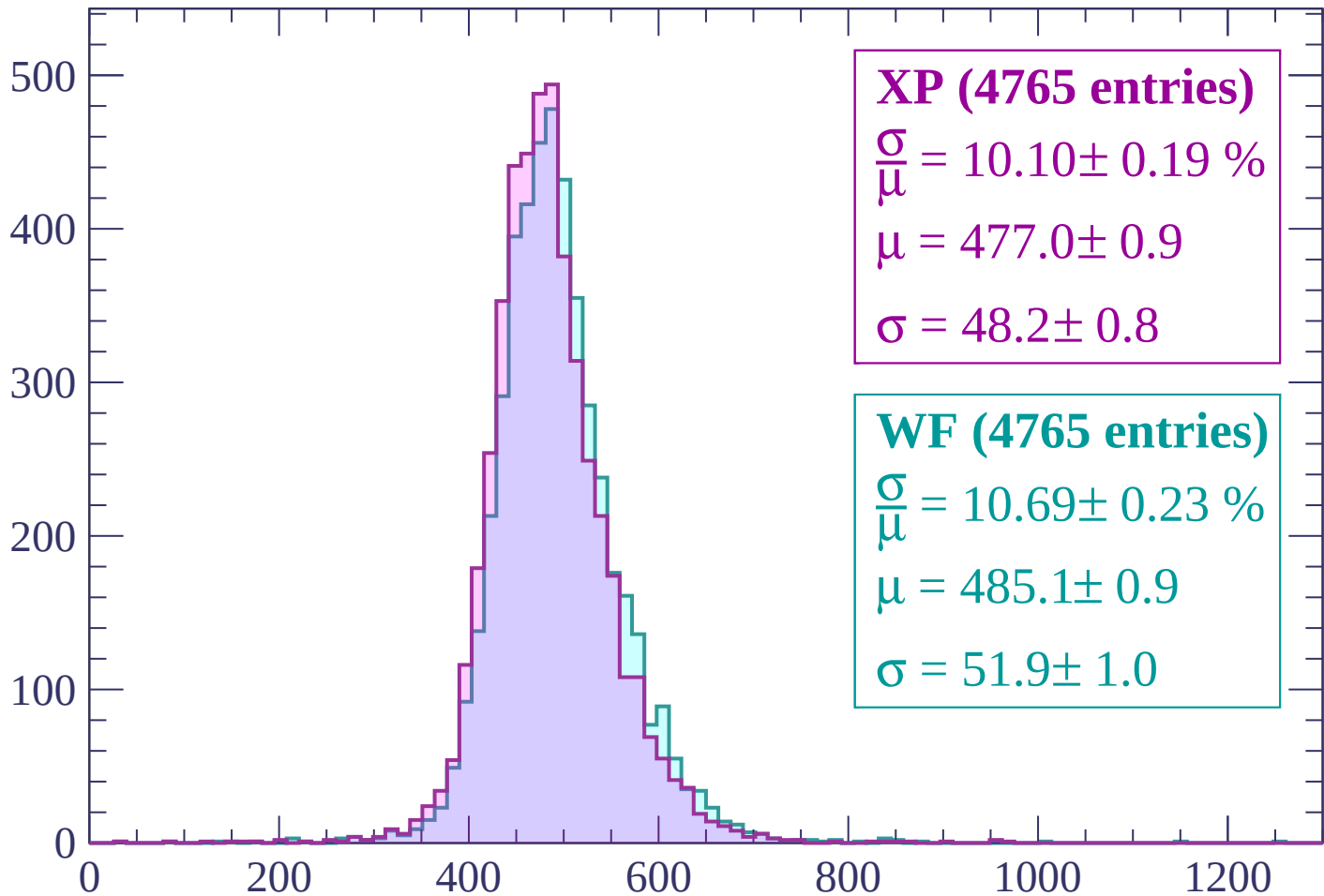
$$\mu = 486.3 \pm 1.0$$

$$\sigma = 52.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $2160 < p < 2220$

Count



XP (4765 entries)

$$\frac{\sigma}{\mu} = 10.10 \pm 0.19 \%$$

$$\mu = 477.0 \pm 0.9$$

$$\sigma = 48.2 \pm 0.8$$

WF (4765 entries)

$$\frac{\sigma}{\mu} = 10.69 \pm 0.23 \%$$

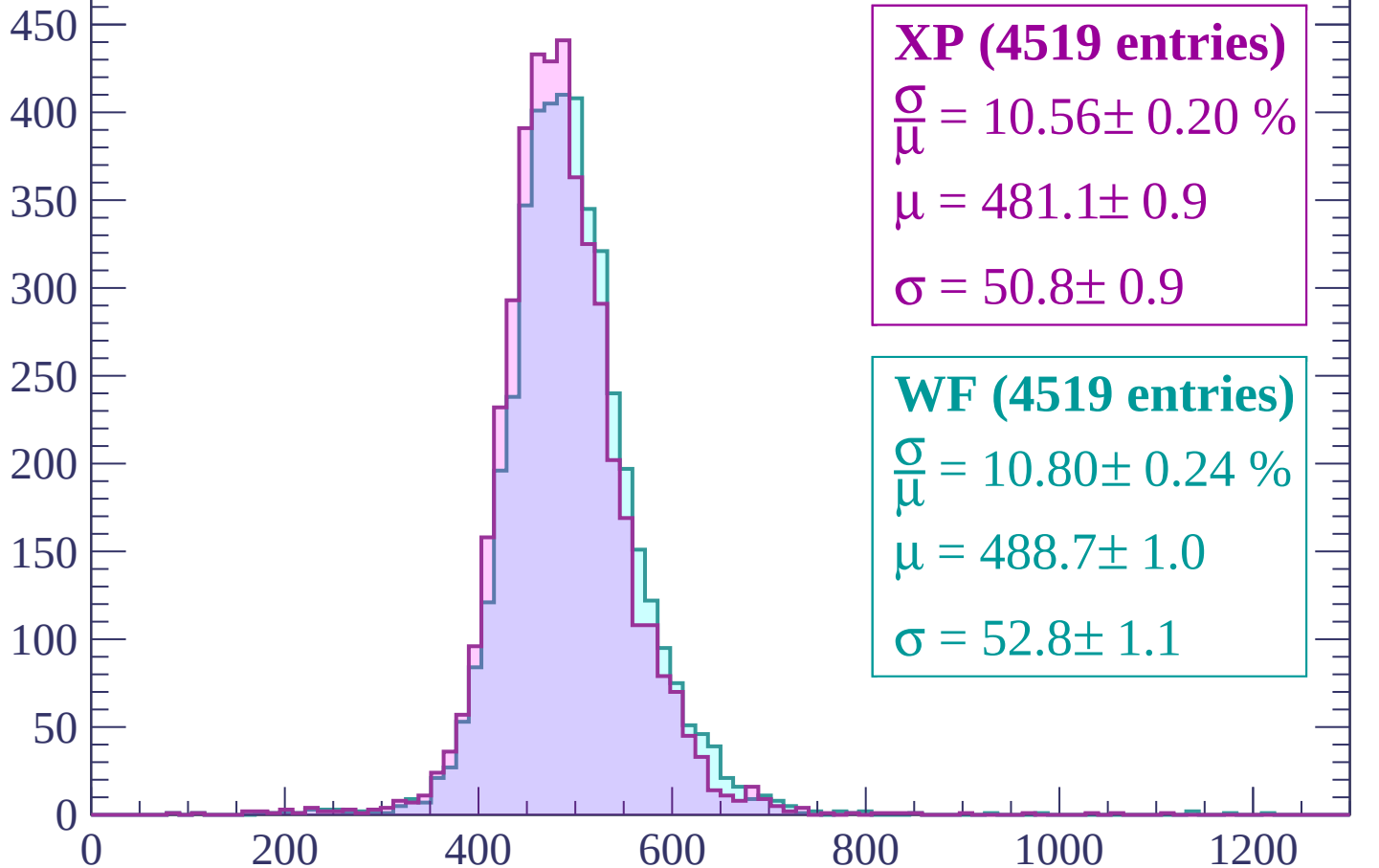
$$\mu = 485.1 \pm 0.9$$

$$\sigma = 51.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $2220 < p < 2280$

Count



XP (4519 entries)

$\frac{\sigma}{\mu} = 10.56 \pm 0.20 \%$

$\mu = 481.1 \pm 0.9$

$\sigma = 50.8 \pm 0.9$

WF (4519 entries)

$\frac{\sigma}{\mu} = 10.80 \pm 0.24 \%$

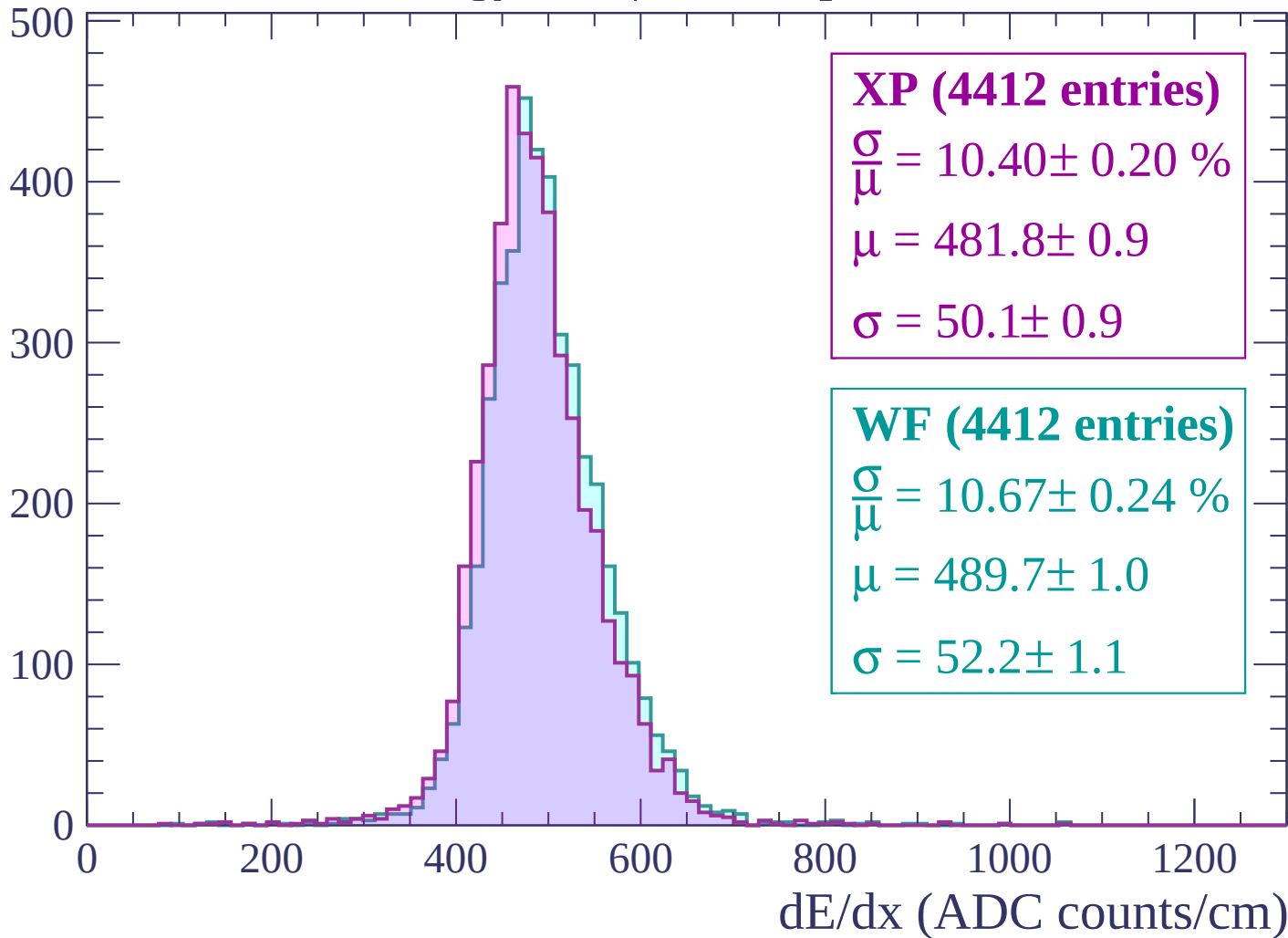
$\mu = 488.7 \pm 1.0$

$\sigma = 52.8 \pm 1.1$

dE/dx (ADC counts/cm)

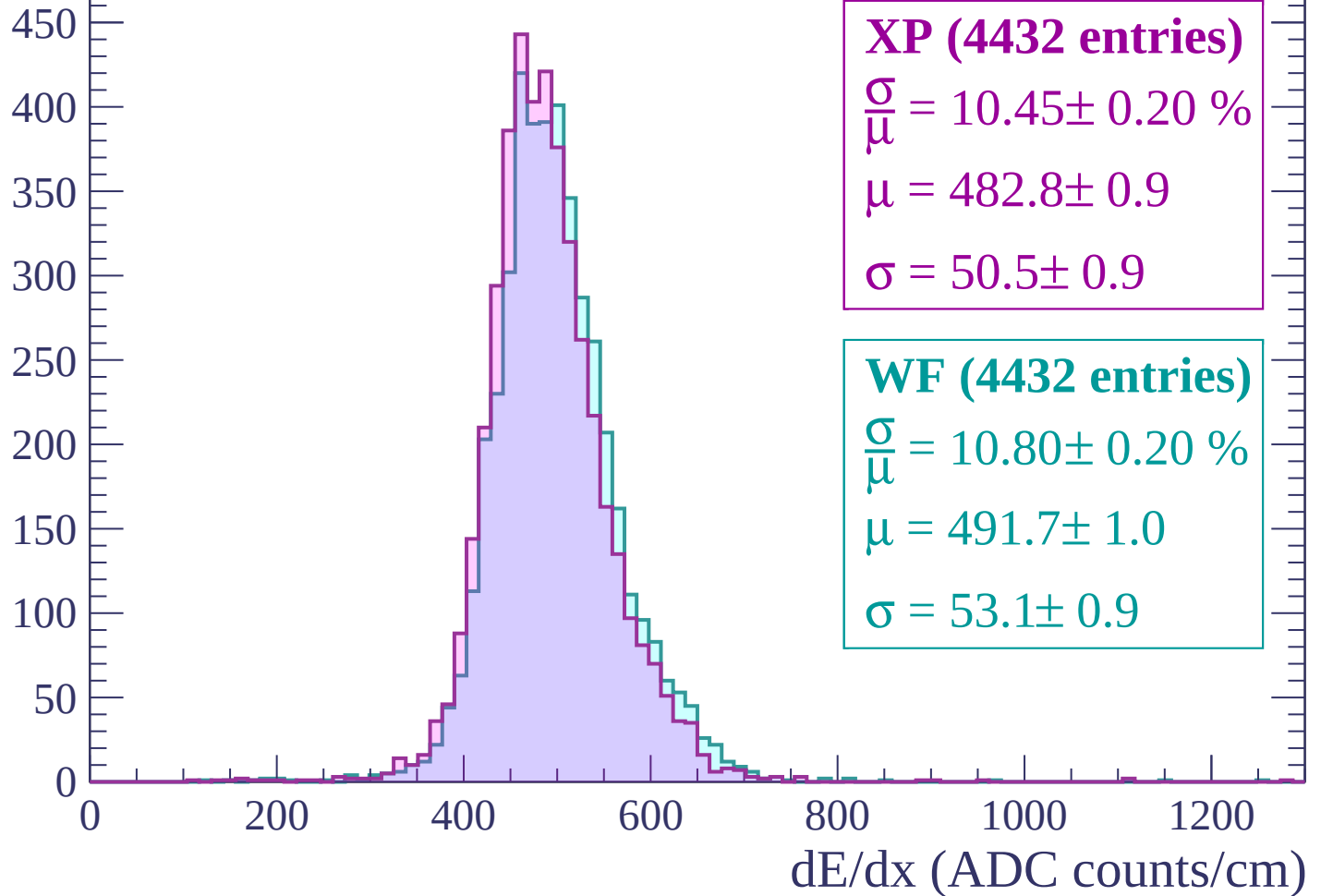
Energy loss | $2280 < p < 2340$

Count



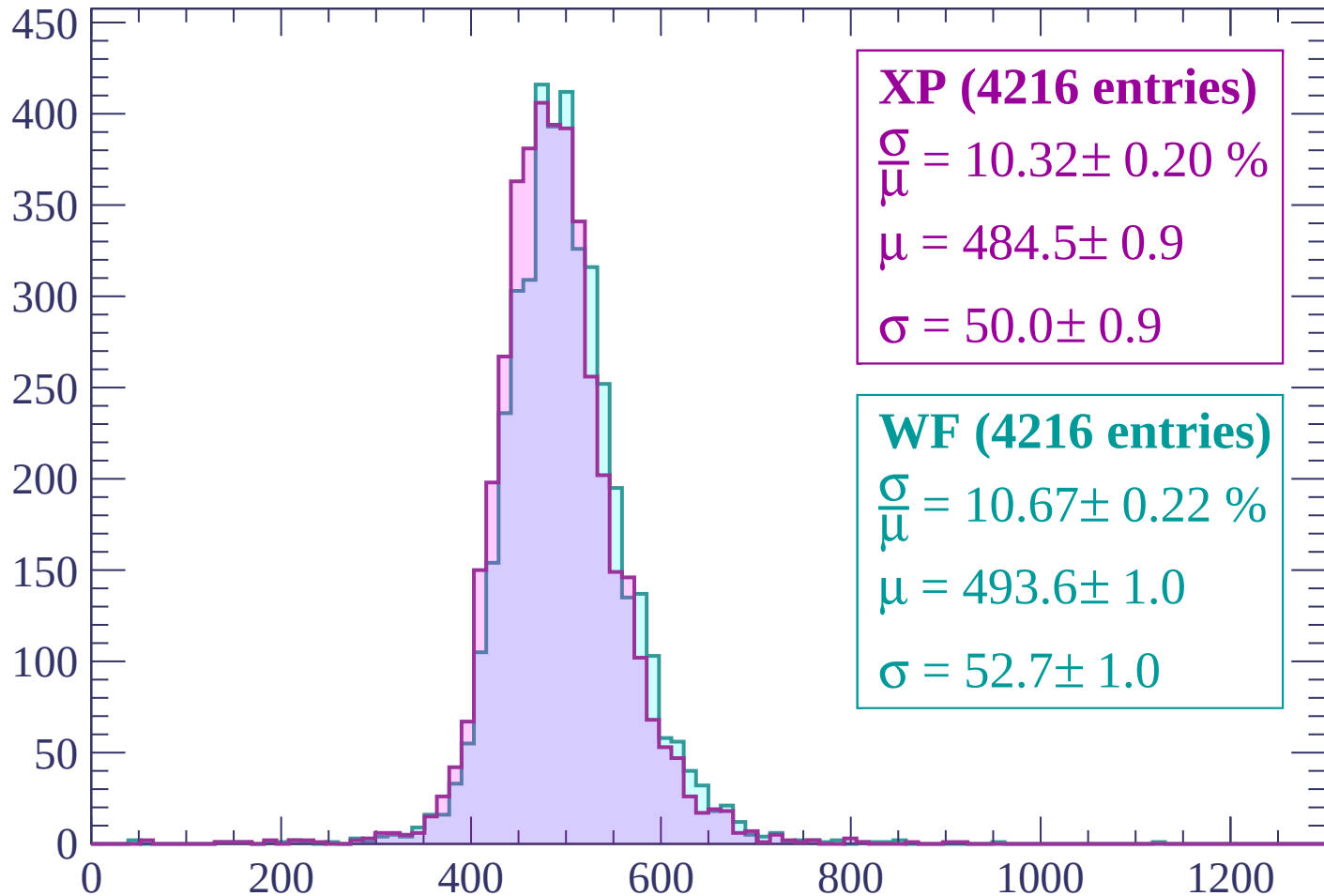
Energy loss | $2340 < p < 2400$

Count



Energy loss | $2400 < p < 2460$

Count



XP (4216 entries)

$$\frac{\sigma}{\mu} = 10.32 \pm 0.20 \%$$

$$\mu = 484.5 \pm 0.9$$

$$\sigma = 50.0 \pm 0.9$$

WF (4216 entries)

$$\frac{\sigma}{\mu} = 10.67 \pm 0.22 \%$$

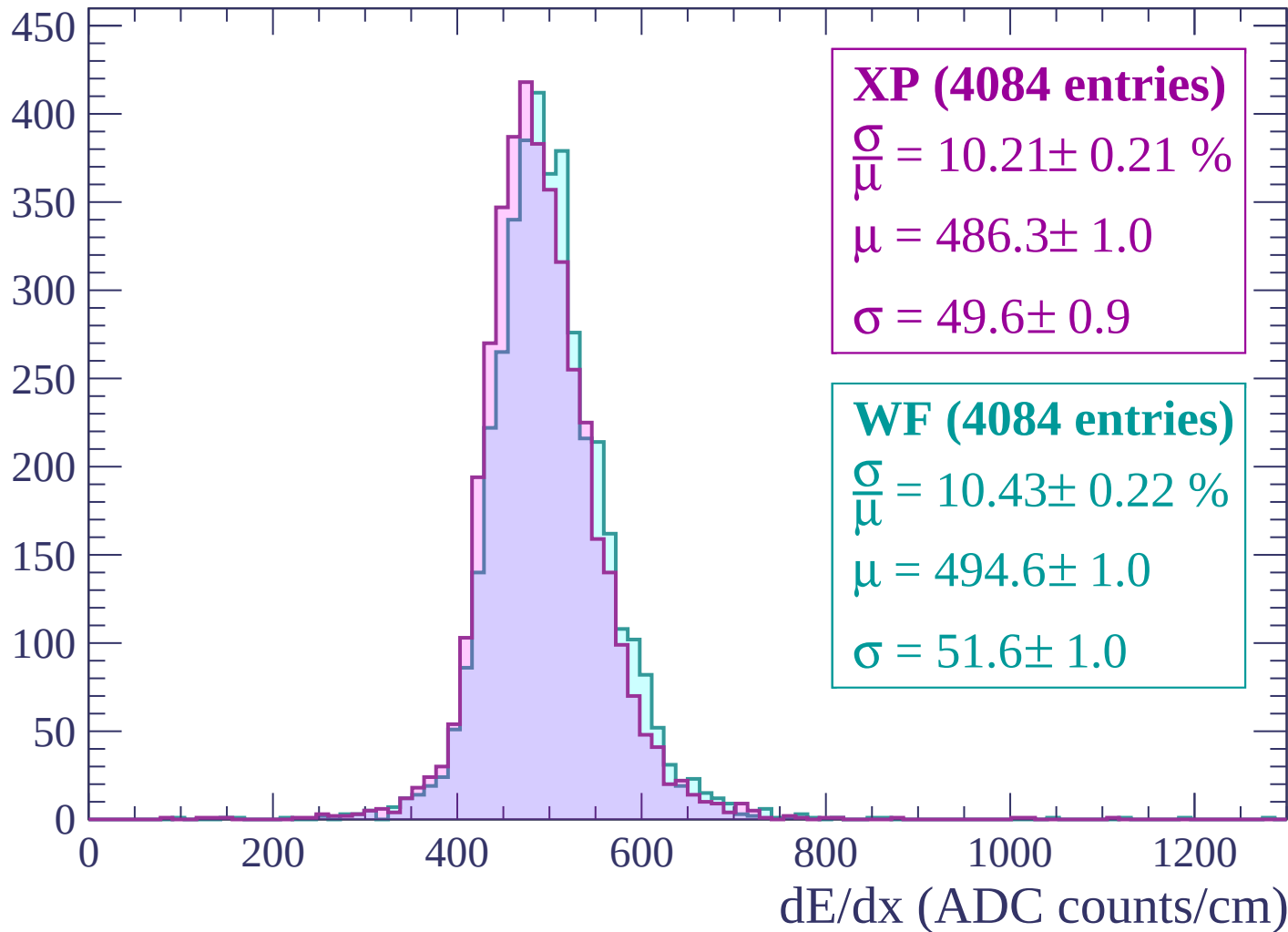
$$\mu = 493.6 \pm 1.0$$

$$\sigma = 52.7 \pm 1.0$$

dE/dx (ADC counts/cm)

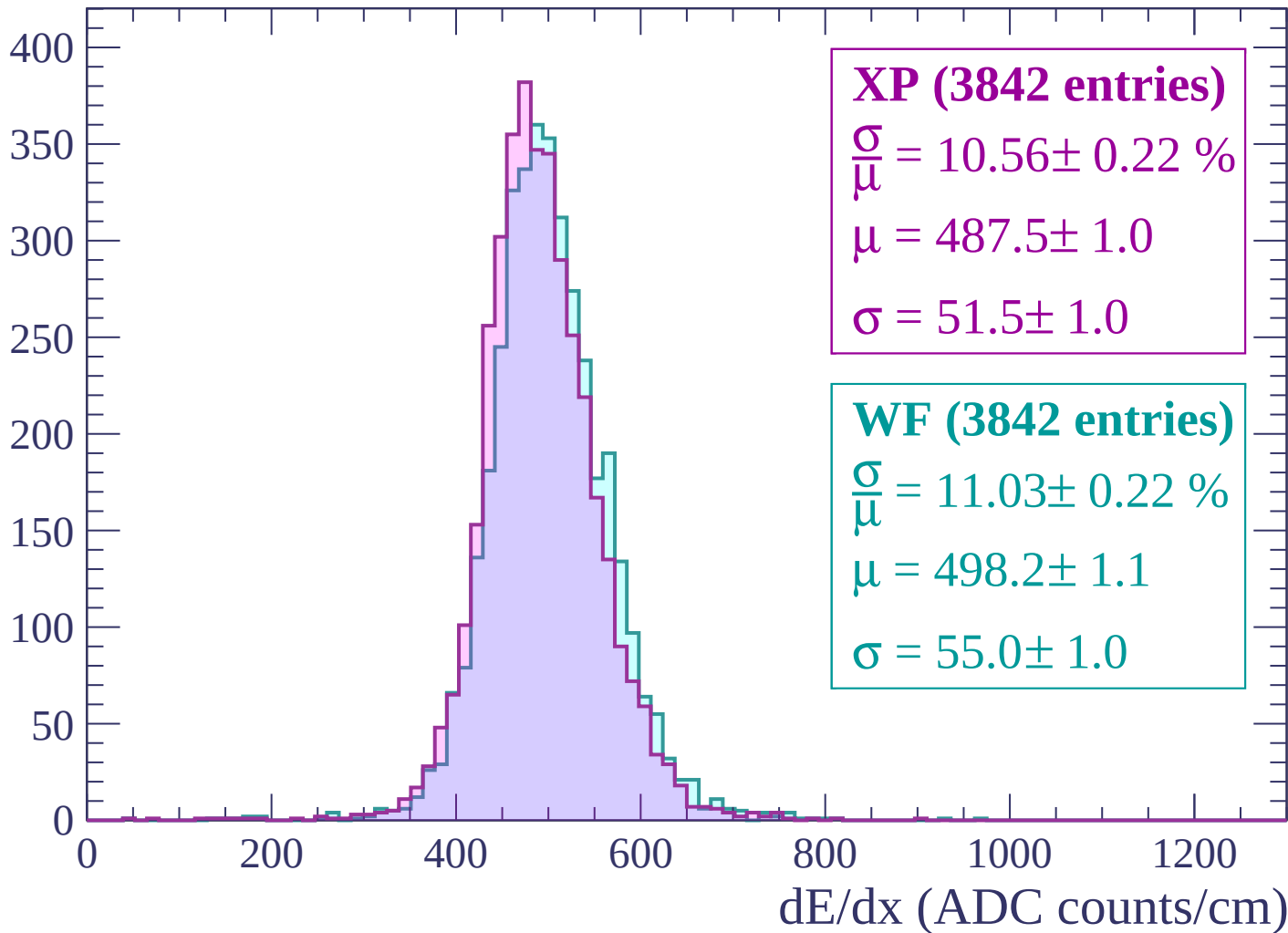
Energy loss | $2460 < p < 2520$

Count



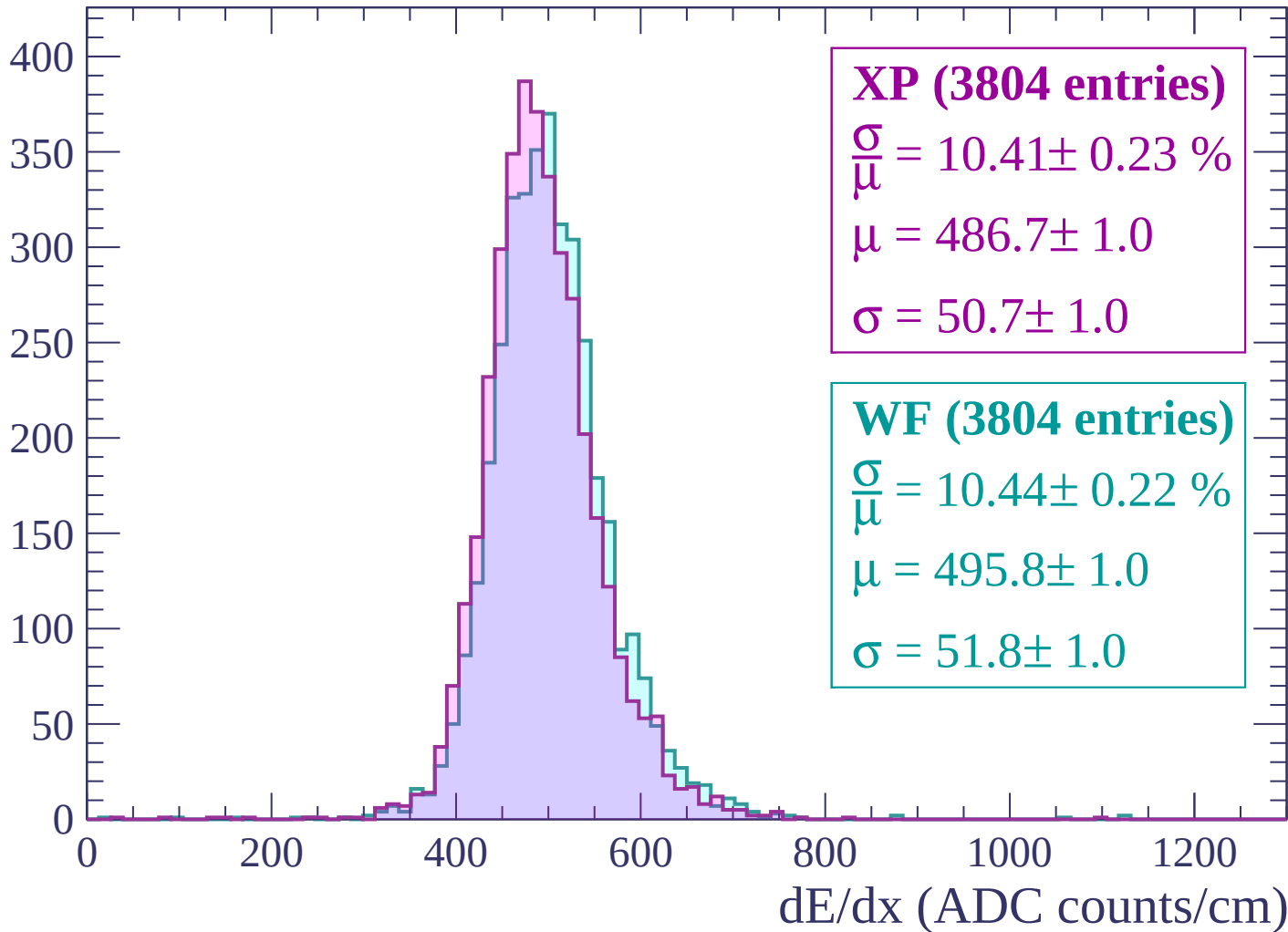
Energy loss | $2520 < p < 2580$

Count



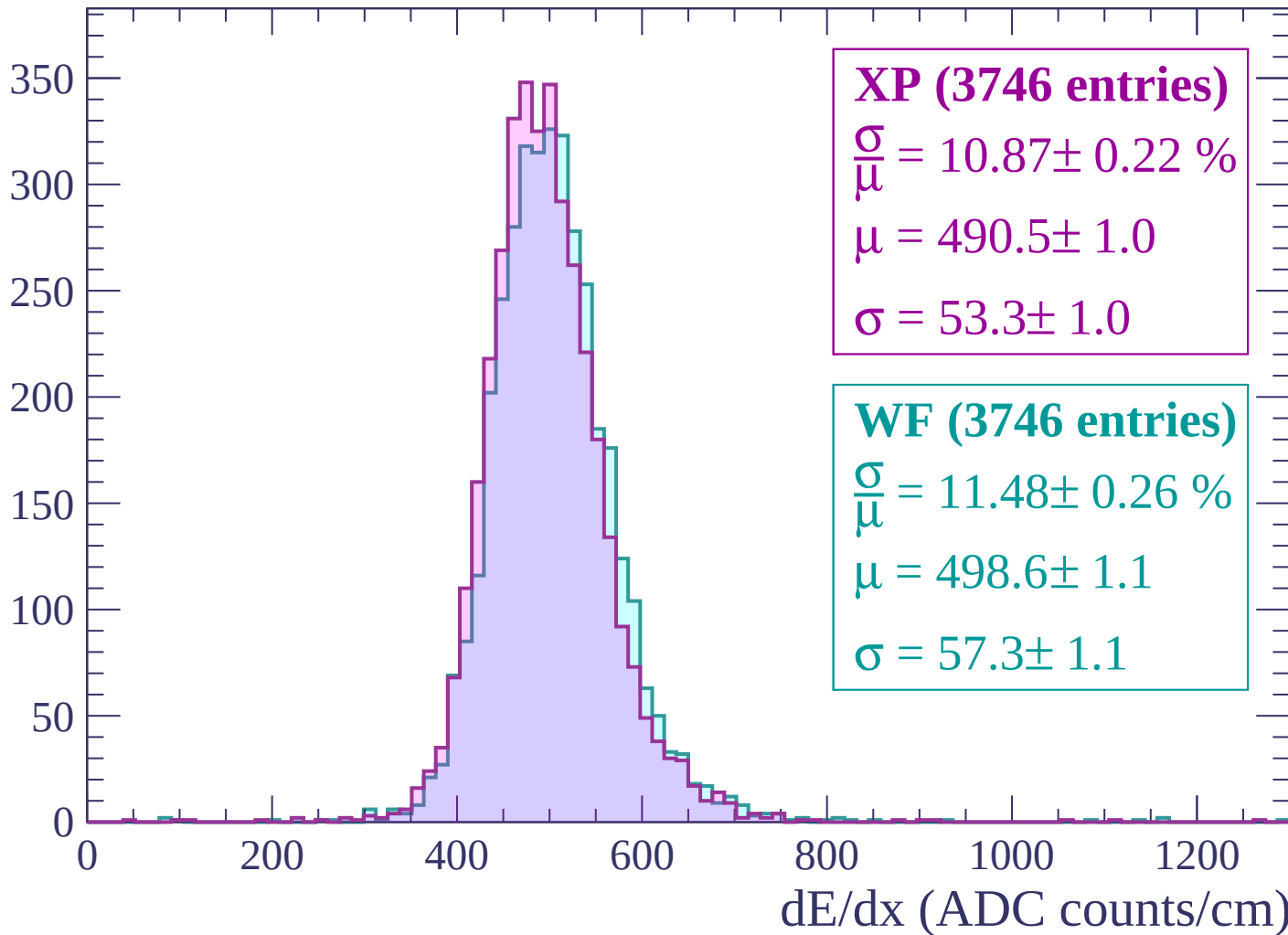
Energy loss | $2580 < p < 2640$

Count



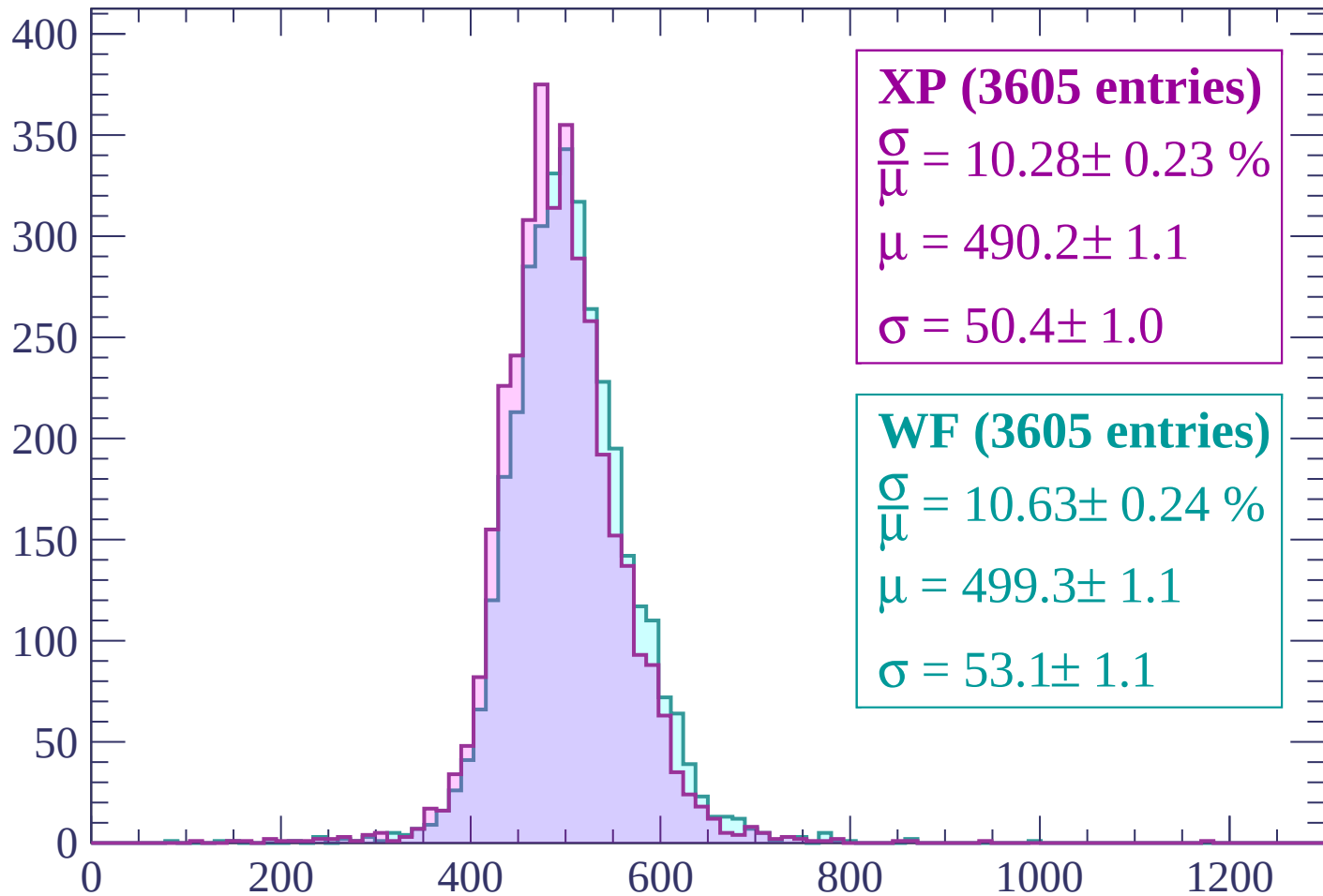
Energy loss | $2640 < p < 2700$

Count



Energy loss | $2700 < p < 2760$

Count



XP (3605 entries)

$\frac{\sigma}{\mu} = 10.28 \pm 0.23 \%$

$\mu = 490.2 \pm 1.1$

$\sigma = 50.4 \pm 1.0$

WF (3605 entries)

$\frac{\sigma}{\mu} = 10.63 \pm 0.24 \%$

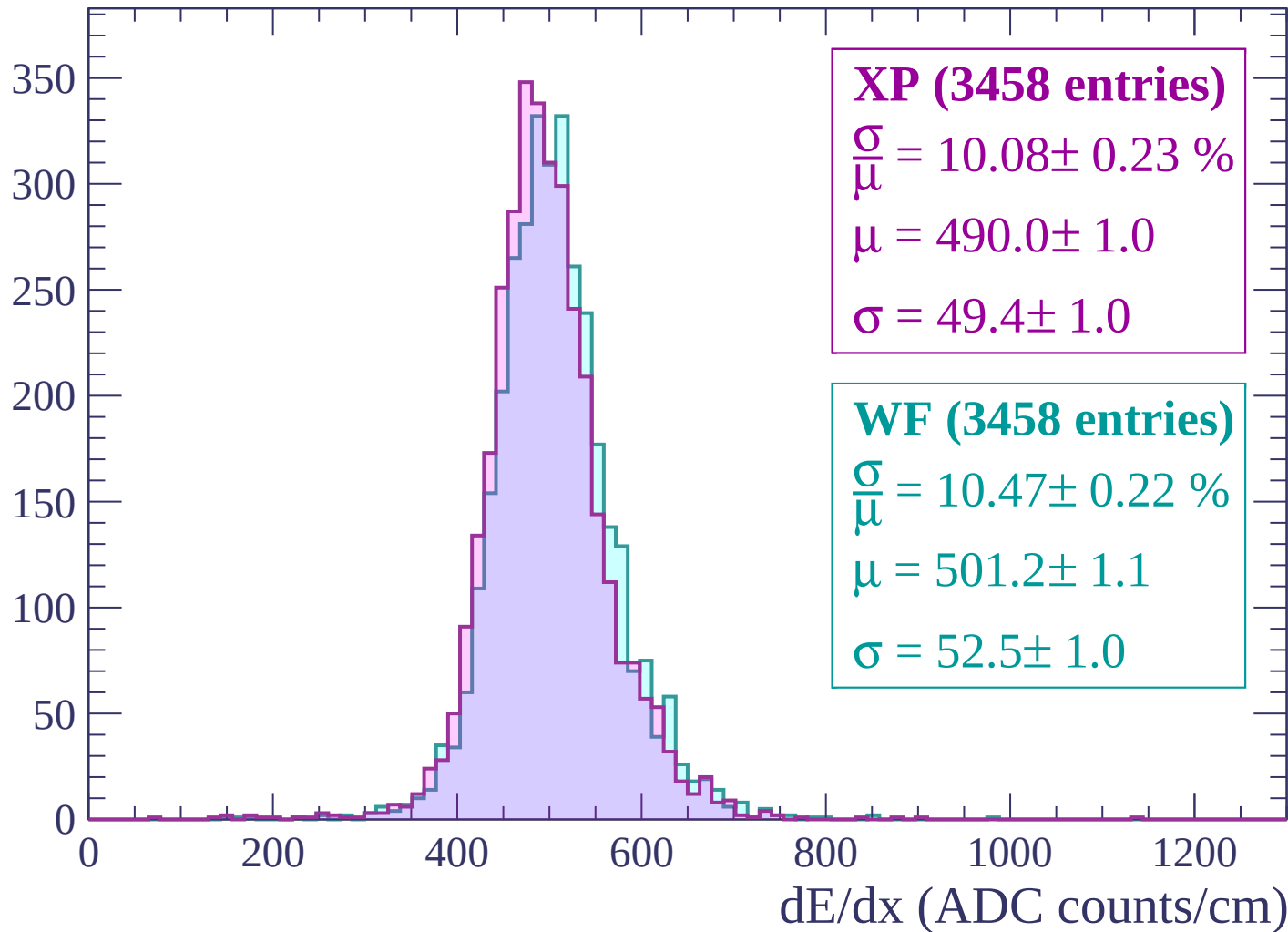
$\mu = 499.3 \pm 1.1$

$\sigma = 53.1 \pm 1.1$

dE/dx (ADC counts/cm)

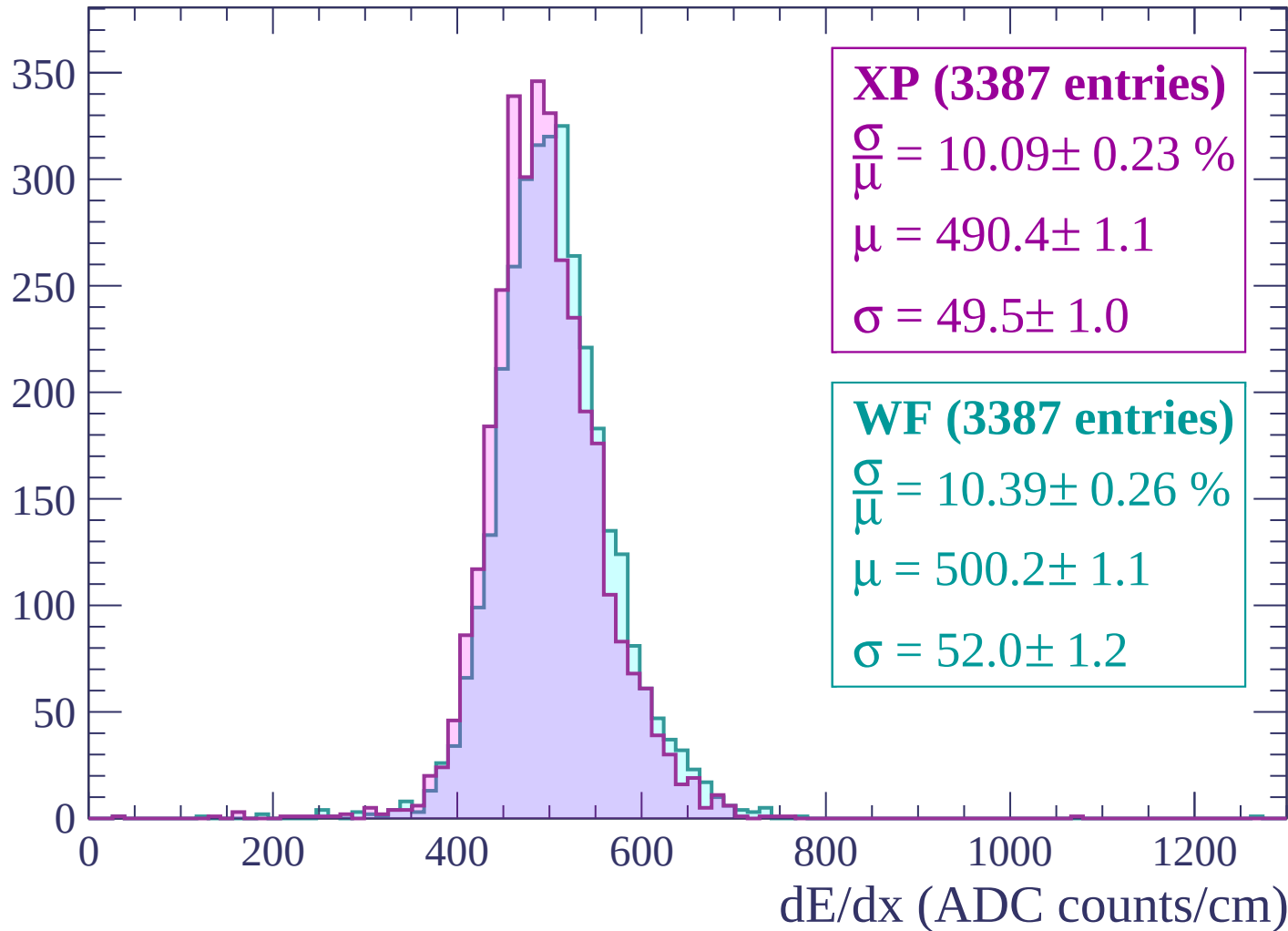
Energy loss | $2760 < p < 2820$

Count



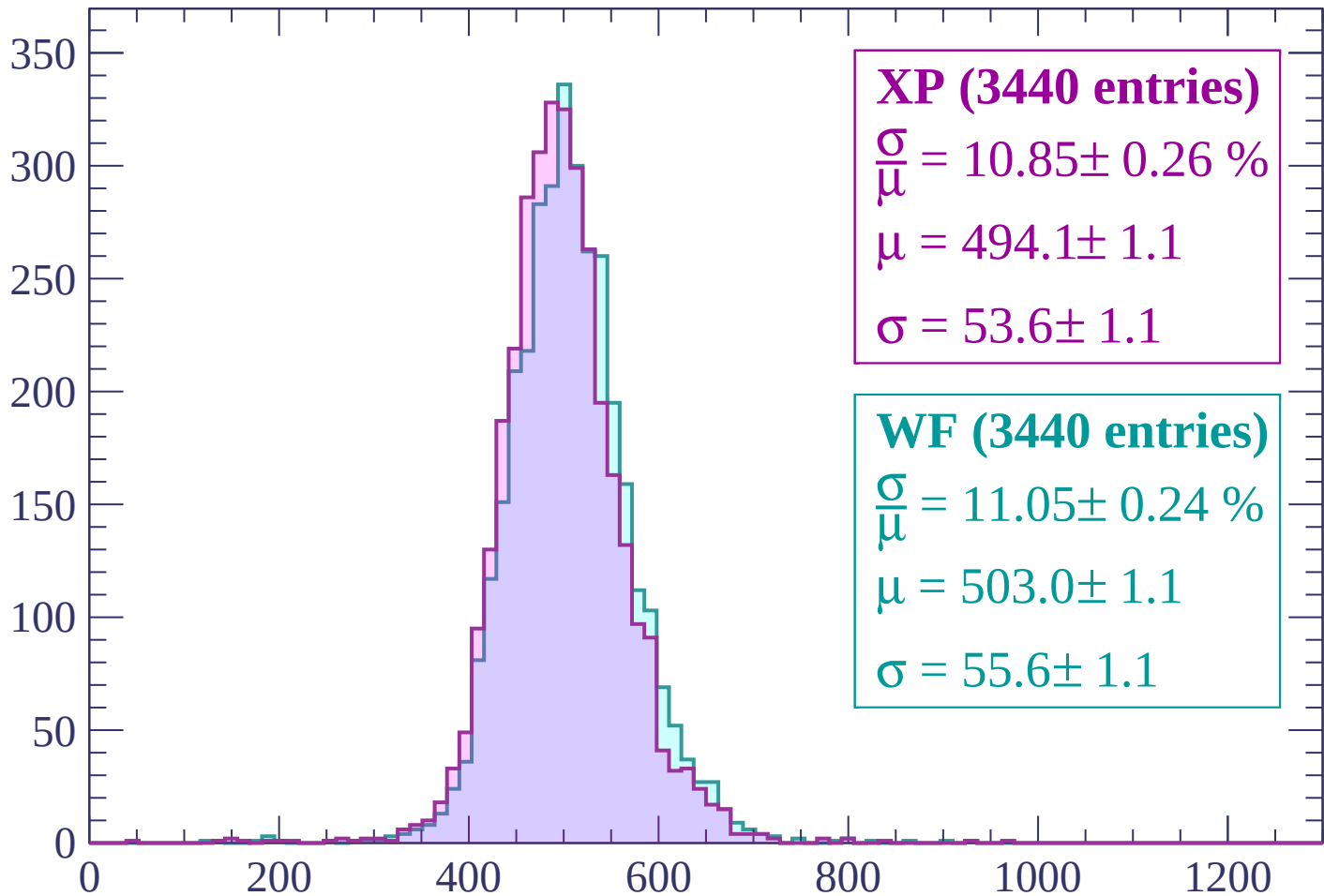
Energy loss | $2820 < p < 2880$

Count



Energy loss | $2880 < p < 2940$

Count



XP (3440 entries)

$\frac{\sigma}{\mu} = 10.85 \pm 0.26 \%$

$\mu = 494.1 \pm 1.1$

$\sigma = 53.6 \pm 1.1$

WF (3440 entries)

$\frac{\sigma}{\mu} = 11.05 \pm 0.24 \%$

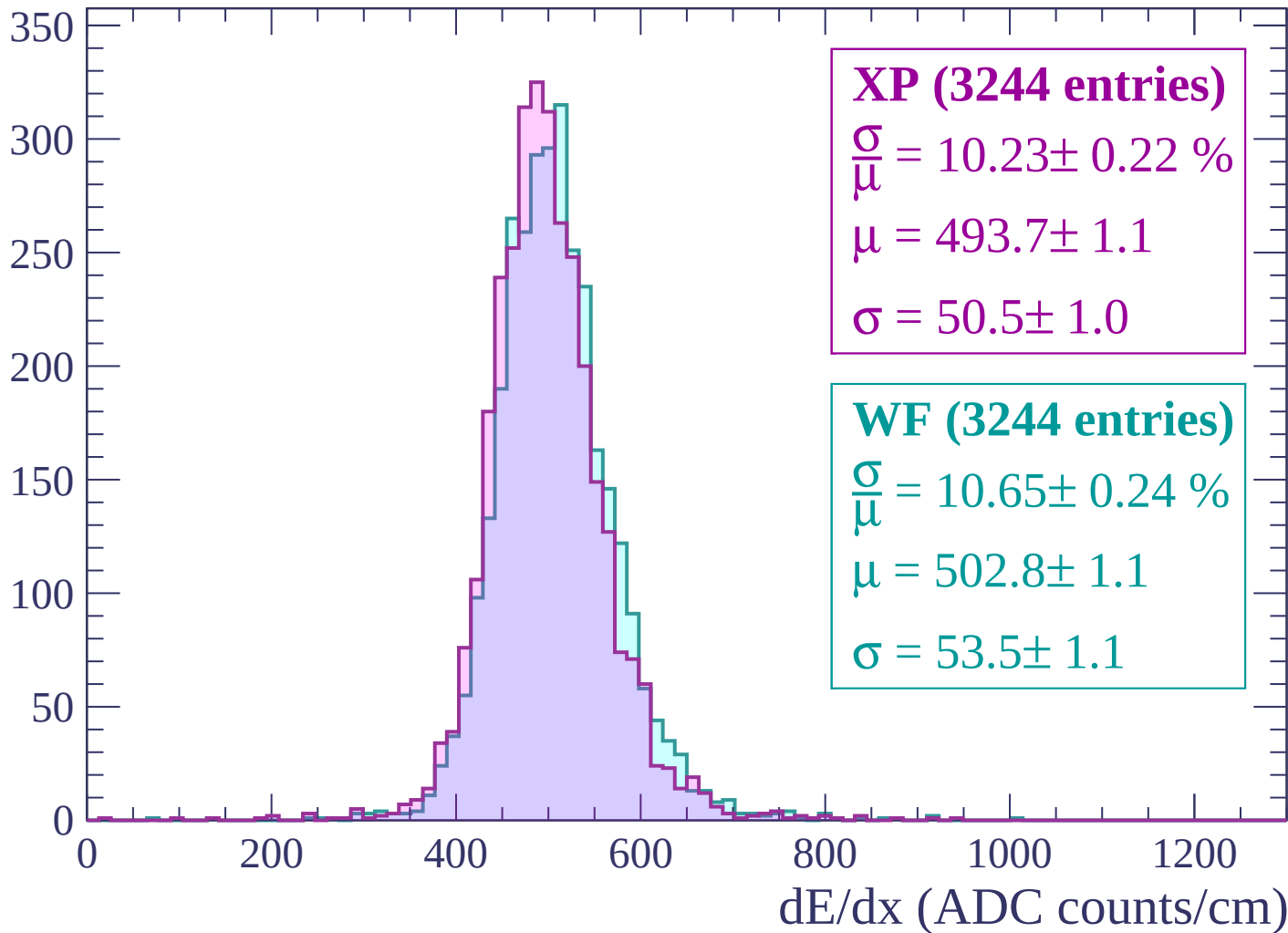
$\mu = 503.0 \pm 1.1$

$\sigma = 55.6 \pm 1.1$

dE/dx (ADC counts/cm)

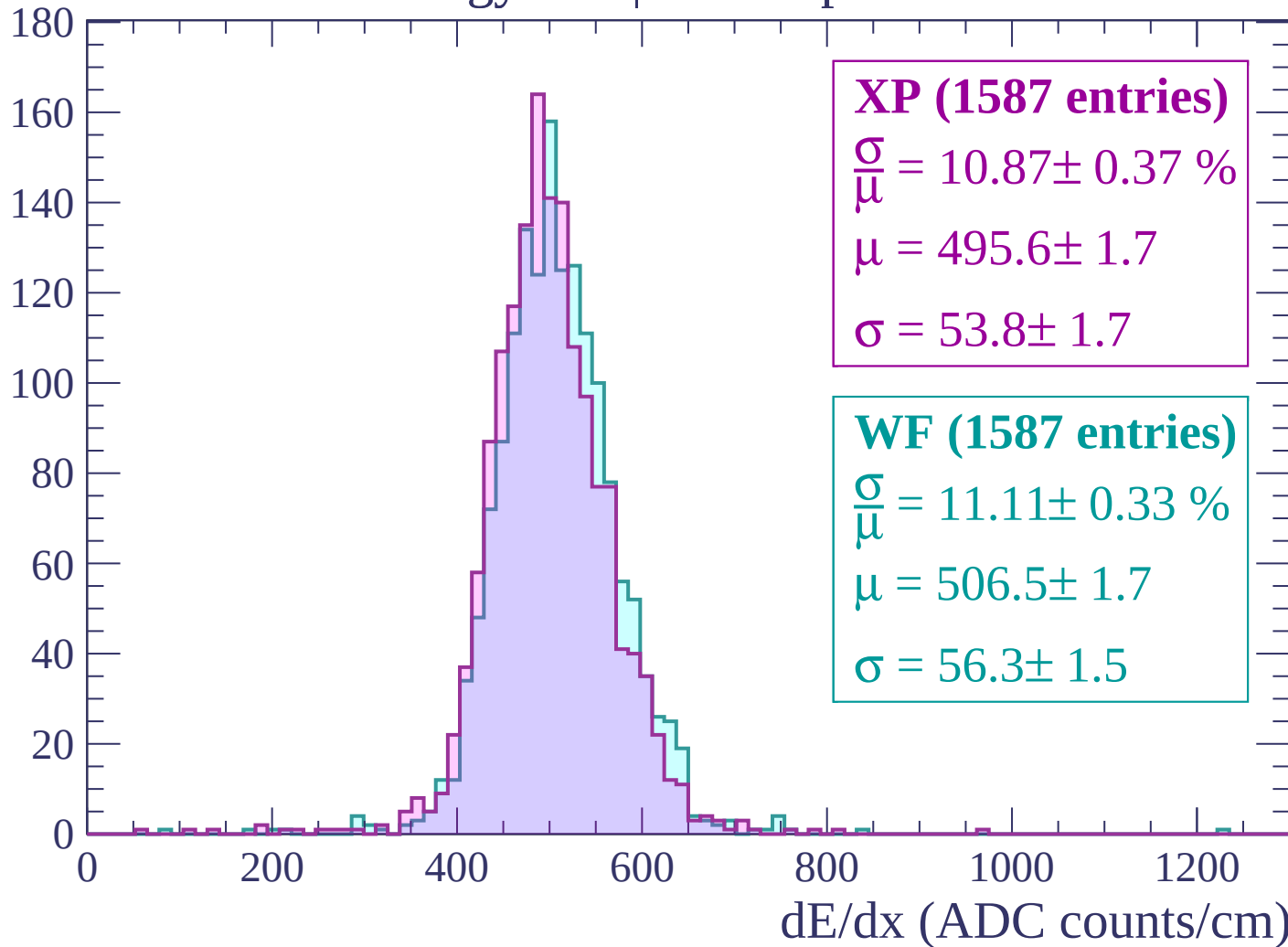
Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count



Energy loss | $-90 < \theta < -88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-88 < \theta < -86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-86 < \theta < -84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

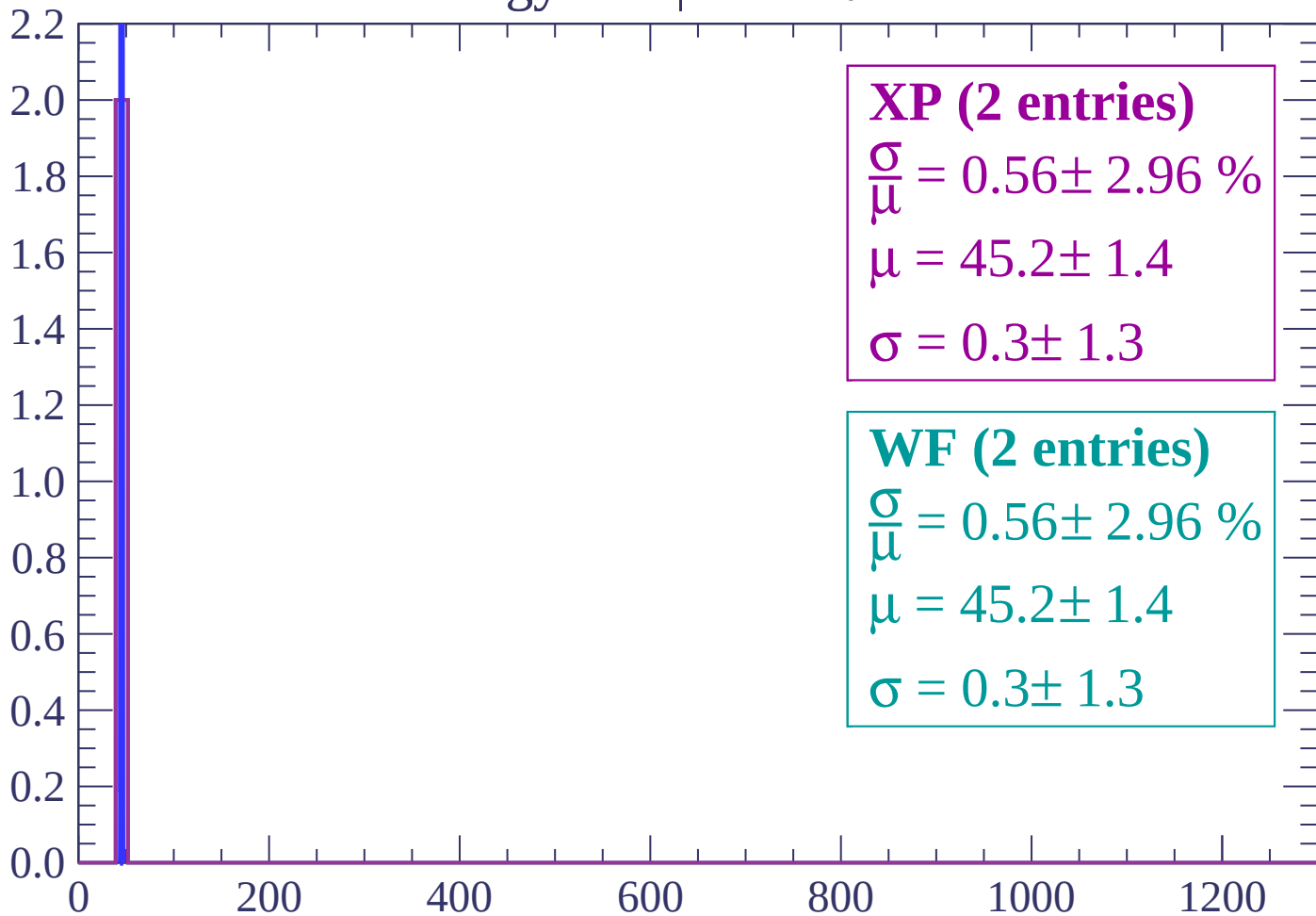
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-84 < \theta < -82$

Count



XP (2 entries)

$\frac{\sigma}{\mu} = 0.56 \pm 2.96 \%$

$\mu = 45.2 \pm 1.4$

$\sigma = 0.3 \pm 1.3$

WF (2 entries)

$\frac{\sigma}{\mu} = 0.56 \pm 2.96 \%$

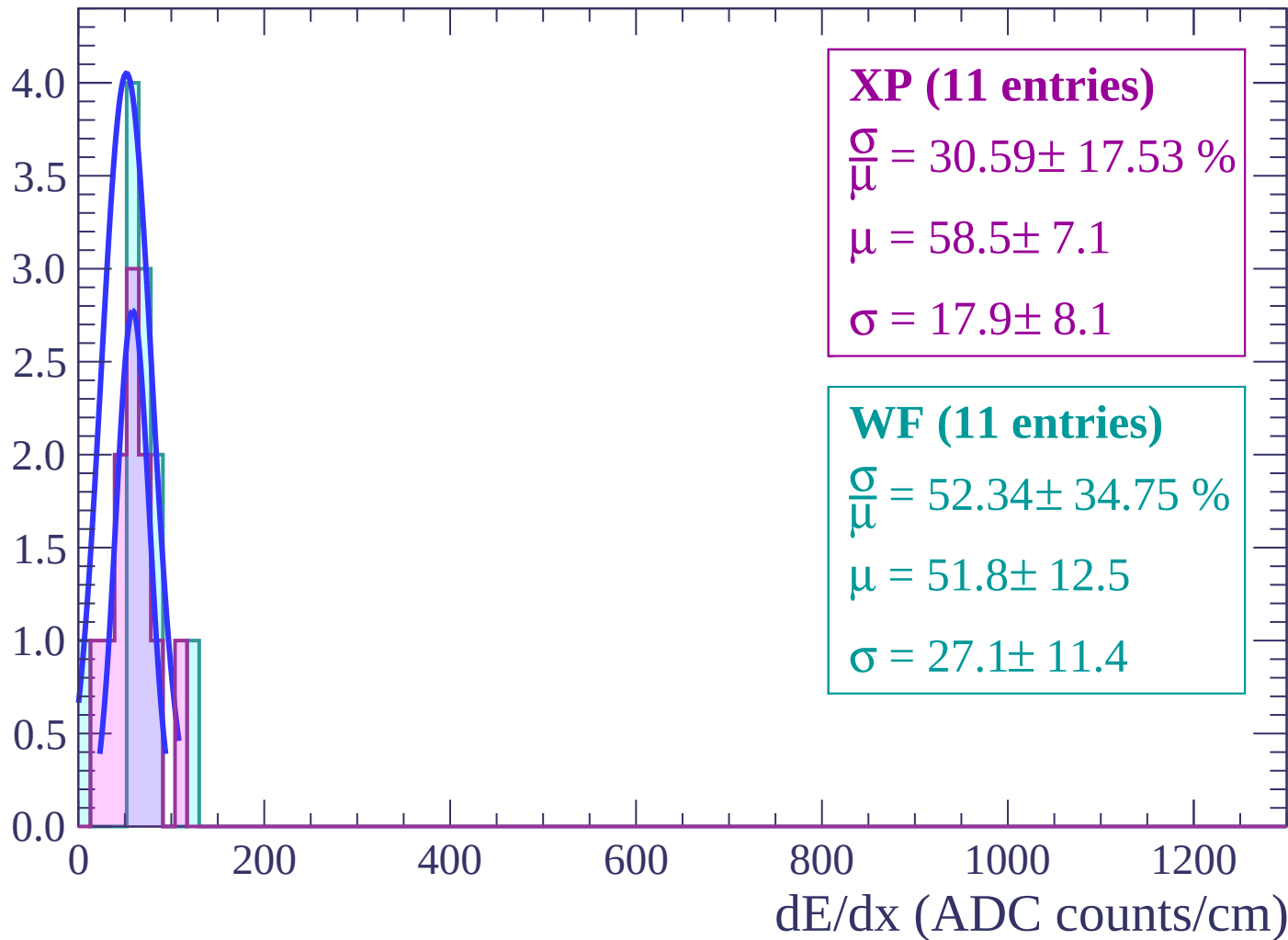
$\mu = 45.2 \pm 1.4$

$\sigma = 0.3 \pm 1.3$

dE/dx (ADC counts/cm)

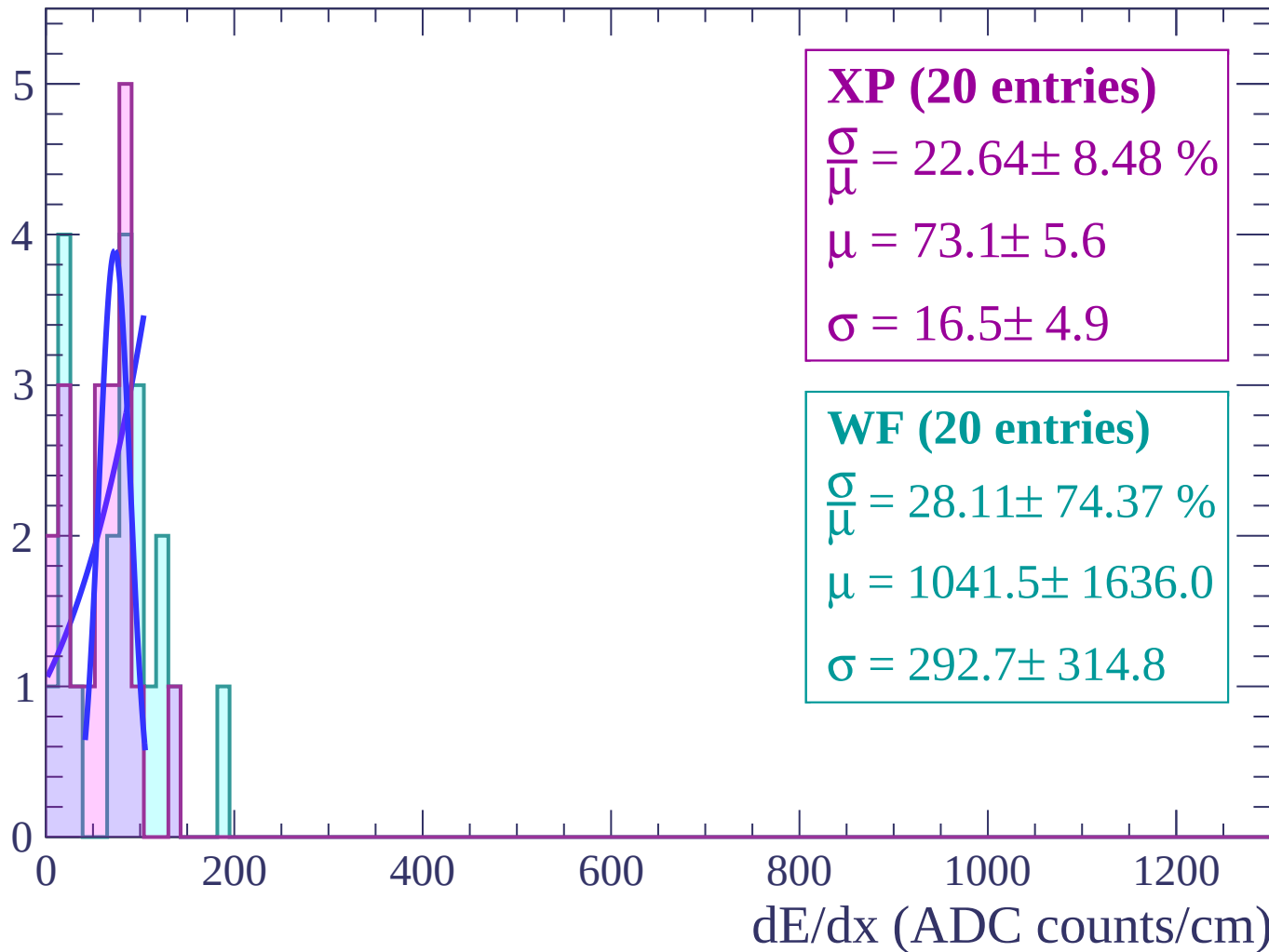
Energy loss | $-82 < \theta < -80$

Count



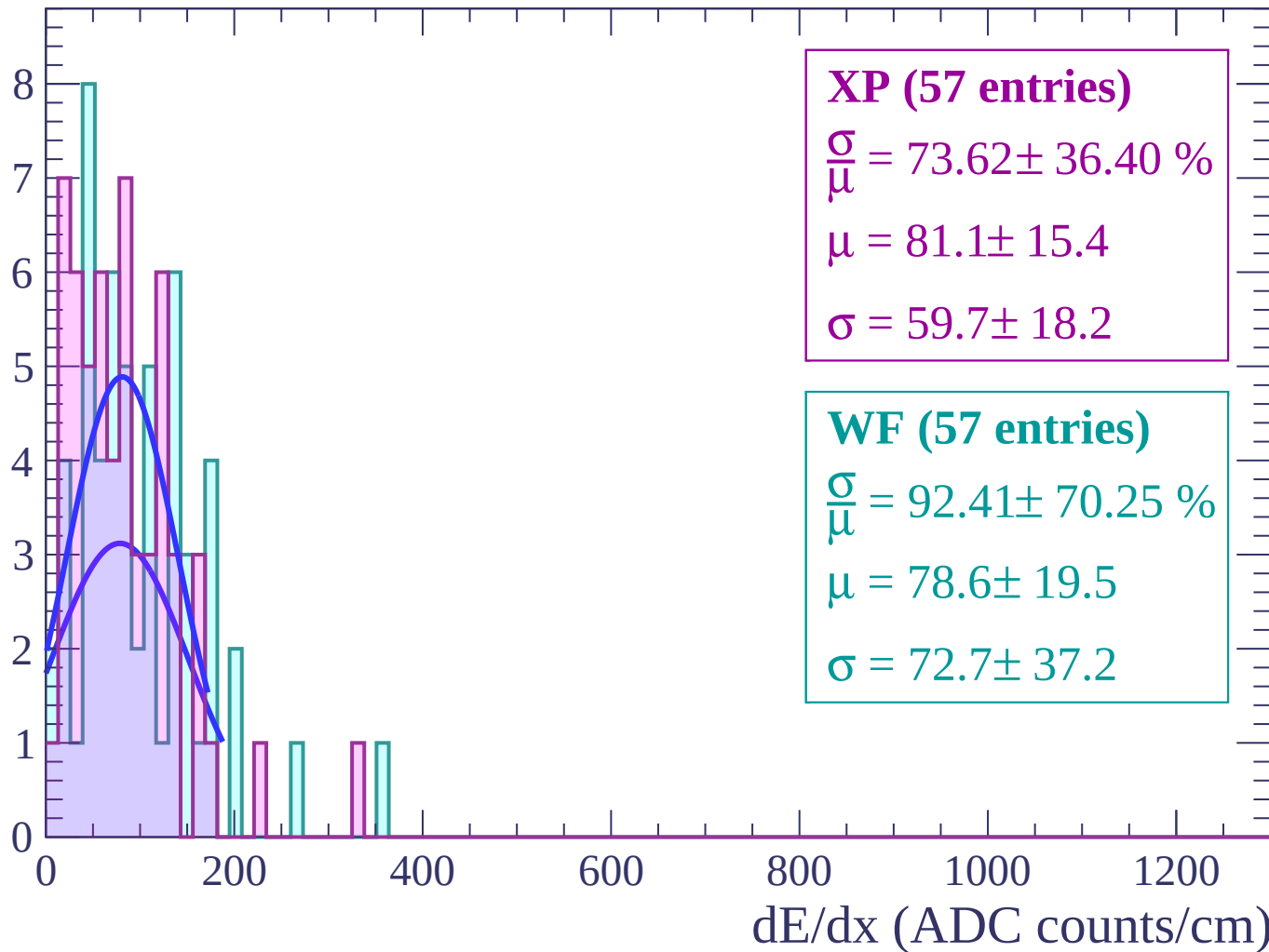
Energy loss | $-80 < \theta < -78$

Count



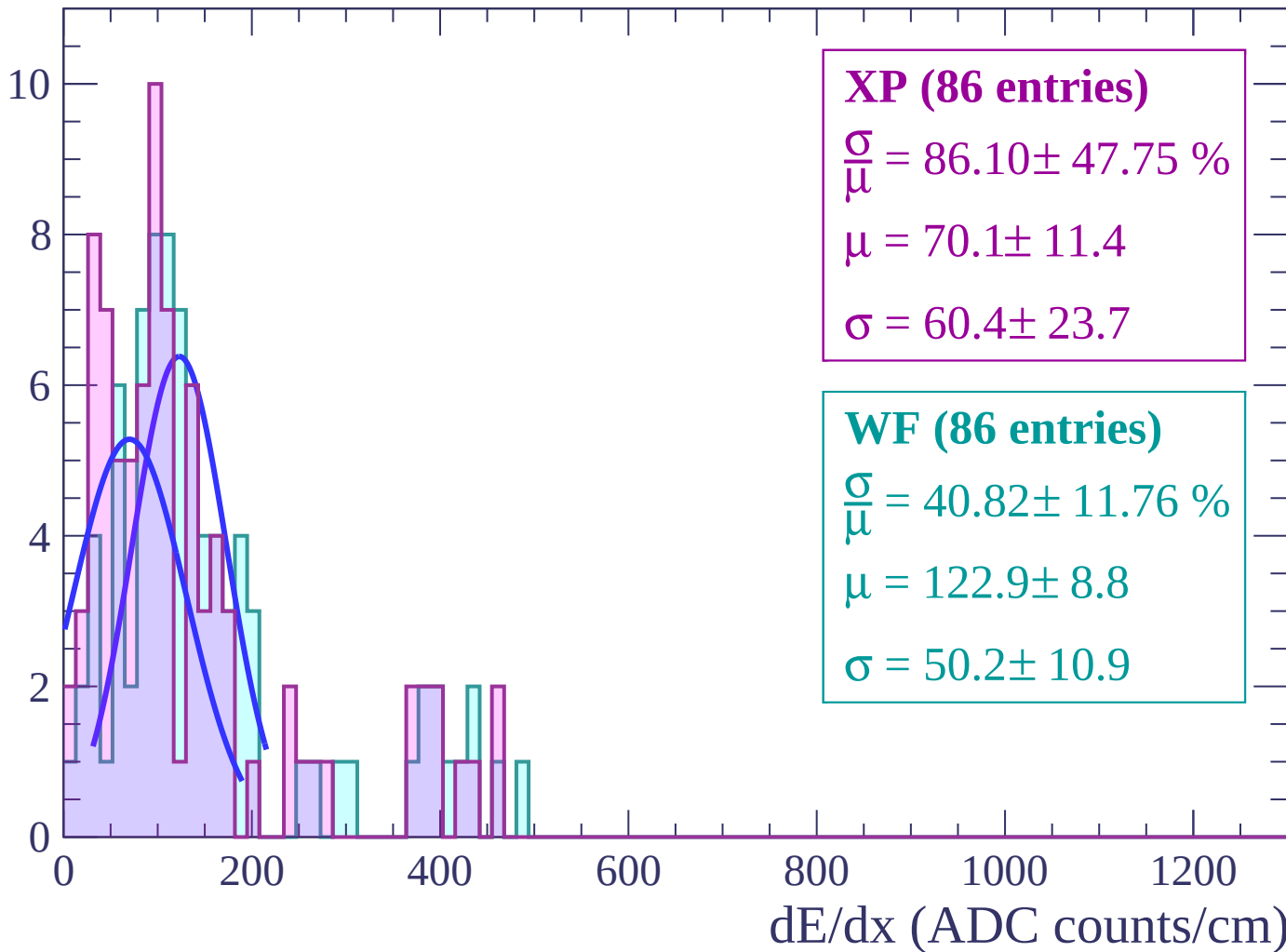
Energy loss | $-78 < \theta < -76$

Count



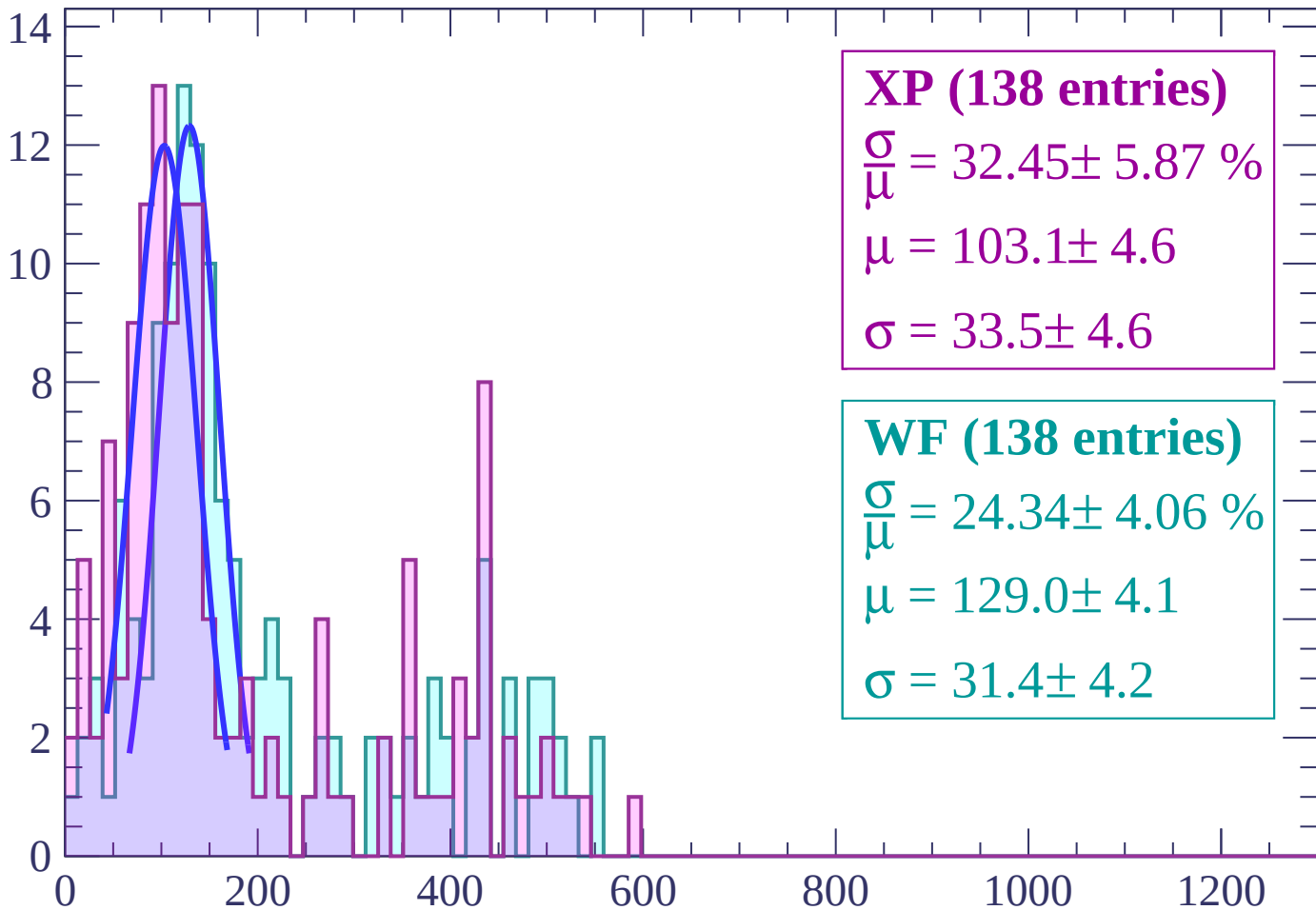
Energy loss | $-76 < \theta < -74$

Count



Energy loss | $-74 < \theta < -72$

Count



XP (138 entries)

$$\frac{\sigma}{\mu} = 32.45 \pm 5.87 \%$$

$$\mu = 103.1 \pm 4.6$$

$$\sigma = 33.5 \pm 4.6$$

WF (138 entries)

$$\frac{\sigma}{\mu} = 24.34 \pm 4.06 \%$$

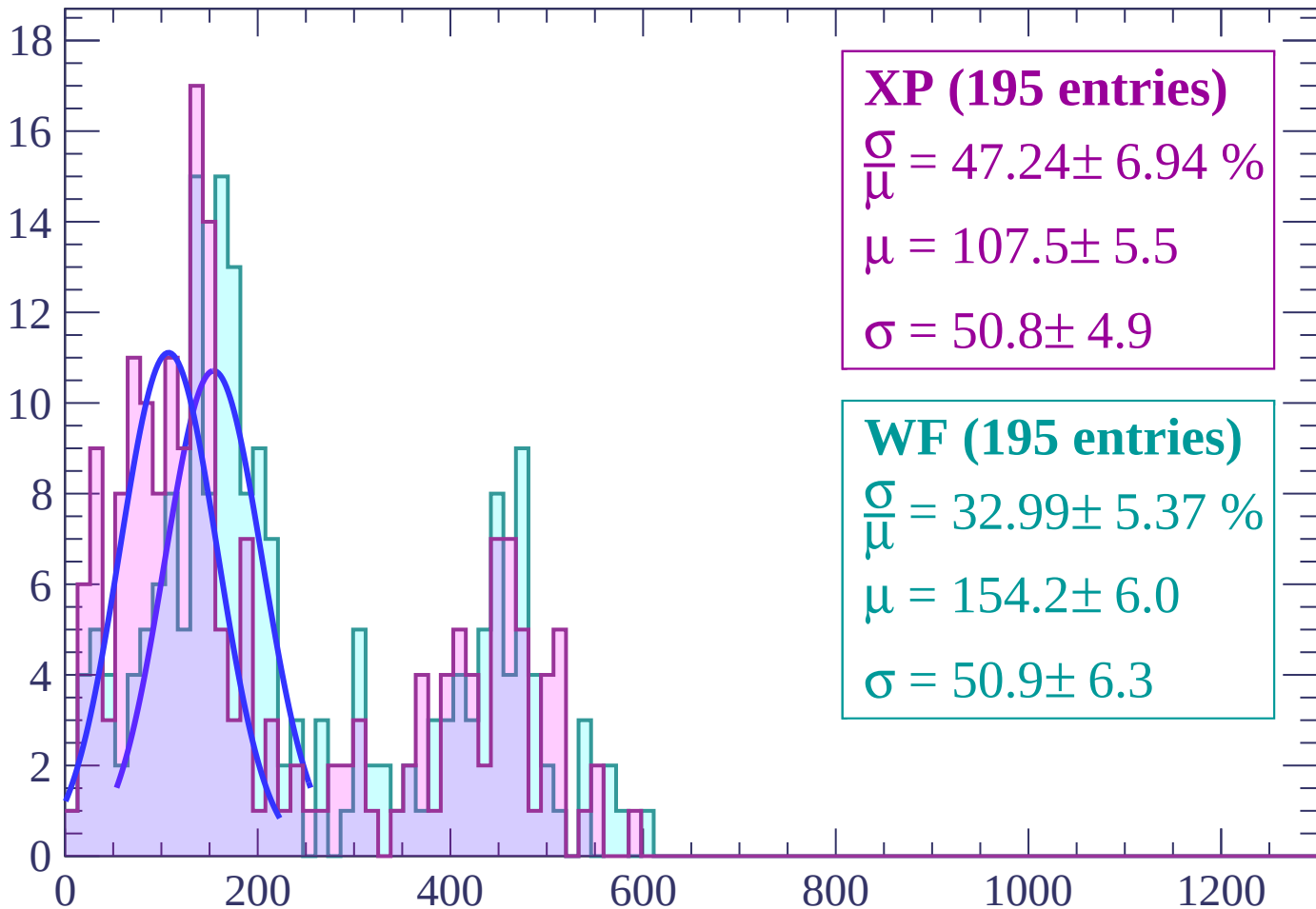
$$\mu = 129.0 \pm 4.1$$

$$\sigma = 31.4 \pm 4.2$$

dE/dx (ADC counts/cm)

Energy loss | $-72 < \theta < -70$

Count



XP (195 entries)

$$\frac{\sigma}{\mu} = 47.24 \pm 6.94 \%$$

$$\mu = 107.5 \pm 5.5$$

$$\sigma = 50.8 \pm 4.9$$

WF (195 entries)

$$\frac{\sigma}{\mu} = 32.99 \pm 5.37 \%$$

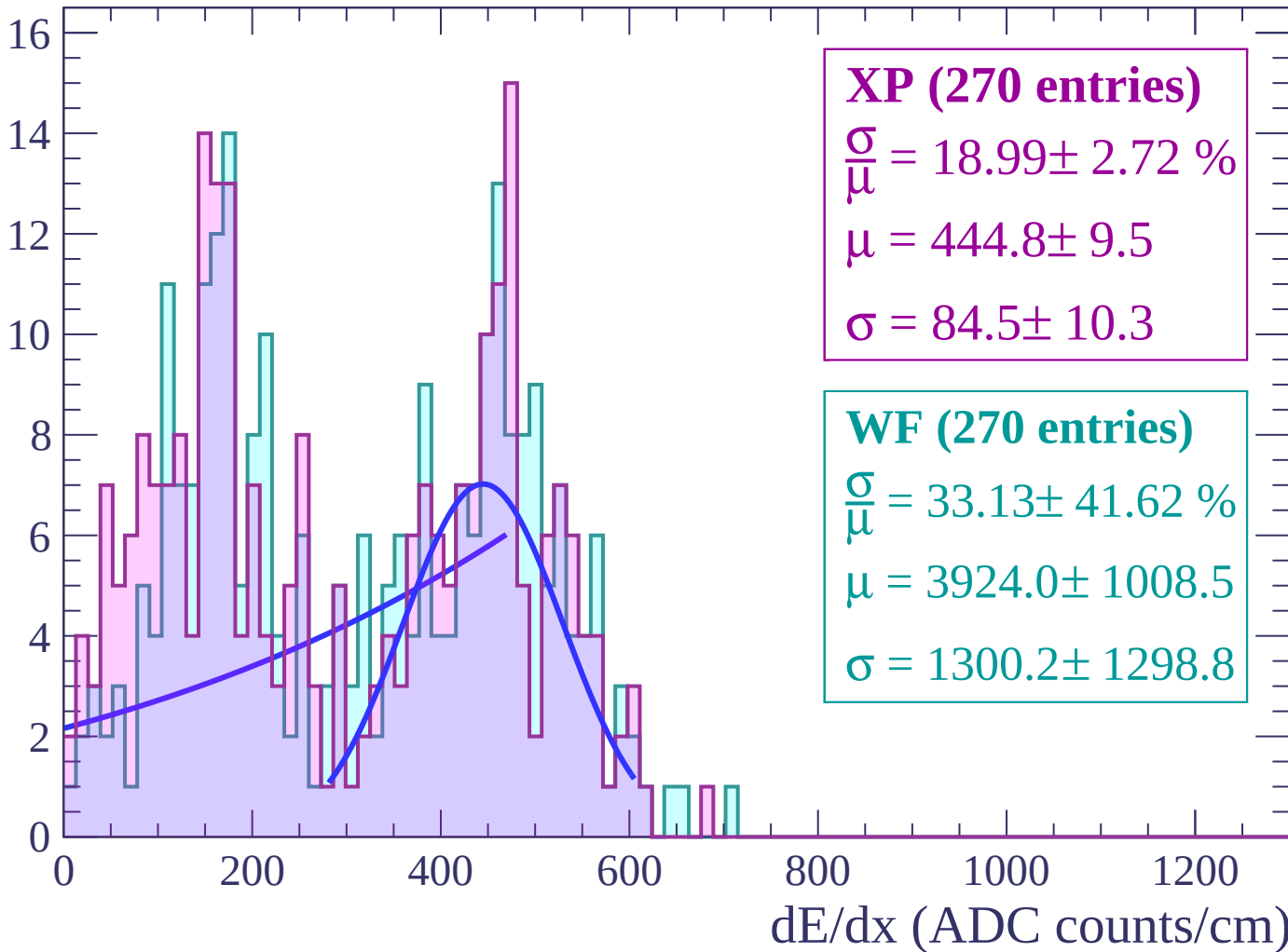
$$\mu = 154.2 \pm 6.0$$

$$\sigma = 50.9 \pm 6.3$$

dE/dx (ADC counts/cm)

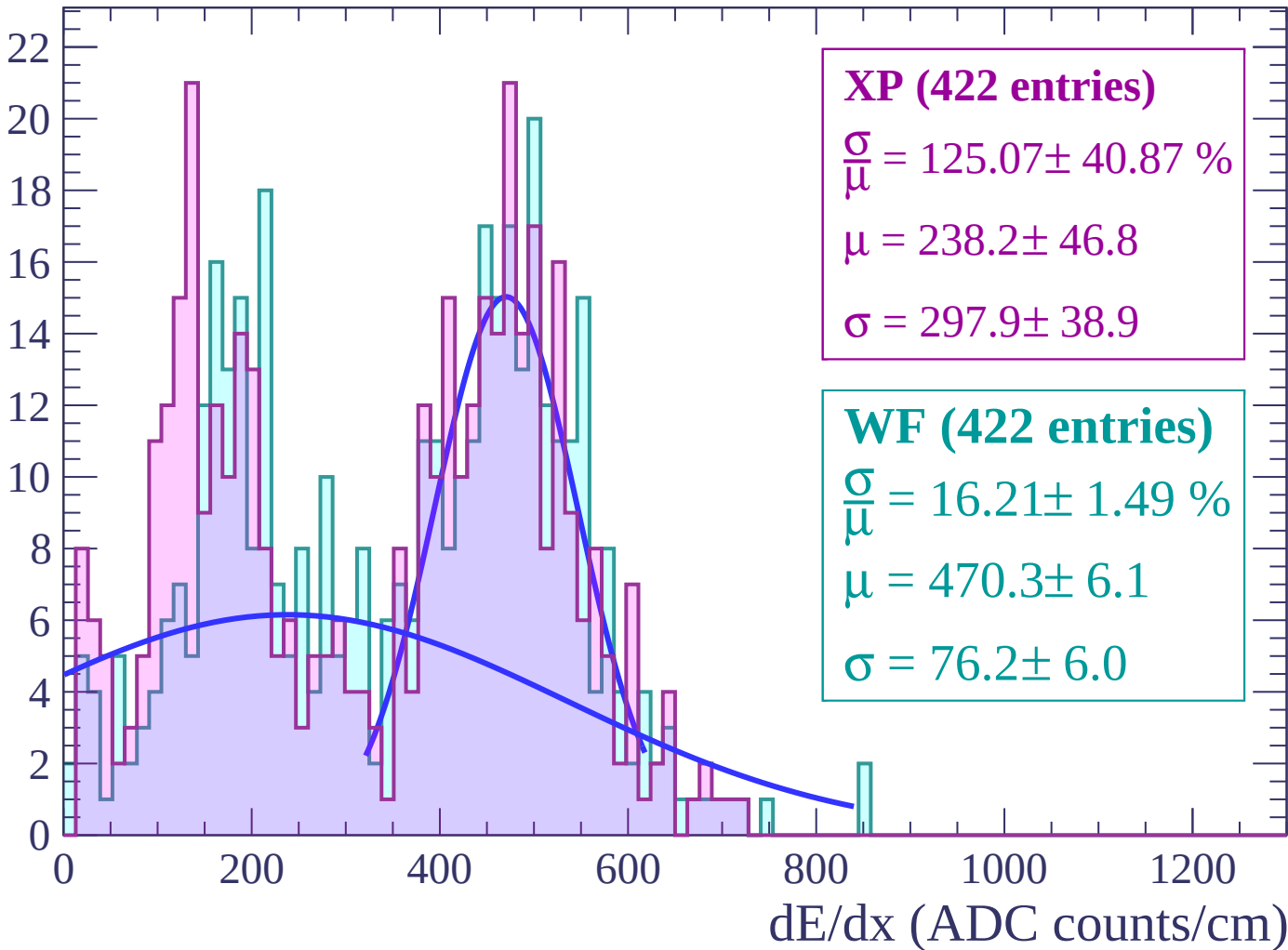
Energy loss | $-70 < \theta < -68$

Count



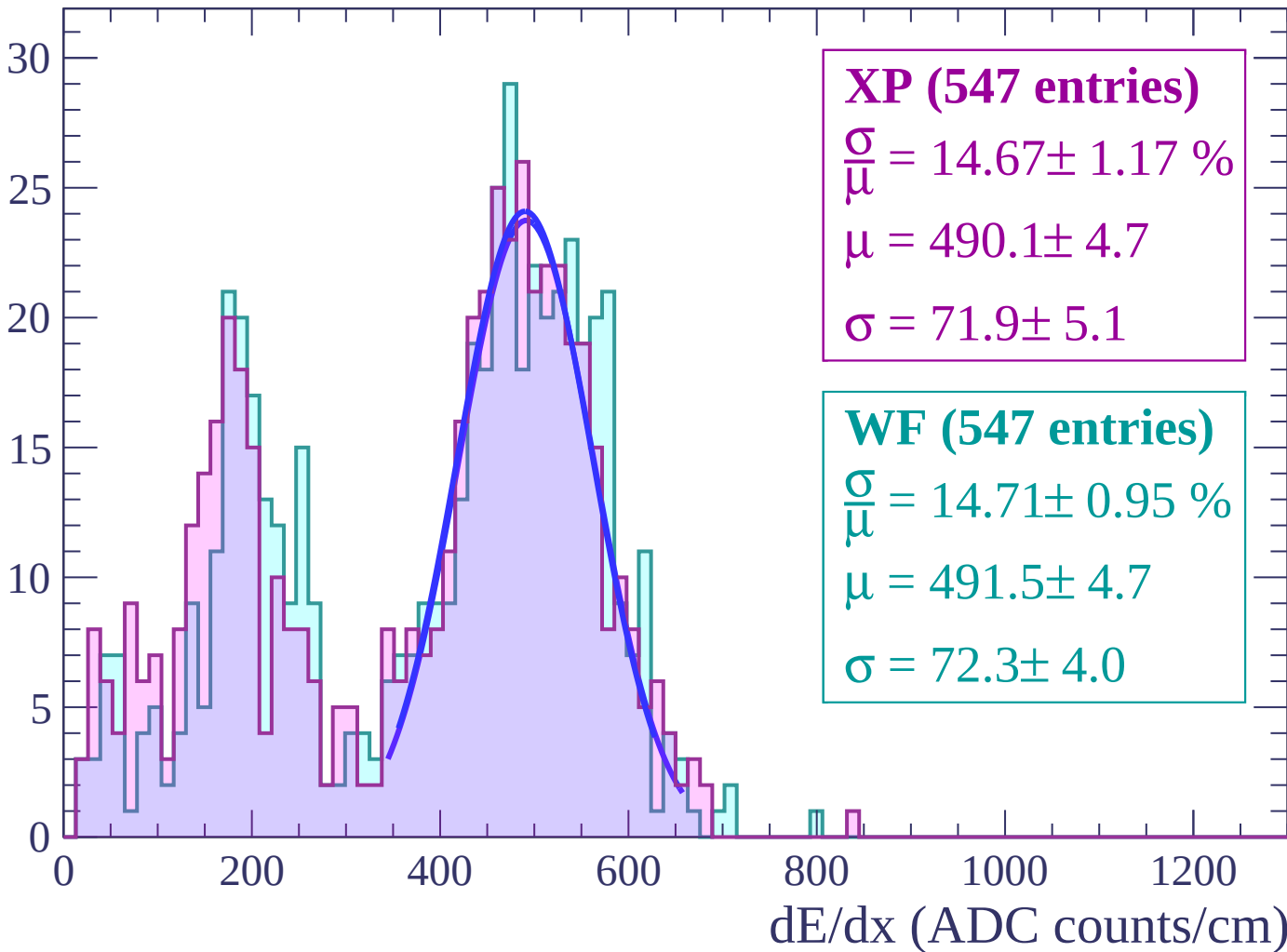
Energy loss | $-68 < \theta < -66$

Count



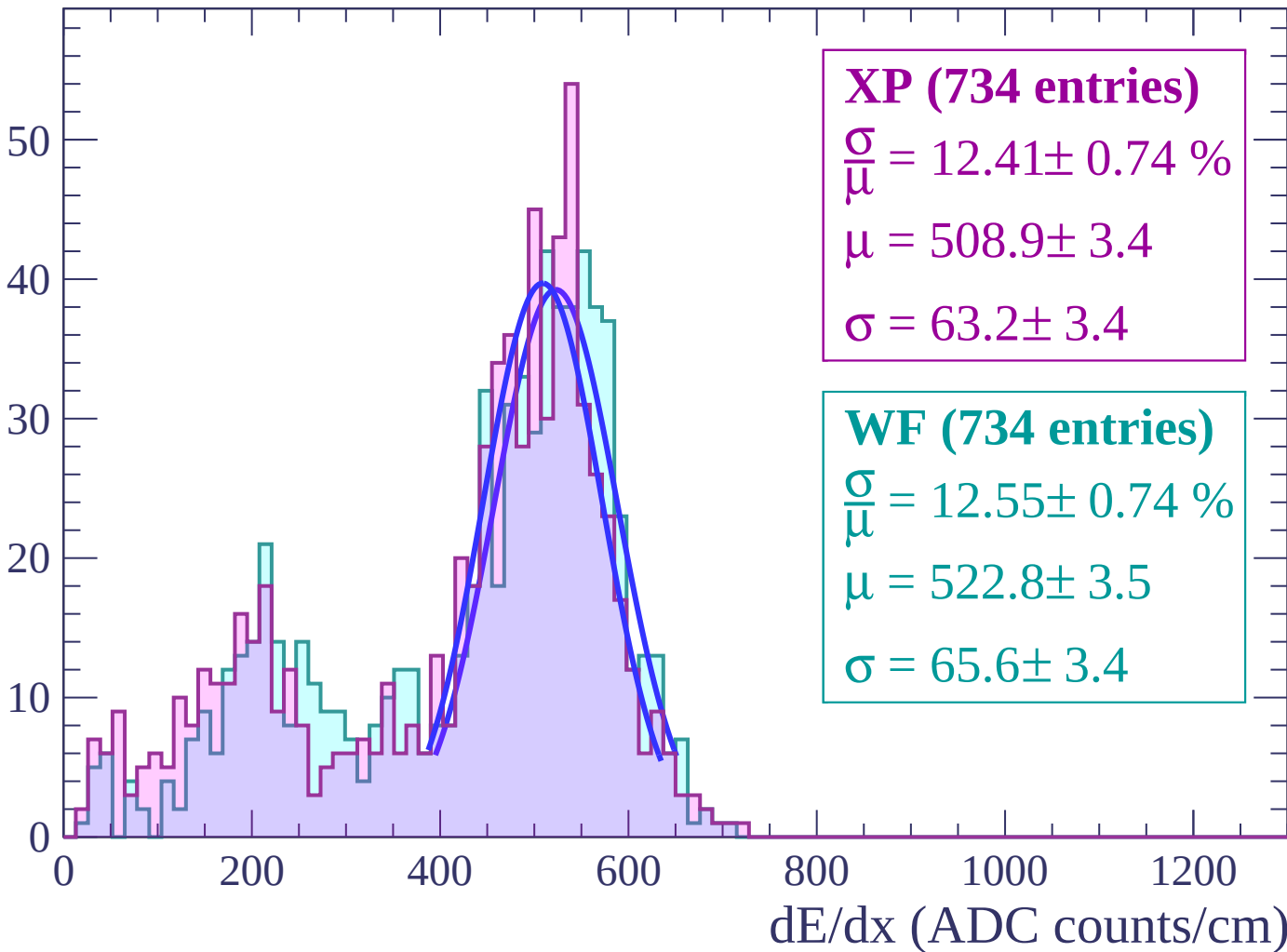
Energy loss | $-66 < \theta < -64$

Count



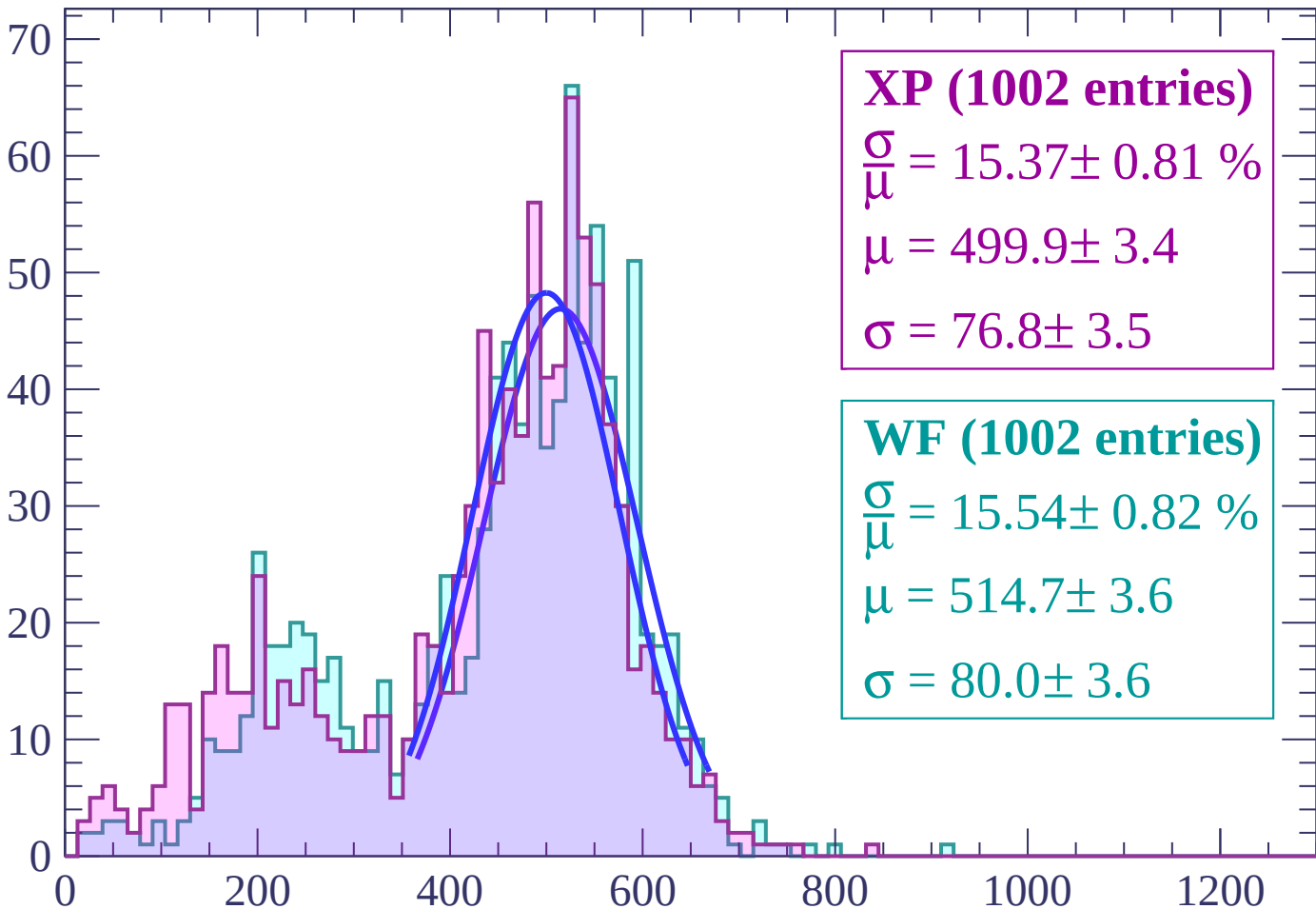
Energy loss | $-64 < \theta < -62$

Count



Energy loss | $-62 < \theta < -60$

Count



XP (1002 entries)

$\frac{\sigma}{\mu} = 15.37 \pm 0.81 \%$

$\mu = 499.9 \pm 3.4$

$\sigma = 76.8 \pm 3.5$

WF (1002 entries)

$\frac{\sigma}{\mu} = 15.54 \pm 0.82 \%$

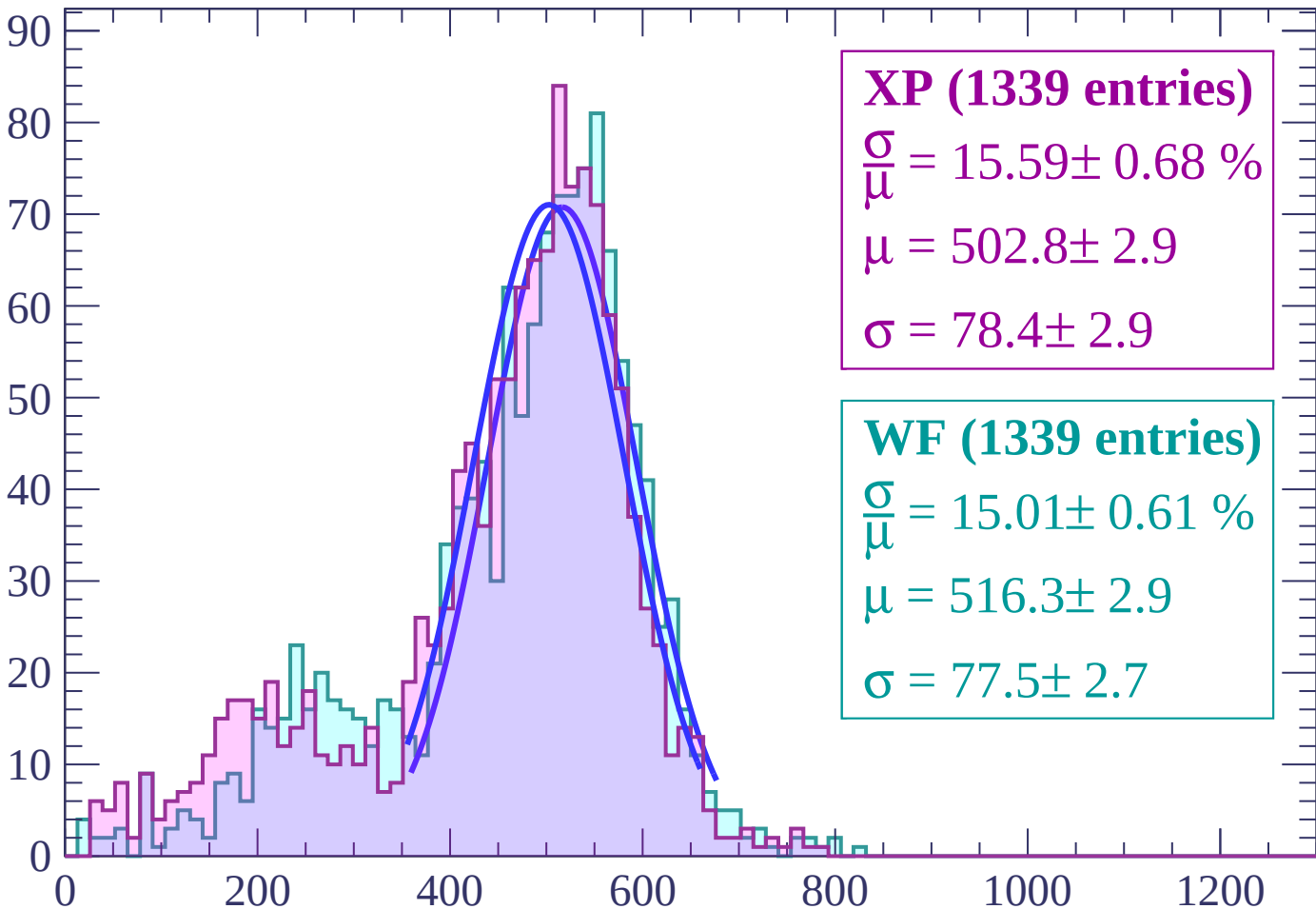
$\mu = 514.7 \pm 3.6$

$\sigma = 80.0 \pm 3.6$

dE/dx (ADC counts/cm)

Energy loss | $-60 < \theta < -58$

Count



XP (1339 entries)

$$\frac{\sigma}{\mu} = 15.59 \pm 0.68 \%$$

$$\mu = 502.8 \pm 2.9$$

$$\sigma = 78.4 \pm 2.9$$

WF (1339 entries)

$$\frac{\sigma}{\mu} = 15.01 \pm 0.61 \%$$

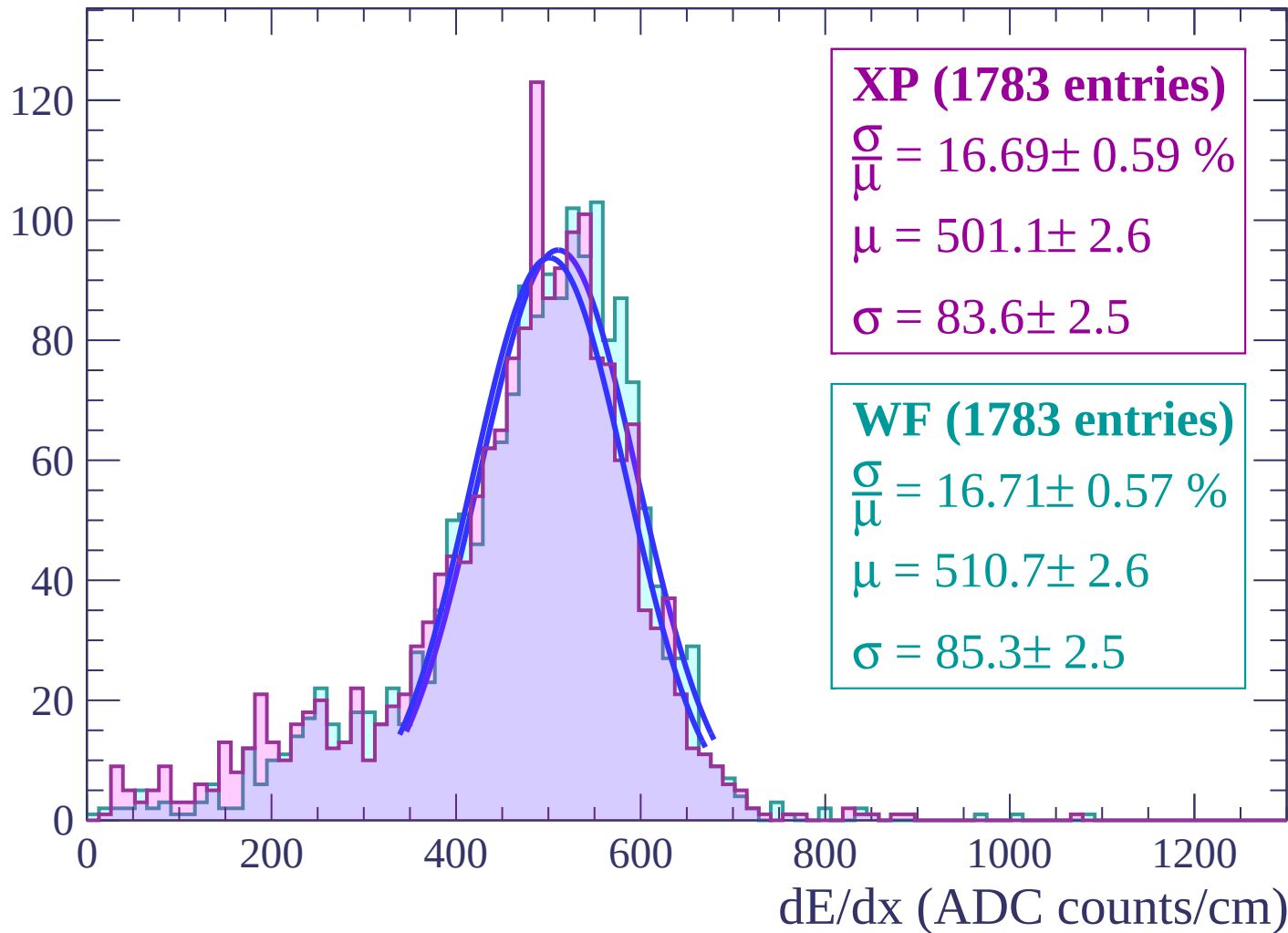
$$\mu = 516.3 \pm 2.9$$

$$\sigma = 77.5 \pm 2.7$$

dE/dx (ADC counts/cm)

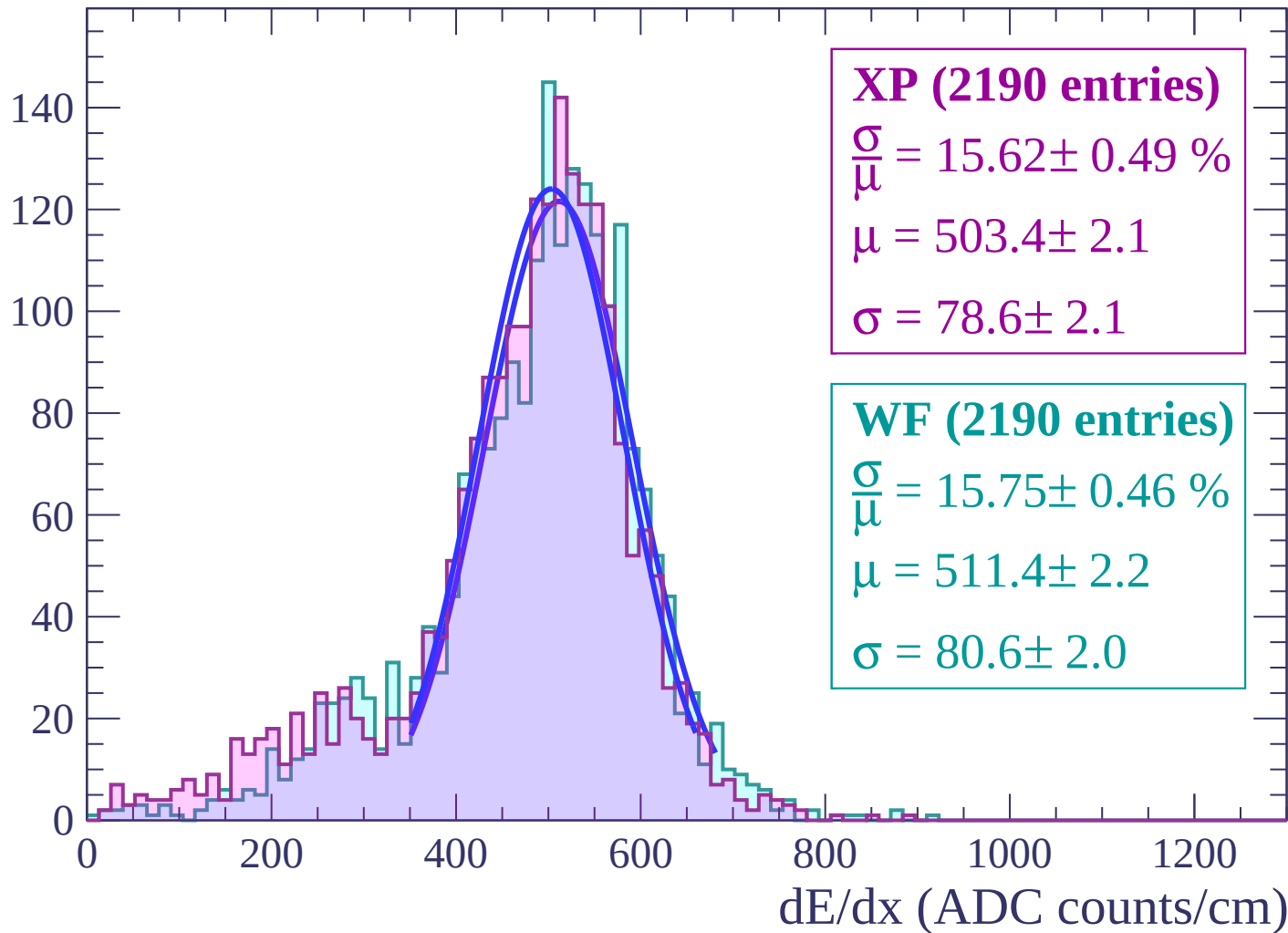
Energy loss | $-58 < \theta < -56$

Count

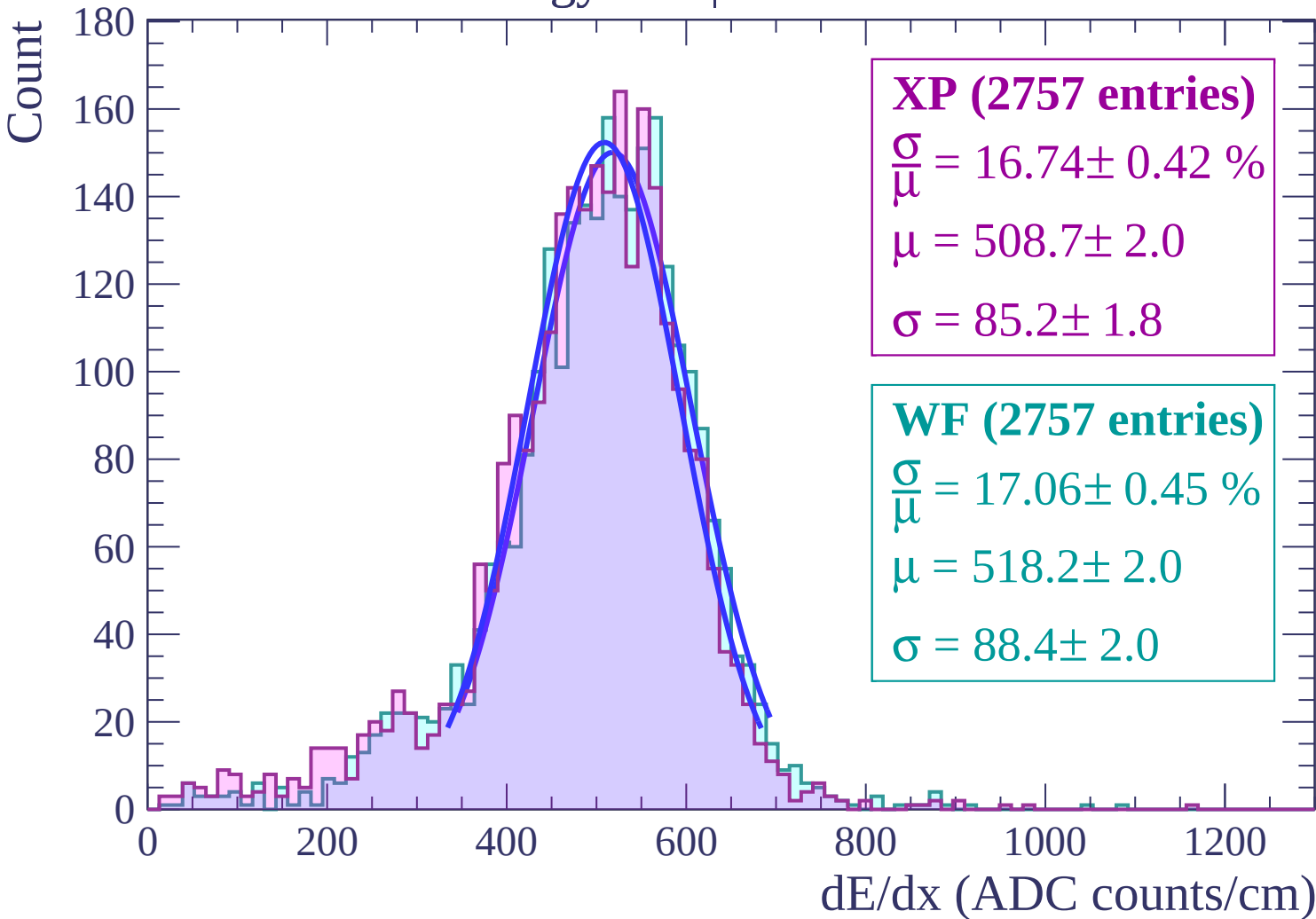


Energy loss | $-56 < \theta < -54$

Count

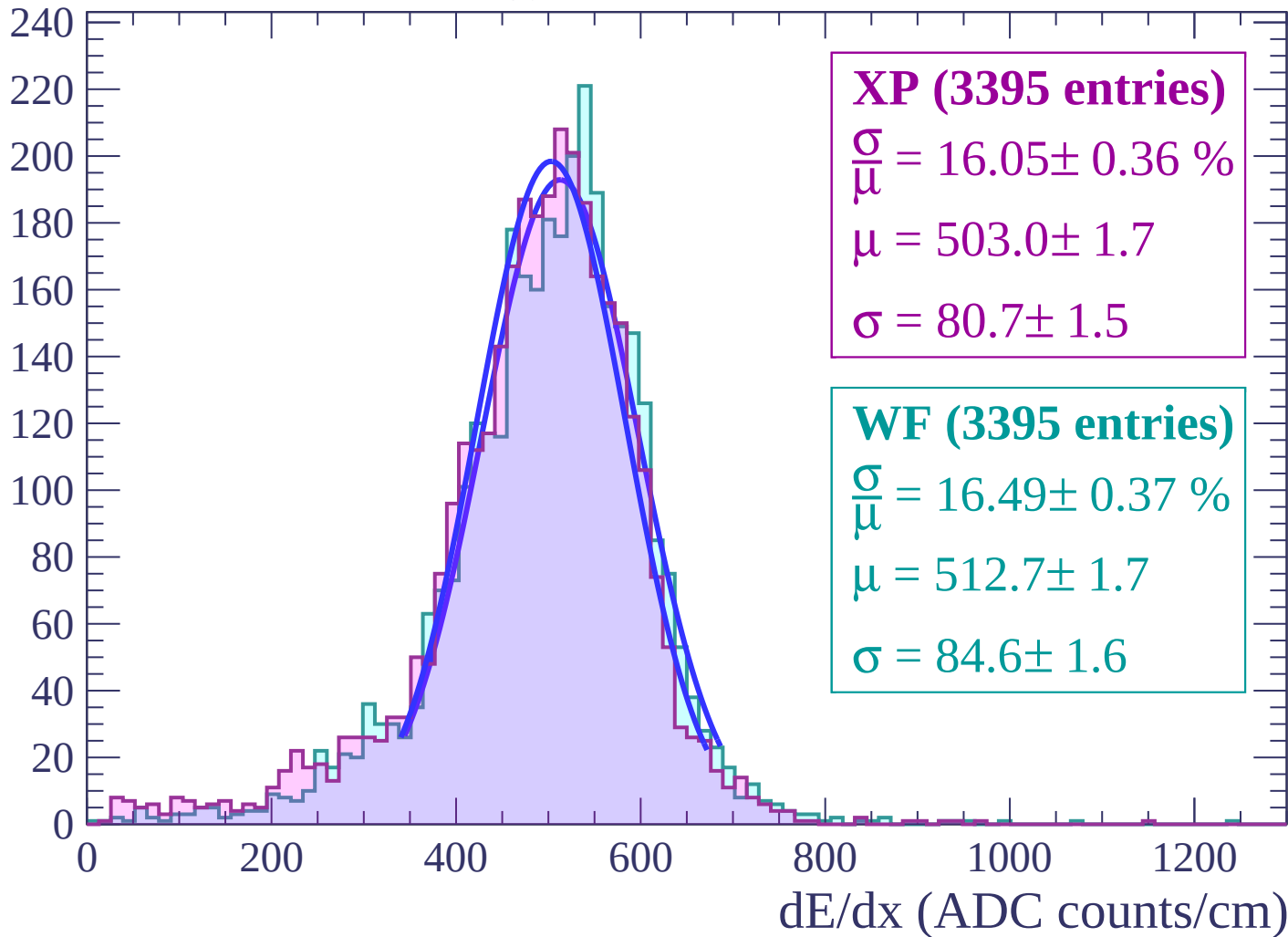


Energy loss | $-54 < \theta < -52$



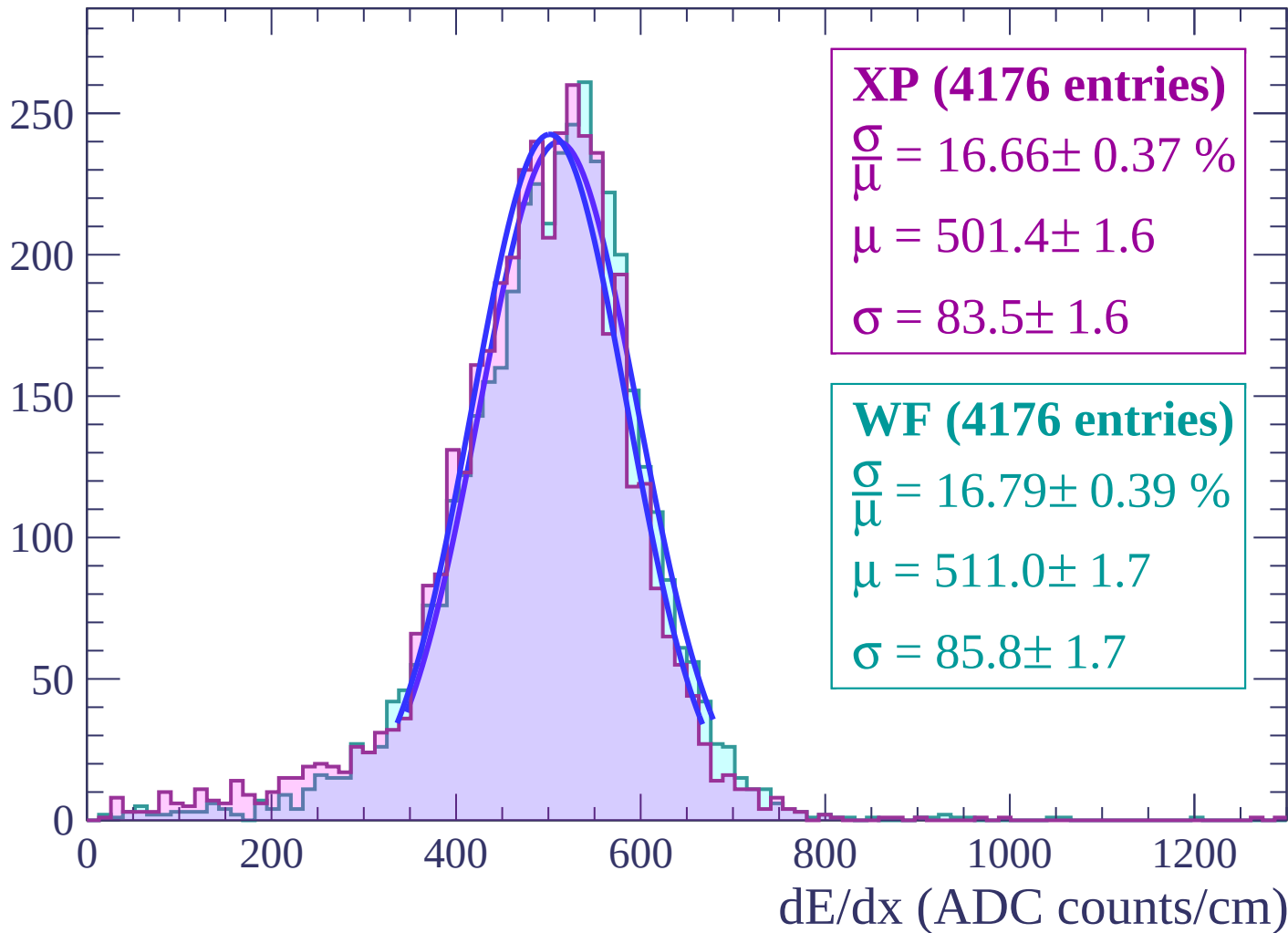
Energy loss | $-52 < \theta < -50$

Count



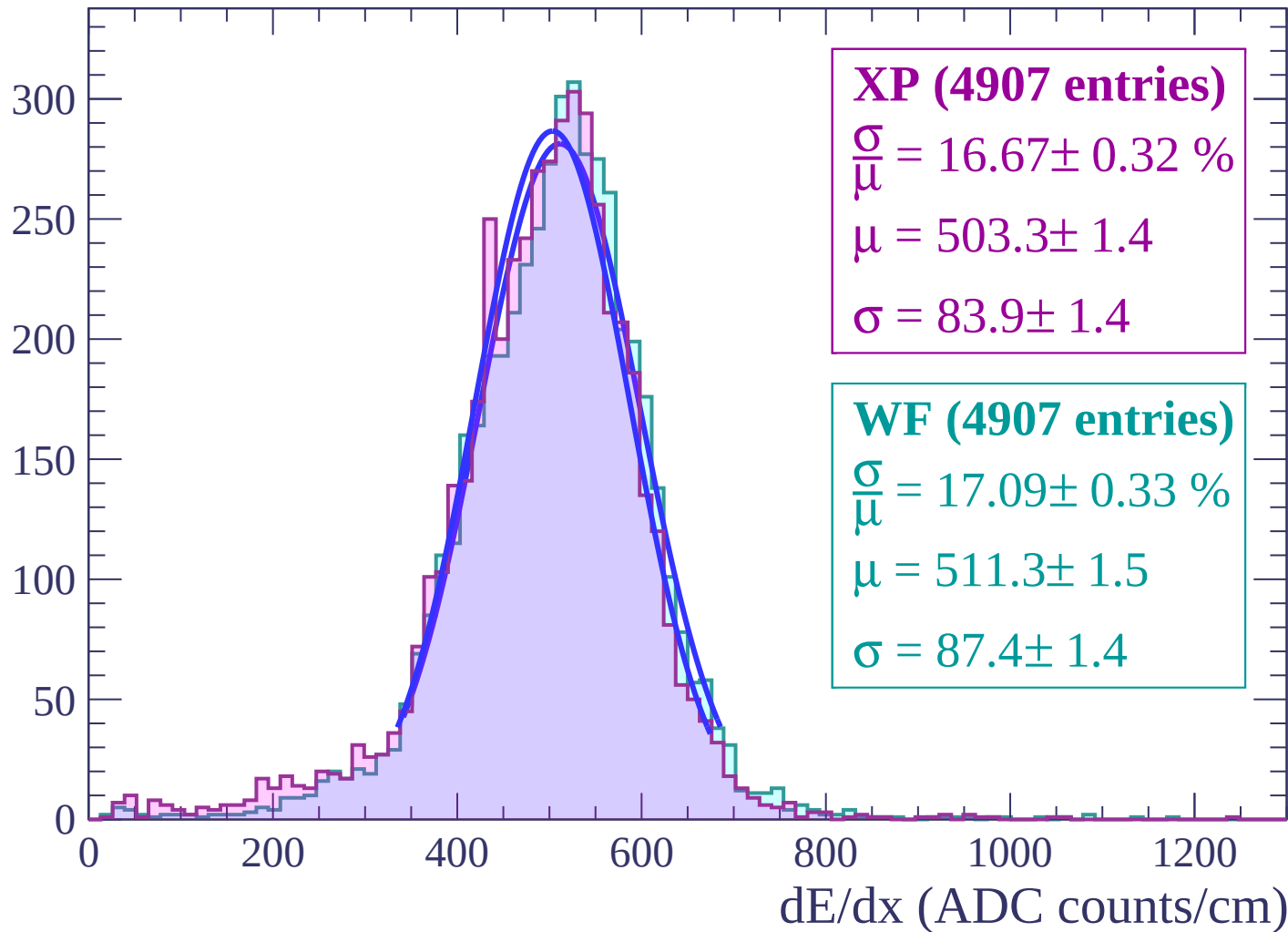
Energy loss | $-50 < \theta < -48$

Count



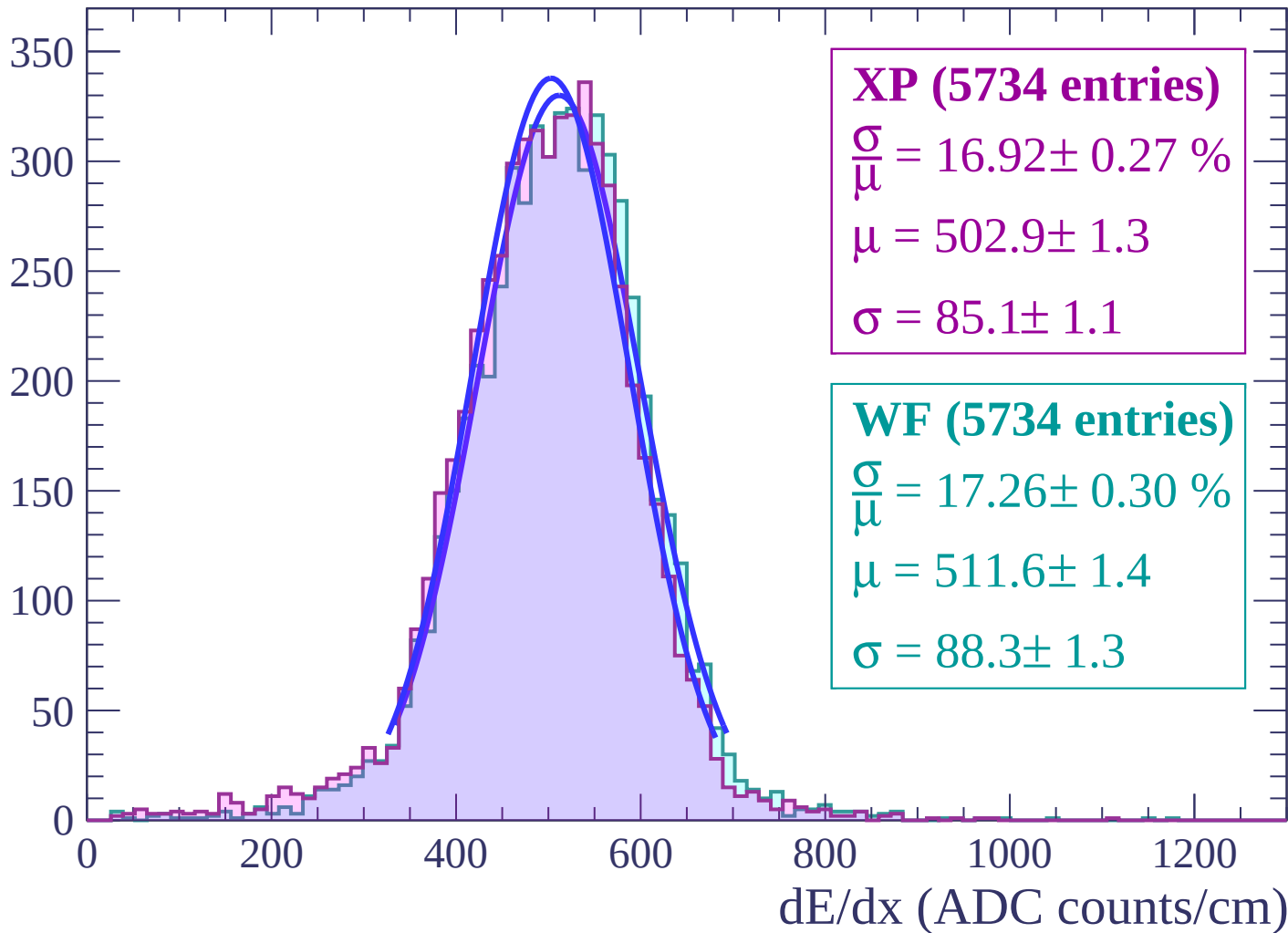
Energy loss | $-48 < \theta < -46$

Count



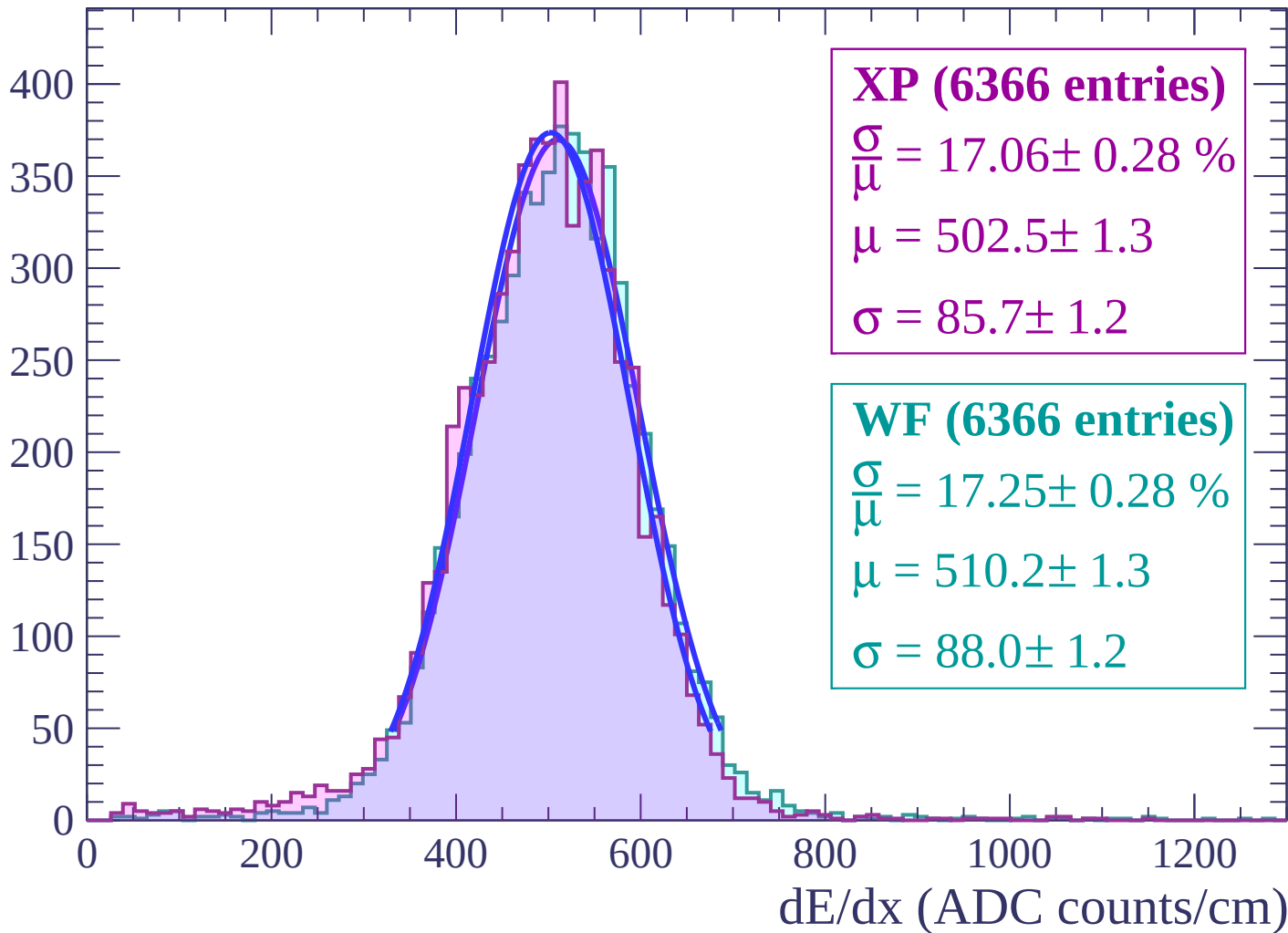
Energy loss | $-46 < \theta < -44$

Count



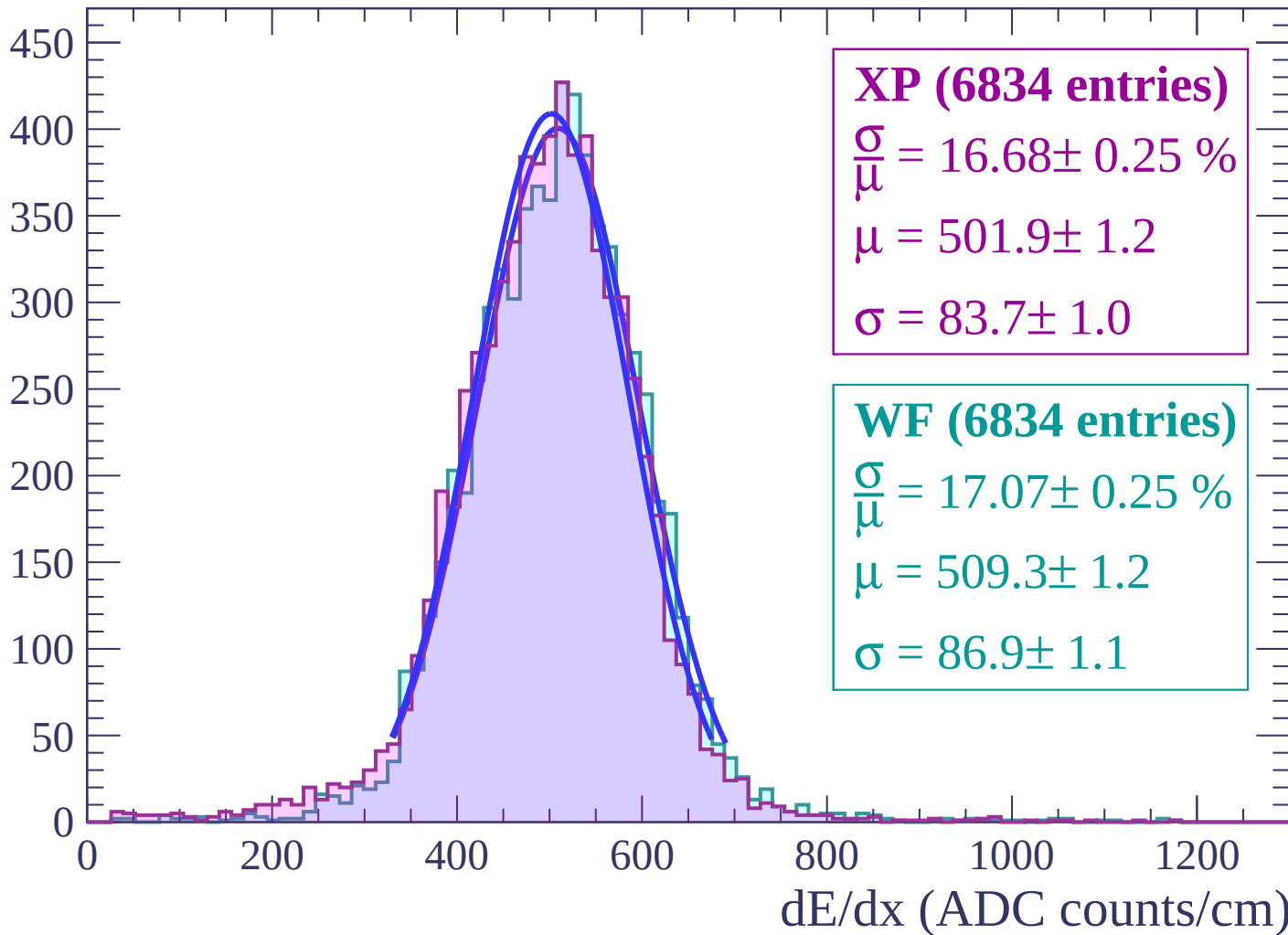
Energy loss | $-44 < \theta < -42$

Count



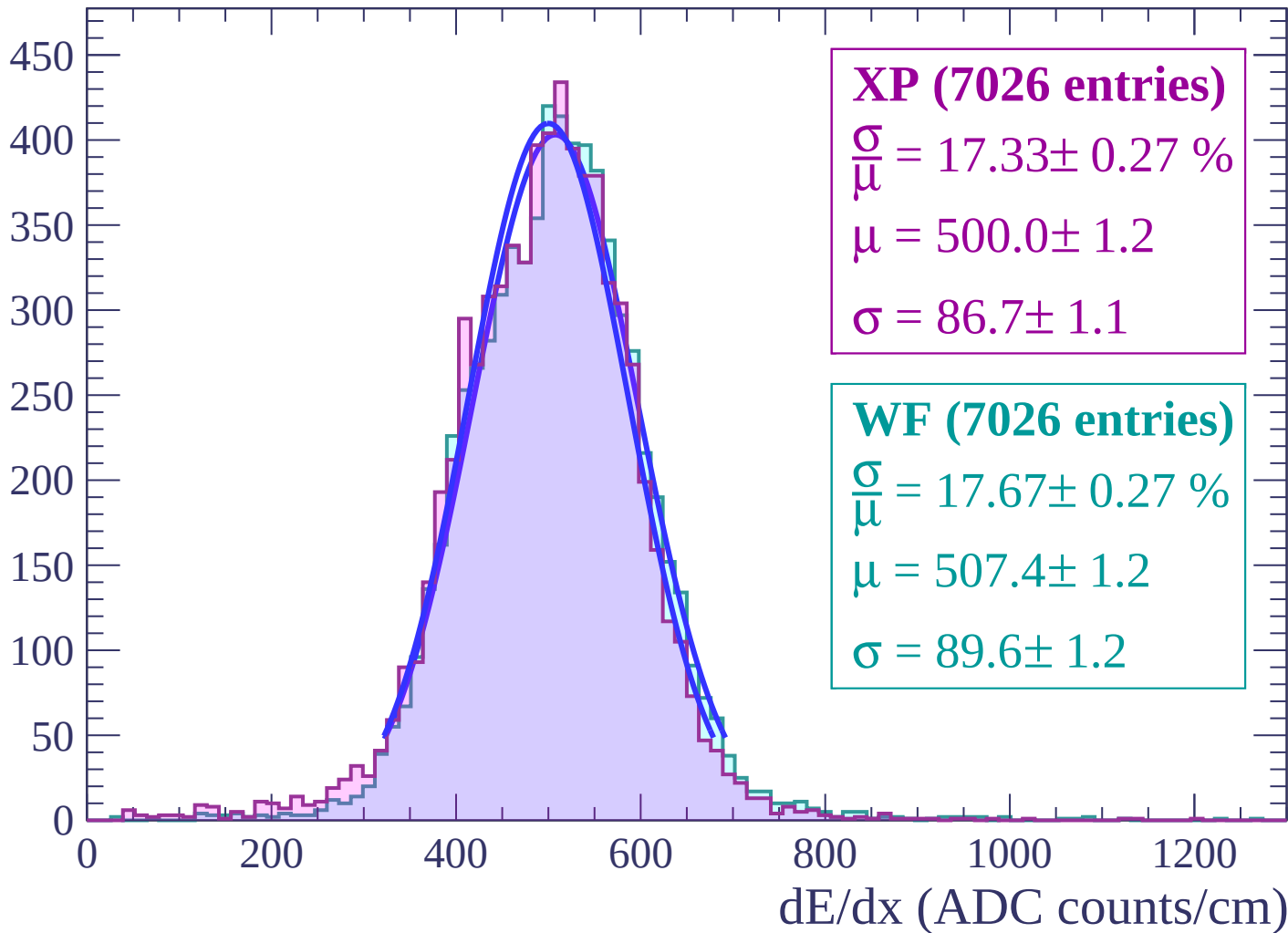
Energy loss | $-42 < \theta < -40$

Count



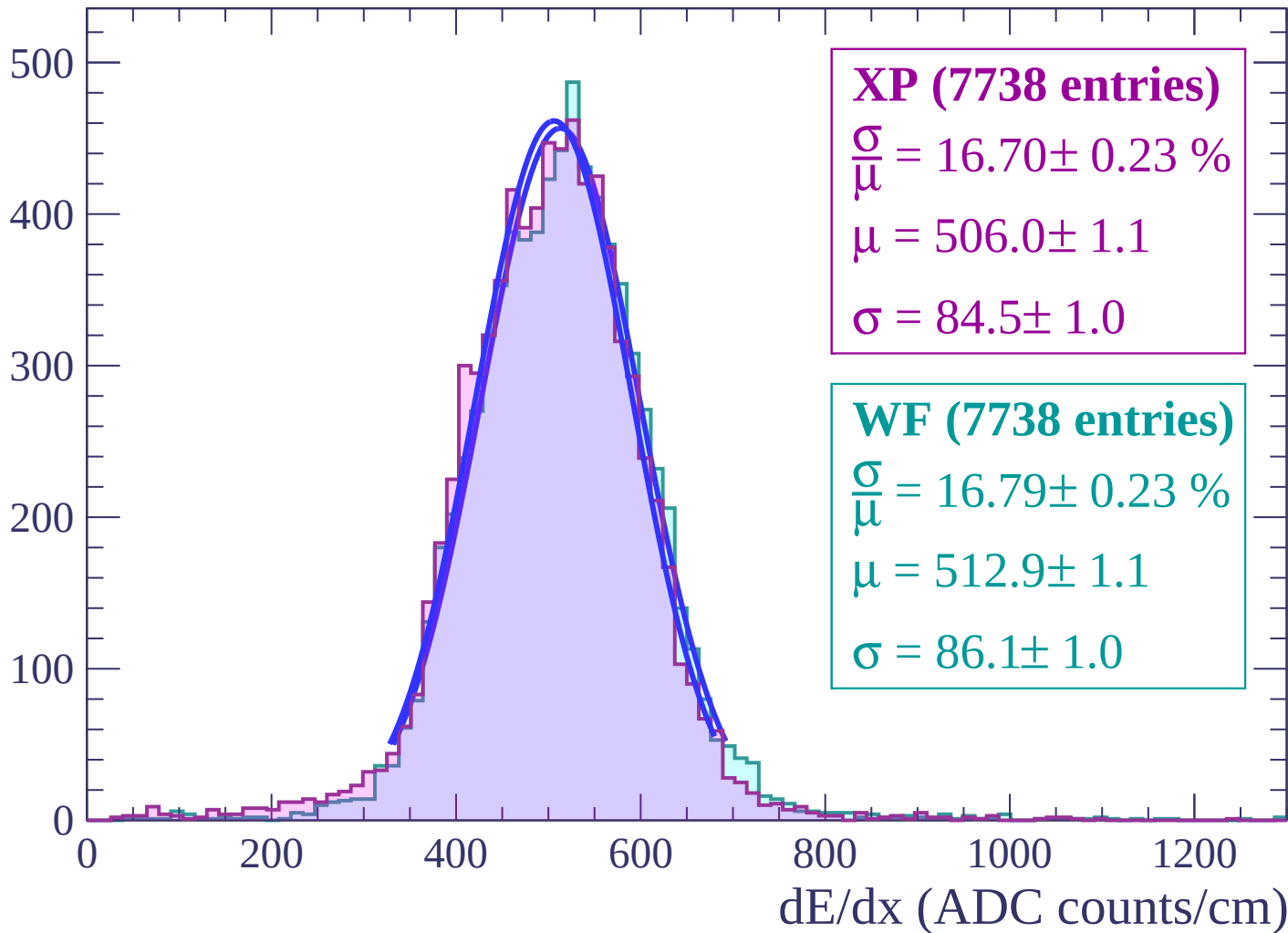
Energy loss | $-40 < \theta < -38$

Count



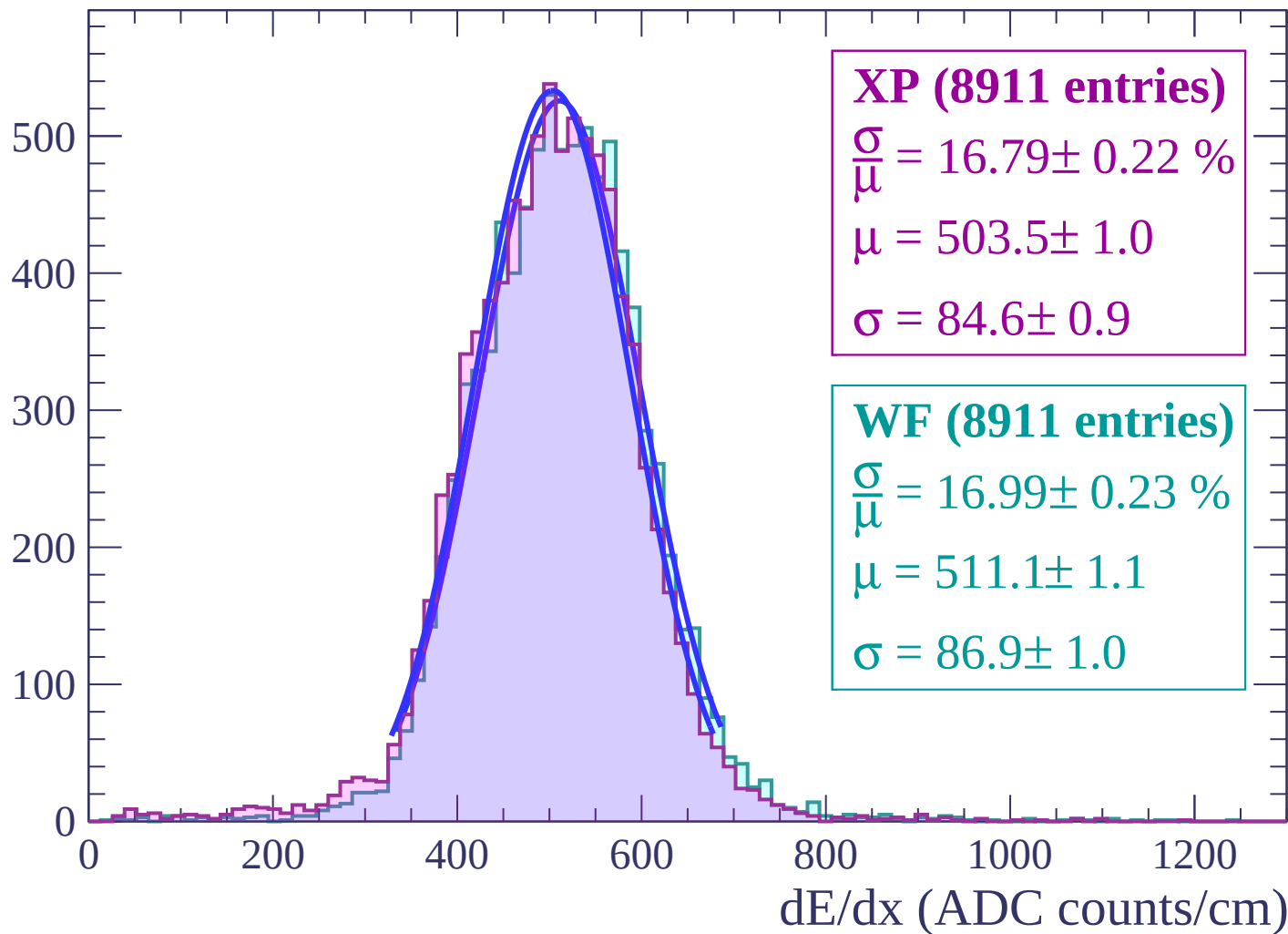
Energy loss | $-38 < \theta < -36$

Count

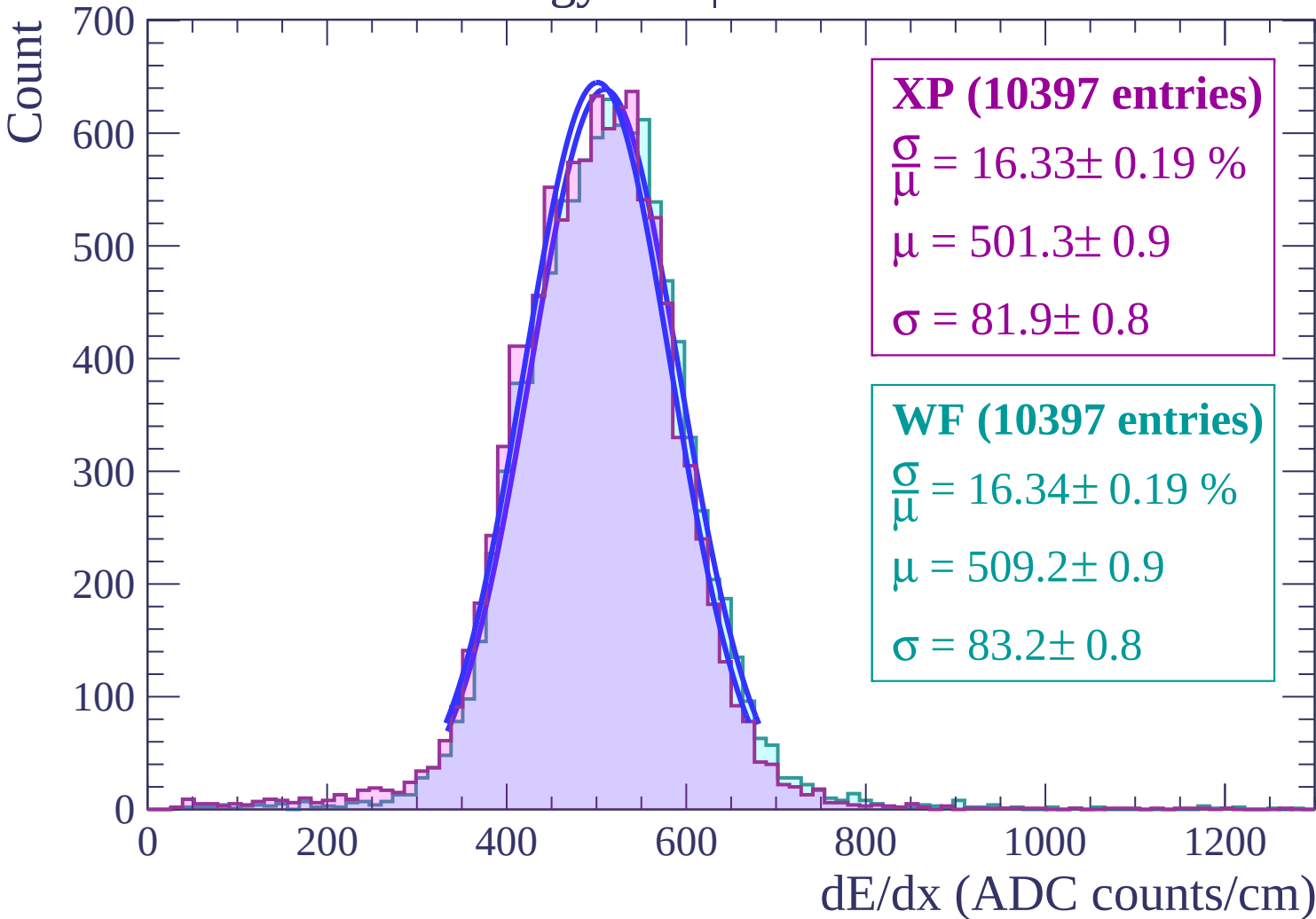


Energy loss | $-36 < \theta < -34$

Count

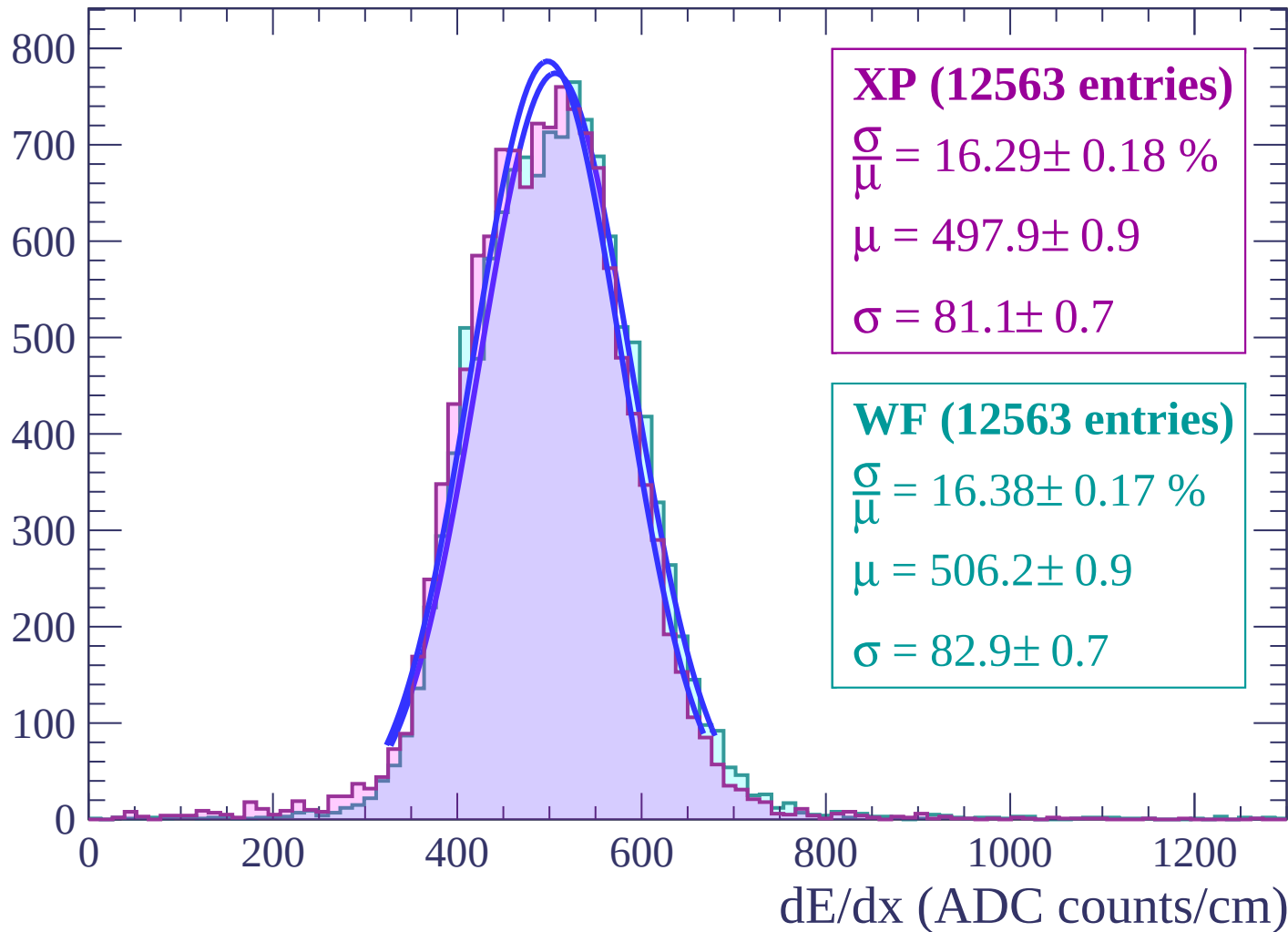


Energy loss | $-34 < \theta < -32$



Energy loss | $-32 < \theta < -30$

Count



Energy loss | $-30 < \theta < -28$

Count

1000

800

600

400

200

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (15565 entries)

$\frac{\sigma}{\mu} = 16.19 \pm 0.15 \%$

$\mu = 492.1 \pm 0.8$

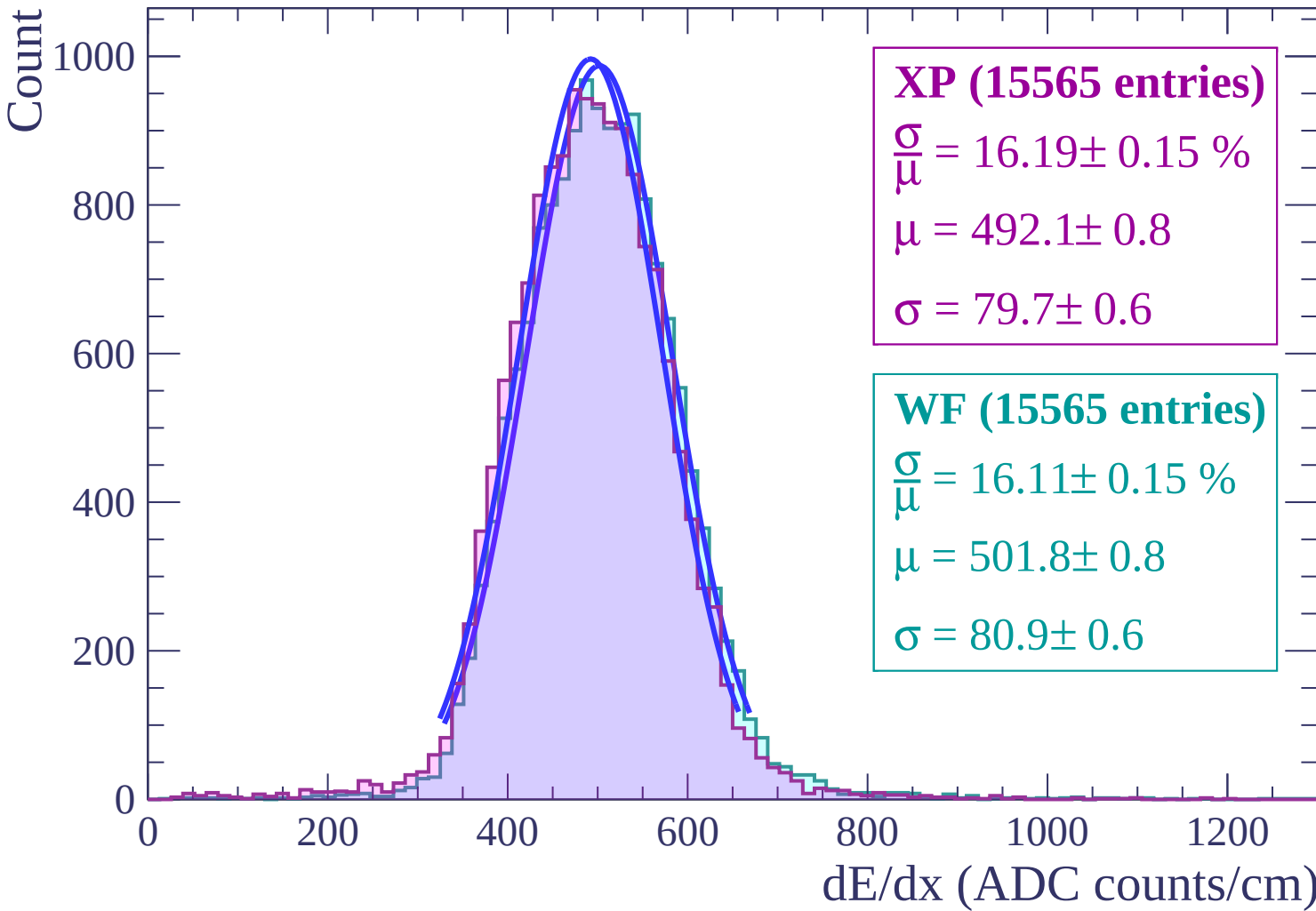
$\sigma = 79.7 \pm 0.6$

WF (15565 entries)

$\frac{\sigma}{\mu} = 16.11 \pm 0.15 \%$

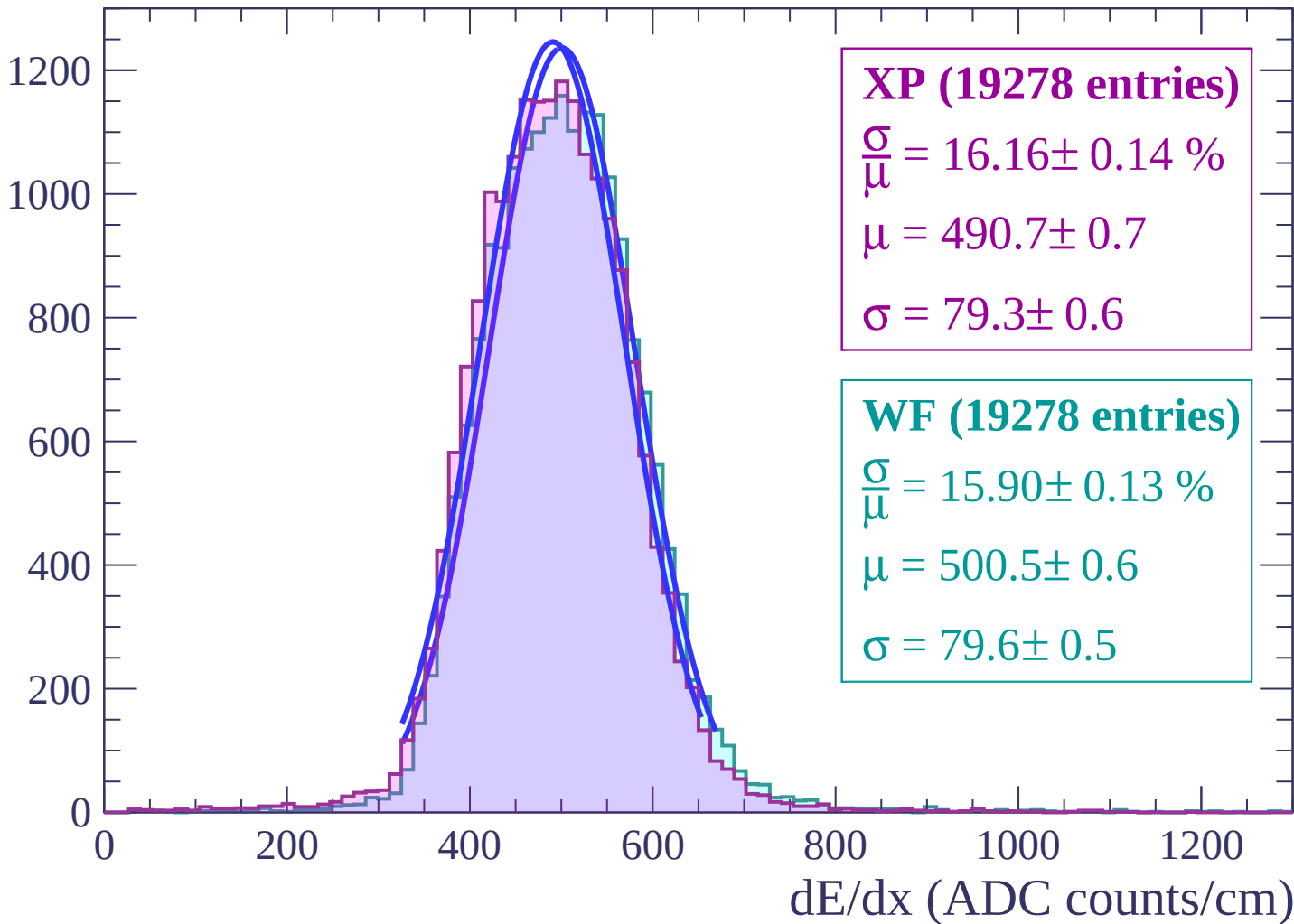
$\mu = 501.8 \pm 0.8$

$\sigma = 80.9 \pm 0.6$



Energy loss | $-28 < \theta < -26$

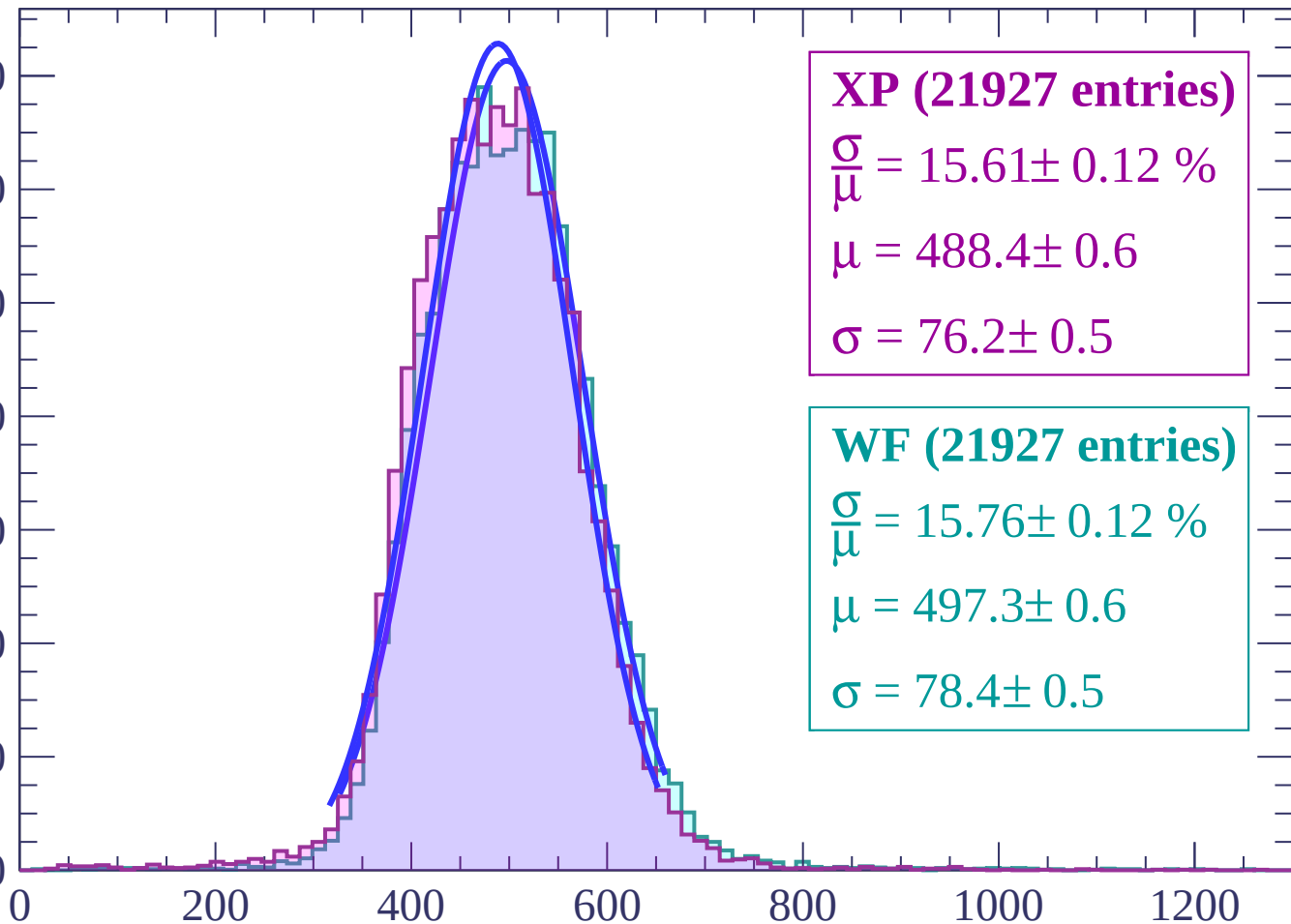
Count



Energy loss | $-26 < \theta < -24$

Count

1400
1200
1000
800
600
400
200
0



XP (21927 entries)

$\frac{\sigma}{\mu} = 15.61 \pm 0.12 \%$

$\mu = 488.4 \pm 0.6$

$\sigma = 76.2 \pm 0.5$

WF (21927 entries)

$\frac{\sigma}{\mu} = 15.76 \pm 0.12 \%$

$\mu = 497.3 \pm 0.6$

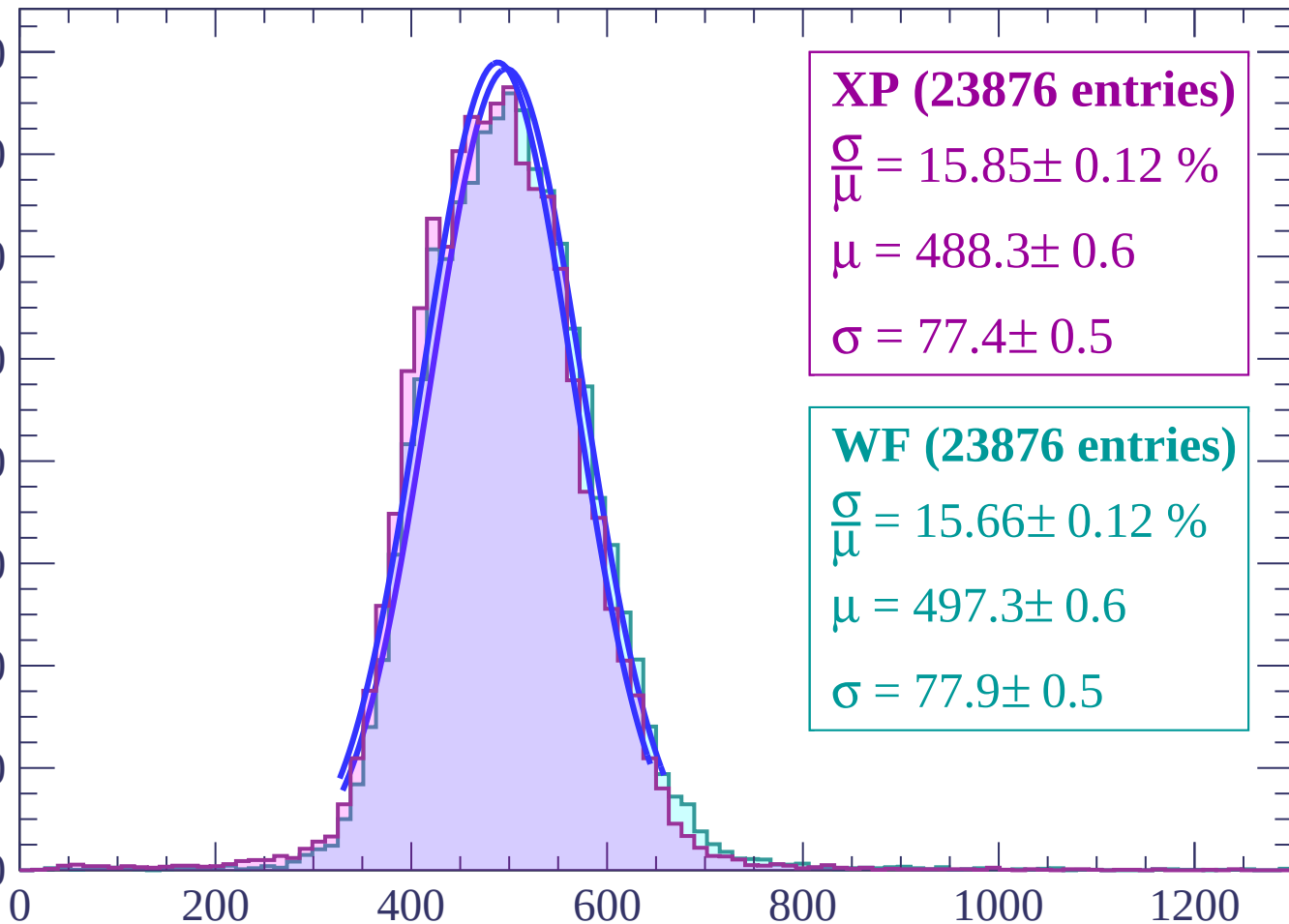
$\sigma = 78.4 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-24 < \theta < -22$

Count

1600
1400
1200
1000
800
600
400
200
0



XP (23876 entries)

$\frac{\sigma}{\mu} = 15.85 \pm 0.12 \%$

$\mu = 488.3 \pm 0.6$

$\sigma = 77.4 \pm 0.5$

WF (23876 entries)

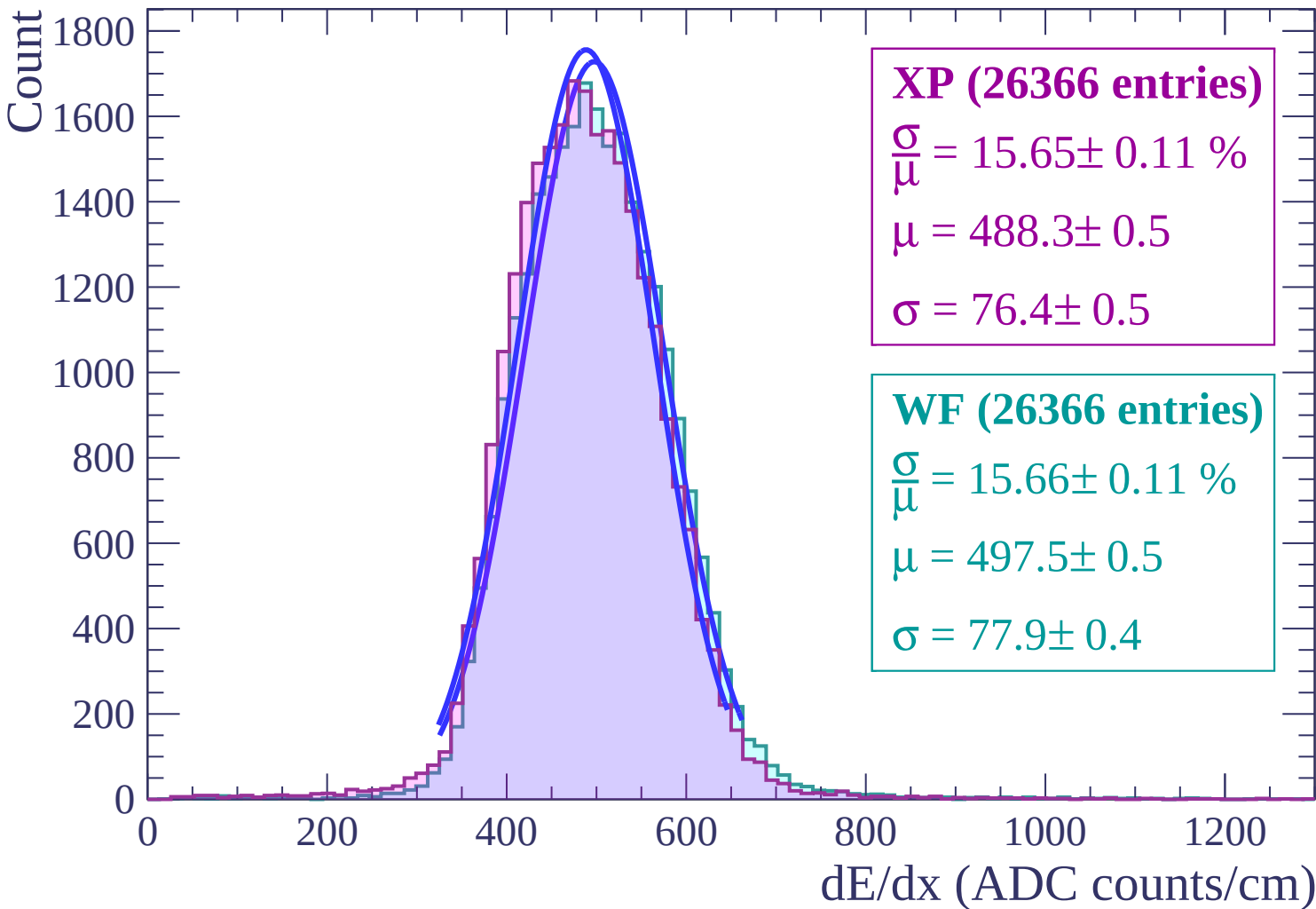
$\frac{\sigma}{\mu} = 15.66 \pm 0.12 \%$

$\mu = 497.3 \pm 0.6$

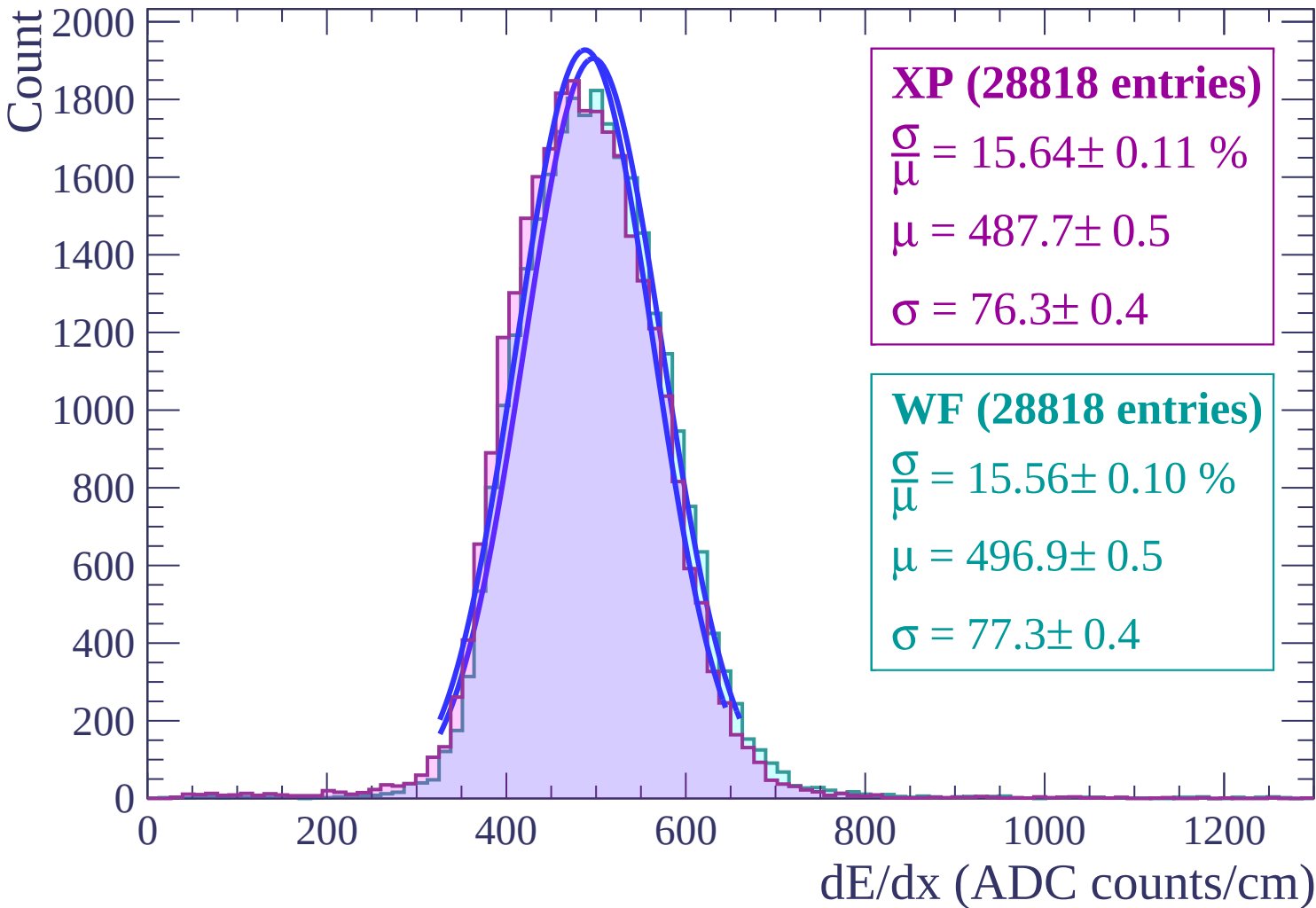
$\sigma = 77.9 \pm 0.5$

dE/dx (ADC counts/cm)

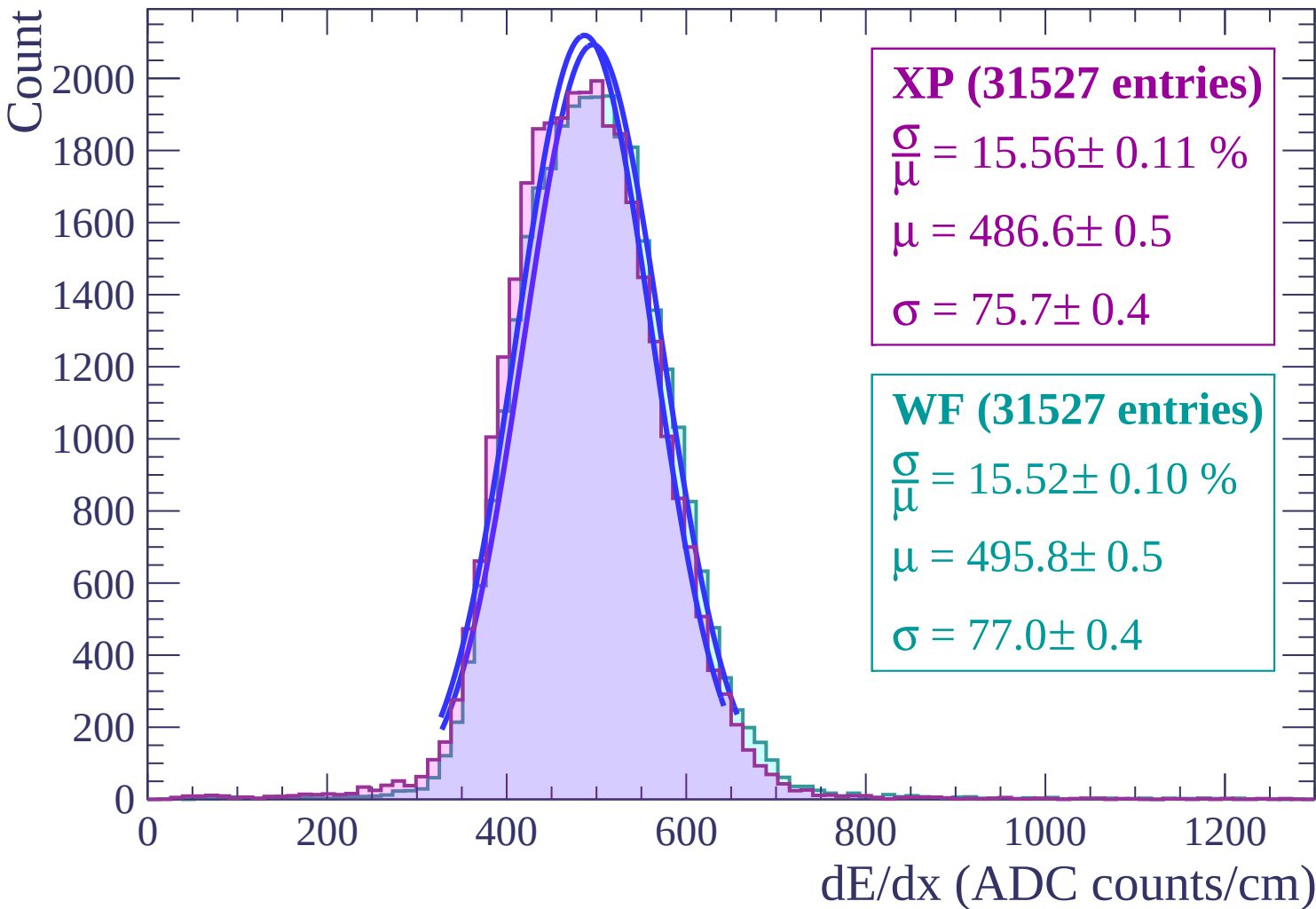
Energy loss | $-22 < \theta < -20$



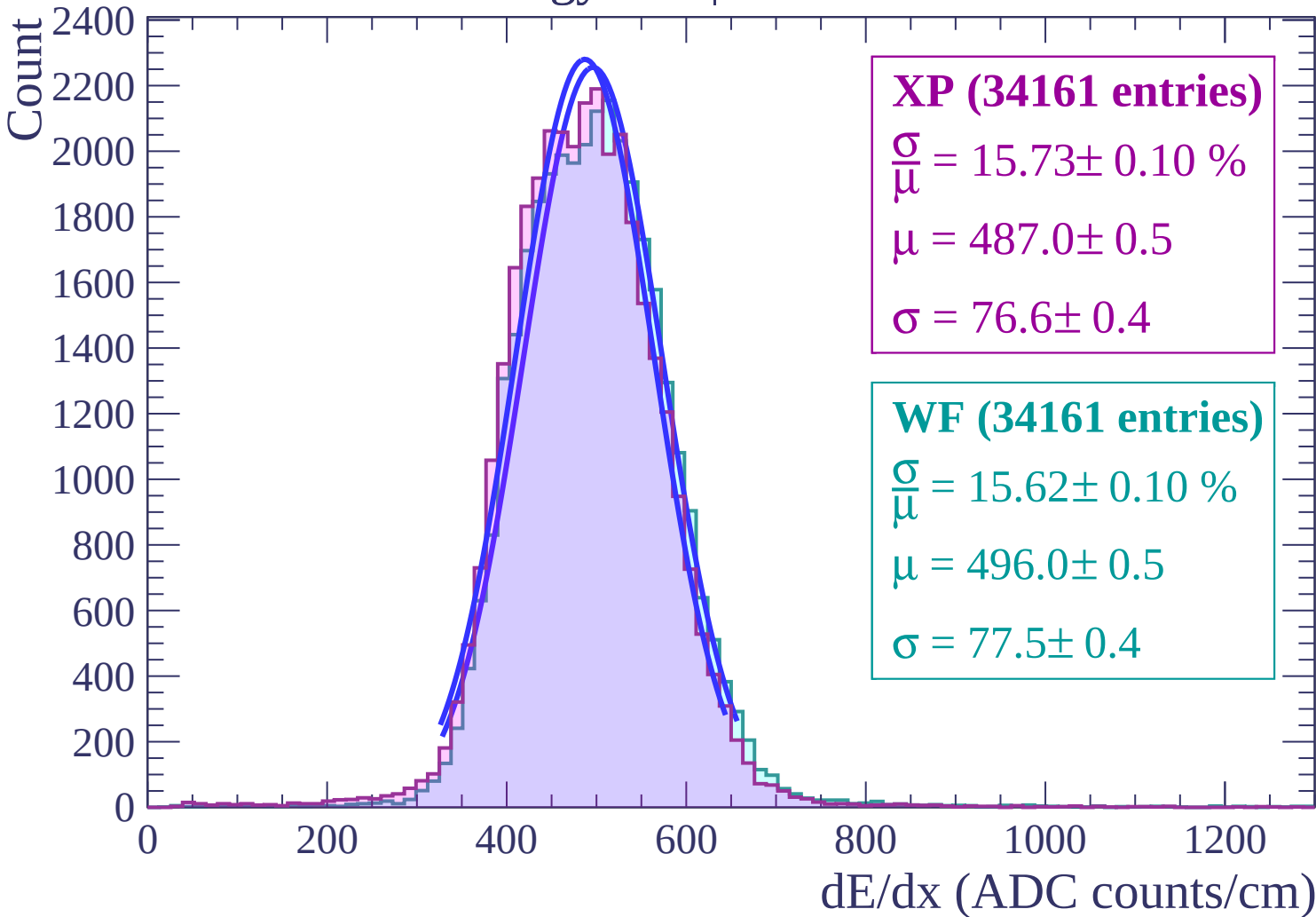
Energy loss | $-20 < \theta < -18$



Energy loss | $-18 < \theta < -16$

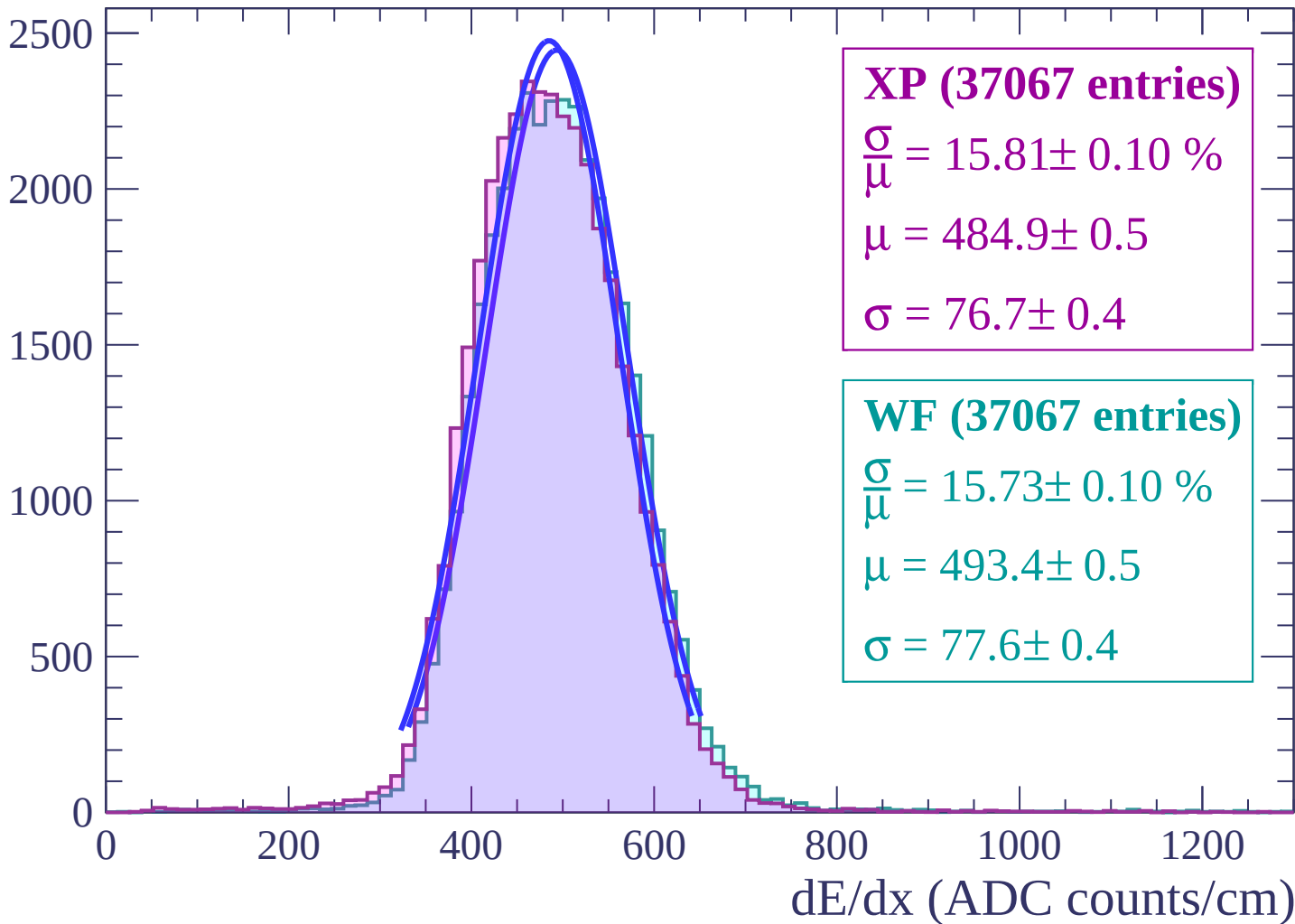


Energy loss | $-16 < \theta < -14$



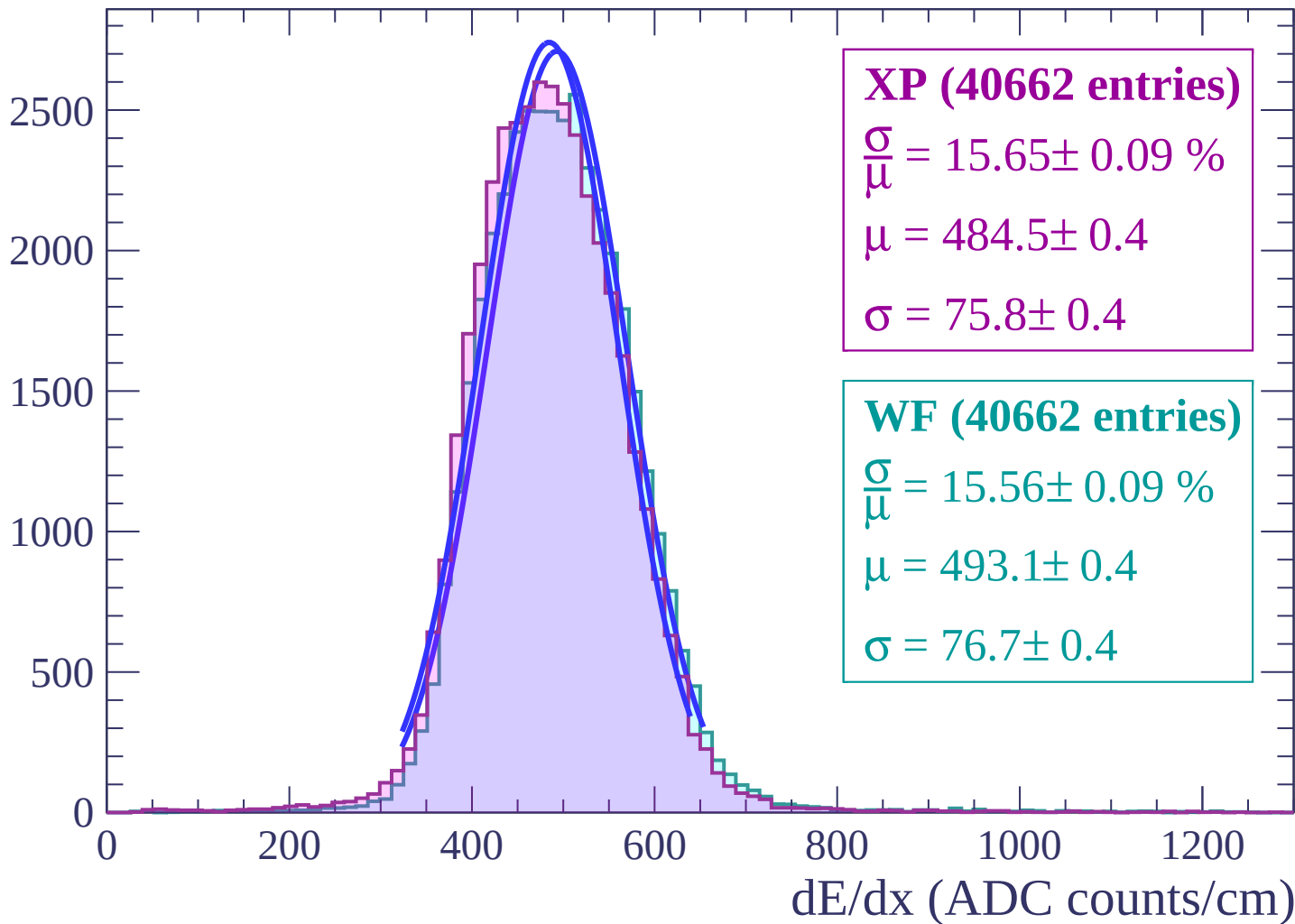
Energy loss | $-14 < \theta < -12$

Count

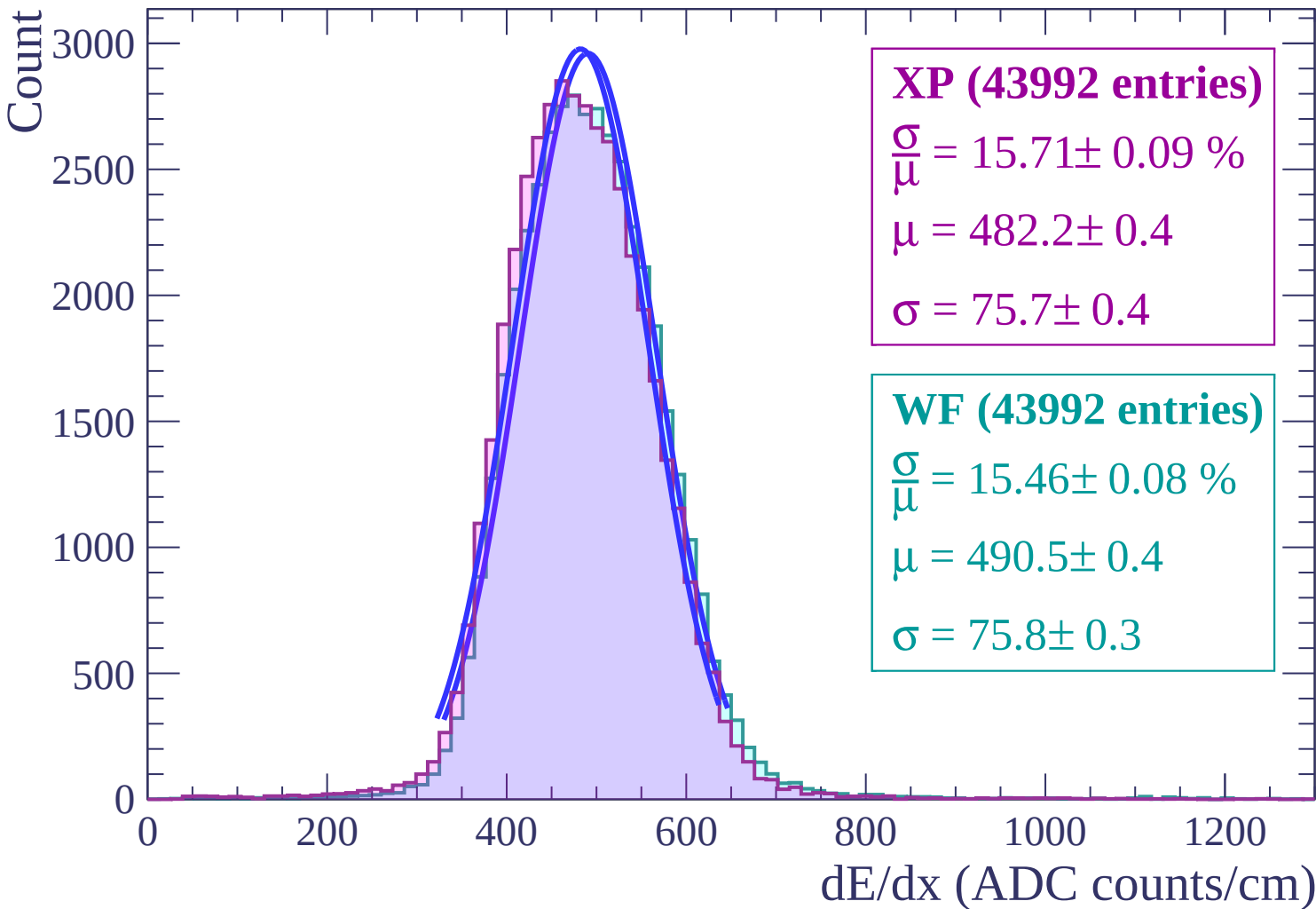


Energy loss | $-12 < \theta < -10$

Count

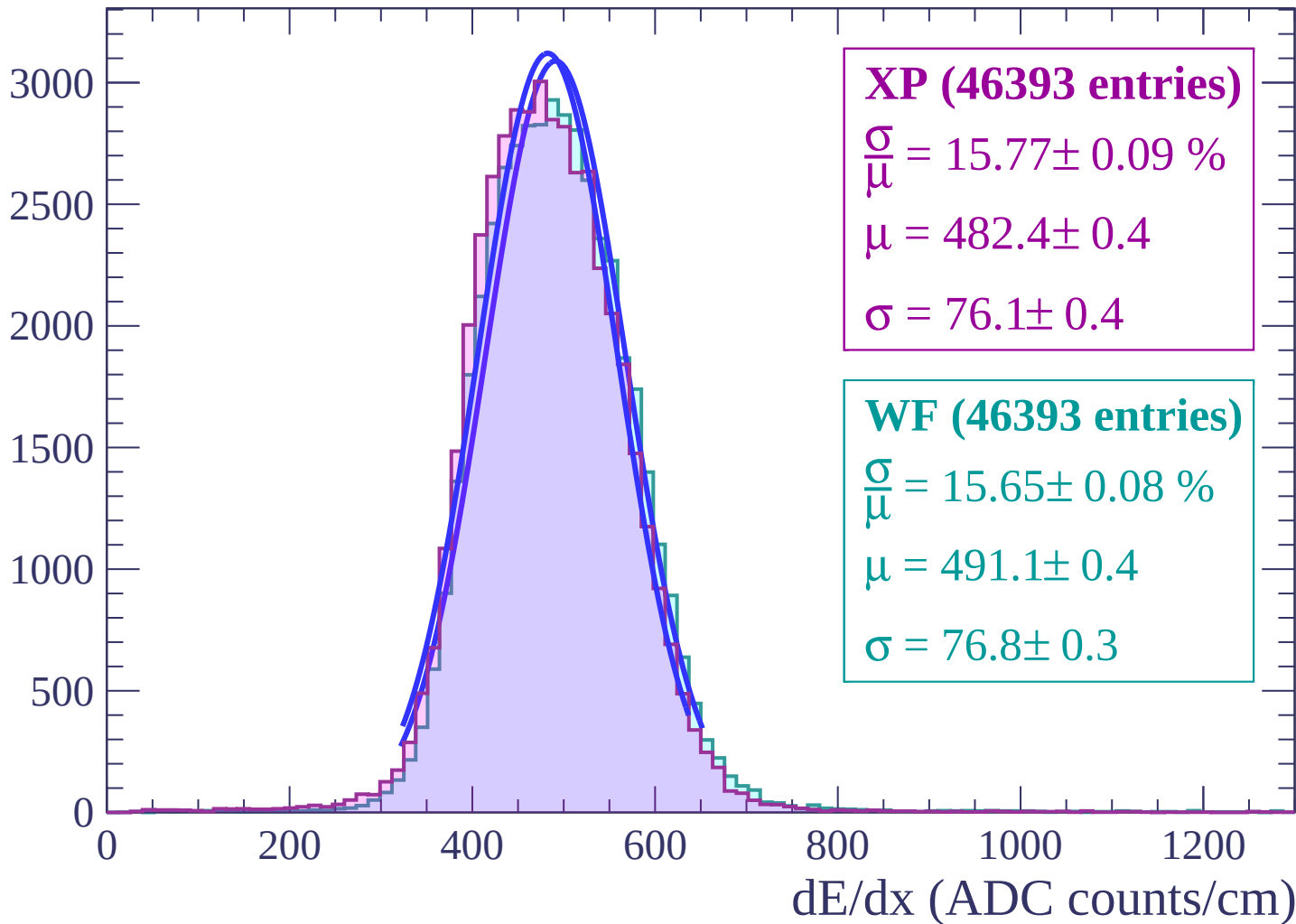


Energy loss | $-10 < \theta < -8$

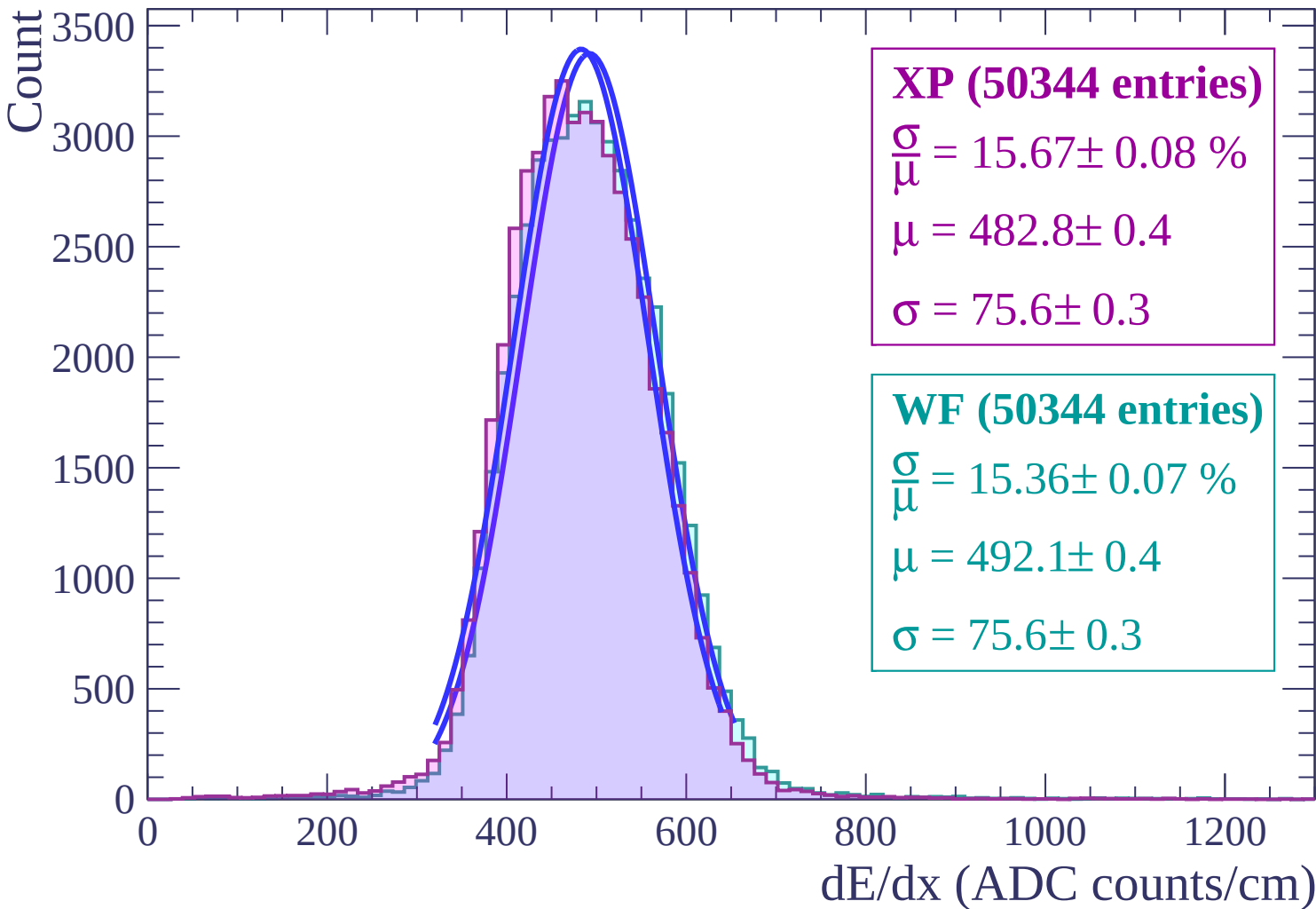


Energy loss | $-8 < \theta < -6$

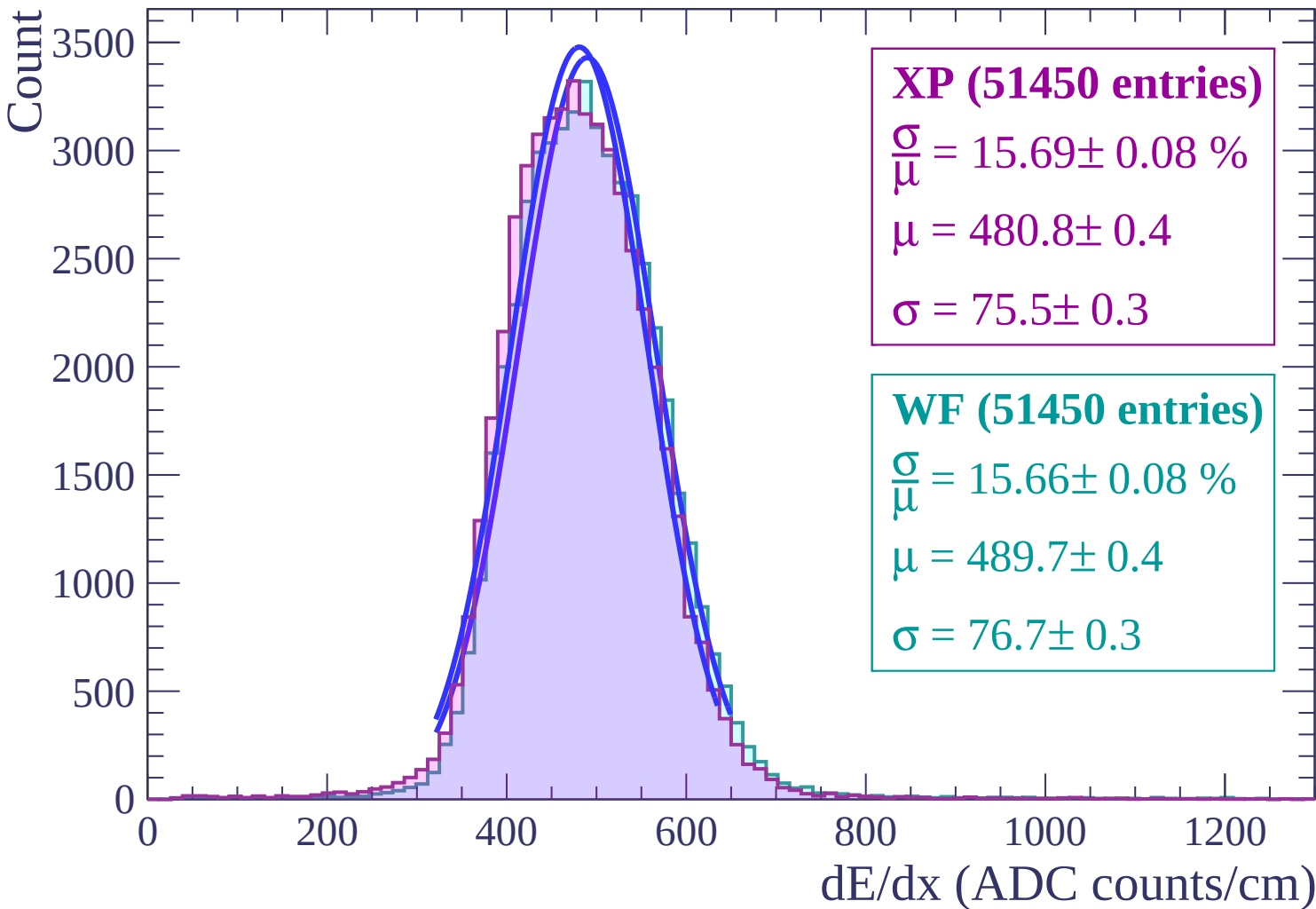
Count



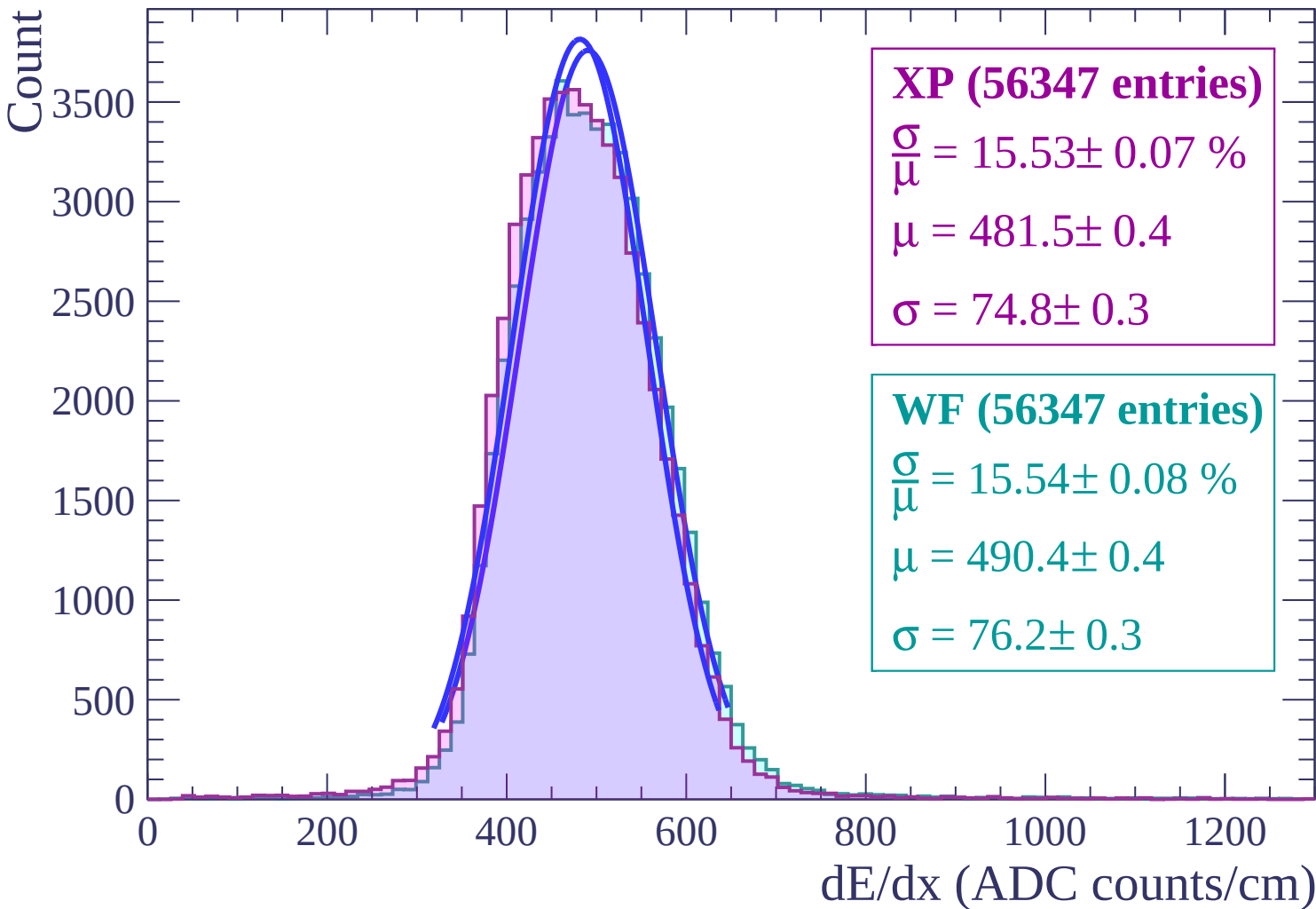
Energy loss | $-6 < \theta < -4$



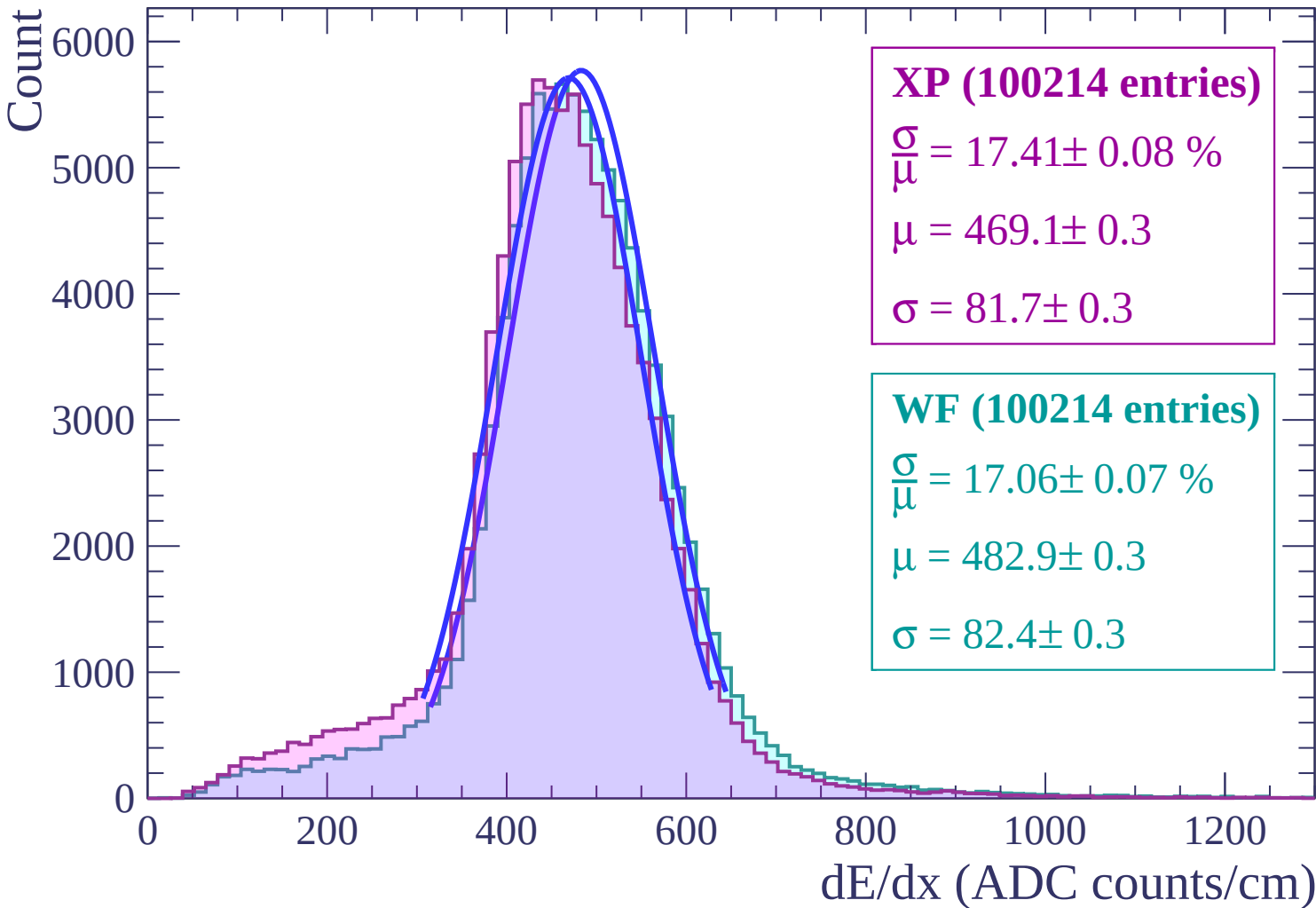
Energy loss | $-4 < \theta < -2$



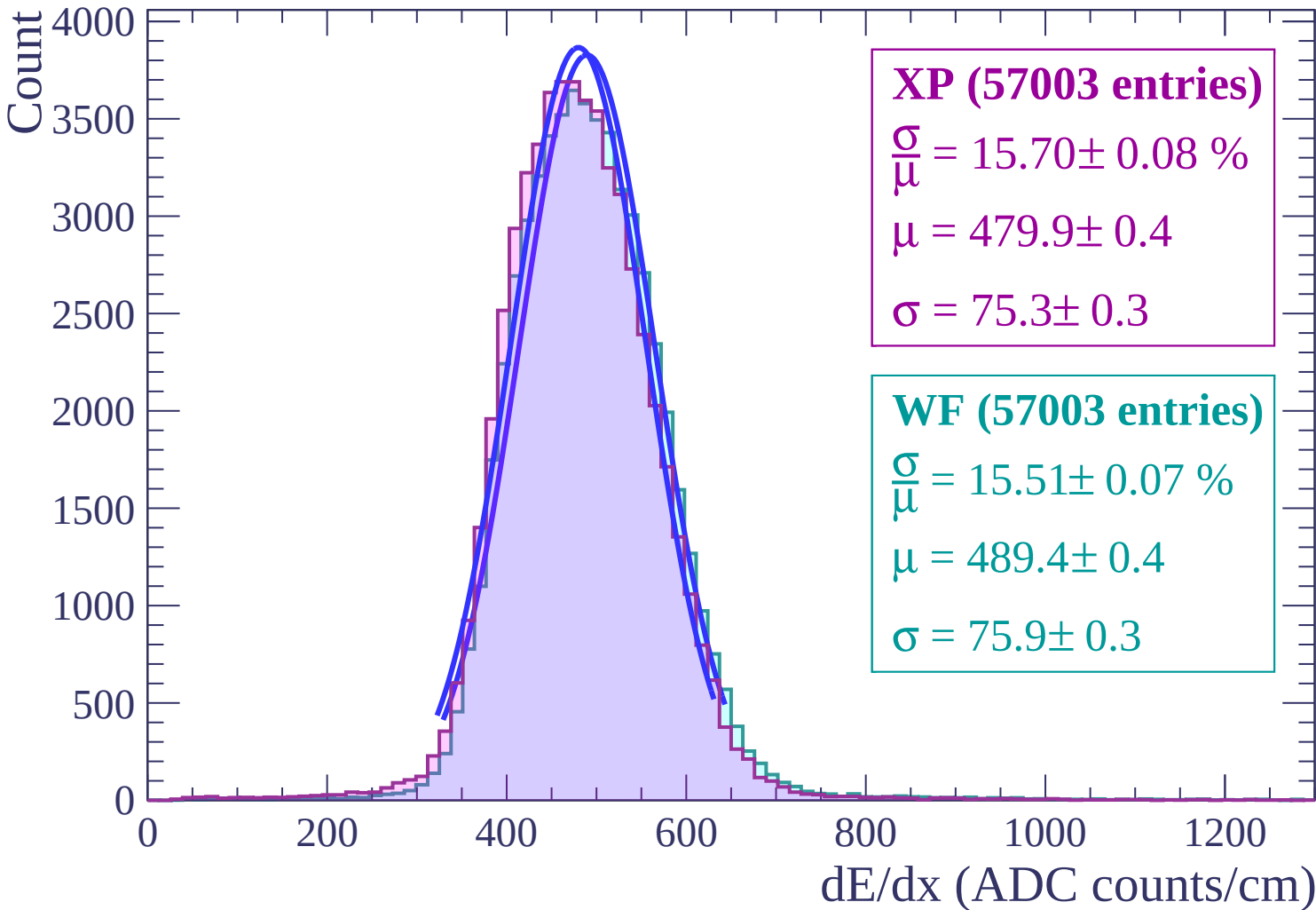
Energy loss $-2 < \theta < 0$



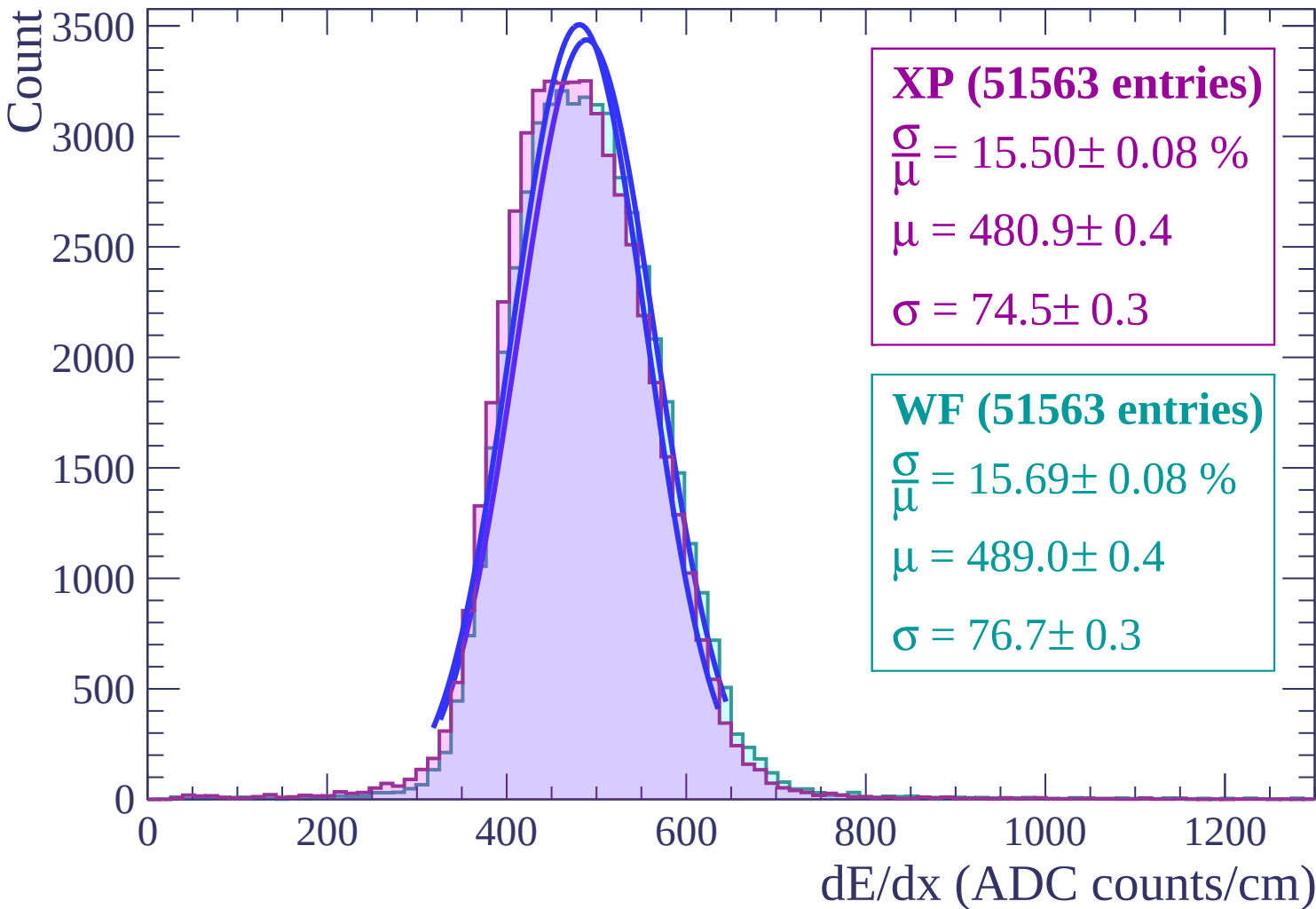
Energy loss | $0 < \theta < 2$



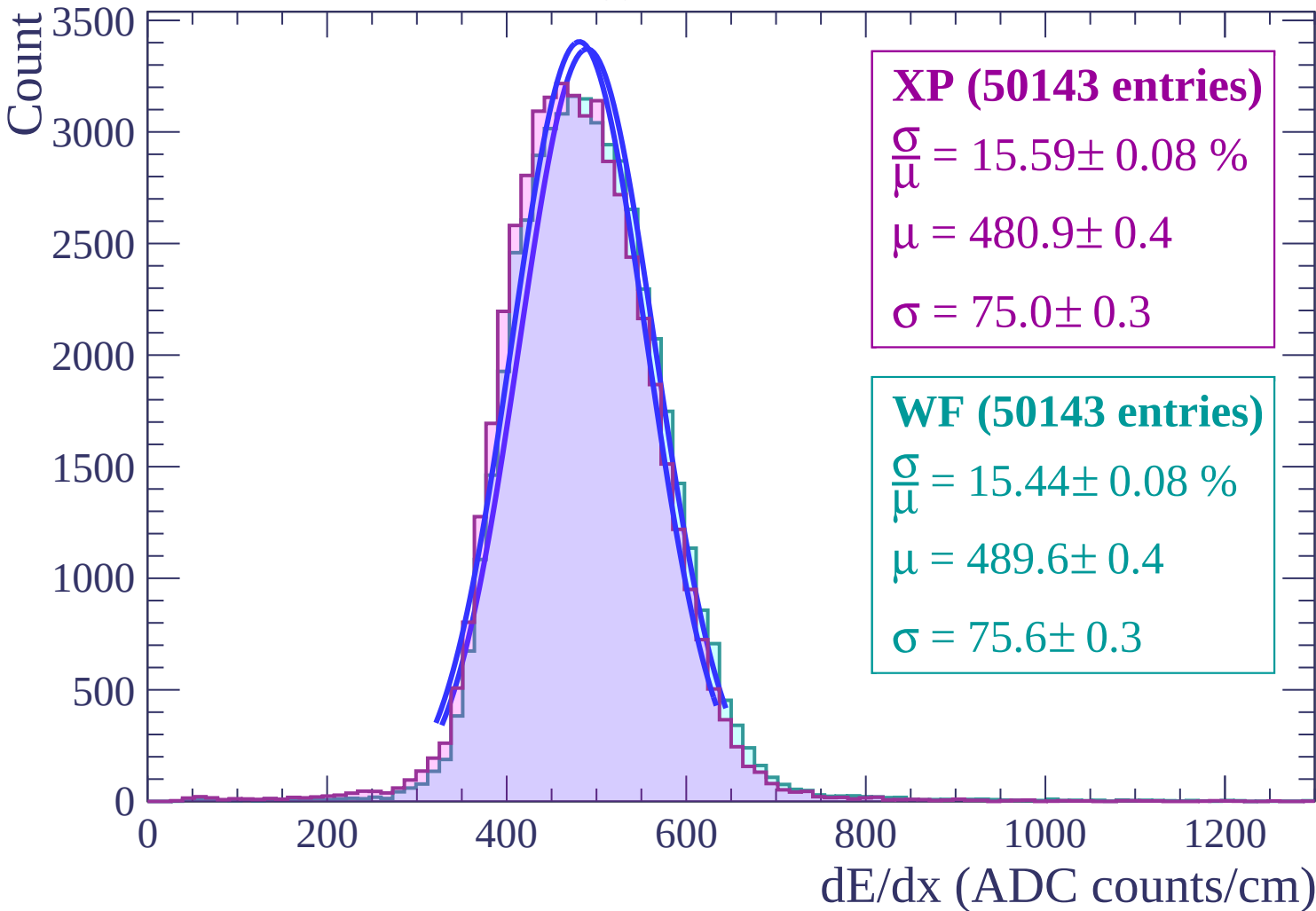
Energy loss | $2 < \theta < 4$



Energy loss | $4 < \theta < 6$

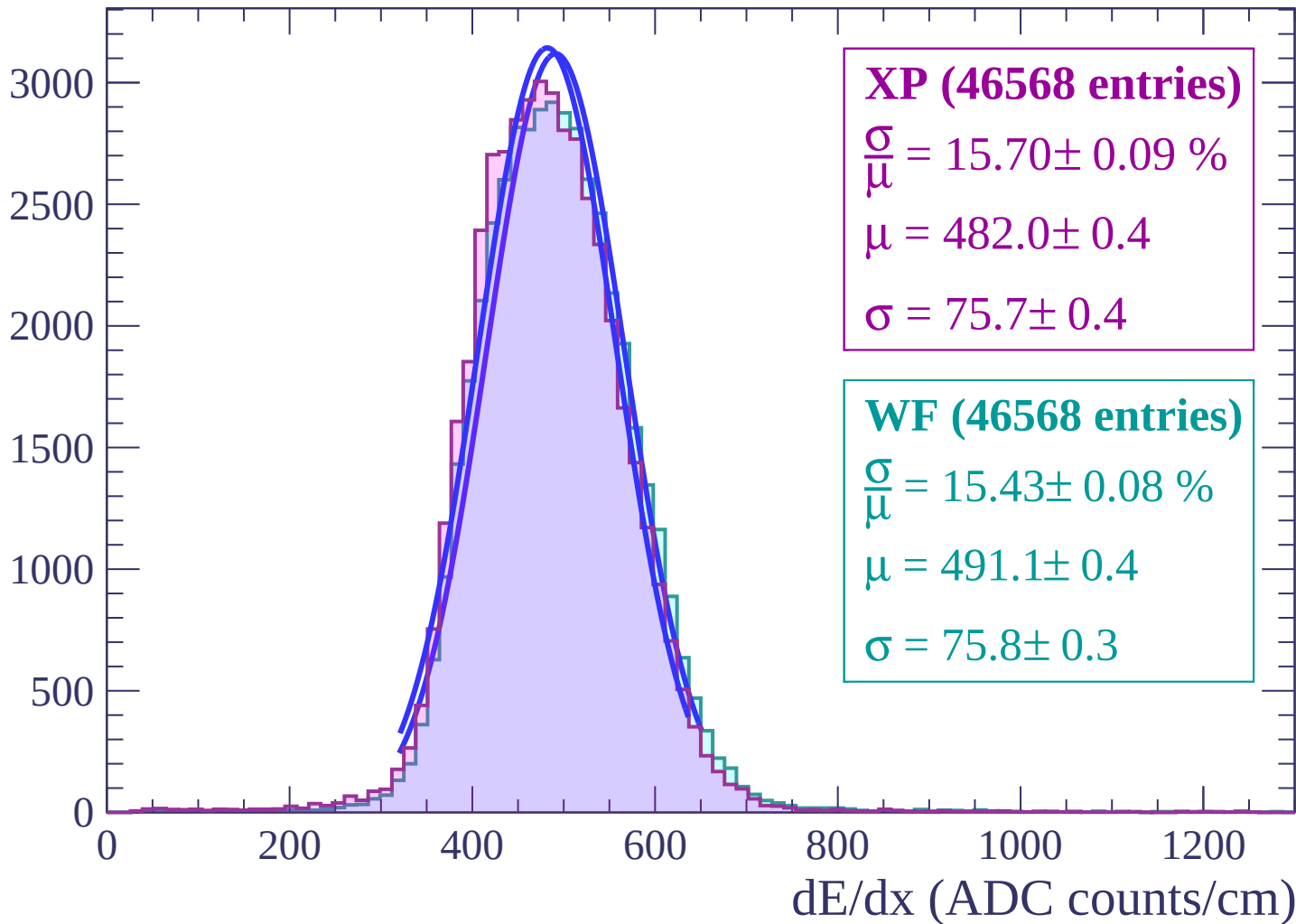


Energy loss | $6 < \theta < 8$



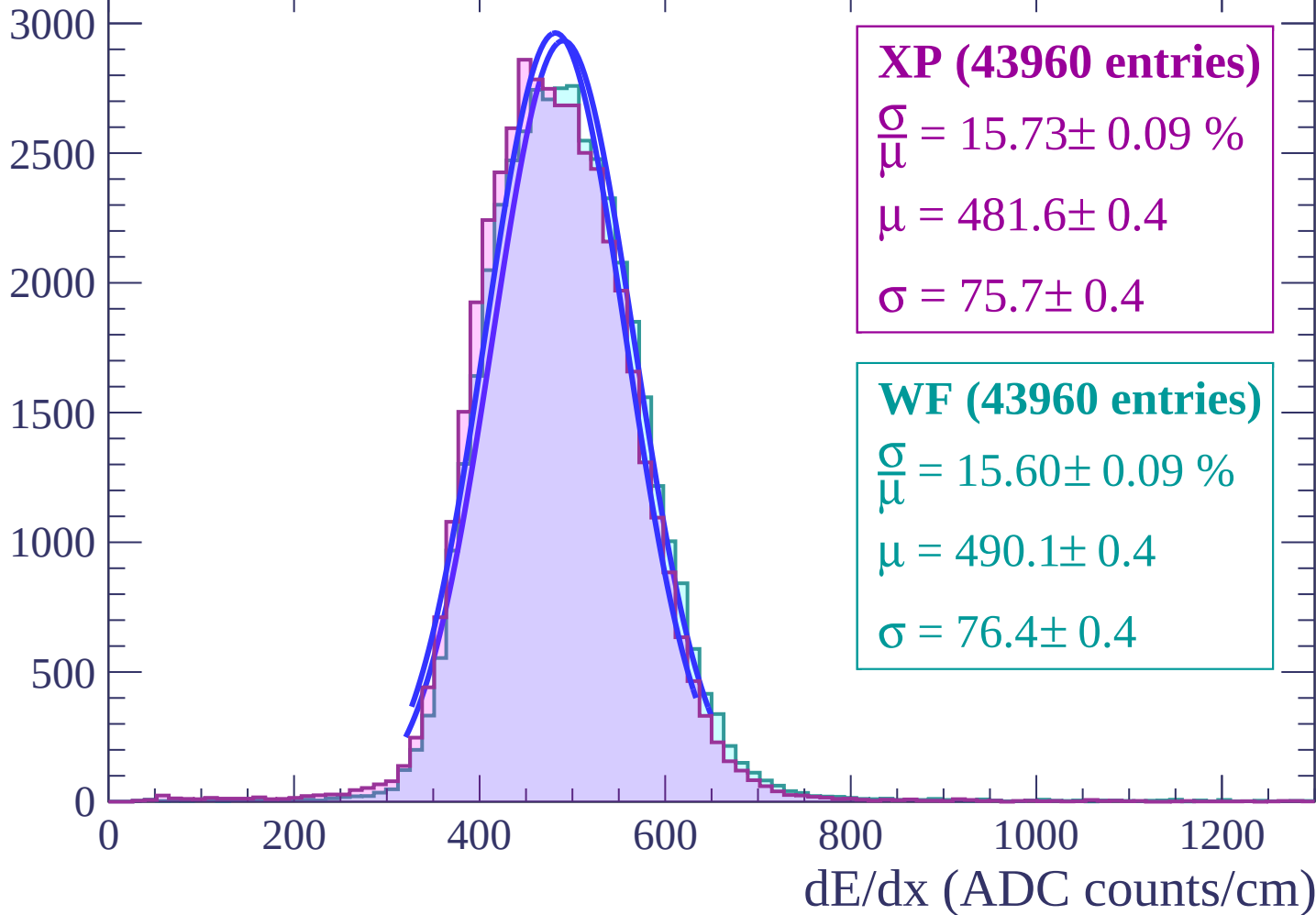
Energy loss | $8 < \theta < 10$

Count

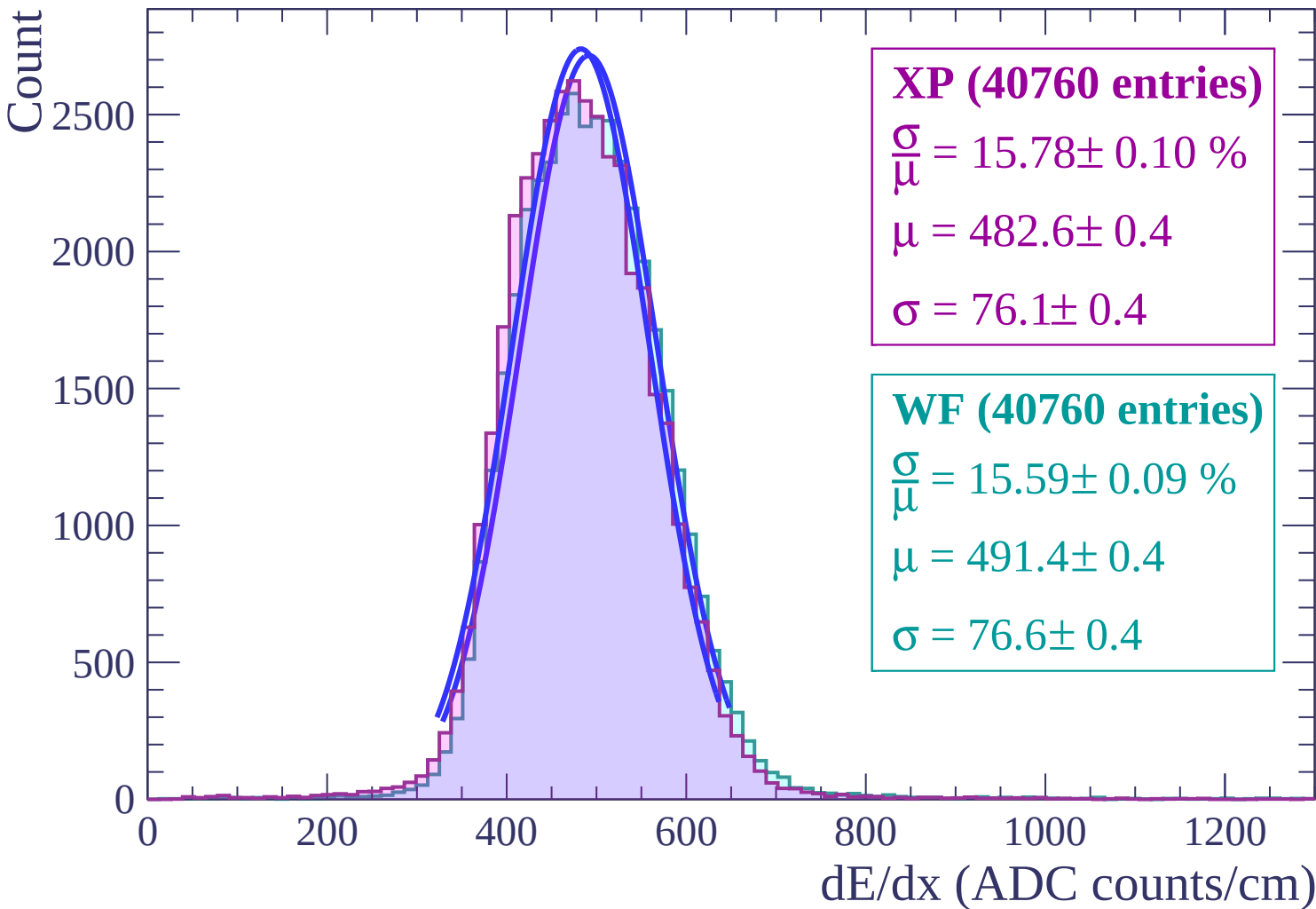


Energy loss | $10 < \theta < 12$

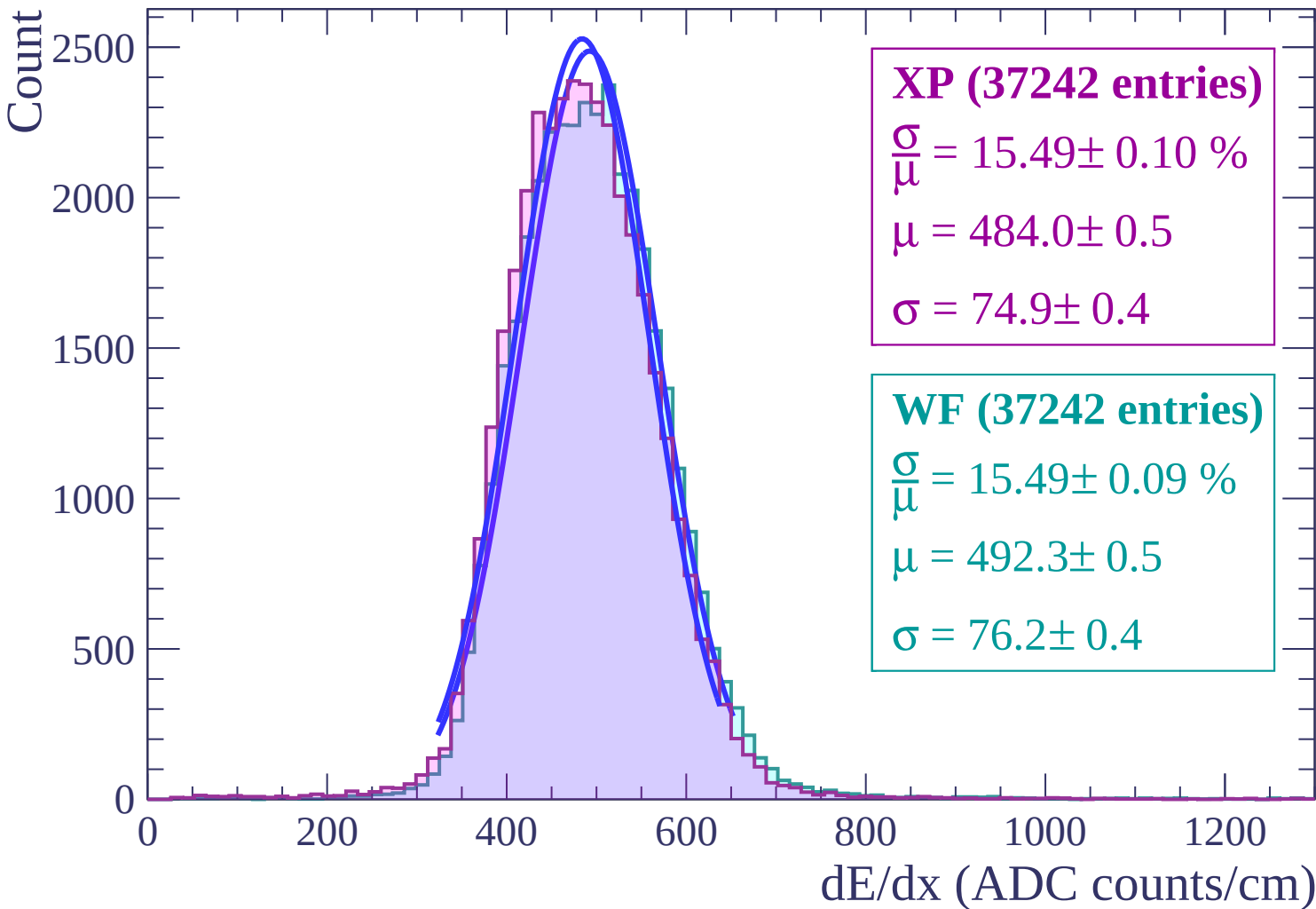
Count



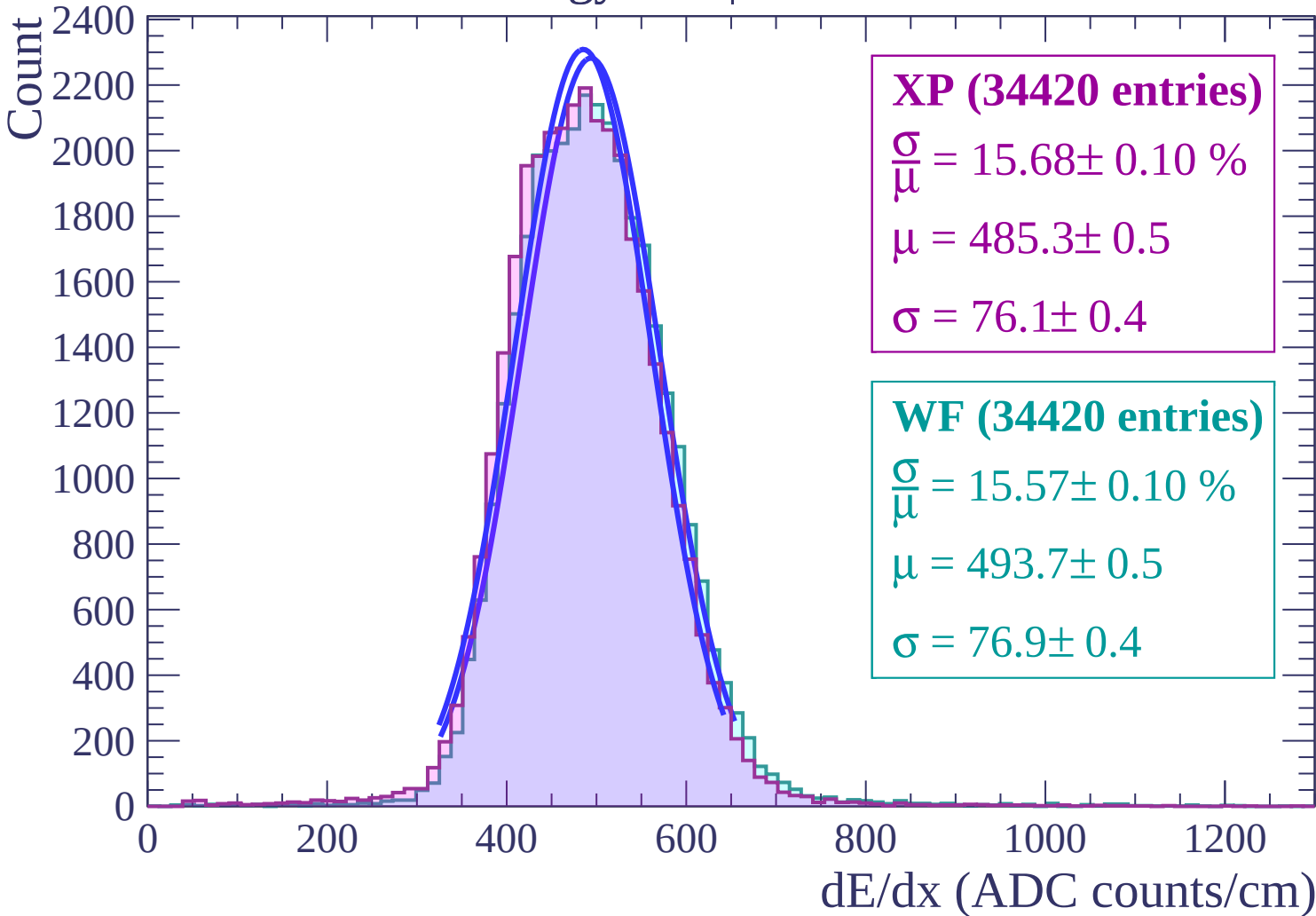
Energy loss | $12 < \theta < 14$



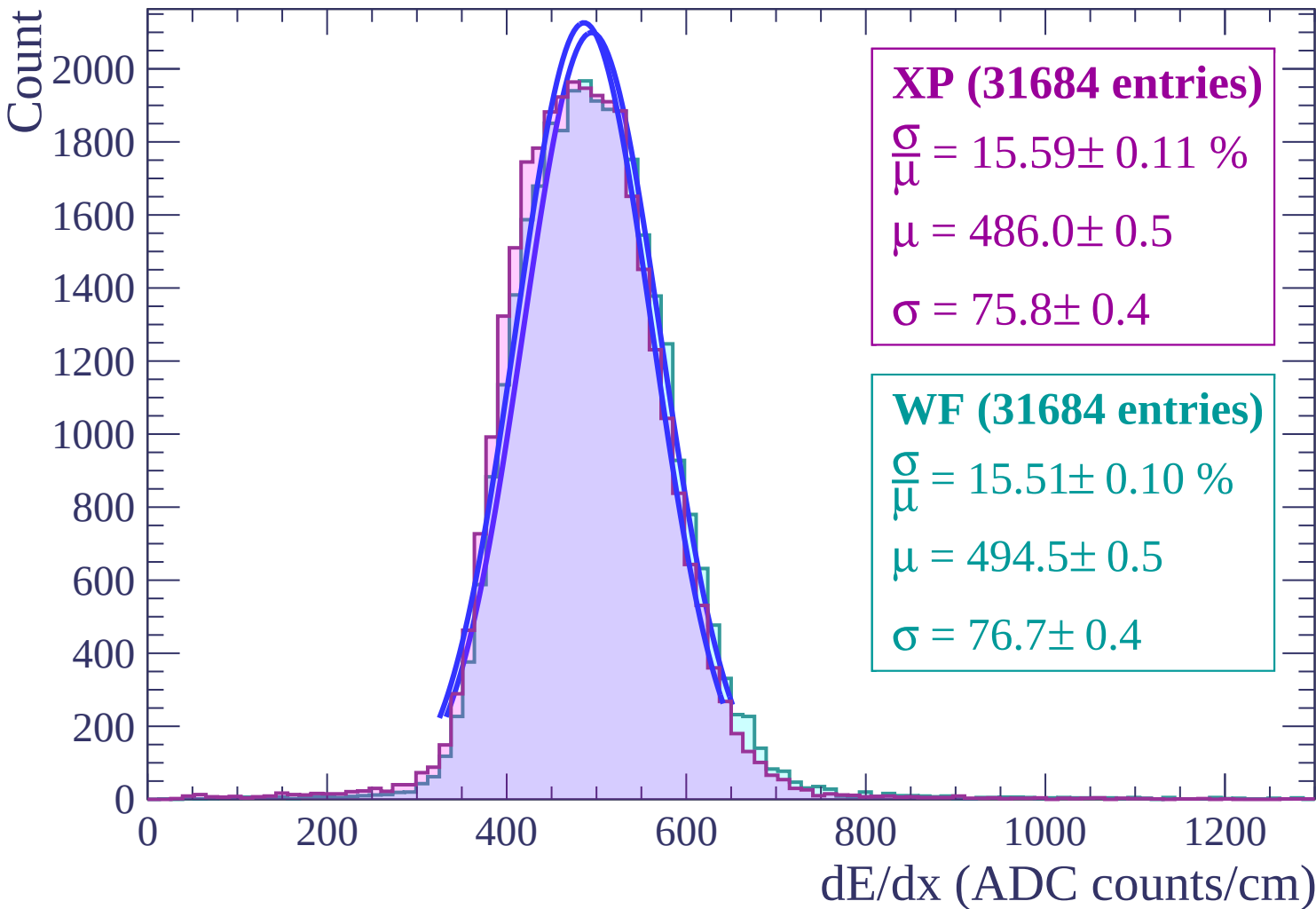
Energy loss | $14 < \theta < 16$



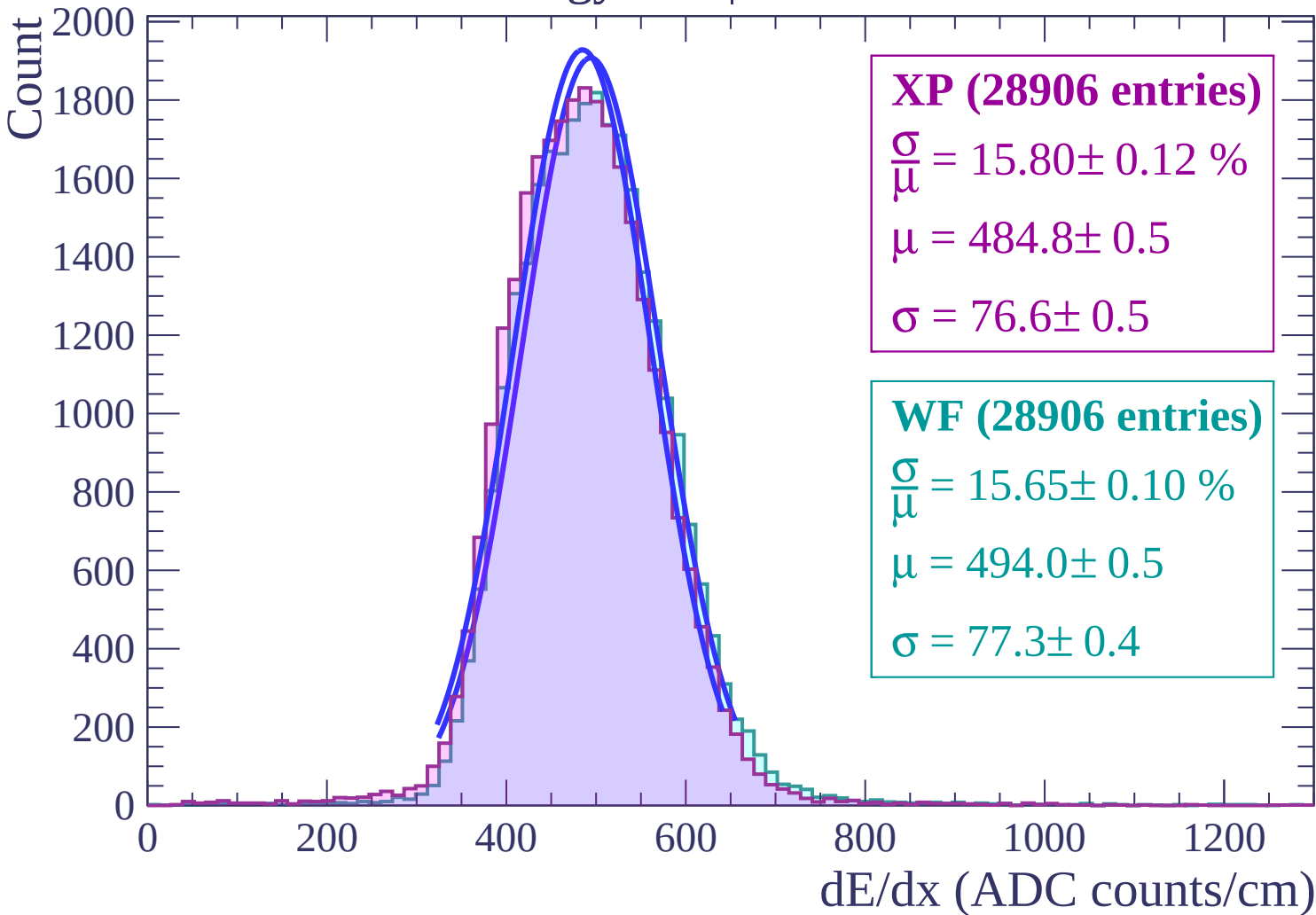
Energy loss | $16 < \theta < 18$



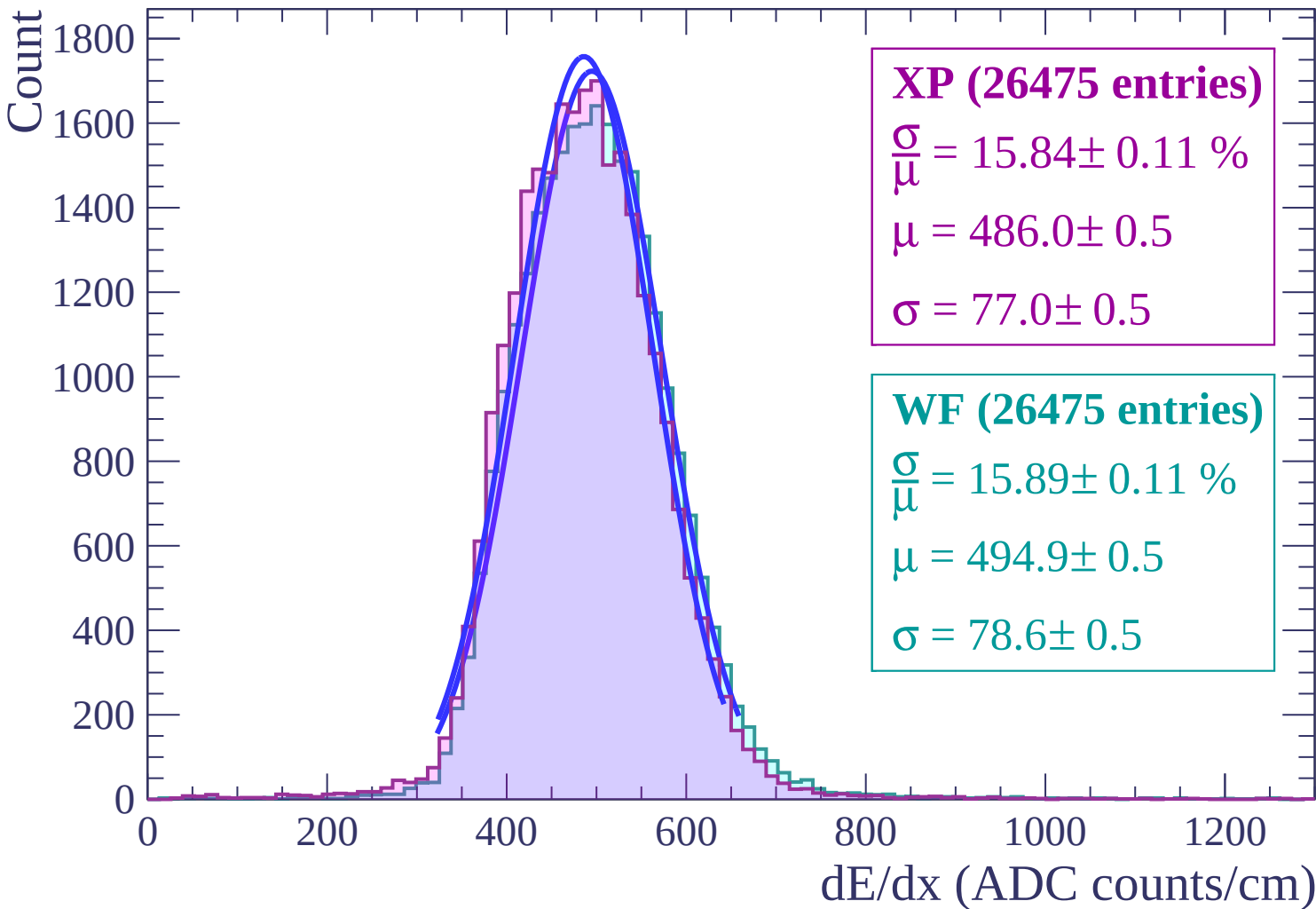
Energy loss | $18 < \theta < 20$



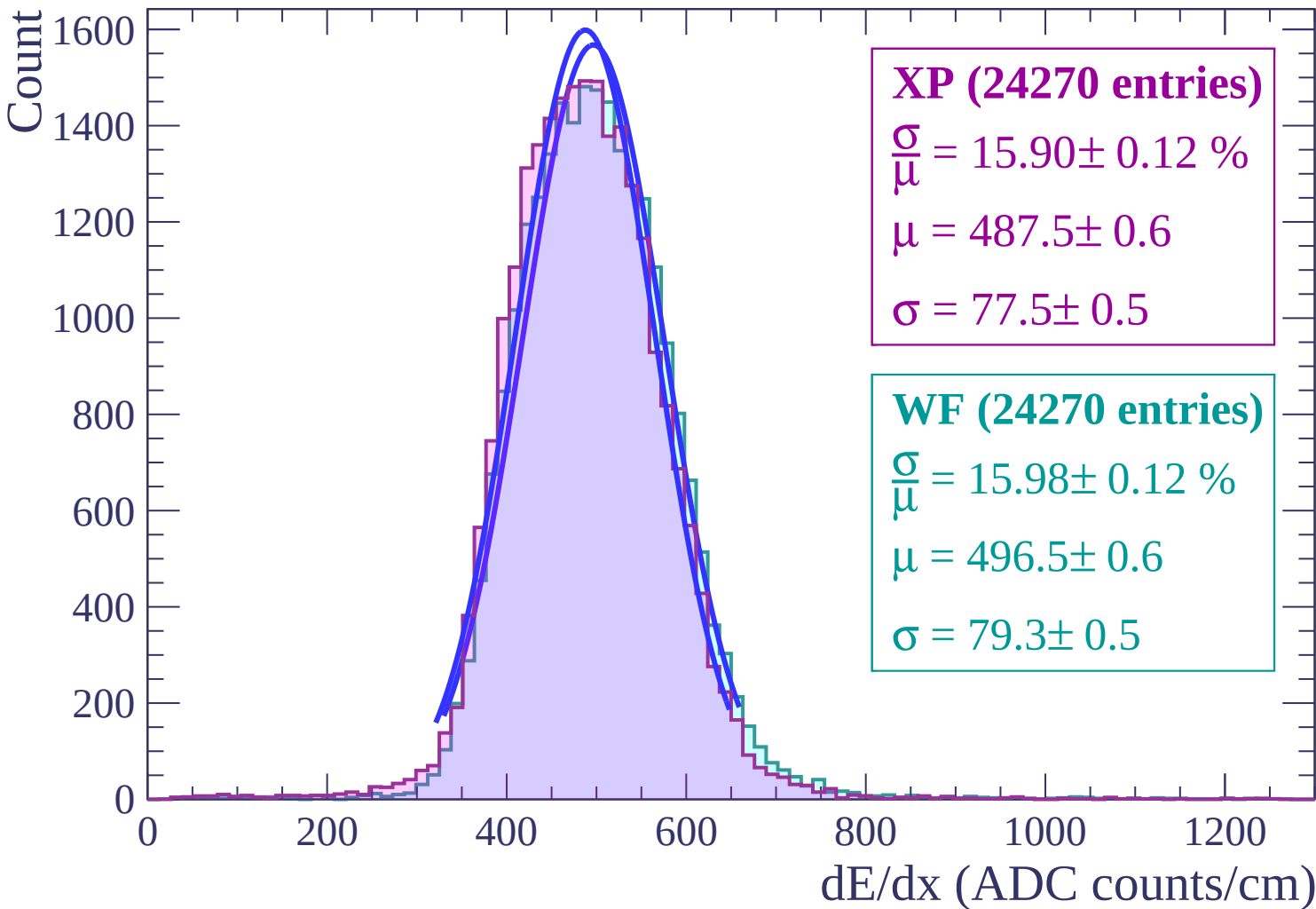
Energy loss | $20 < \theta < 22$



Energy loss | $22 < \theta < 24$

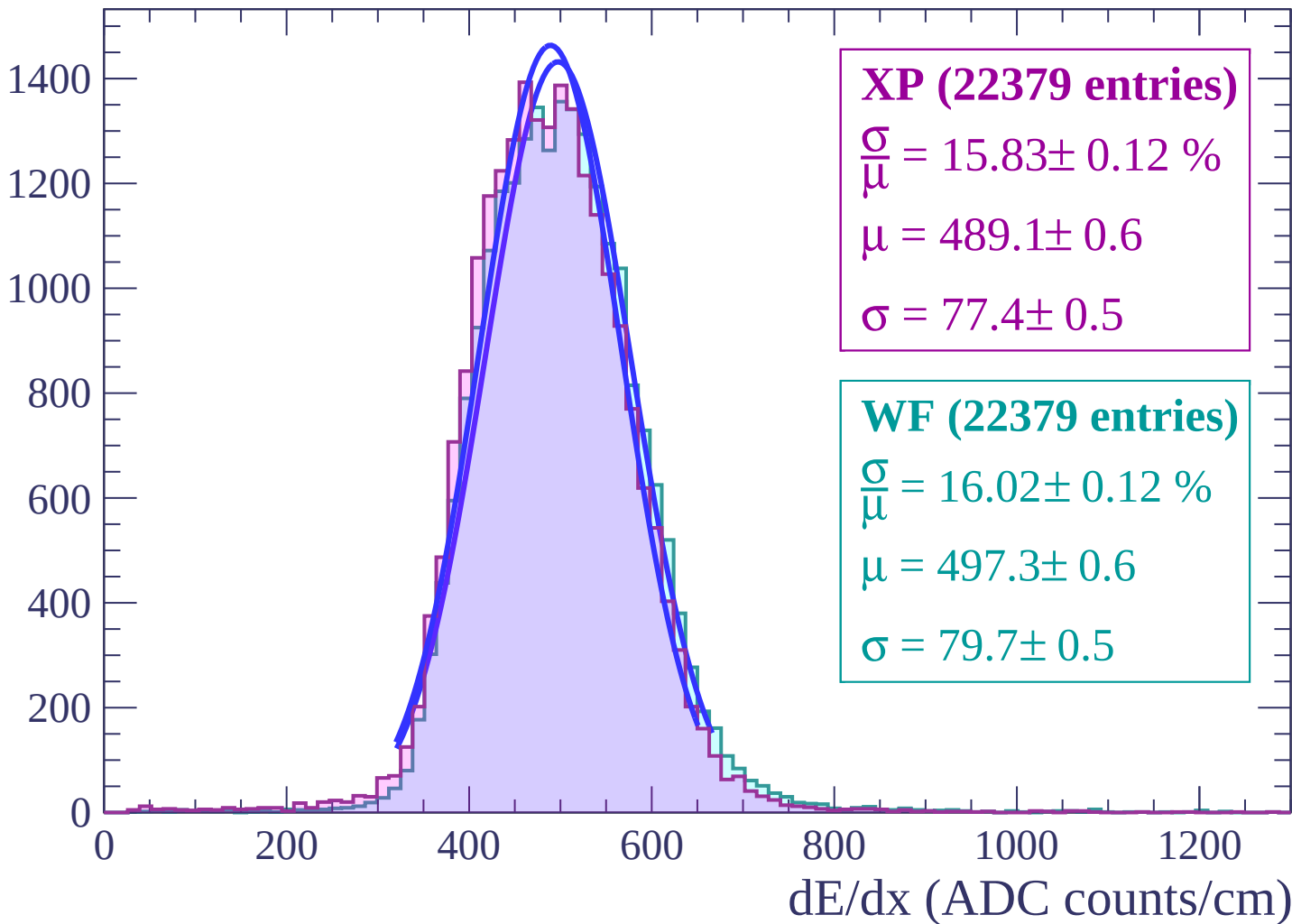


Energy loss | $24 < \theta < 26$



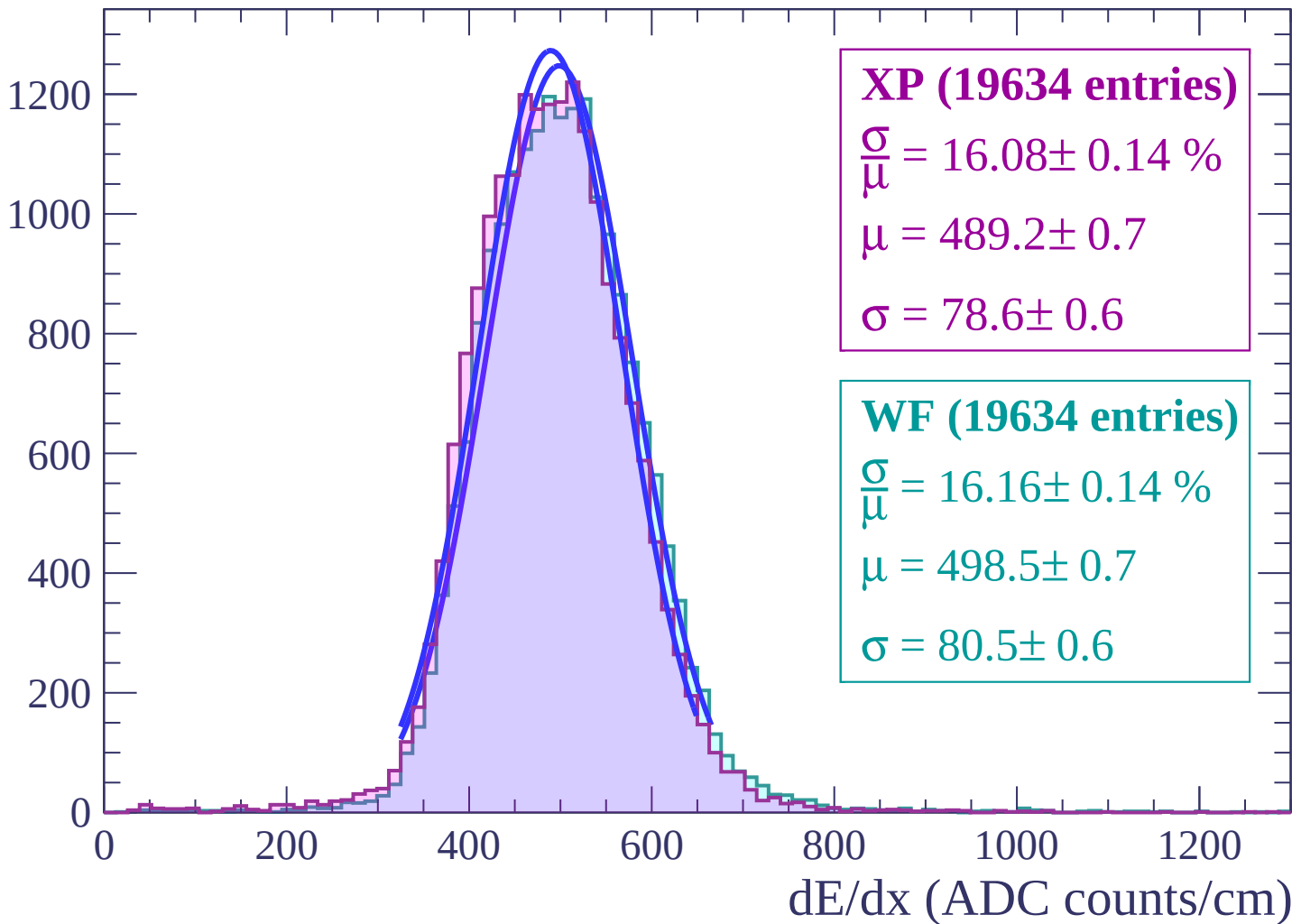
Energy loss | $26 < \theta < 28$

Count



Energy loss | $28 < \theta < 30$

Count



Energy loss | $30 < \theta < 32$

Count

1000

800

600

400

200

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (15894 entries)

$\frac{\sigma}{\mu} = 16.17 \pm 0.14 \%$

$\mu = 493.8 \pm 0.8$

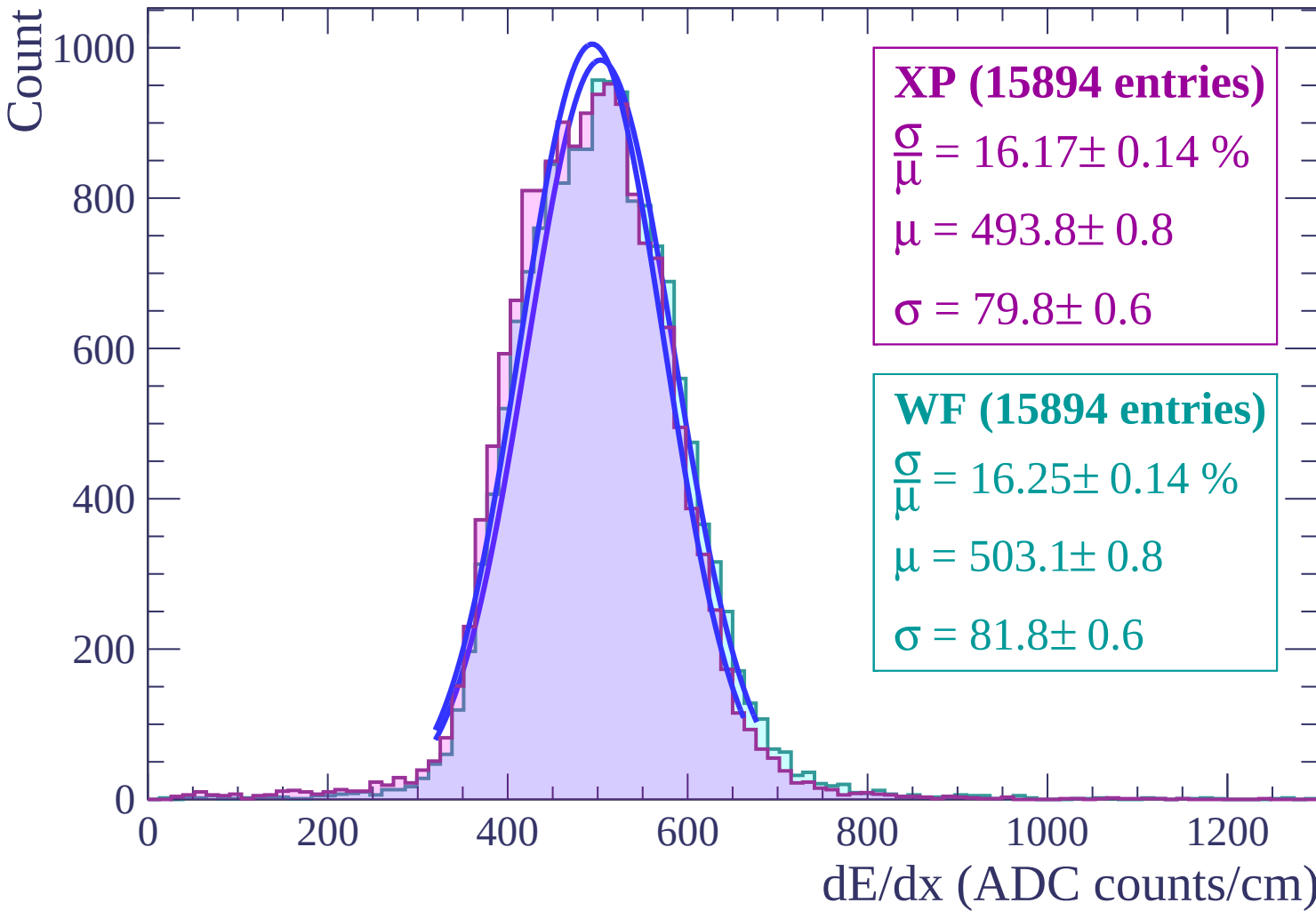
$\sigma = 79.8 \pm 0.6$

WF (15894 entries)

$\frac{\sigma}{\mu} = 16.25 \pm 0.14 \%$

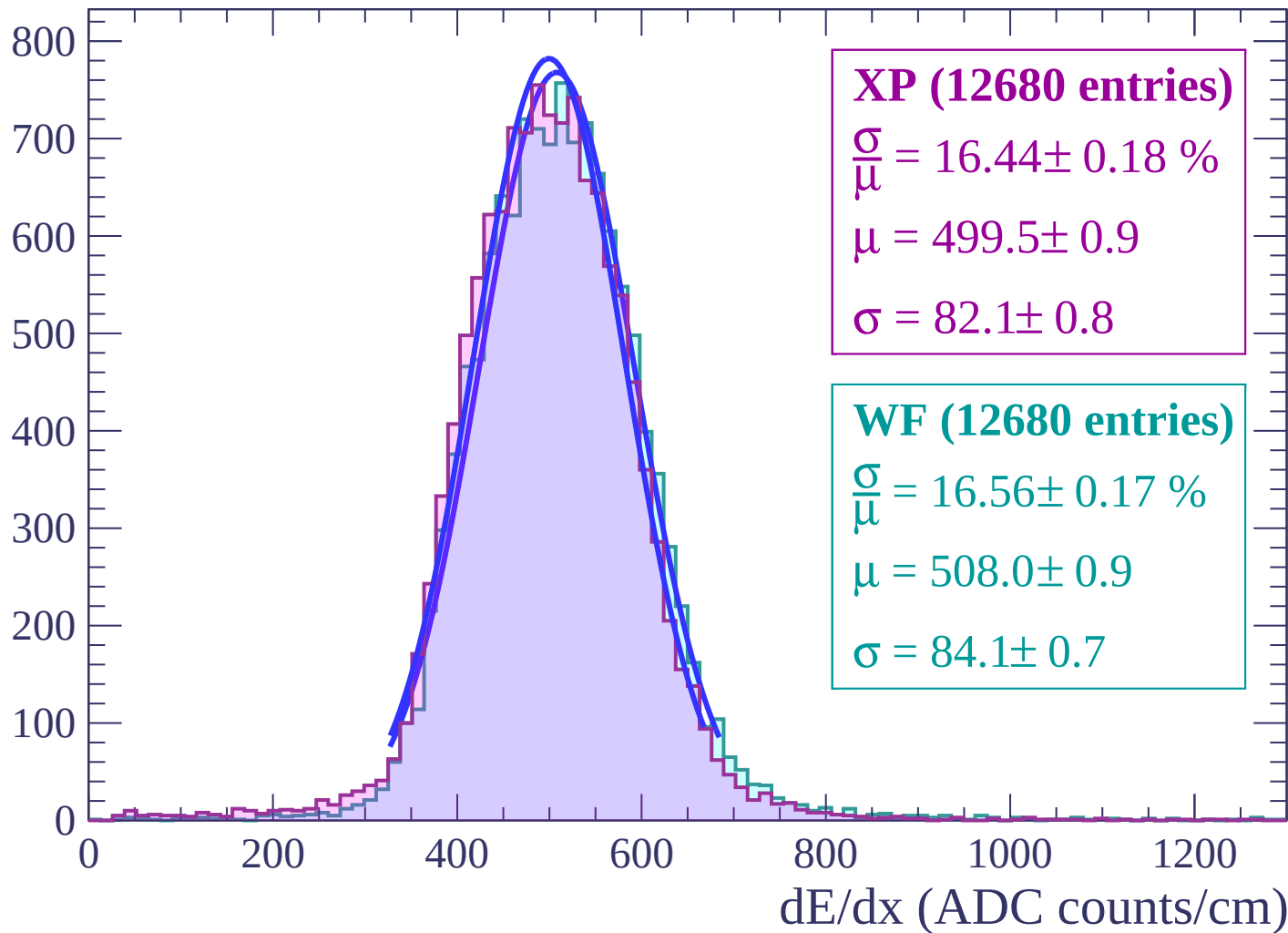
$\mu = 503.1 \pm 0.8$

$\sigma = 81.8 \pm 0.6$



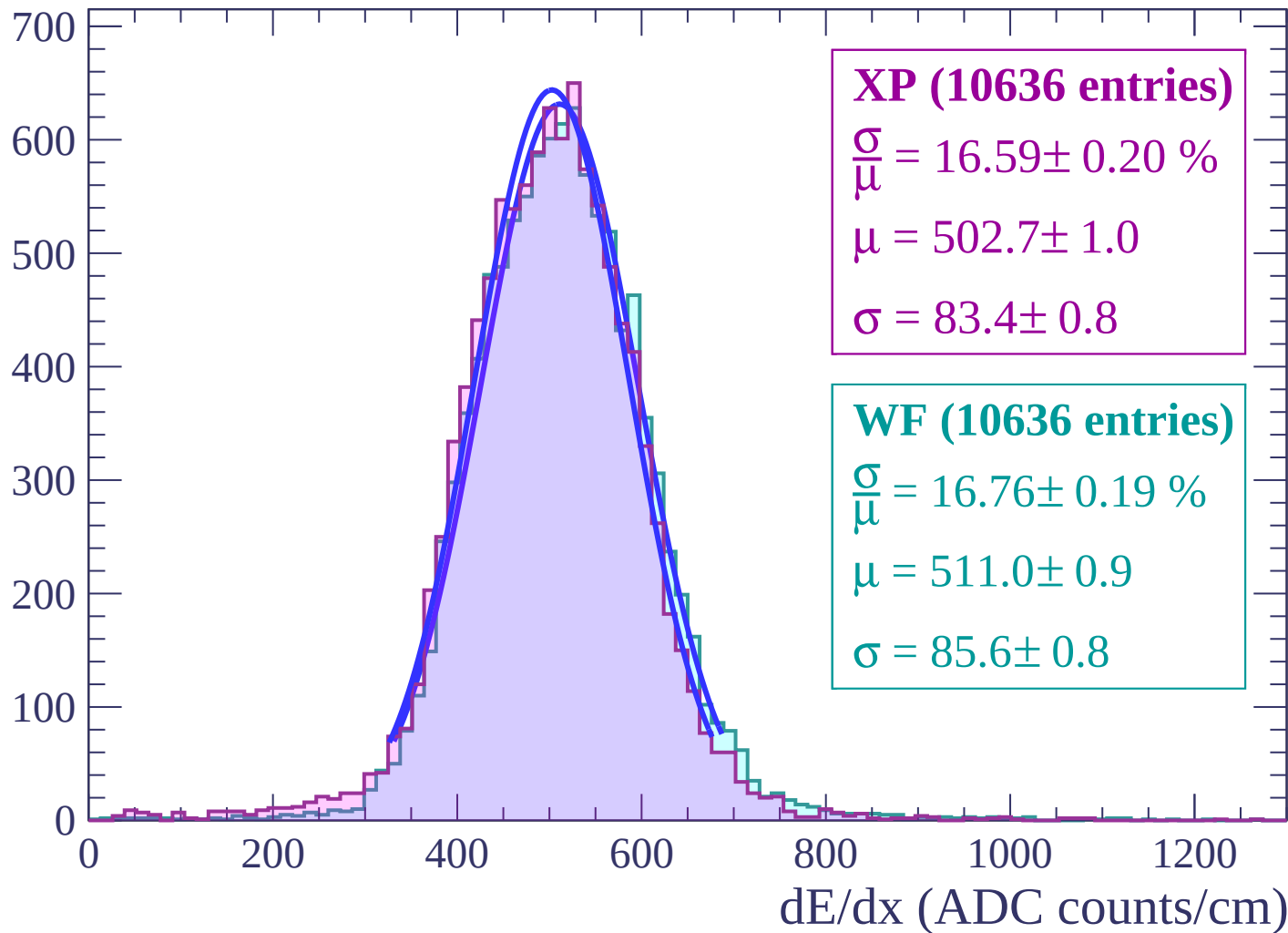
Energy loss | $32 < \theta < 34$

Count



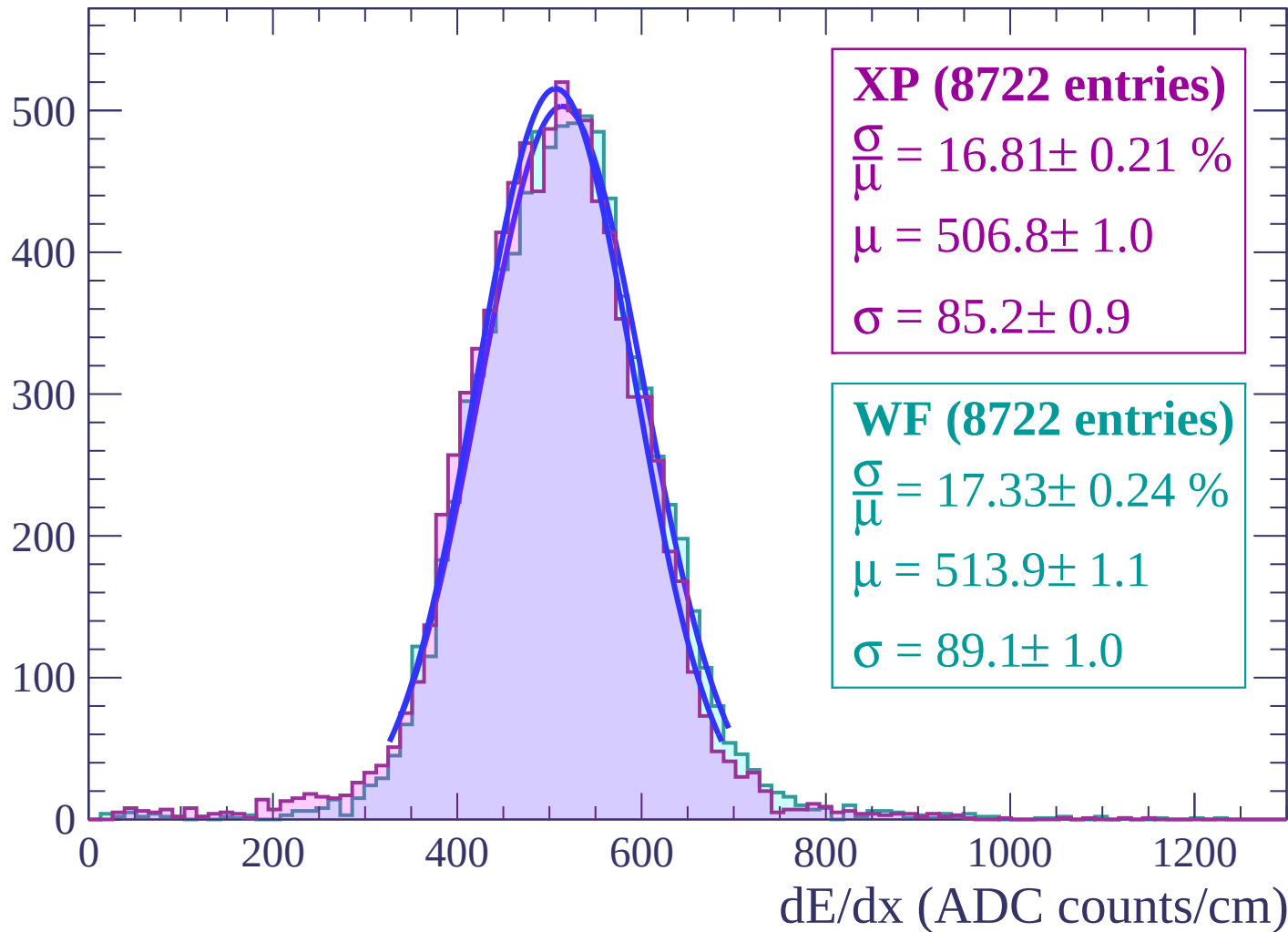
Energy loss | $34 < \theta < 36$

Count



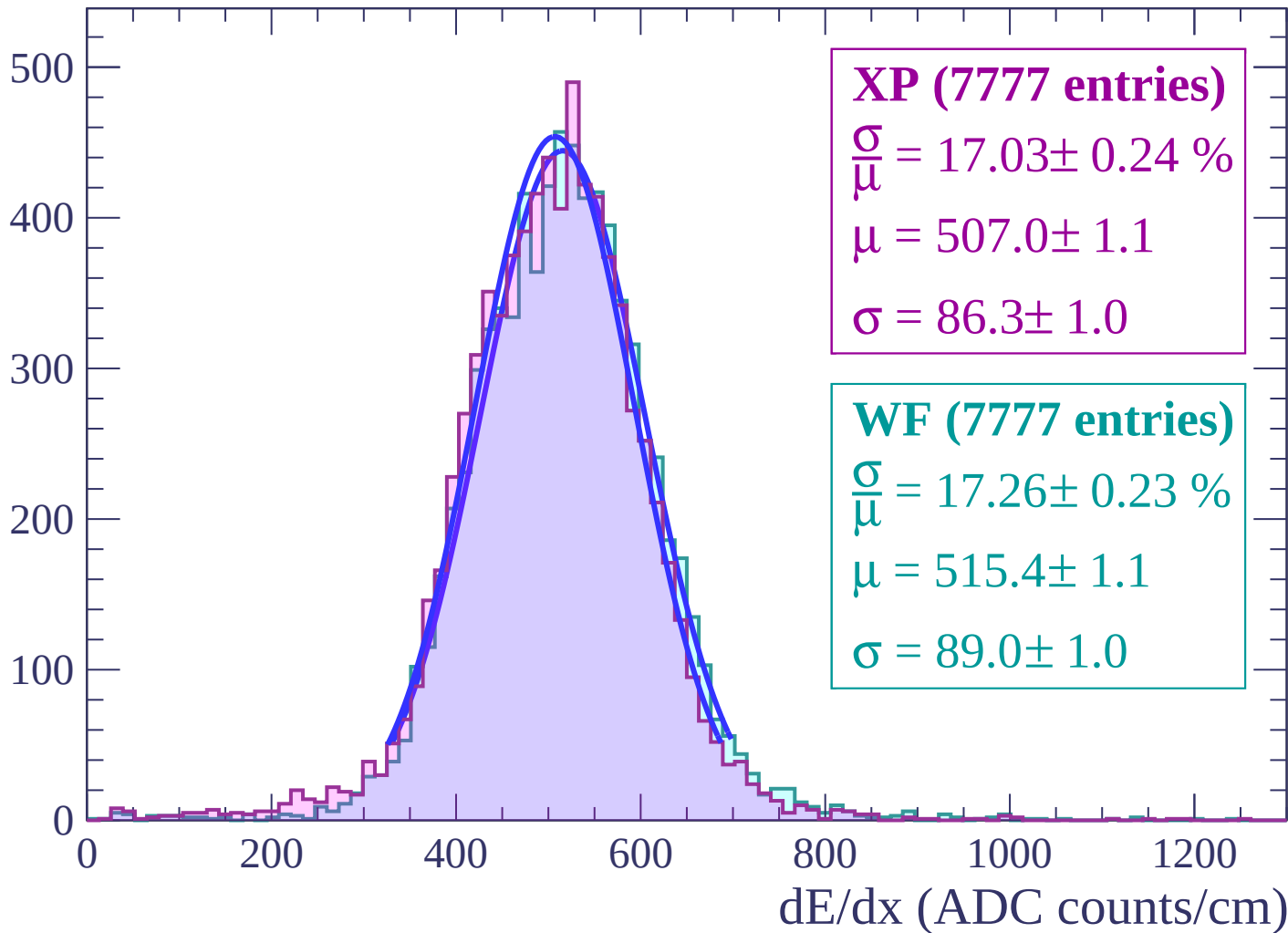
Energy loss | $36 < \theta < 38$

Count



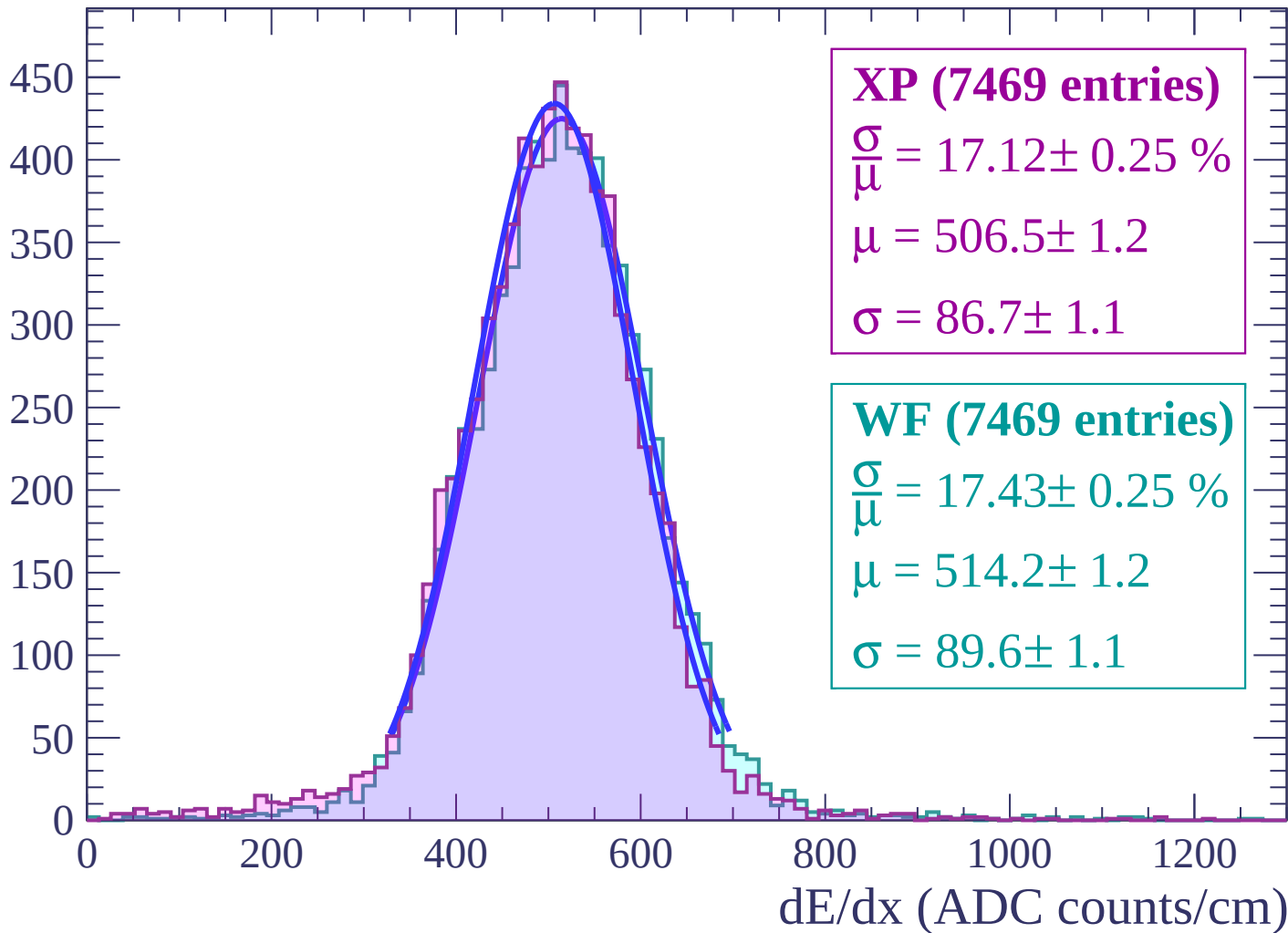
Energy loss | $38 < \theta < 40$

Count



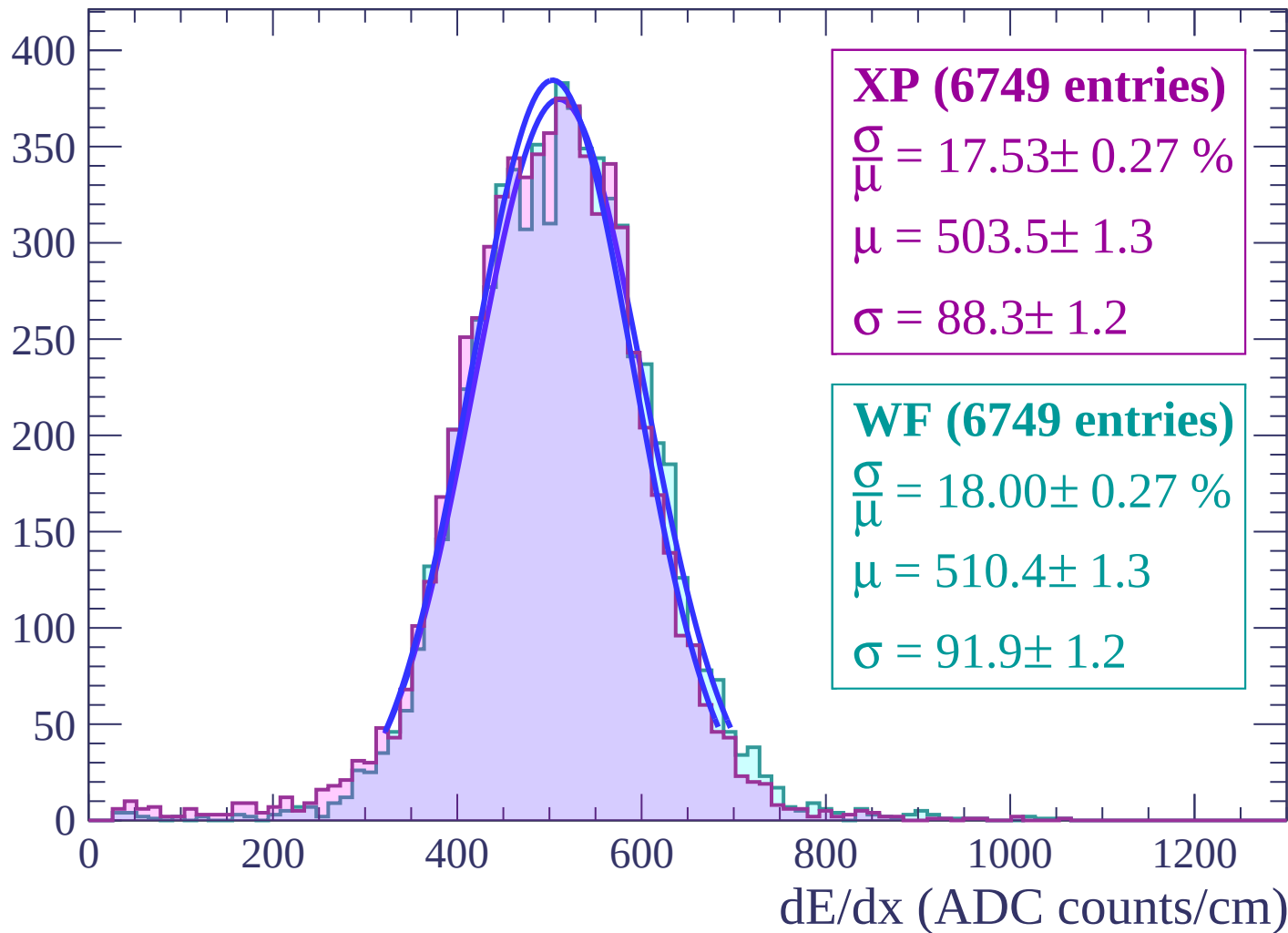
Energy loss | $40 < \theta < 42$

Count



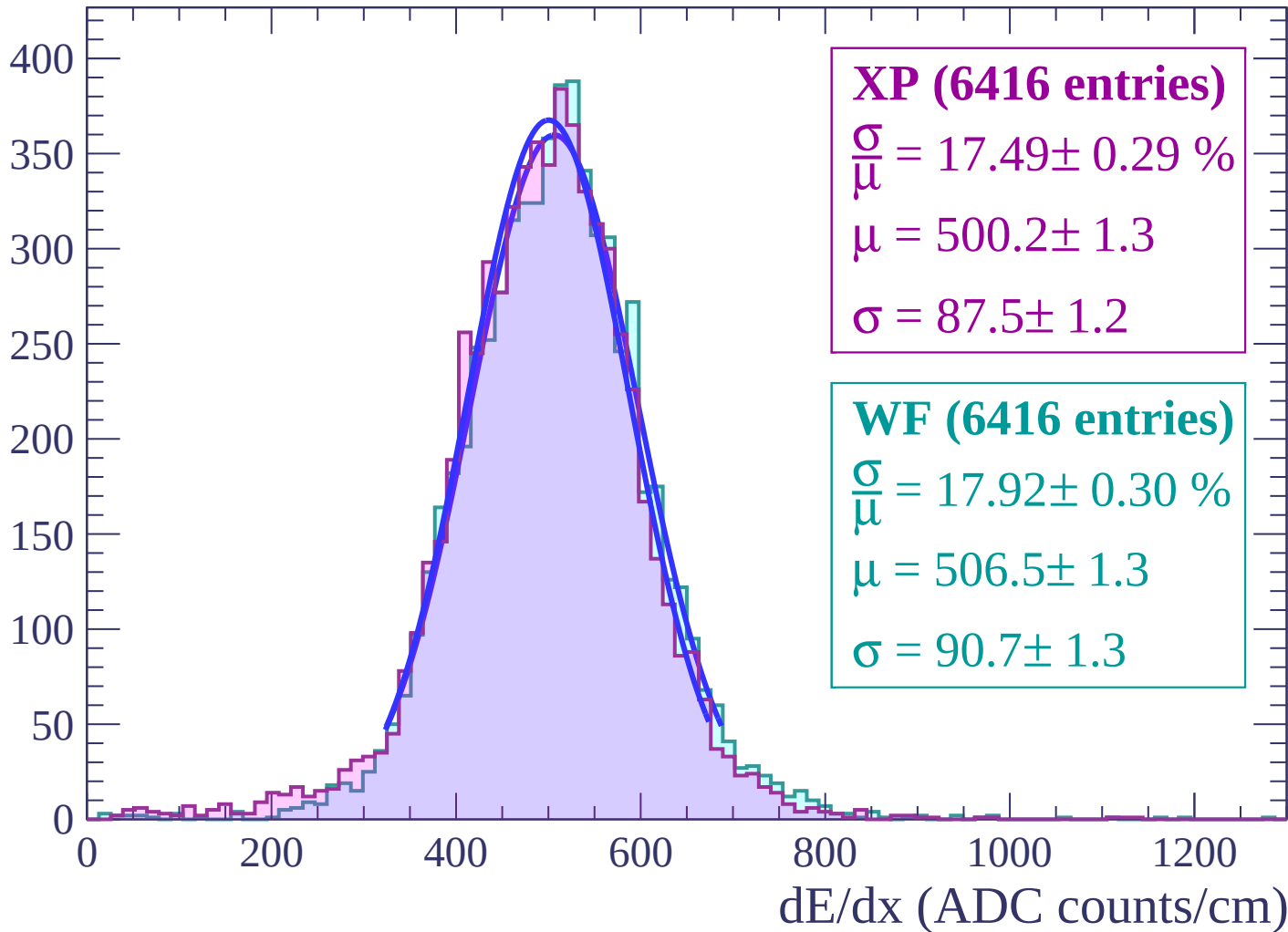
Energy loss | $42 < \theta < 44$

Count



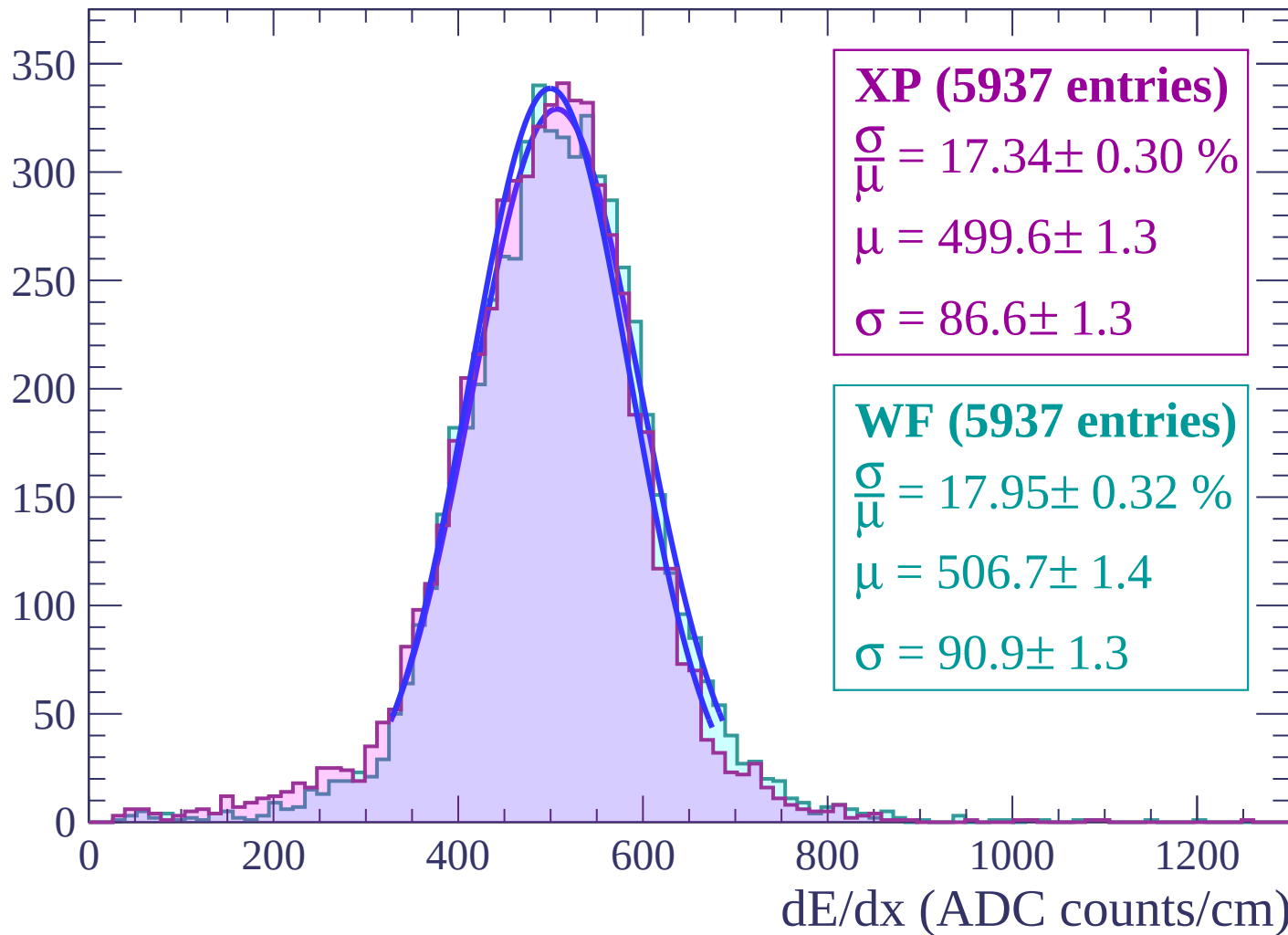
Energy loss | $44 < \theta < 46$

Count



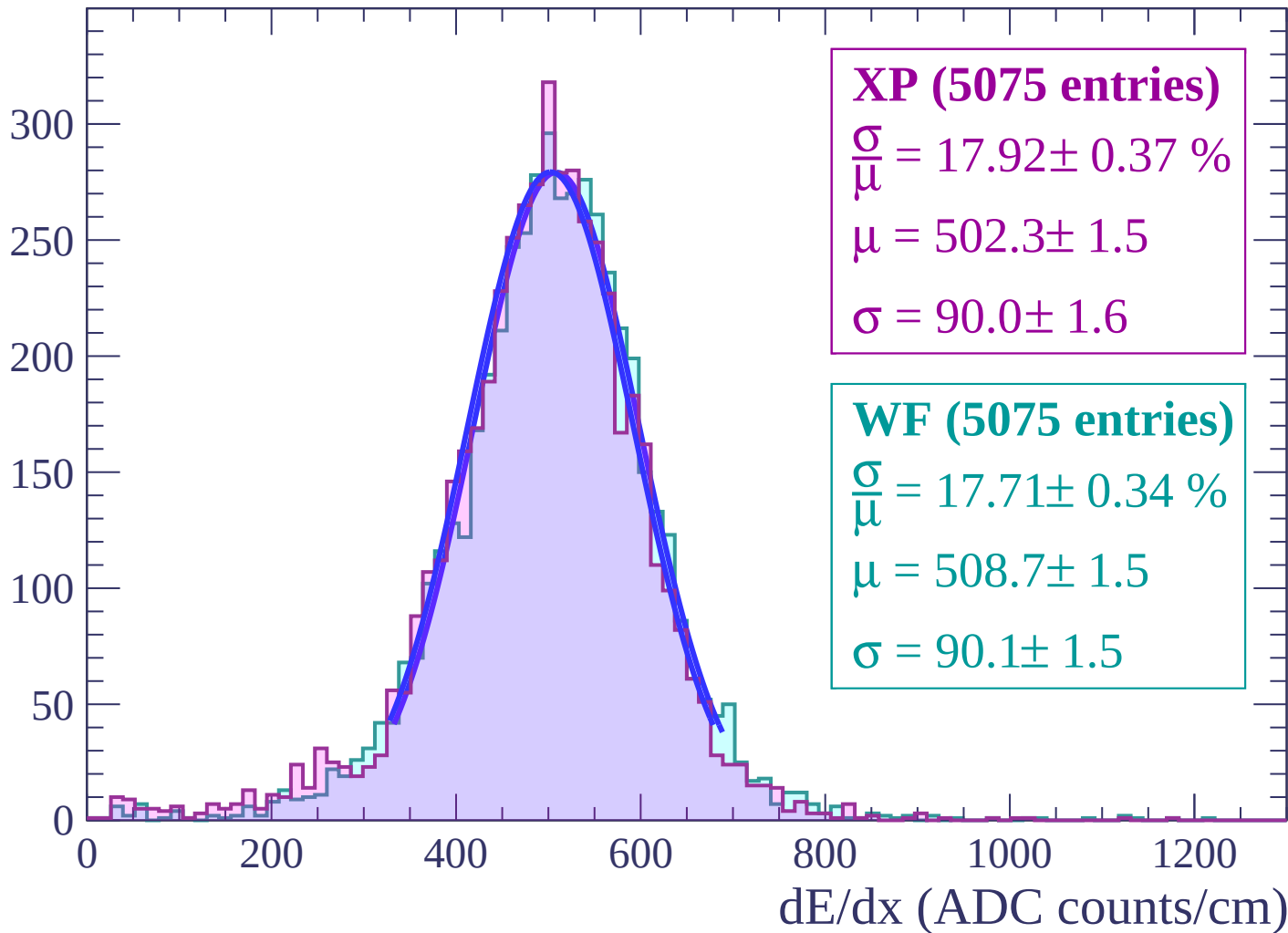
Energy loss | $46 < \theta < 48$

Count



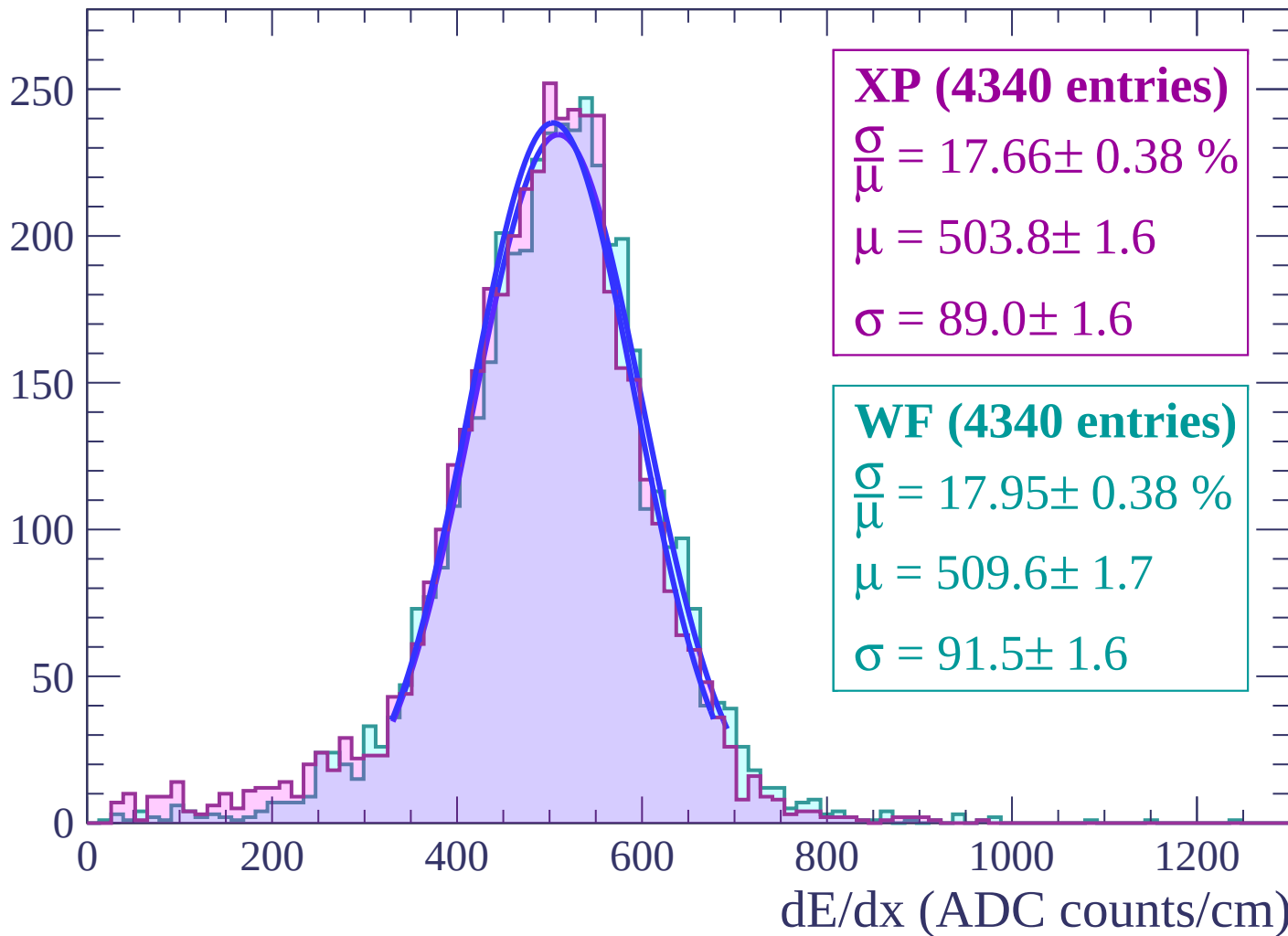
Energy loss | $48 < \theta < 50$

Count



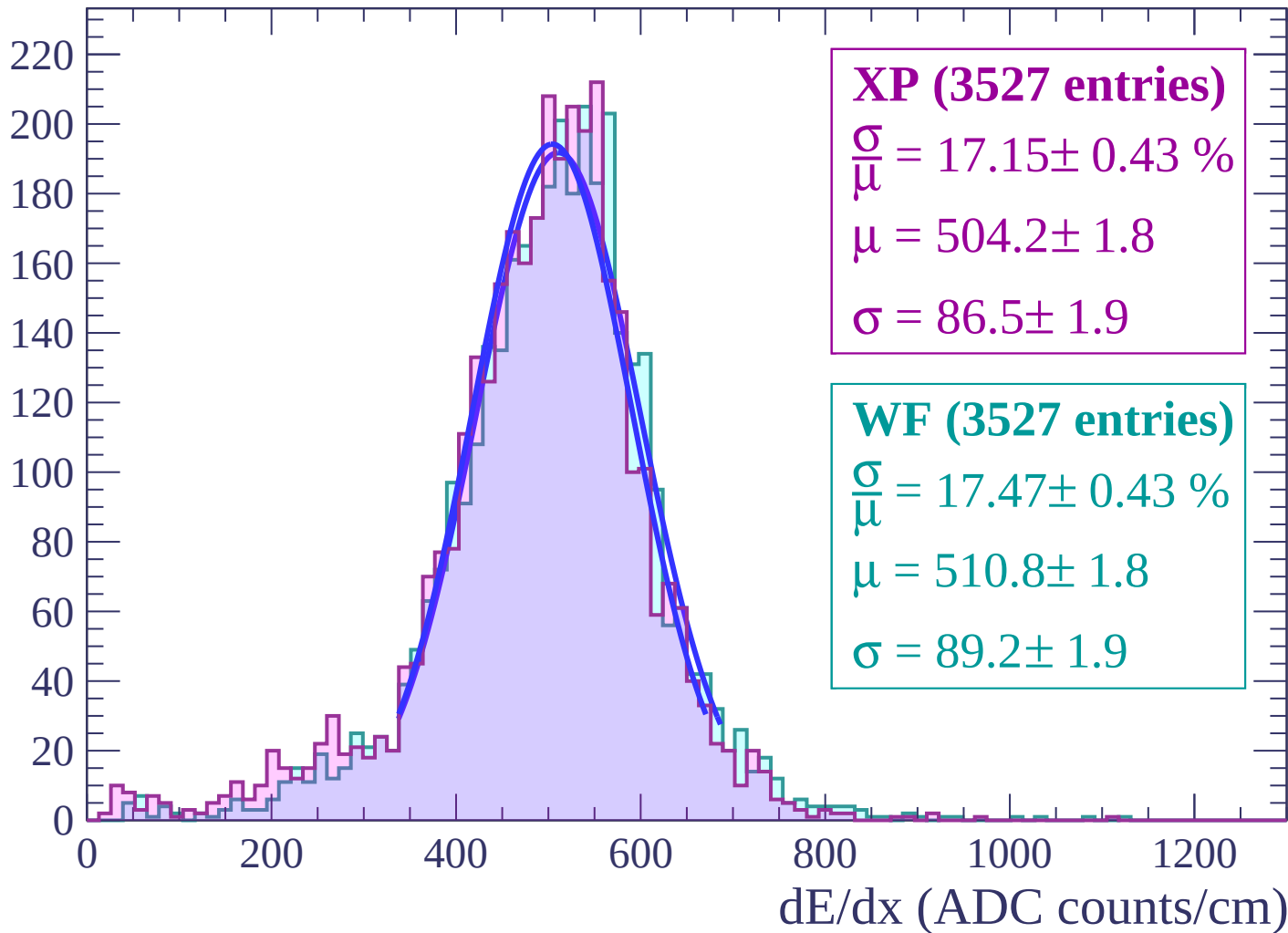
Energy loss | $50 < \theta < 52$

Count



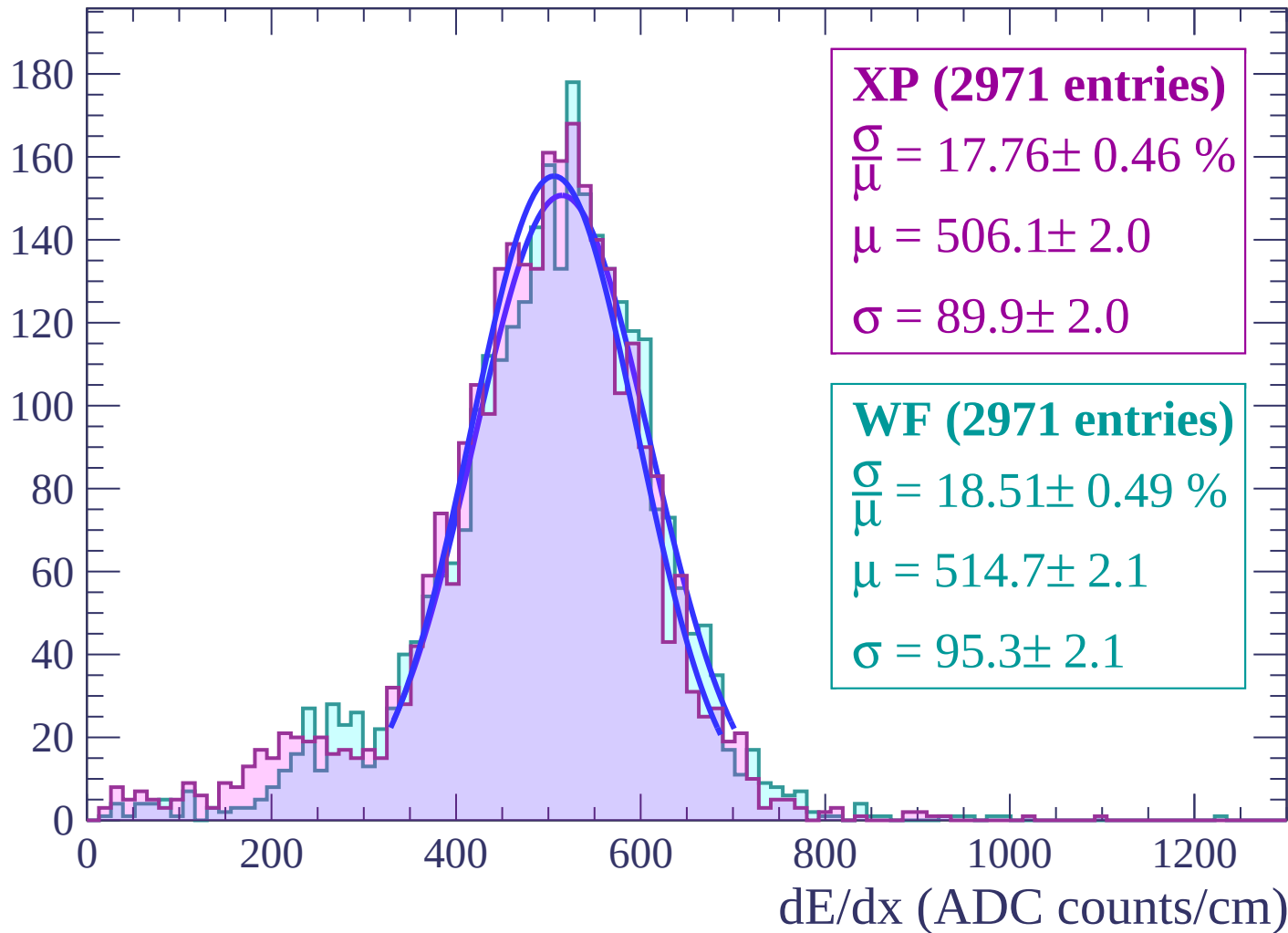
Energy loss | $52 < \theta < 54$

Count



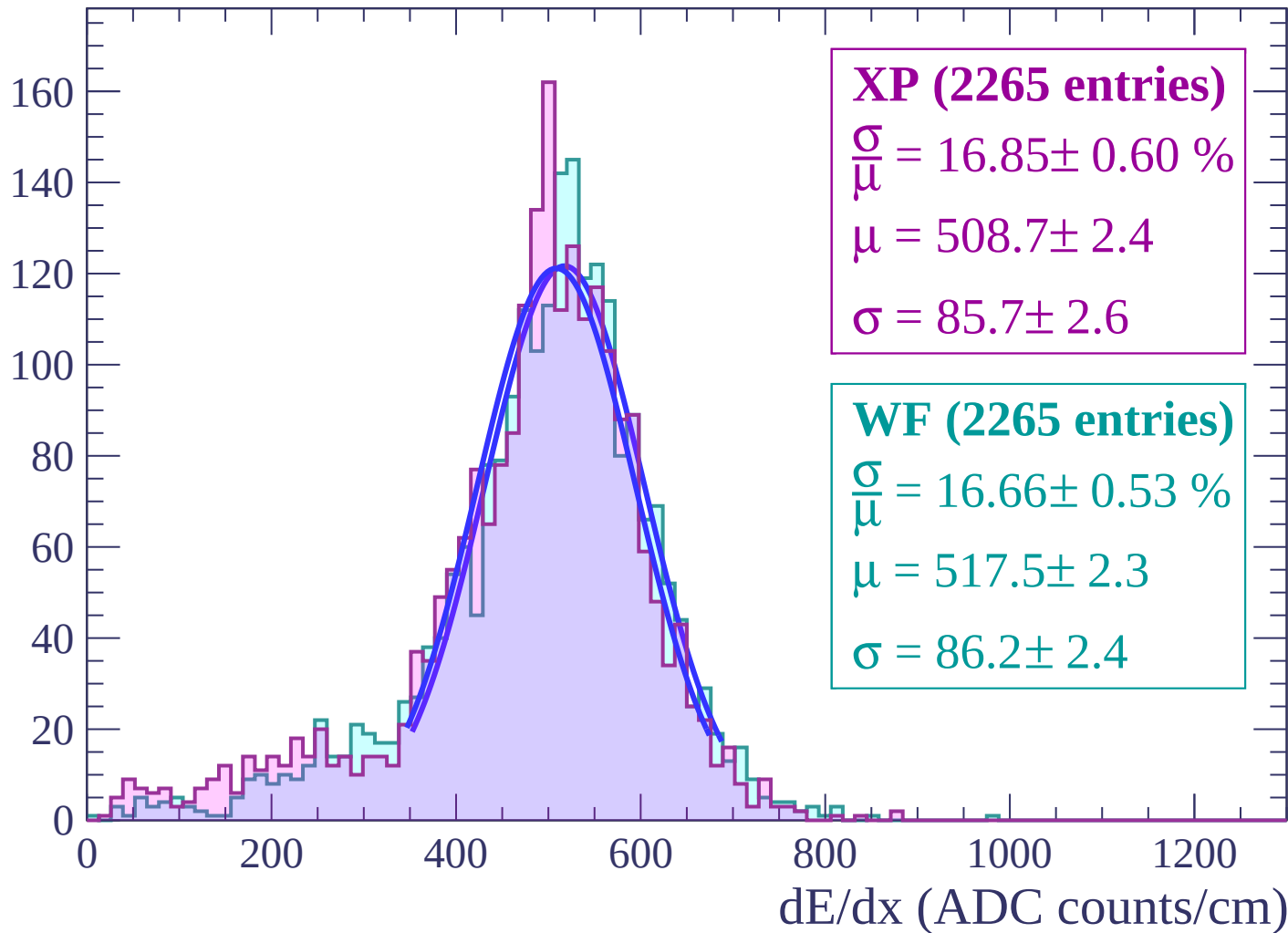
Energy loss | $54 < \theta < 56$

Count



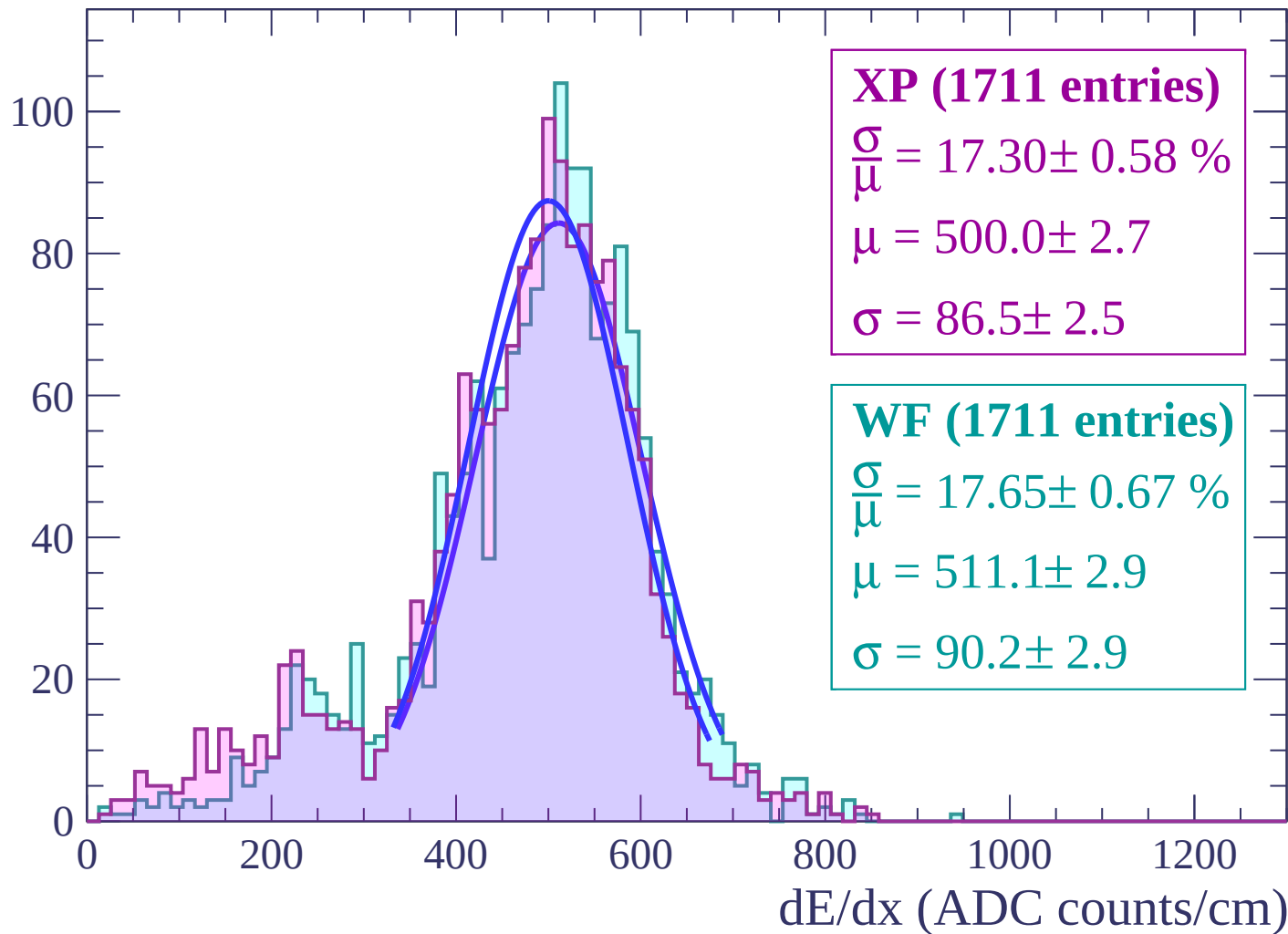
Energy loss | $56 < \theta < 58$

Count



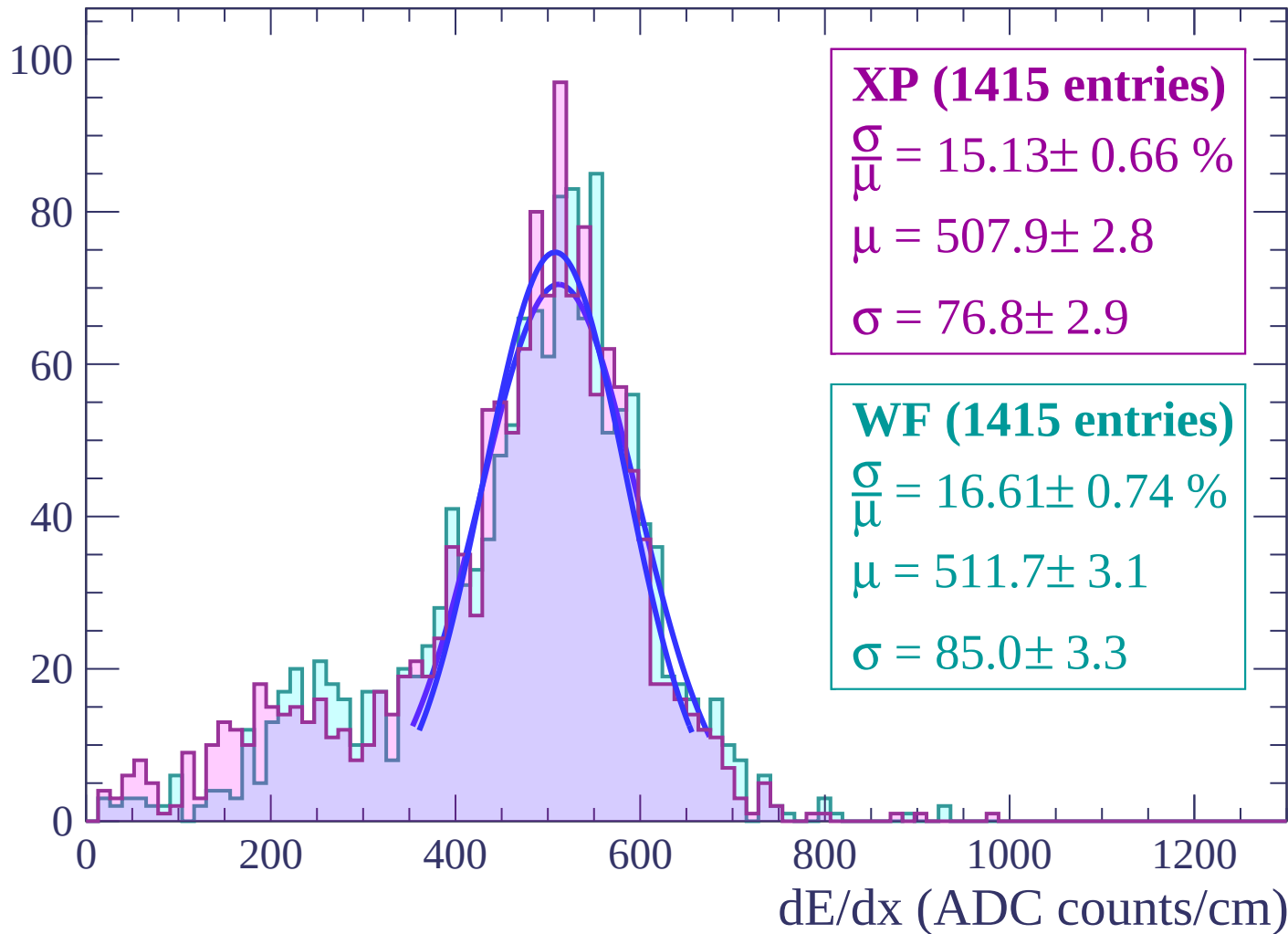
Energy loss | $58 < \theta < 60$

Count



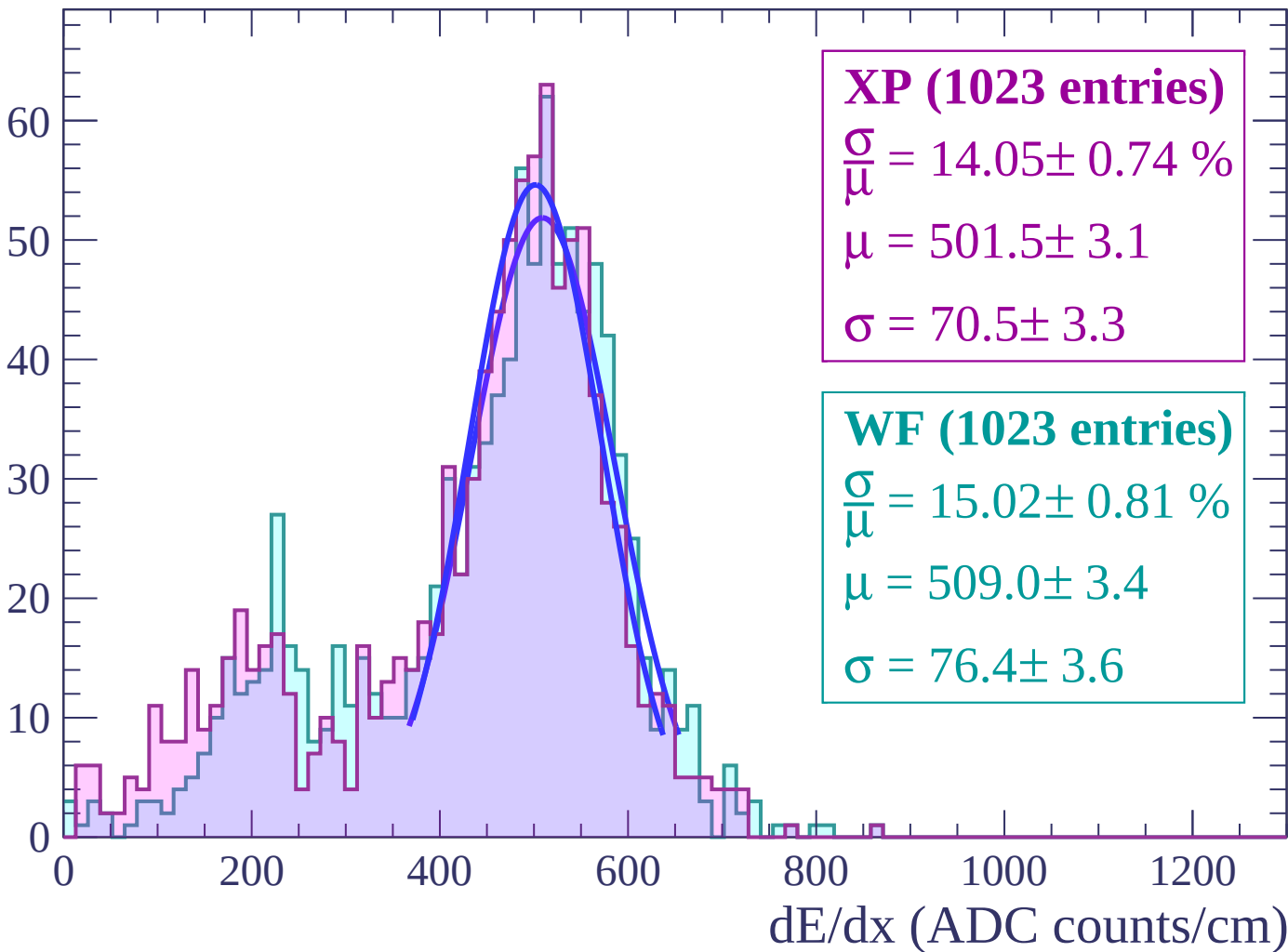
Energy loss | $60 < \theta < 62$

Count



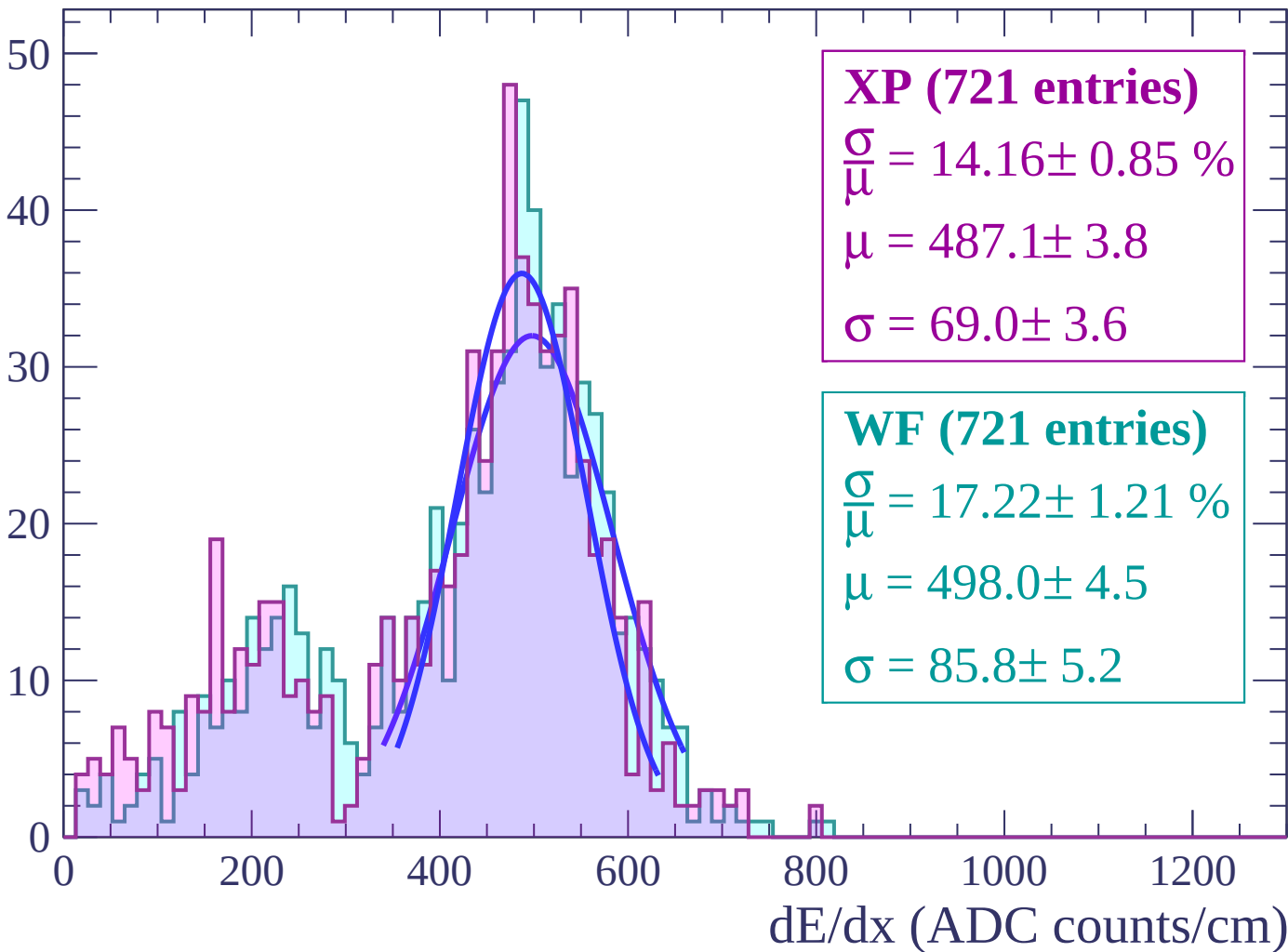
Energy loss | $62 < \theta < 64$

Count



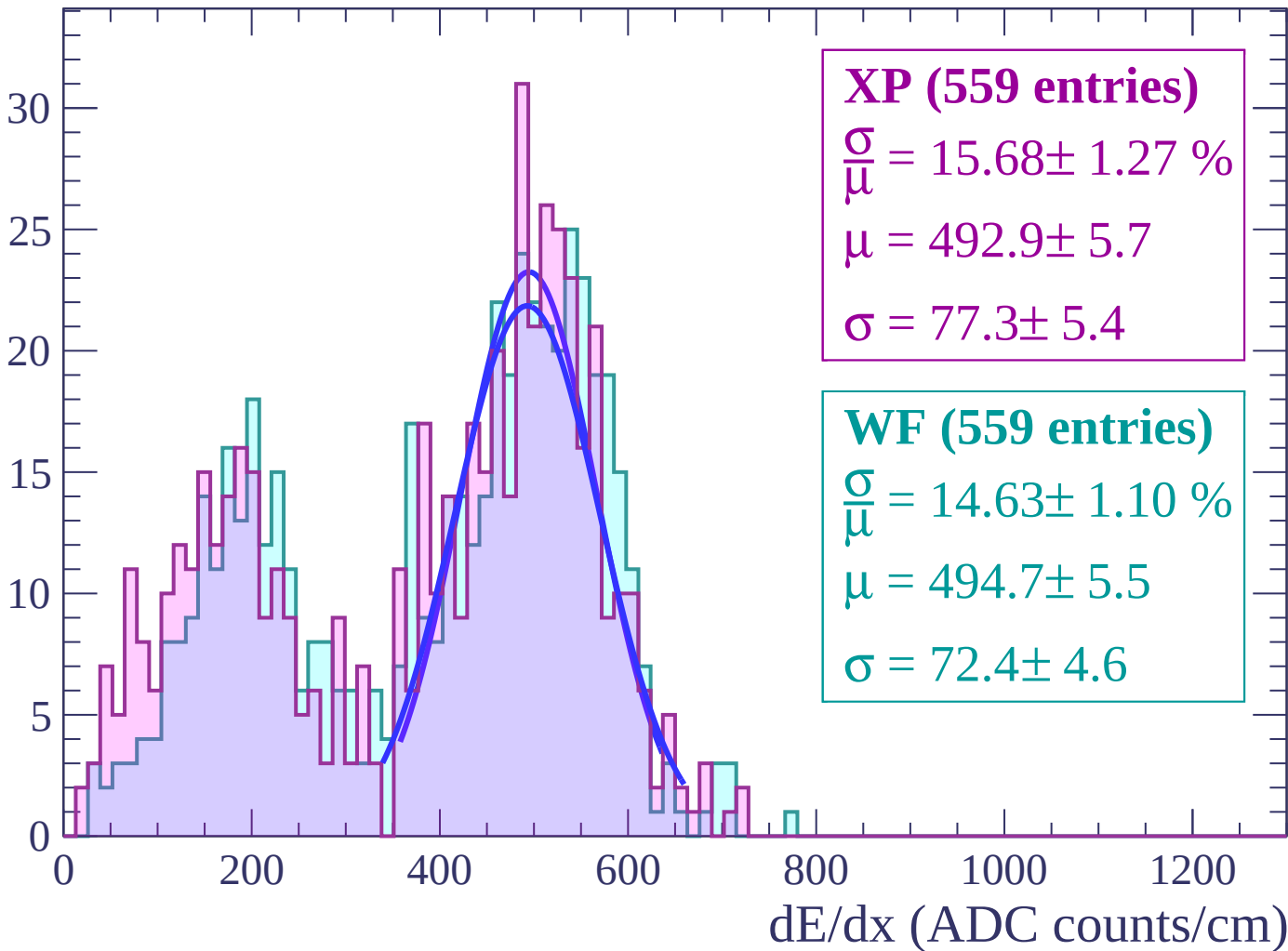
Energy loss | $64 < \theta < 66$

Count



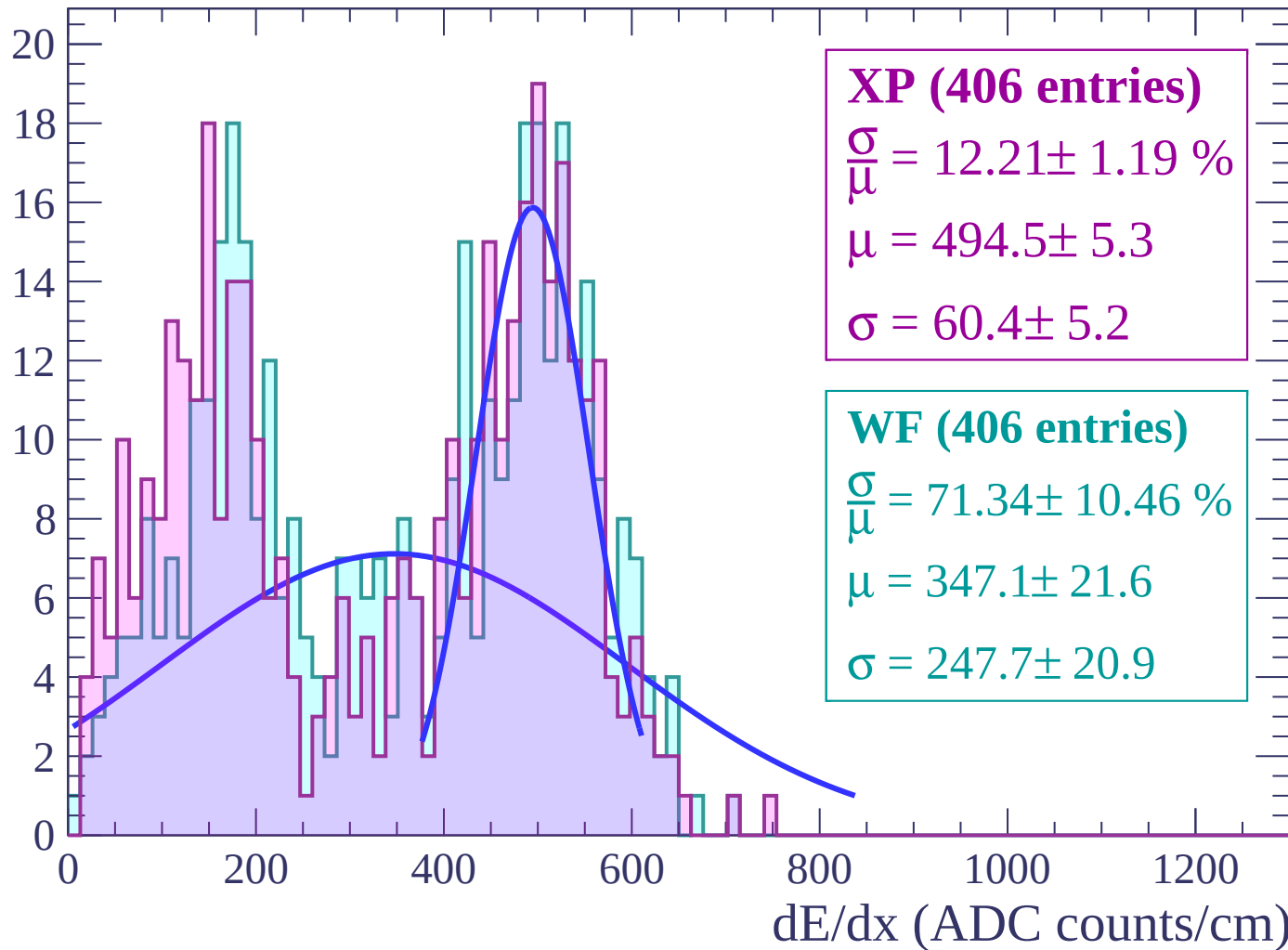
Energy loss | $66 < \theta < 68$

Count



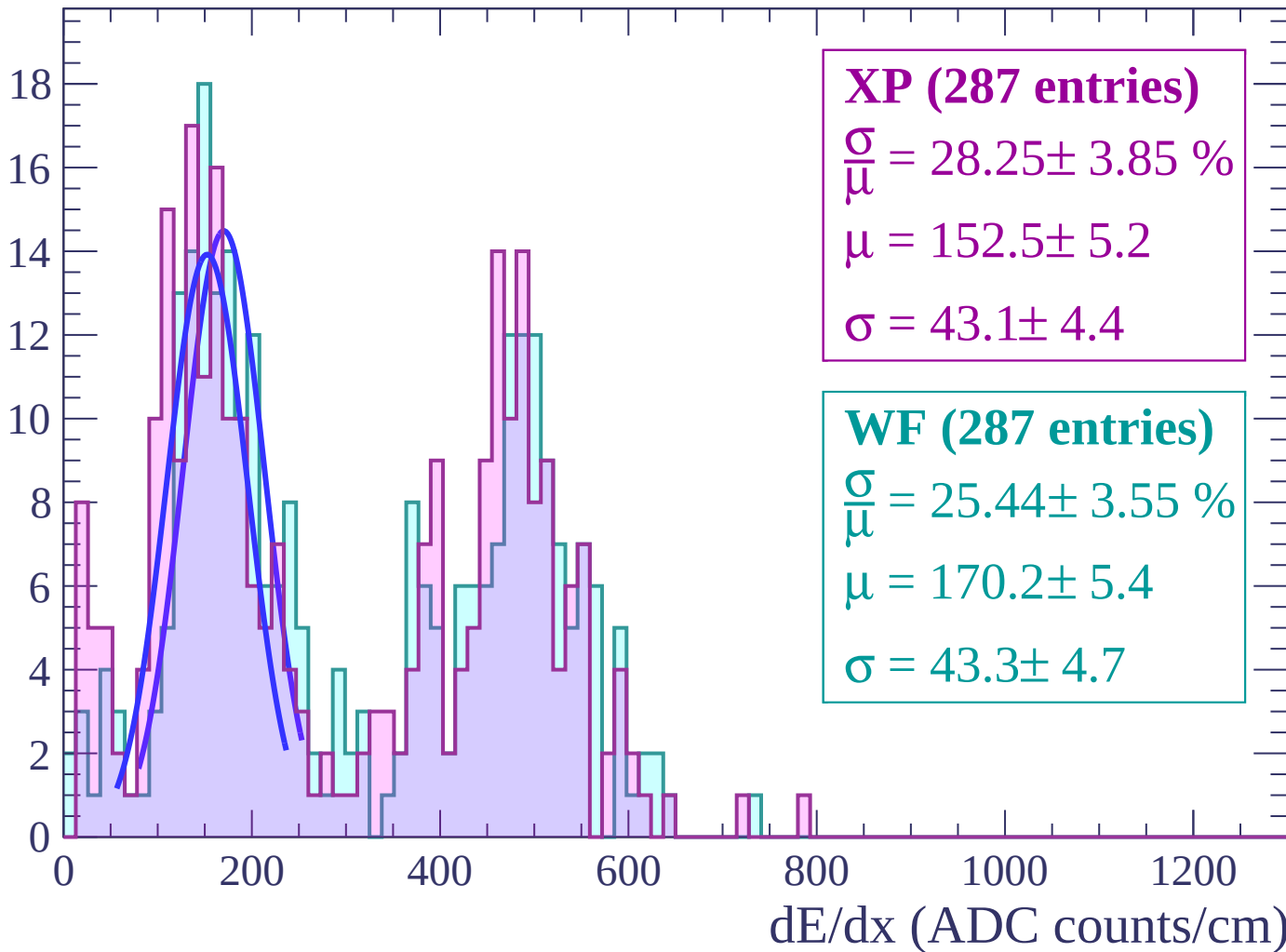
Energy loss | $68 < \theta < 70$

Count



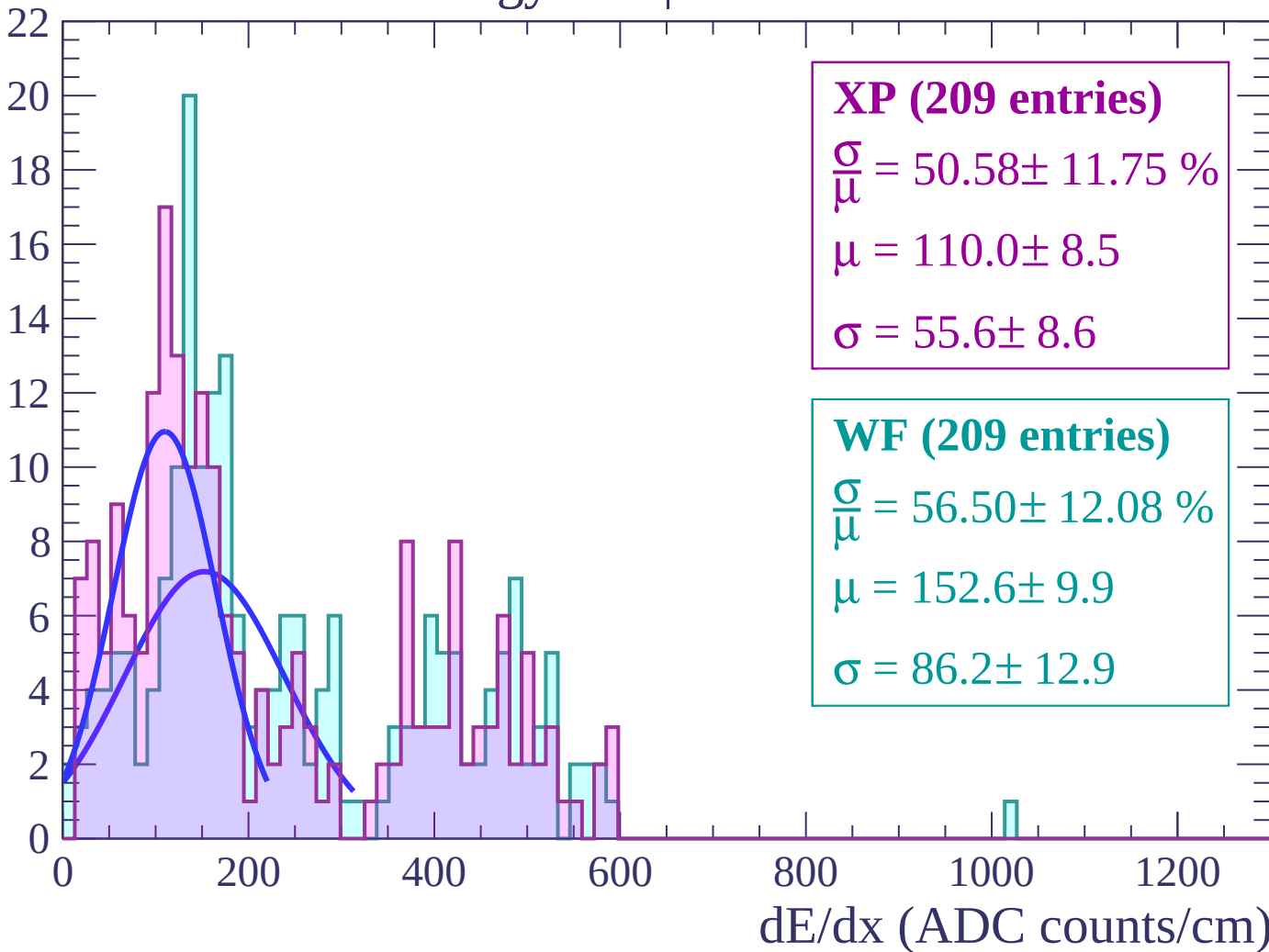
Energy loss | $70 < \theta < 72$

Count



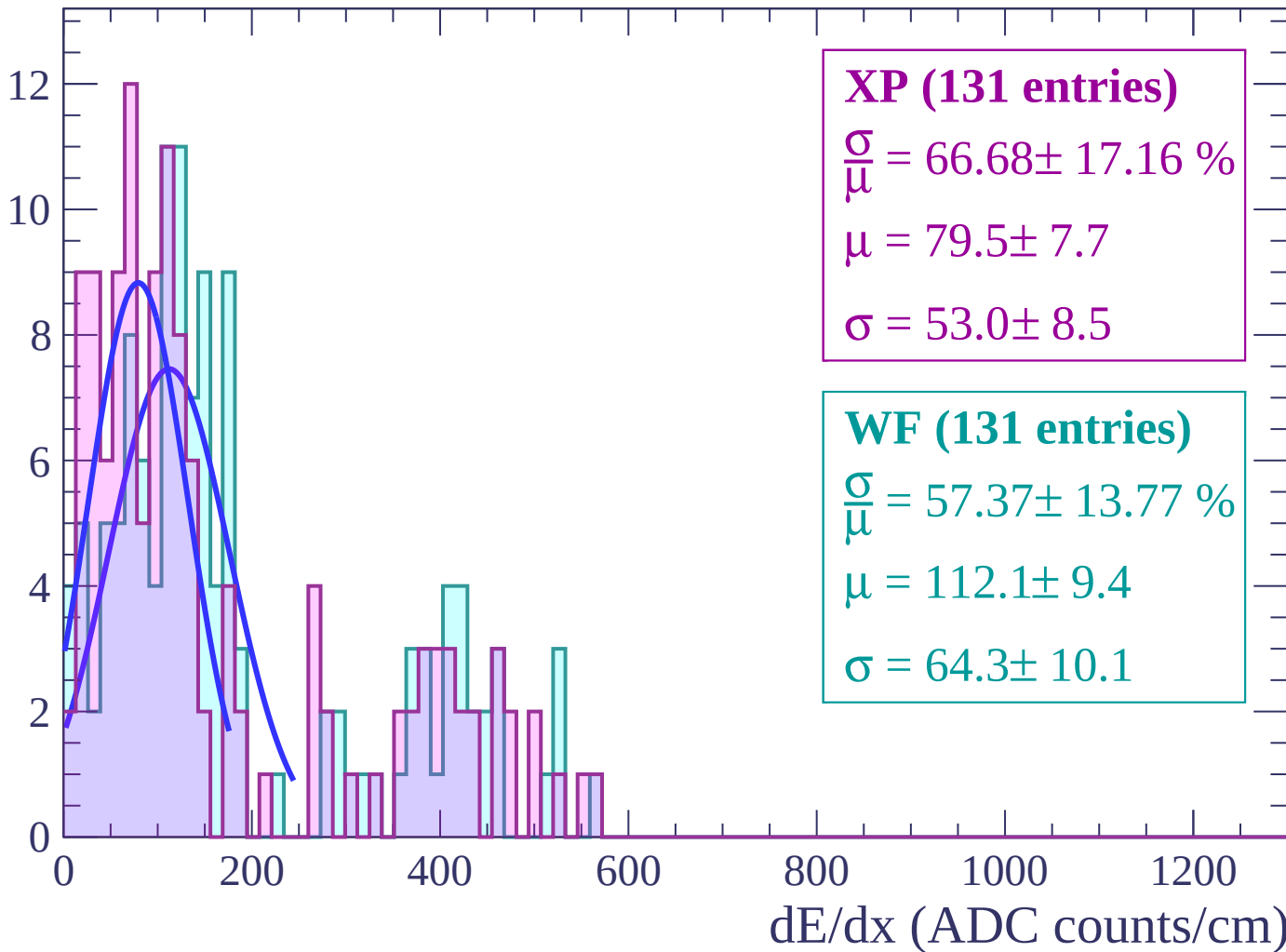
Energy loss | $72 < \theta < 74$

Count



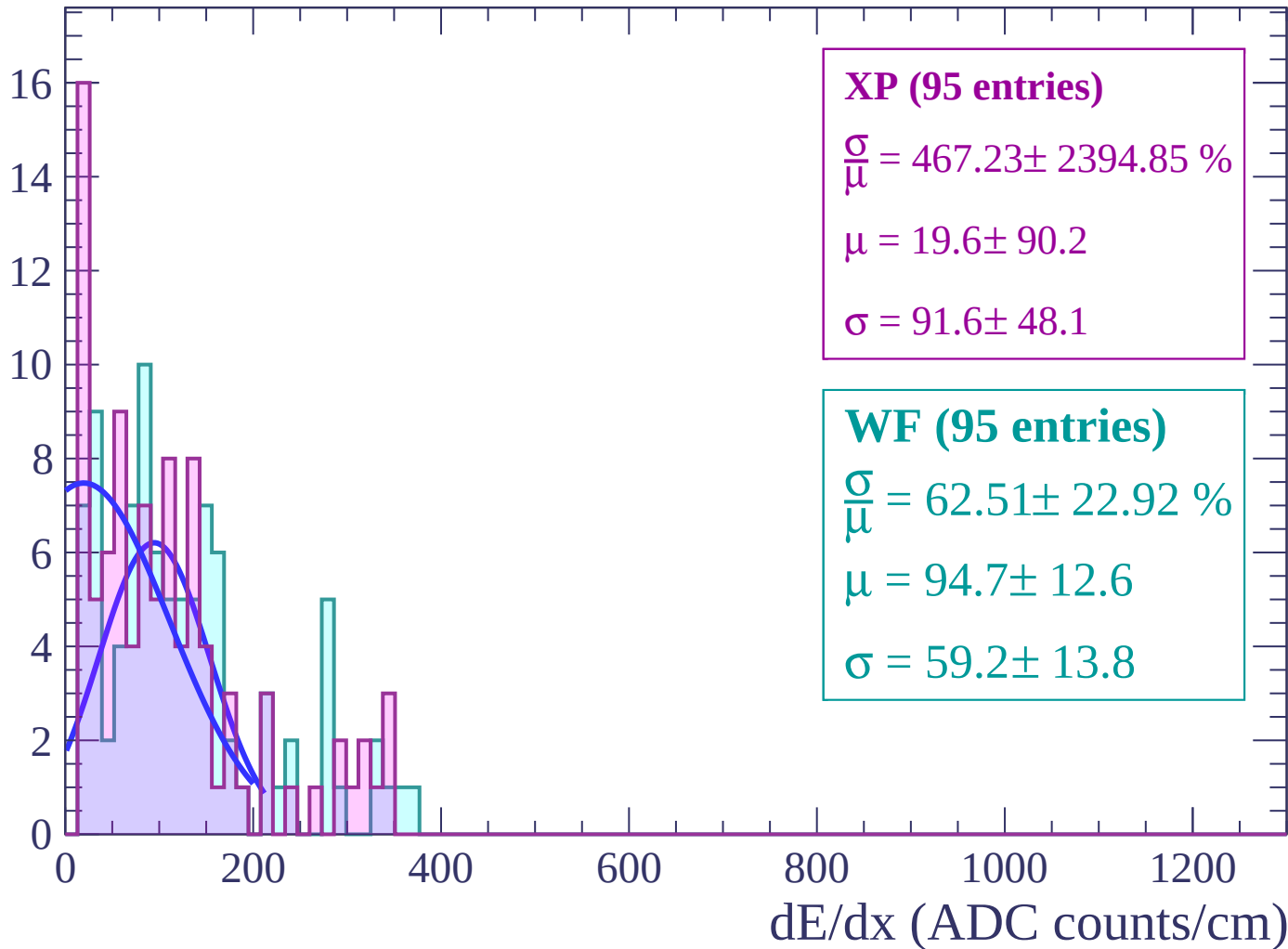
Energy loss | $74 < \theta < 76$

Count



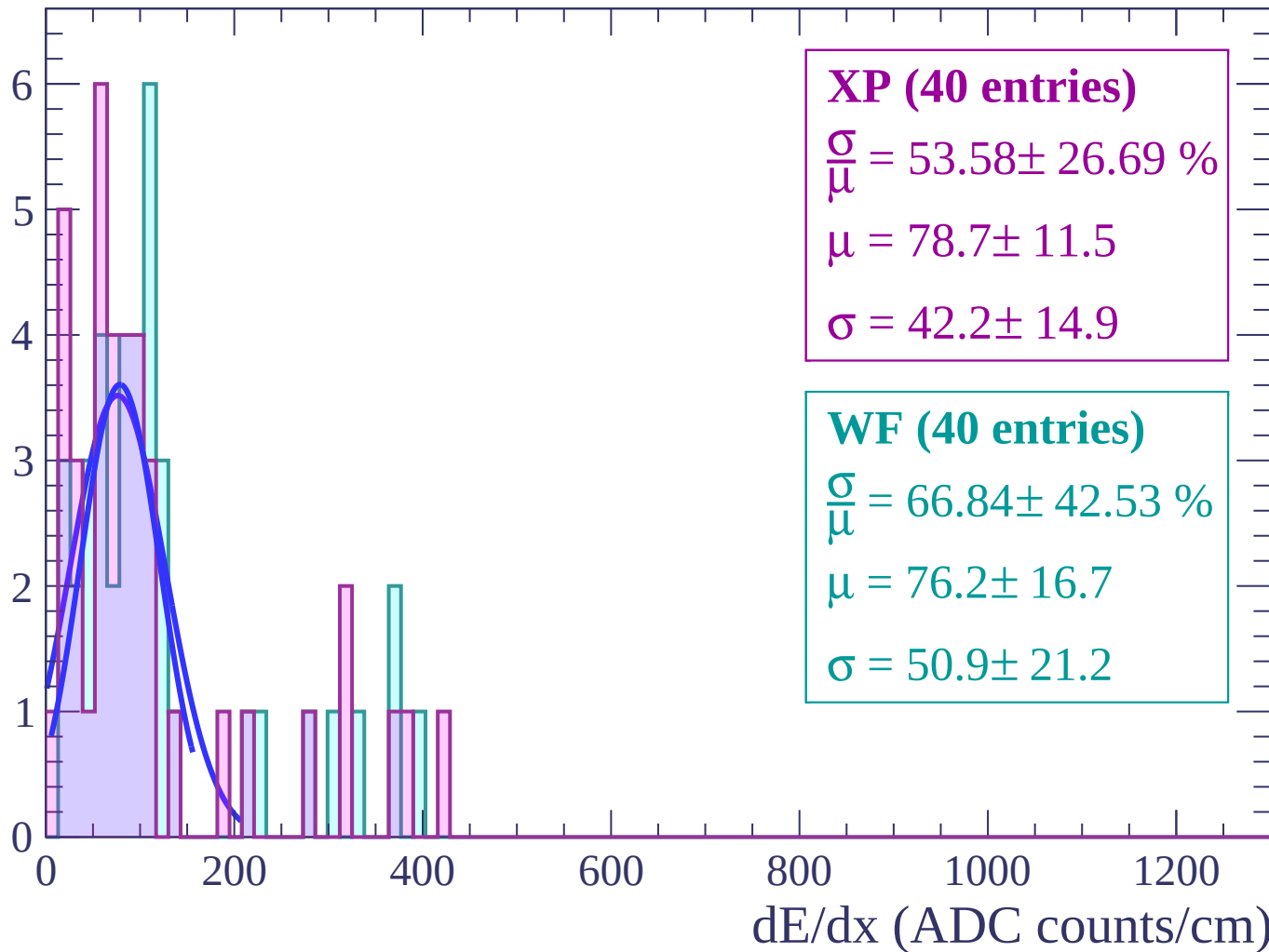
Energy loss | $76 < \theta < 78$

Count



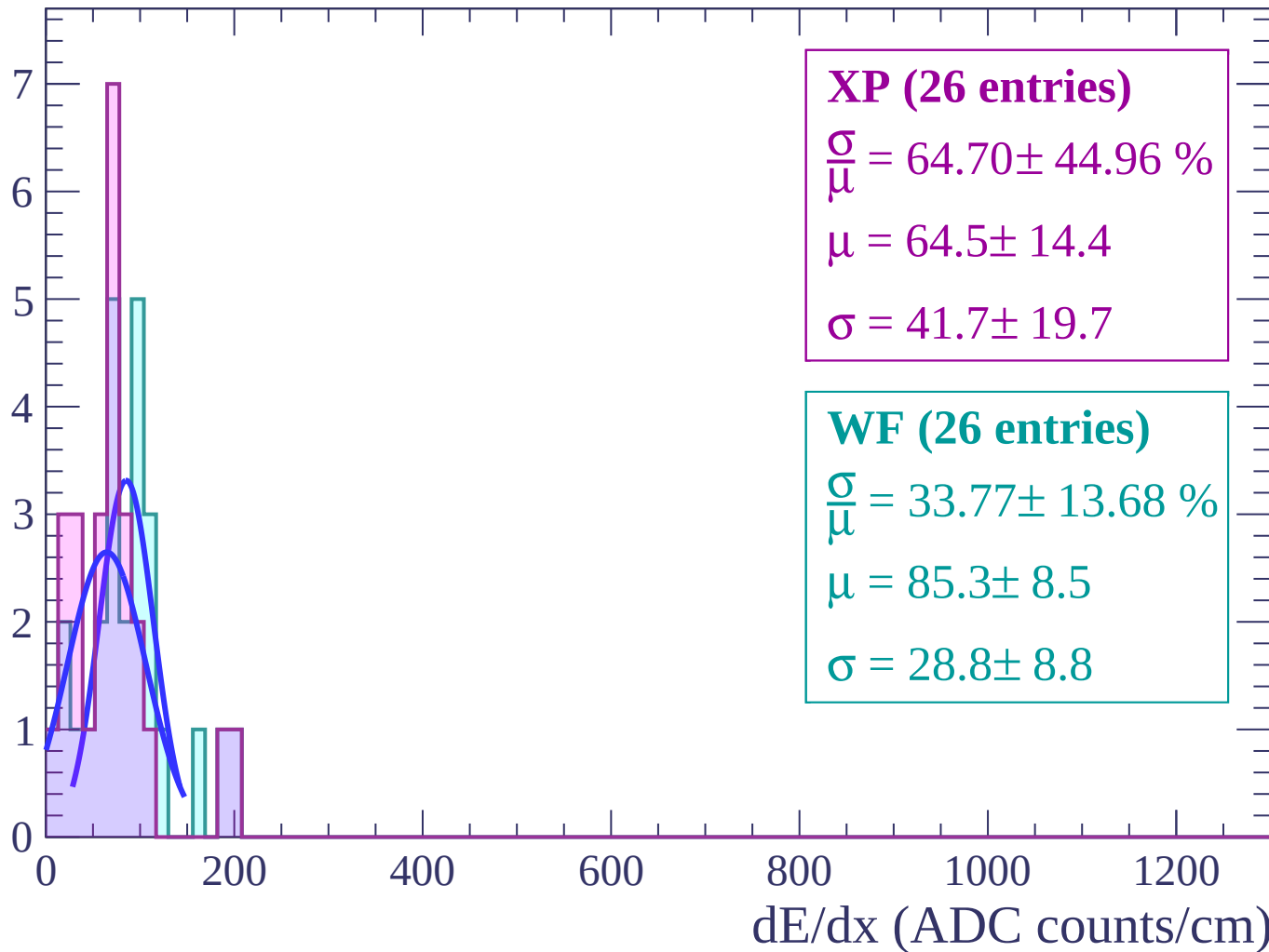
Energy loss | $78 < \theta < 80$

Count



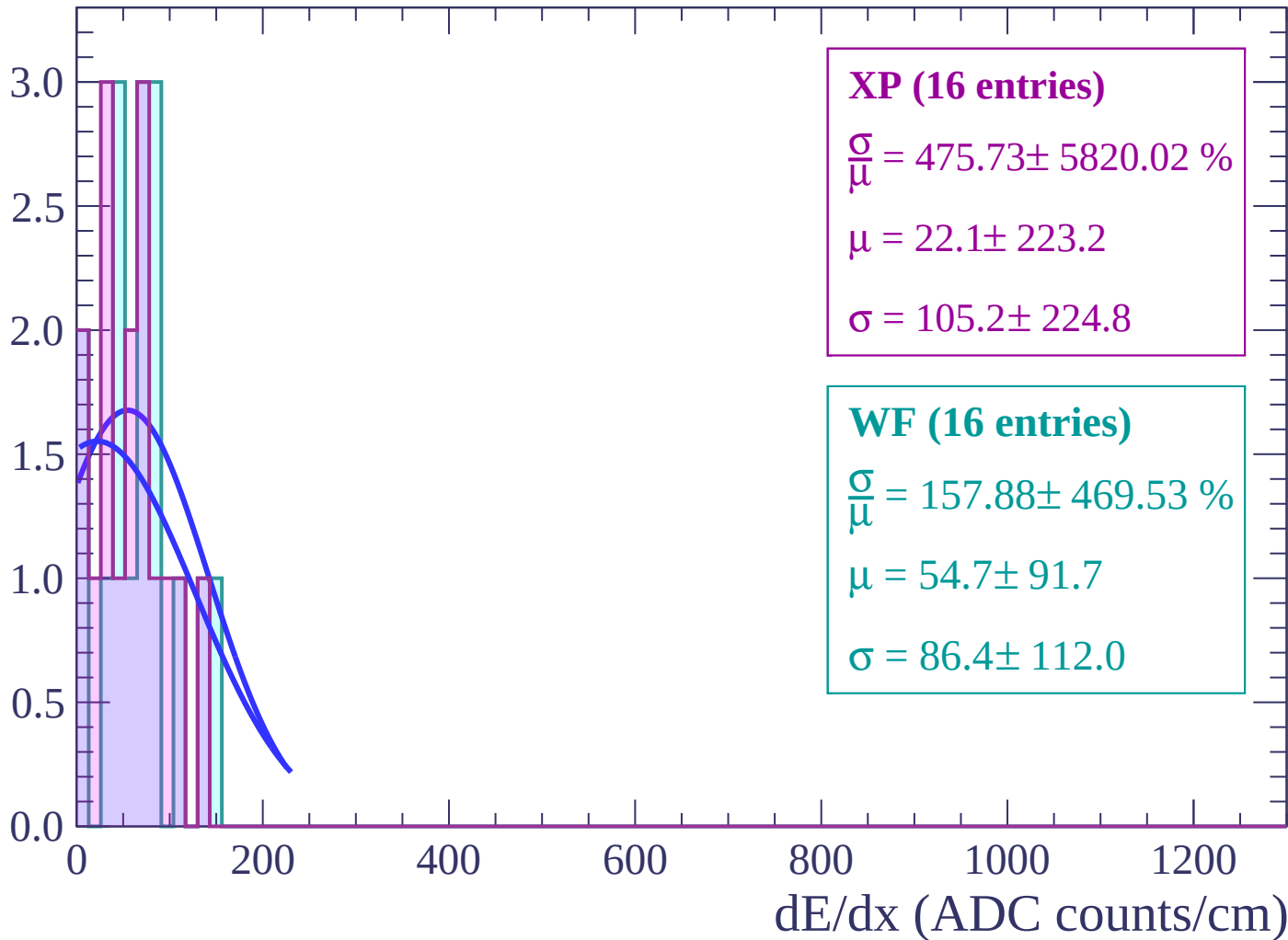
Energy loss | $80 < \theta < 82$

Count



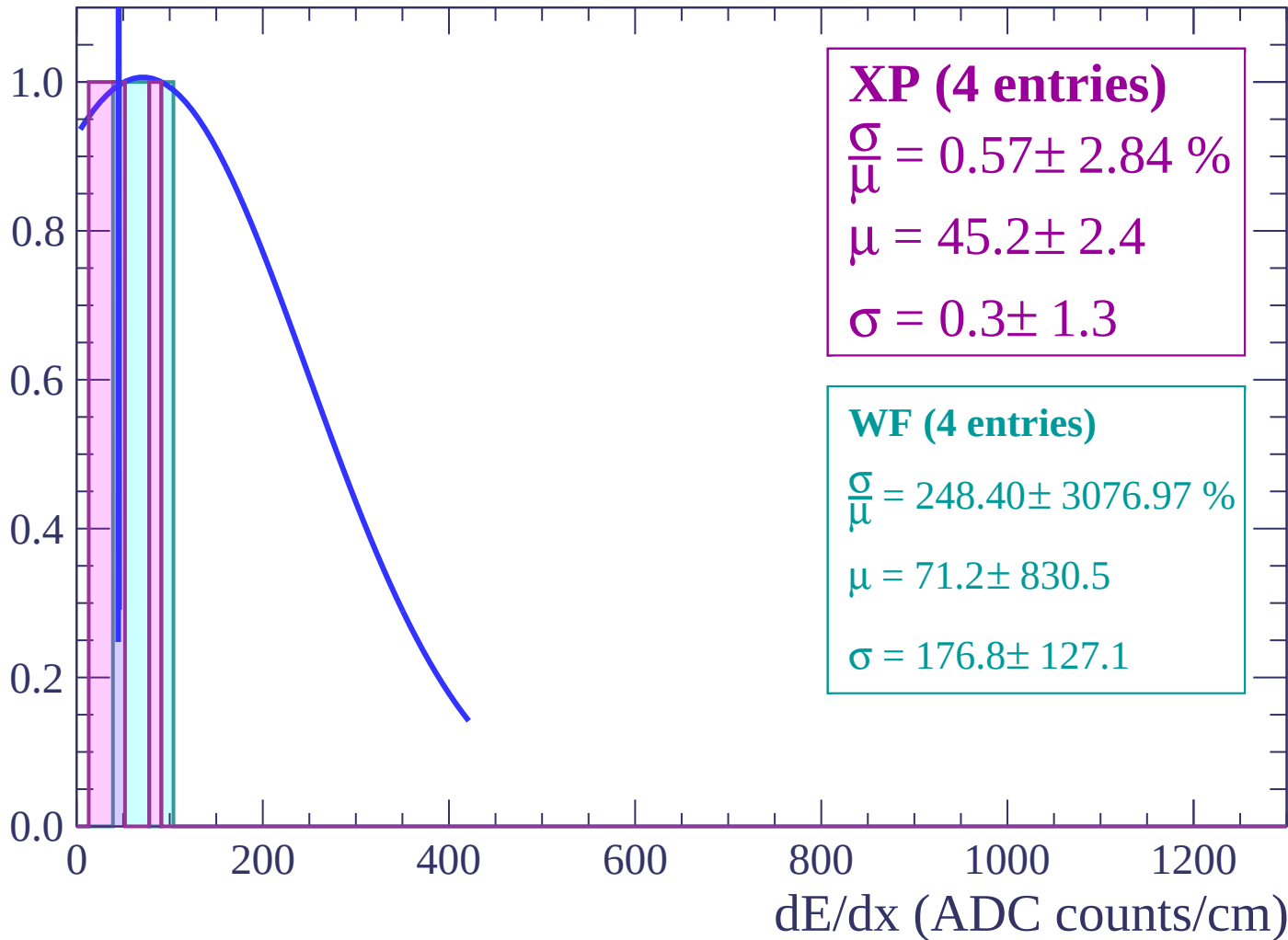
Energy loss | $82 < \theta < 84$

Count



Energy loss | $84 < \theta < 86$

Count



Energy loss | $86 < \theta < 88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $88 < \theta < 90$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $90 < \theta < 92$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

